



Voting & Civic Participation in the Alabama Gulf Coast

Client: Southern Poverty Law Center (SPLC)

IConsultancy Capstone
Section 105 Spring 2026

May 18, 2026

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About This Report

This report presents the findings of eleven student consulting teams working on behalf of the Southern Poverty Law Center (SPLC) through the INST490 iConsultancy capstone program at the University of Maryland's College of Information. All work was completed January through May during the Spring 2026 semester. The analyses and deliverables in this report were produced by undergraduate students in their final semester of study, working under faculty supervision and in direct consultation with SPLC staff.

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The Clients

The Southern Poverty Law Center (SPLC) is a nonprofit civil rights organization headquartered in Montgomery, Alabama. Its work spans litigation, policy analysis, data-driven research, and public education aimed at addressing systemic inequities affecting marginalized communities. The teams represented in this report worked with staff from three SPLC units: the Data and Research Team; the Democracy—Voting Rights Litigation Team and Policy Team; and the SPLC Alabama State Office.

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The Project

SPLC commissioned this work to support descriptive, historical analysis of civic participation and voting access across two regions of Alabama: the Gulf Coast counties (this report) and the Black Belt counties (Section 103, a companion report). The Gulf Coast project focused on seven counties: Baldwin, Clarke, Conecuh, Escambia, Mobile, Monroe, and Washington. All analyses examined conditions affecting Black and Latinx communities in particular, using publicly available or

client-approved data sources. All findings are descriptive and based on publicly available data; they do not constitute legal analysis or advocacy.

How the Teams Worked

The project was structured so that each team addressed a distinct dimension of civic participation access, and the teams' work is designed to be read in relation to one another. GC-DIG built and documented the core data infrastructure that most other teams drew from. GC-VOTE and GC-PAD conducted the primary quantitative analyses of turnout trends and physical access to polling locations. GC-TRAC, GC-AACS, and GC-BAND examined compounding structural barriers: transportation access, ADA compliance at polling locations, and broadband availability. GC-CPRI synthesized indicators from those teams into a composite risk index ranking counties by civic participation risk. GC-VIZ translated data across all teams into an interactive client-facing dashboard. GC-CCP developed community personas grounding the quantitative findings in human context. GC-PORT translated findings into materials suitable for policy and organizational decision-making. GC-CPAD produced public-facing civic process materials designed for community members navigating participation for the first time.

How to Read This Report

Each team's section is self-contained and can be read independently. Every section opens with a one-page brief summarizing what the team studied, its key findings, the deliverables it produced, significant challenges encountered, and connections to other teams. The full report narrative follows, including methods, detailed findings, limitations, and recommendations. Cross-references to other teams are included where teams' findings directly inform one another.

Readers approaching this report for the first time are encouraged to begin with GC-DIG (data infrastructure) and GC-VOTE (turnout trends) before moving to the access-focused sections, as those two sections establish the empirical foundation the rest of the project builds on. A companion report covering the Black Belt counties is available separately through the Section 103 iConsultancy team at the University of Maryland's College of Information.

Regional Voting Data Integration & Governance

Gulf Coast Counties Voter Participation and Access Analysis

GC-DIG

Colin Sullivan (Data Analysis and Documentation), Carter Kinch (Project Manager), Lin Han (Data visualization), John Valencia (Data Analysis & Cleaning), Cidel Tetteh (Recommendations and conclusion)

Southern Poverty Law Center (SPLC)

INST490 - Integrated Capstone for Information Science (Section 0105)

May 18, 2026

Team Brief

Team 1: GC-DIG

Key Findings

- Finding 1: Black voter turnout fell an average of 6.7 percent across all seven Gulf Coast counties from 2020 to 2024, while white voter turnout rose an average of 0.7 percent over the same period. This divergence represents a meaningful and consistent pattern across the region
 - Analysis of ballots cast by race (General_Race_Combined.xlsx) against active registered voters by race (Registration_Combined.xlsx) shows a consistent pattern of declining black voter participation between the 2020 and 2024 Alabama general elections. A decline that was present in all seven counties which ranged from 4.6 percentage points (Monroe) to 10.1 percentage points (Escambia). Oppositely, white voter turnout remained relatively stable, with slight gains in Baldwin, Clarke, Conecuh, and Monroe counties offset by marginal declines in Escambia, Mobile, and Washington counties.
 - The magnitude of divergence is notable in the overall turnout context. Total ballots cast across the seven counties were nearly identical in 2020 (350,388) and 2024 (349,763), suggesting that turnout was stable while the racial composition of participants shifted. This pattern may reflect changes in voter outreach, polling place access, registration list management, or other structural factors.
- Finding 2: Hispanic registrants have the highest rate of inactive voters of any racial group in the region at 19.7 percent in 2024, followed by Asian (14.1%) and American Indian registrants (13.2%). All minority group inactive rates rose between 2020 and 2022 before showing partial recovery in 2024
 - Inactive status is placed upon a voter by their county board of registrars, typically because they have not voted in recent cycles or have not responded to correspondence from the board. Inactive voters are still eligible to vote, but may encounter additional obstacles at the polling place. A high percentage of inactive voters may indicate that a group's records are more frequently flagged, due to

- frequent address changes, housing instability, or other administrative practices, rather than low engagement exclusively.
- Hispanic registrants show the highest inactive rate (19.7%), substantially higher than the rate for white registrants (12.4%). This is significant given the low overall registration numbers for Hispanic residents in these counties. The elevated inactive rates for Hispanic, Asian, and American Indian registrants in the region warrant further investigation into whether administrative processes are applied uniformly across all demographics.
 - Finding 3: Polling place density does not predict voter turnout across the seven Gulf Coast counties. Voters per polling place ranged from 610 in Monroe County to 5,310 in Baldwin County, an 8.7-fold difference in nominal access. The correlation between voters per polling place and 2024 turnout was weak and positive ($r = 0.361$), the opposite direction of what a simple access argument would predict.
 - Analysis of polling place locations (AI_Polls_Flood_SLED shapefile) against active registered voters in 2024 (Registration_Combined.xlsx) and total ballots cast (General_Race_Combined.xlsx) shows that counties with the fewest voters per polling place did not record the highest turnout.
 - Monroe County, with 610 voters per polling place, recorded a turnout of 68.1 percent, while Baldwin County, with 5,310 voters per polling place, recorded the region's highest turnout at 72.0 percent. Conecuh and Escambia tied for the lowest turnout at 61.7 percent despite having very different access ratios of 688 and 1,010 voters per polling place respectively.
 - This pattern suggests that polling place counts alone are an insufficient measure of voter access. Factors such as geographic distance to the polling place, transportation availability, polling hours, precinct assignment, and demographic composition likely have a stronger influence on whether registered voters cast ballots than the raw number of polling places available in a county. Future analysis should pair this data with travel-time or distance-based access metrics rather than rely on county-level counts.
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Deliverables Produced

OReadMe File:  OREADME

Folder overview with project description, file inventory, access instructions, and team contact information.

Data Dictionary:  GCDIG-GulfCoastVoting-DataDictionary-v1.docx

Documentation of every variable across all GC-DIG datasets including source, type, definition, limitations, and naming conventions.

Final Report:  GCDIG-GulfCoastVoting_Final_Report.docx

Comprehensive documentation of methodology, findings, and recommendations produced for SPLC.

Access Analysis:  GCDIG-GulfCoastVoting-AccessAnalysis-v1.xlsx

County-level analysis combining polling place counts, registered voters, ballots cast, voters per polling place ratio, and 2024 turnout percentage. Supports Finding 3.

Turnout Change Visual:  GCDIG-GulfCoastVoting-Fig2-TurnoutChange.png

Change in voter turnout rates by race between 2020 and 2024.

Race Analysis:  GCDIG-GulfCoastVoting-ElectionRaceData-v1.xlsx

Cleaned general election ballots cast by race for all seven counties for 2020 and 2024.

Race Analysis Visual:  GCDIG-GulfCoastVoting-Fig1-TurnoutByRace.png

Average voter turnout rates by race for 2020 and 2024.

Race Analysis Visual 2:  GCDIG-GulfCoastVoting-Fig3-InactiveByRace.png

Average inactive voter registration rates by race in 2024.

Registration Analysis:  GCDIG-GulfCoastVoting-RegistrationData-v1.xlsx

Cleaned and standardized voter registration data (active and inactive) by race for all seven Gulf Coast counties across 2020, 2022, and 2024.

Access vs Turnout Visual:  GCDIG-GulfCoast-Voting-Fig4-AccessVsTurnout.png

Scatter plot of voters per polling place against 2024 turnout for the seven counties.

Shapefiles Folder:  GCDIG-GulfCoastVoting-PollingPlaces_ShapeFiles

Geographic point data for all polling places in the seven Gulf Coast counties. Includes precinct name, address, county, and coordinates.

Challenges

We encountered several data quality challenges throughout the project.

Gathering data from different sources, we found inconsistencies in formatting across datasets. The shapefiles used mixed-case county names while the election and registration datasets used all-uppercase. Python scripts using pandas were written to standardize fields across all datasets before merging.

The Alabama Secretary of State reports votes cast in categories that do not map cleanly to Census classifications, including “fed_registered,” “korean,” and “not_identified.” These categories were retained in the integrated dataset rather than removed, and are noted as a limitation of cross-source comparison

Our team avoided language suggesting voter apathy. Observed turnout declines and inactivity disparities may reflect structural or administrative factors like polling place access, transportation barriers or registration maintenance. Our team focused on identifying measurable patterns rather than assigning blame. In addition, our team worked under a limited timeline while integrating multiple election cycles and geographic datasets. This was hard because of lack of communication at first but it got better as the semester progressed.

Connections to Other Teams

Team 8 (GC-VIZ) used our voter file data directly in their interactive dashboard and choropleth maps. Their final report explicitly states that when applied to Team 1-DIG voter file data, the GC-VIZ dashboard showed Mobile County’s total ballots cast dropped by approximately 5,000 from 2024 and that the share of ballots cast by white voters increased by about 1.5 percentage points. Their choropleth maps demonstrate the demographic distribution of white voters across the Gulf Coast counties in 2020 and 2024, with Mobile County’s white share rising from 63.67 percent to 65.15 percent and the Black share falling from 33.48 percent to 31.85 percent. Their Voter Turnout by Race Over Time chart, which compares ballots cast by demographic group across the 2020 and 2024 general elections, is built on the General_Race_Combined dataset produced by GC-DIG.

GC-DIG Data can also be used in a multitude of other analyses and teams such as:

Team 2 (GC-VOTE) focuses on voter turnout and participation trends across the Gulf Coast. The Registration_Combined and General_Race_Combined datasets provide the standardized year-over-year data for their trend analysis.

Team 3 (GC-PAD) analyzes polling place access and distance using network-based travel time. GC-DIG's polling places shapefile and the registered voter counts per county are direct inputs to this analysis. GC-PAD's findings on travel time and GC-DIG's Finding 3 on voters per polling place address the same access question from different angles.

Team 4 (GC-TRAC) examines transportation barriers to civic participation. GC-DIG's polling place locations support GC-TRAC's analysis of which polling sites sit within reach of public transit, and their findings on transit coverage support GC-DIG's recommendation that future access analysis should incorporate travel-time and transportation metrics rather than rely on per-voter ratios alone.

Team 5 (GC-AACS) analyzes ADA compliance at polling locations. GC-DIG's polling places shapefile provides the location set their compliance analysis can be layered onto.

Team 6 (GC-BAND) identified low broadband access in Clarke, Escambia, Monroe, and Washington counties. These are the same counties where GC-DIG documented elevated inactive registration rates, suggesting a potential connection between digital access and sustained voter engagement worth further investigation.

Teams 10 (GC-PORT) and 11 (GC-CPAD) translate technical findings into policy briefs, decision trees, and audience-facing materials. GC-DIG's findings on turnout decline, inactive registration disparity, and polling place access are available for inclusion in those communication products where they support the broader voter access narrative.

Final Report and Deliverables

Project Context

Team 1 (GC-DIG) is the Data Integration and Governance team within the eleven-team regional analysis for the Southern Poverty Law Center. On a broad level, the project examines voting rights, access, and participation across Alabama's Gulf Coast Counties. Each team contributes to a specialized goal such as polling access (GC-PAD), turnout and registration analysis (GC-TRAC) and many others.

GC-DIG's role is responsible for collecting, cleaning, and standardizing the existing voter registration, election participation, and geographic datasets for other teams to use as a baseline during their analysis. As a result, our team has produced cleaned datasets such as the combined

registration data for all counties and the combined race statistics that followed. We relied on the SPLC directly to source our data and were referred to the Redistricting Data Hub, a very helpful resource.

Our findings are intended to be used as both a standalone observation for the SPLC, but also as inputs for other teams to have a deeper analysis.

Methods

Data Sources:

Alabama Secretary of State:

The Alabama Secretary of State provided election results and voter registration data. These datasets included county-level turnout totals, ballots cast by race, and active and inactive voter registration counts for the 2020 and 2024 general elections. These formed the foundation for turnout analysis and registration status comparisons.

Redistricting Data Hub:

The hub provided geographic boundary files and election-related geographic information. These datasets were important because Alabama went through substantial redistricting changes following the 2020 census including court-ordered adjustments to congressional maps for the 2024 and 2026 election cycles.

SPLC Client Data:

The SPLC offered polling place shapefiles and geographic datasets related to voting accessibility and flood-zone analysis. These files supported polling place mapping and accessibility analysis.

ICPSR and Supplemental Election Data:

Election datasets from ICPSR and related repositories were reviewed for validation. These datasets helped confirm broader election trends and supported cross-source consistency checks.

Analytical Approach:

Our team combined voter registration and election turnout datasets from 2020 and 2024 to compare voting participation patterns across seven Gulf Coast Alabama counties. We calculated turnout rates by comparing ballots cast with active registered voters for each racial group and county.

We also analyzed inactive voter registration rates and polling place access ratios to better understand differences in voter engagement and voting access across counties and demographic

groups. The goal of the analysis was to identify broader regional patterns and disparities over time instead of trying to prove direct cause and effect relationships.

Tools Used:

Our team mainly used Python for data cleaning, analysis, and visualization throughout the project. We used the pandas library to merge and organize voter registration and election datasets from different counties and election years. Matplotlib was used to create the charts and visualizations included in the report, such as turnout comparisons, turnout changes by race, inactive voter registration rates, and polling place access ratios. Microsoft Excel was also used for quick data inspection and validation before running analysis in Python. Using Python helped our team keep calculations and visualizations consistent across the project.

Challenges Faced and How They Were Handled:

Inconsistent County-Level Reporting:

One of the biggest challenges involved inconsistent formatting across county-level election datasets. Counties used different naming conventions, formatting structures, racial labels, and registration categories. Our team addressed these problems by developing standardized variable naming structures and cleaning workflows before analysis began.

Boundary Changes:

Alabama experienced substantial post 2020 redistricting changes that complicated geographic comparisons between election years. Certain structures changed between 2020 and 2024, limiting longitudinal analysis at geographic levels. To reduce the inconsistency our team focused on county-level comparisons where geographic continuity could be maintained more reliably.

Geographic File Limitations:

Polling place shapefiles sometimes had incomplete metadata or inconsistent projections. Geographic files were validated using QGIS and geopandas before integration into the broader analysis pipeline.

Time Constraints:

Our team worked under a limited timeline while integrating multiple election cycles and geographic datasets. This was hard because of lack of communication at first but it got better as the semester progressed.

Ethical Interpretation:

Our team avoided language suggesting voter apathy or unsupported casual explanations. Observed turnout declines and inactivity disparities may reflect structural or administrative factors

like polling place access, transportation barriers or registration maintenance. Our team focused on identifying measurable patterns rather than assigning blame.

Deliverables and Findings

GC-DIG produced four core deliverables that supported both standalone analysis and downstream use by other teams in Section 105: a combined registration dataset, a combined election race dataset, a polling places shapefile filtered to the seven Gulf Coast counties, and a polling place access analysis. These deliverables enabled the three findings described below:

Key Finding 1:
Black voter turnout fell an average of 6.7 percentage points across all seven Gulf Coast counties from 2020 to 2024, while white voter turnout rose an average of 0.7 percentage points over the same period. This divergence represents a meaningful and consistent pattern across the region

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Analysis of ballots cast by race (General_Race_Combined.xlsx) against active registered voters by race (Registration_Combined.xlsx) shows a consistent pattern of declining Black voter participation between the 2020 and 2024 Alabama general elections. The decline was present in all seven counties and ranged from 4.6 percentage points in Monroe to 10.1 percentage points in Escambia. White voter turnout remained relatively stable, with slight gains in Baldwin, Clarke, Conecuh, and Monroe counties offset by marginal declines in Escambia, Mobile, and Washington counties.

The magnitude of divergence is notable in the overall turnout context. Total ballots cast across the seven counties were nearly identical in 2020 (350,388) and 2024 (349,763), suggesting that overall turnout was stable while the racial composition of participants shifted. This pattern may reflect changes in voter outreach, polling place access, registration list management, or other structural factors.

County	Black Turnout 2020	Black Turnout 2024	Change	White Turnout 2020	White Turnout 2024	Change
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BALDWIN	58.8%	51.3%	-7.5 pts	73.6%	74.0%	+0.4 pts
CLARKE	67.0%	61.2%	-5.8 pts	77.3%	77.6%	+0.3 pts
CONECUH	59.4%	53.4%	-6.0 pts	62.5%	69.0%	+6.5 pts
ESCAMBIA	58.0%	47.8%	-10.1 pts	68.2%	67.6%	-0.7 pts
MOBILE	64.4%	57.7%	-6.7 pts	71.2%	69.5%	-1.7 pts
MONROE	63.0%	58.4%	-4.6 pts	74.4%	75.4%	+1.0 pts
WASHINGTON	63.5%	57.6%	-5.9 pts	73.5%	72.3%	-1.2 pts
Average			-6.7 pts			+0.7 pts

Table 1. Black and White voter turnouts by county, 2020 and 2024. Calculated as ballots cast by race divided by active registered voters of that race. Source: Alabama Secretary of State; team calculations.

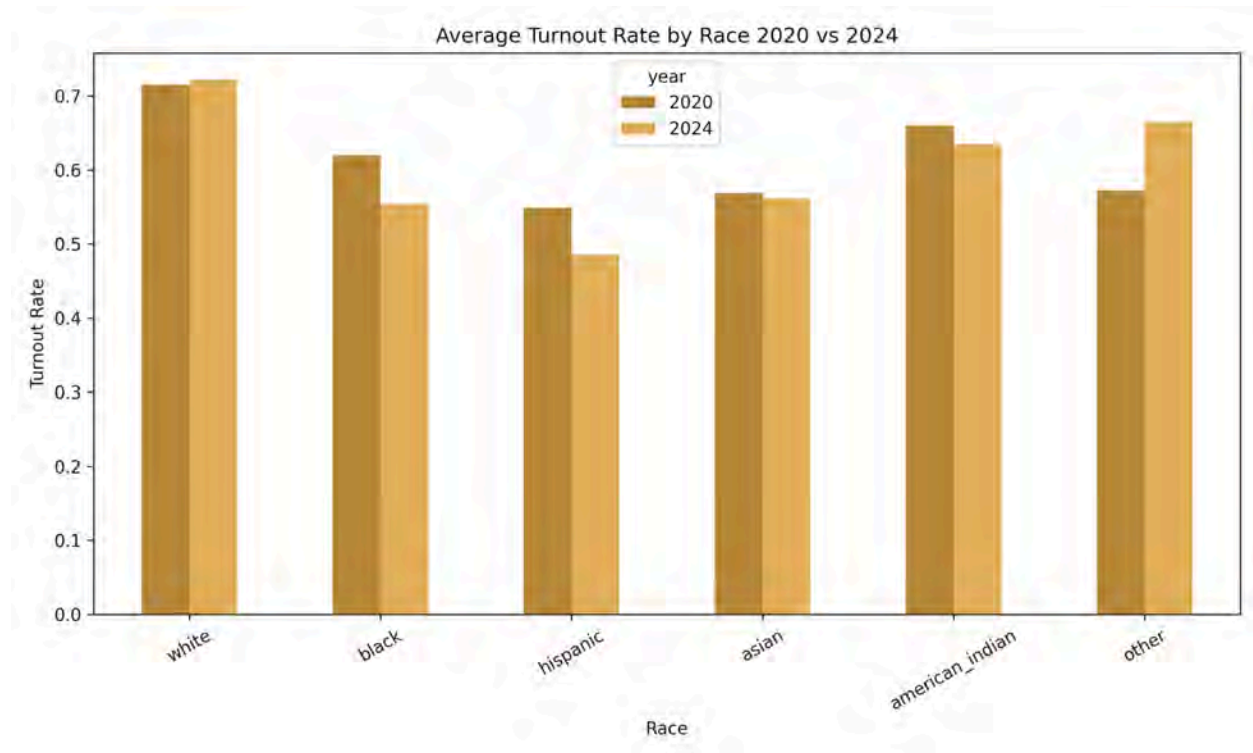


Figure 1. Average Turnout by Race, comparing 2020 and 2024 across all seven Gulf Coast counties. Source: General_Race_Combined.xlsx, Registration_Combined.xlsx; team analysis

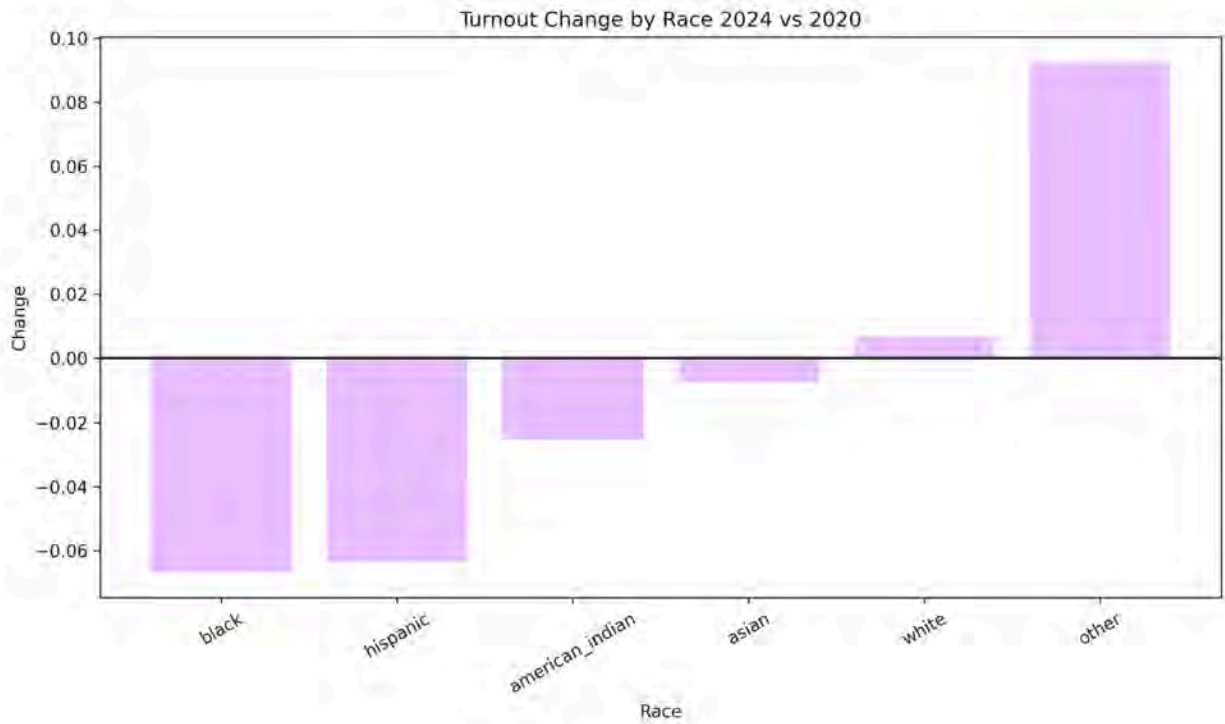


Figure 2. Turnout Change by Race (2020 vs. 2024). Positive values indicate increased turnout; negative values indicate decline. Source: General_Race_Combined.xlsx, Registration_Combined.xlsx; team analysis

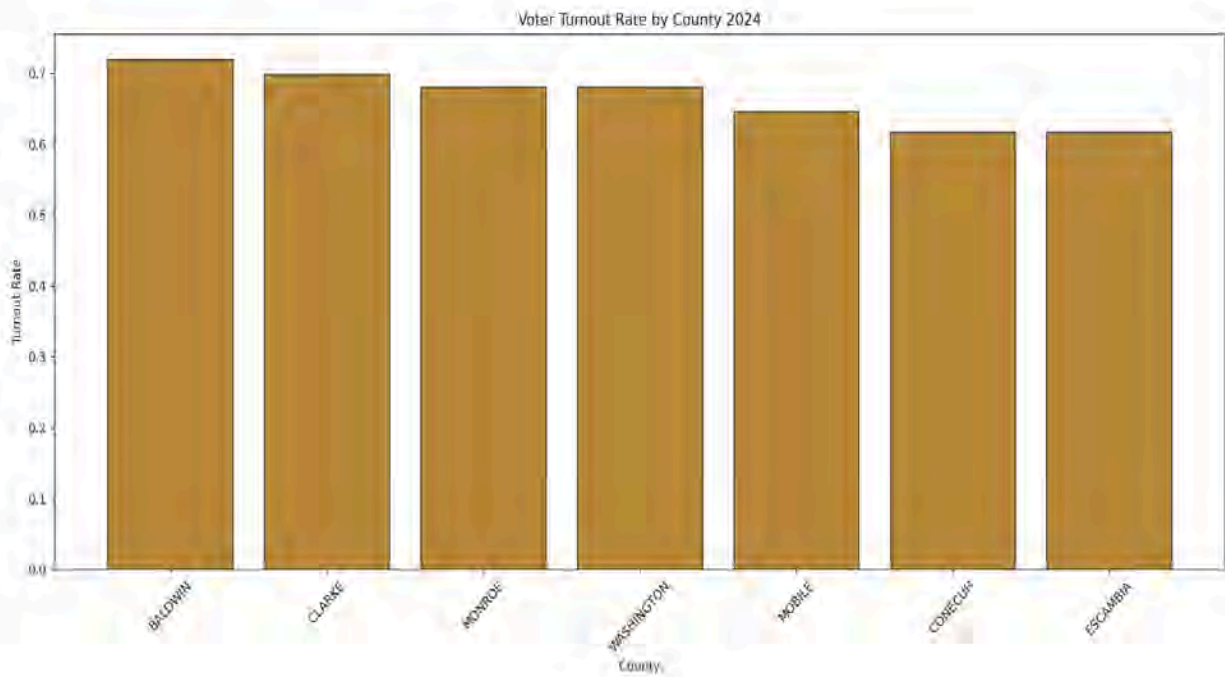


Figure 3. Voter Turnout Rate by County in 2024, ranked from highest (Baldwin, 72.0%) to lowest (Escambia and Conecuh, approx. 61.7%). Source: General_Race_Combined.xlsx, Registration_Combined.xlsx; team analysis

Key Finding 2:

Hispanic registrants have the highest rate of inactive voters of any racial group in the region at 19.7 percent in 2024, followed by Asian (14.1%) and American Indian registrants (13.2%). All minority group inactive rates rose between 2020 and 2022 before showing partial recovery in 2024.

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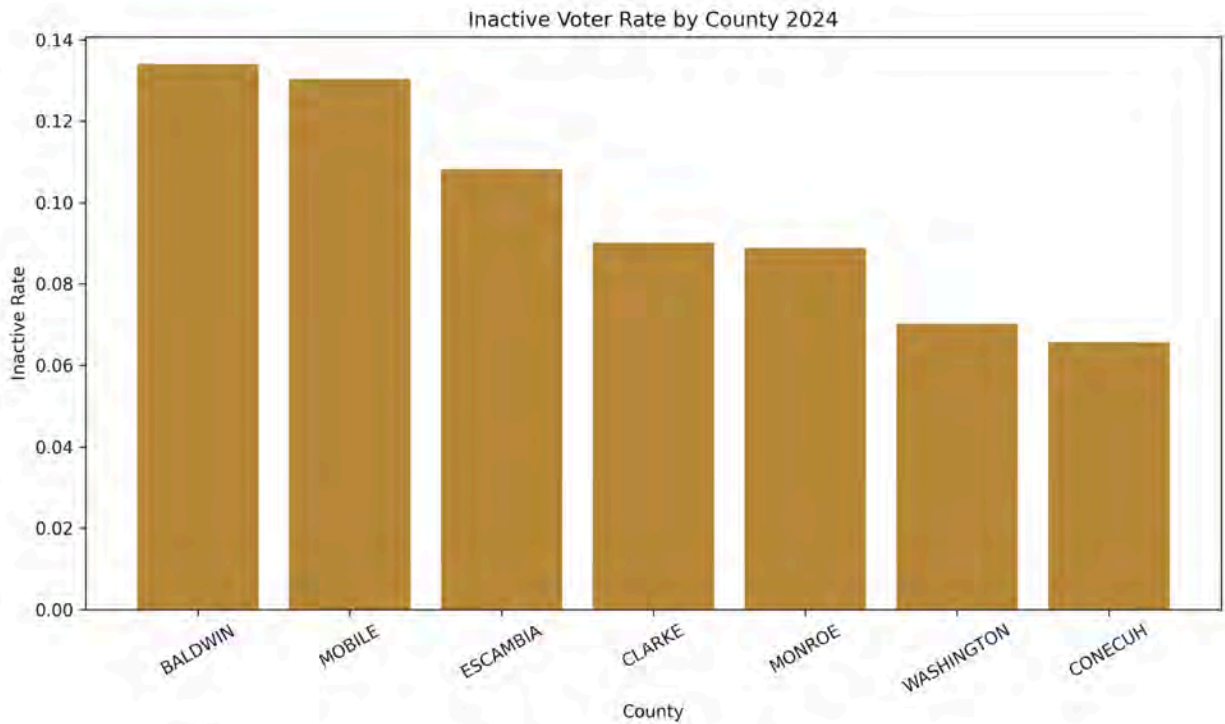


Figure 4. Average inactive voter registration rates by race in 2024 across all seven Gulf Coast counties. Source: *Registration_Combined.xlsx*; team analysis.

This finding is supported by registration data that is also positioned for use by Team 6 (GC-BAND), whose broadband access analysis identifies the same rural counties (Clarke, Escambia, Monroe, Washington) where inactive registration rates were elevated. Team 7 (GC-CPRI) may incorporate registration variables into their composite civic participation risk index.

Key Finding 3:

Polling place density does not predict voter turnout across the seven Gulf Coast counties. Voters per polling place ranged from 610 in Monroe County to 5,310 in Baldwin County, an 8.7-fold difference in nominal access. The correlation between voters per polling place and 2024 turnout was weak and positive ($r = 0.361$), the opposite direction of what a simple access argument would predict.

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Analysis of polling place locations (AI_Polls_Flood_SLED shapefile) against active registered voters in 2024 (Registration_Combined.xlsx) and total ballots cast (General_Race_Combined.xlsx) shows that counties with the fewest voters per polling place did not record the highest turnout. Monroe County, with 610 voters per polling place, recorded a turnout of 68.1 percent, while Baldwin County, with 5,310 voters per polling place, recorded the region's highest turnout at 72.0 percent. Conecuh and Escambia tied for the lowest turnout at 61.7 percent despite having very different access ratios of 688 and 1,010 voters per polling place respectively.

This pattern suggests that polling place counts alone are an insufficient measure of voter access. Factors such as geographic distance to the polling place, transportation availability, polling hours, precinct assignment, and demographic composition likely have a stronger influence on whether registered voters cast ballots than the raw number of polling places available in a county. Future analysis should pair this data with travel-time or distance-based access metrics rather than rely on county-level counts.

County	Polling Places	Voters per Polling Place	2024 Turnout
Monroe	26	610	68.1%
Conecuh	14	688	61.7%
Clarke	23	819	69.9%
Washington	14	956	68.0%
Escambia	27	1,010	61.7%
Mobile	73	4,302	64.6%
Baldwin	37	5,310	72.0%

Deliverable 4:

In addition to the three findings above, GC-DIG produced standardized datasets that function as shared infrastructure for Section 105. Each dataset is documented in the GC-DIG Data Dictionary and is available in the team's Client Deliverables folder.

Registration_Combined.xlsx: Cleaned voter registration data (active and inactive) by race for all seven Gulf Coast counties across 2020, 2022, and 2024. Used by GC-VIZ and GC-VOTE for analysis of registration patterns and turnout calculations.

General_Race_Combined.xlsx: Cleaned general election ballots cast by race for all seven counties for 2020 and 2024. 2022 ballots cast by race were not published by the Alabama Secretary of State and are documented as a known limitation. Used by GC-VIZ as the basis for their Voter Turnout by Race Over Time chart and by GC-VOTE for trend analysis.

Polling Places Shapefile (Al_Polls_Flood_SLED): Geographic point data for 214 polling places filtered to the seven Gulf Coast counties. Includes precinct name, address, county, and coordinates. Used by GC-PAD for travel time network analysis, GC-TRAC for transit accessibility analysis, and GC-AACS for ADA compliance analysis.

Polling Place Access Analysis (GCDIG-GulfCoastVoting-AccessAnalysis-v1.xlsx): County-level synthesis dataset combining polling place counts, registered voters, ballots cast, voters per polling place ratio, and 2024 turnout percentage. Supports Finding 3.

Data Dictionary (GCDIG-GulfCoastVoting-DataDictionary-v1.docx): Documentation of every variable across all GC-DIG datasets including source, type, definition, limitations, and naming conventions. Provides the standardized reference used across the section.

Recommendations

The recommendations below are organized into two parts. The first addresses what SPLC may want to investigate, prioritize, or build on directly from the findings in this report. The second is directed at future capstone teams who continue this work, covering data sources, methods, and natural extensions of the analysis. Both sets are grounded in what the data actually supports. Where the analysis identifies a pattern without establishing cause, the recommendation shows our limitations rather than overstating findings.

Recommendations for the client (SPLC)

Escambia County saw the largest single-county decline in Black voter turnout of any county in the region, falling from 58.0 percent to 47.8 percent in 2024. Combined with a mid-range access ratio and no clear offsetting gain in other demographic groups, this decline warrants targeted investigation. SPLC's legal and community engagement teams should examine whether polling place changes, transportation barriers, or specific voter list maintenance in Escambia disproportionately affected Black voters between cycles. Conducting on-site, interactive investigations along with analysis and findings provided by GC-PAD, GC-TRAC, and other relevant teams who worked on this region.

The 6.7 percent average decline in Black voter turnout represents a pattern consistent across all counties. This regional consistency suggests a systemic trend rather than a county-level

phenomenon. SPLC may want to support qualitative research with community organizations, particularly in Baldwin, Escambia, and Mobile, to understand voter perception and structural barriers that may have changed between 2020 and 2024.

The team also identified higher inactive voter registration rates among Hispanics compared to white voters throughout the region. Hispanic voters experienced the highest average inactive registration rate at around 19.7% in 2024. SPLC may want to further investigate whether registration maintenance procedures, language barriers, outreach limitations or administrative practices have an effect on these disparities. While inactive registration status doesn't prevent participation, elevated inactivity rates may indicate populations more vulnerable to registration instability.

SPLC should also continue expanding regional geographic accessibility analysis. The polling place and access-ratio datasets created through this project offer a framework for evaluating county-level voting accessibility. But, additional precinct-level analysis might help identify more localized accessibility disparities. Geographic factors like transportation infrastructure, polling place relocation, and travel distance may be important in rural counties or communities with underserved populations. Future analysis having to do with transportation and ADA compliance datasets could offer stronger insight into structural voting barriers across the Gulf Coast region.

Moving on, another important recommendation would be to improve long-term election data standardization and governance practices. One of the biggest challenges during this project was inconsistent county-level reporting structures, racial classifications, and naming conventions across election years. SPLC may benefit from keeping standardized datasets and geographic identifiers that can be reused across election cycles and research initiatives. Developing a centralized source for Gulf Coast election and demographic analysis would improve future longitudinal analysis and reduce repetitive cleaning and integration work.

To finish up here, the findings of our team should be interpreted carefully as the analyses conducted were not designed to establish causal conclusions about voter inactivity. The findings identify disparities and regional patterns that call for more investigation and not definitive evidence of voter intent or apathy. Collaboration between SPLC, community organizations, and future research teams will be important for developing an understanding of the structural and geographic factors that have to do with voter participation.

Recommendations for future project teams

Additional analysis utilizing precinct-level data is a valuable expansion upon this research. This team only worked with county level data for the seven Gulf Coast counties. Disaggregating turnout and registration data to the precinct level would enable identification of specific precincts where disparities are most prevalent.

Future teams may benefit from additional election cycles beyond 2020 and 2024 to better evaluate whether the turnout and registration patterns identified in this project represent long-term trends or election changes. Including election data before 2020, as well as future election cycles could make longitudinal analysis stronger and offer more context for understanding shifts in voter participation across the region. Another recommendation would be to standardize county names, racial categories, and geographic identifiers as a good amount of time was spent cleaning inconsistent naming conventions and formatting differences across datasets. If you establish standardized identifiers early it would reduce integration challenges and improve workflow.

Future teams should also maintain separate raw data and cleaned data folders throughout the process. Keeping original dataset separate from transformed files improves reproducibility, reduces accidental modifications of source data, and makes validation easier when inconsistencies arise later in the analysis process.

For future teams that work on geographic or GIS analysis, validating shapefile projections and metadata before starting spatial analysis is recommended. The project went through inconsistencies in geographic files that required correction before polling place accessibility analysis could be completed in a reliable way. The future teams should document projection systems, boundary assumptions, and geographic limitations carefully.

Another continuation of this project that would be of value would be to integrate broadband access or socioeconomic datasets for analyses as these variables could offer stronger understanding of the structural and geographic factors that influence turnout and registration disparities throughout the Gulf Coast. To finish up here, future project teams should approach turnout and participation analysis carefully and responsibly too. The findings in this project identified disparities and regional patterns but they don't establish voter intent. You have to keep a distinction between descriptive analysis and causal interpretation so research remains responsible and grounded in the data.

Conclusions

GC-DIG set out to collect, integrate and standardize voter registration and election data for seven counties on Alabama's gulf coast to support SPLC's voting rights analysis. By producing two standardized and well-structured datasets covering 2020, 2022 and 2024 general election data and a standardized geographic file which were used to conduct data analysis and create supporting visualizations.

The analysis generated three key findings with direct implications for SPLC's advocacy in the region. Black voter turnout declined by an average of 6.7 percent across all seven counties between 2020 and 2024, a pattern that white voter turnout did not share. Hispanic registrants have the highest inactive registration rate in the region at nearly 20 percent, a gap that is disproportionate and unexplained by the data alone. Polling place access, when measured as a per-voter ratio, does not predict turnout, showing that counts alone are an insufficient measure for access and that distance, transportation, and demographic context are also meaningful indicators necessary for any access assessment.

These findings are grounded in publicly available data and bound by its limitations. Future teams should treat the work provided as a documented starting point to be expanded upon with further analysis and precinct-level data.

Contact Information

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Deliverables Checklist

OReadMe File:  OREADME

Data Dictionary:  GCDIG-GulfCoastVoting-DataDictionary-v1.docx

Final Report:  GCDIG-GulfCoastVoting_Final_Report.docx

Access Analysis:  GCDIG-GulfCoastVoting-AccessAnalysis-v1.xlsx

Turnout Change Visual:  GCDIG-GulfCoastVoting-Fig1-TurnoutByRace.png

Race Analysis:  GCDIG-GulfCoastVoting-ElectionRaceData-v1.xlsx

Race Analysis Visual:  GCDIG-GulfCoastVoting-Fig2-TurnoutChange.png

Race Analysis Visual 2:  GCDIG-GulfCoastVoting-Fig3-InactiveByRace.png

Registration Analysis:  GCDIG-GulfCoastVoting-RegistrationData-v1.xlsx

Access vs Turnout Visual:  GCDIG-GulfCoast-Voting-Fig4-AccessVsTurnout.png

Shapefiles Folder:  GCDIG-GulfCoastVoting-PollingPlaces_ShapeFiles

Alabama Gulf Coast Voter Turnout & Participation Trends

GC-VOTE

Nana Akuffo (Project Manager), Ryan Yu (Visual Analyst and Data Manager), Frank Okonsky (Visual Analyst and Data Manager), Nivitha Paranthaman (Visual Analyst), Manasvi Hosahalli (Editor and Project Reviewer), Afreen Ahmed (Documentation Lead)

Southern Policy Law Center (SPLC)

INST490 - Integrated Capstone for Information Science Section 105

May 18, 2026

Team Brief

Team 2: GC-VOTE

Key Findings

- **Finding 1:** The analysis indicated that when comparing the Hispanic voting eligible population to its registration, they are most underrepresented in Washington County. This suggests that additional voter outreach, education, and registration efforts may be needed in the area to improve civic participation. For Black eligible voters, that gap is largest in Escambia County.
 - **Finding 2:** The analysis also found turnout and demographic representation differences across several Gulf Coast counties, especially in Escambia and Baldwin counties. Our heatmap visualizations suggest that Escambia and Baldwin counties should be the focus of further mobilization efforts for voting participation. This comes from seeing that these two counties have the largest discrepancies of Black and Hispanic eligible voters who are not only registered, but are actually participating in the elections.
 - **Finding 3:** Across all counties, black voting turnout is substantially behind that of white voters, being approximately ~15% lower across the region, with that gap actually increasing across the 2020 and 2024 election cycles.
 - **Finding 4:** From 2020 to 2024 voter turnout dropped among all major demographic groups significantly in Conecuh, Monroe, and Washington counties.
-

Deliverables Produced

Raw Data (Deliverable 1): ZIP file containing all original raw data collected for this project

<https://drive.google.com/file/d/1vuCsKMIj6ng-wdgKUexRu9rGxaVXML1y/view?usp=sharing>

Cleaned Dataset (Deliverable 2): CSV file containing a cleaned and standardized version of our data

<https://drive.google.com/file/d/1Yjr0ISAxkFUAnsGgCNw8MqUs4ABcoxhK/view?usp=sharing>

Methodology/Assumptions File (Deliverable 3): An in-depth discussion of our analytical methodologies and assumptions in this project

https://docs.google.com/document/d/1b0CvPCwoEGduG7FcFKWhn4DcwwlT_RTh_b2xH9w4kHE/edit?usp=sharing

Interactive Dashboard (Deliverable 4): Tableau dashboard visualizing turnout trends across demographics and the Gulf Coast Region

<https://drive.google.com/file/d/1ptcKATTxOpB99lad5tSpj5lecglCpSZe/view?usp=sharing>

Slide Deck Presentation (Deliverable 5): Google Slideshow summarizing the findings and methodologies across the course section

<https://docs.google.com/presentation/d/1PUaEQZFBaHEBHqHnjIj80Fq9FLFE4t0u8dsuf-Tf8e0/edit?usp=sharing>

Final Report (Deliverable 6): A comprehensive report documenting the project process

<https://docs.google.com/document/d/19xglutkPmVdglJzMgEdaipLf9yWHGwbP5lH338f3Cms/edit?tab=t.s7x3095r7tuh>

Challenges

A major challenge in this analysis was the lack of transparency and documentation surrounding some of our key data sources, particularly from the Alabama Secretary of State, making it somewhat difficult to fully understand and verify the accuracy of the datasets. This was addressed by relying on the credibility of official sources while also acknowledging potential margins of error, particularly in Citizen Voting Age Population (CVAP) estimates, which were derived from American Community Survey data. Another significant challenge was the absence of demographic data specific to the 2nd Congressional District races, along with major total voting participation

disparities between contests. This prevented reliable extrapolation, and as a result, the group chose to leave those data fields blank rather than introducing potentially misleading assumptions.

Connections to Other Teams

Our analysis had a fair bit of thematic overlap with some of the other project groups from this section, however, we did not rely on the outputs from other teams to come to our conclusions. Particularly, GC-DIG, GC-CPRI, and GC-PORT. For all of these teams, voting turnout, registration, and participation were key to developing findings, similarly to us. They may have used the data for different purposes, but the core aspects still remain. For example, GC-DIG used participation and registration data to assess which demographic groups have the highest rate of inactive voters. GC-CPRI, on the other hand, used similar data about activity, participation, and access to make their civic participation composite risk score. Additionally, GC-PORT used some of the findings from our analysis in relation to participation to make some policy-based recommendations.

Final Report and Deliverables

Project Context

This project examined potential inequalities in voter participation across the Alabama Gulf Coast counties through analyzing turnout in the 2020 Presidential and 2024 Presidential, and 2024 2nd Congressional District elections. Within the grand scope of analysis in the class section, our project serves as the core basis of electoral behavior. This was done through our analysis of election and demographic data into findings about who is participating and how civic engagement has varied across communities and election cycles. Specifically, we addressed: drops and gains in voting turnout and registration rates, and comparisons between eligible and participating voters across six racial demographics. Our analysis gave the needed context in the broader scope of the class to make discernments on which groups of people should be focused on in SPLC's civic engagement initiatives.

Methods

For this project, data were primarily collected from the Alabama Secretary of State's election datasets, including county level election results, voter registration statistics, and demographic participation by race, gender, and age. Additional Citizen Voting Age Population (CVAP) estimates were obtained through the Redistricting Data Hub to provide a standardized denominator for turnout calculations. All datasets were imported and processed through Python to be merged and eventually cleaned and standardized into one consolidated dataset. The data cleaning process involved converting numeric fields, removing irrelevant columns, renaming variables for consistency, aligning CVAP variables with the Secretary of State's datasets, and more.

The final consolidated dataset included election results, demographic participation data, voter registration statistics, and CVAP estimates for each county and election year. Turnout rates were ultimately calculated by dividing demographic participation totals by the corresponding CVAP estimate, allowing for estimated turnout comparison across racial demographic groups for eligible voters.

Several assumptions and limitations affected the analysis, though. For example, the project relies heavily on the accuracy of Alabama Secretary of State Data and American Community Survey based CVAP estimates, both of which may contain small margins of error. Additionally, the demographic participation datasets did not distinguish between specific election contests, making it impossible to reliably estimate demographic turnout for the 2024 2nd Congressional District race. Due to this, we felt it was appropriate to leave the demographic fields for those contests blank and not include them in our further analysis.

This description is only a summary of our methodological process for this project. The entirety of this discussion can be referenced in the following section under "Methodology/Assumptions File (Deliverable 3)."

Deliverables and Findings

Raw Data (Deliverable 1): A ZIP file containing all original raw data collected for this project. It contains a total of 24 .csv files spanning 2020 and 2024. This deliverable meets the project goals by assisting SPLC in understanding the complete context of our project through providing a transparent record of the data that we collected. This makes replicating or reusing any of the data significantly easier.

Cleaned Dataset (Deliverable 2): A CSV file containing a cleaned and standardized version of the data collected from the first deliverable. It contains a consistent structure across years, elections, and demographics to ensure the data is readable and prepared for analysis. This deliverable meets the project goals by sharing the exact dataset our analysis and visualizations were based on. This makes reproducibility easier and makes understanding the findings easier. The following image is an example excerpt of what the cleaned dataset looks like, however, it is of note that the true dataset has 38 total columns compared to the 7 features in the image.

YEAR	COUNTY	Contest Title	TOTAL_BALLOTS_CAST	TOTAL_BLACK	REG_BLACK	CVAP_EST_BLACK
2020	BALDWIN	PRESIDENT AND VICE PRESIDENT	110922	6670	12310	14245
2020	CLARKE	PRESIDENT AND VICE PRESIDENT	13239	5460	8434	8355
2020	CONECUH	PRESIDENT AND VICE PRESIDENT	5709	2558	4449	4464
2020	ESCAMBIA	PRESIDENT AND VICE PRESIDENT	15958	4143	7470	8965
2020	MOBILE	PRESIDENT AND VICE PRESIDENT	184949	61924	103017	109210
2020	MONROE	PRESIDENT AND VICE PRESIDENT	10693	4147	6798	6744
2020	WASHINGTON	PRESIDENT AND VICE PRESIDENT	8918	1952	3103	3199
2024	BALDWIN	PRESIDENT AND VICE PRESIDENT	122501	6117	12774	14780
2024	CLARKE	PRESIDENT AND VICE PRESIDENT	11975	4749	7877	7727
2024	CLARKE	UNITED STATES REPRESENTATIVE	2650			
2024	CONECUH	PRESIDENT AND VICE PRESIDENT	5552	2175	3942	4014
2024	CONECUH	UNITED STATES REPRESENTATIVE	5063			
2024	ESCAMBIA	PRESIDENT AND VICE PRESIDENT	15005	3310	7136	8503
2024	MOBILE	PRESIDENT AND VICE PRESIDENT	176426	56195	100097	109030
2024	MOBILE	UNITED STATES REPRESENTATIVE	103165			
2024	MONROE	PRESIDENT AND VICE PRESIDENT	9841	3582	6341	6291
2024	MONROE	UNITED STATES REPRESENTATIVE	9707			
2024	WASHINGTON	PRESIDENT AND VICE PRESIDENT	8463	1675	2991	2862
2024	WASHINGTON	UNITED STATES REPRESENTATIVE	8368			

Methodology/Assumptions File (Deliverable 3): A document addressing all of the analytical methodologies used throughout the course of the project. Including but not limited to data sources, cleaning, assumptions, and limitations. This document goes into depth about the methodologies and assumptions, allowing for increased interpretability of the findings and the data sets.

1. Project Overview

We are addressing potential inequalities in voting participation by analyzing voter turnout in the Alabama Gulf Coast counties. This project aims to identify patterns that may suggest unequal access to civic participation, helping SPLC identify where further investigations should be placed. We are specifically going to be assessing the 2020 Presidential, 2024 Presidential, and 2024 2nd Congressional District elections.

2. Data Sources

Much of the data was sourced from the Alabama Secretary of State's website of Election Data, which provides direct links to .xls datasets. The following is a list of the specific data sources and the variables used from the Alabama Secretary of State in this analysis:

- 2020 General Election Results - Precinct Level
 - Contest Title, Candidate, (All Precinct Columns Were Totals To Create a New column called Total Votes)
 - Because the data was collected at a precinct level, each county had its own dataset, so adding the precincts together to create total_votes was the best way to find that data
- Voter Registration Statistics - 2020
 - County, Asian, American Indian, Black, Hispanic, White, Other, Not Identified, Total Active
- 2020 General Election Participation by Gender
 - County, Total Male, Total Female, Total Not Identified
- 2020 General Election Participation by Race
 - County, Total Asian, Total American Indian, Total Black, Total Hispanic, Total Other, Total White, Total Not Identified
- 2020 General Election Participation by Age
 - County, Age 18-29, Age 30-39, Age 40-49, Age 50-59, Age 60-69, Age 70-79, Age 80-89, Age 90-99, Age 100+
- 2024 General Election Results - Precinct Level
 - Contest Title, Candidate, (All Precinct Columns Were Totals To Create a New column called Total Votes)
 - Because the data was collected at a precinct level, each county had its own dataset, so adding the precincts together to create total_votes was the best way to find that data
- Voter Registration Statistics - 2024
 - County, Asian, American Indian, Black, Hispanic, White, Other, Not Identified, Total Active
- 2024 General Election Participation by Gender
 - County, Total Male, Total Female, Total Not Identified
- 2024 General Election Participation by Race
 - County, Total Asian, Total American Indian, Total Black, Total Hispanic, Total Other, Total White, Total Not Identified
- 2024 General Election Participation by Age

- County, Age 18-29, Age 30-39, Age 40-49, Age 50-59, Age 60-69, Age 70-79, Age 80-89, Age 90-99, Age 100+

A known limitation coming from this data is the fact that the demographic datasets do not differentiate between participation in specific election races. This topic will be further discussed later in this report.

With all of this data, the only aspect remaining was to find a metric for Voting Eligible Population, as previously discussed with the clients to be our basis for turnout calculations. To do this, we sourced data from the Redistricting Data Hub, specifically two data sets provided as .csvs:

- Alabama County CVAP Data (2020)
- Alabama County CVAP Data (2024)

CVAP in this instance refers to “Citizen Voting Age Population” and the data is based on estimates made by the American Community Survey. In regards to the variables we pulled from this data, we only looked at the demographic columns with the “CVAP_EST” prefix. As noted by the Citizen Voting Age Population (CVAP) Special Tabulation documentation, this refers to “The rounded estimate of the total number of United States citizens 18 years of age or older for that geographic area and group.” The following is a list of the specific variables pulled from these datasets:

- CVAP_TOT20 - CVAP Estimate for Total
- CVAP_AIA20 - CVAP Estimate for American Indian or Alaska Native Alone or In Combination
- CVAP_ASN20 - CVAP Estimate for Asian Alone or In Combination
- CVAP_BLK20 - CVAP Estimate for Black or African American Alone or In Combination
- CVAP_WHT20 - CVAP Estimate for White Alone
- CVAP_2OM20 - CVAP Estimate for Two or More Race Responses
- CVAP_HSP20 - CVAP Estimate for Hispanic or Latino

A known limitation of these datasets is that the data is based on information coming from the American Community Survey. While the survey is an official government process done by the U.S. Census Bureau, because it is only a survey sent to 3.5 million households, the populations of these demographic groups need to be estimated to a certain extent.

3. Data Ingestion & Storage

Each dataset was imported into Python using the Pandas library. To consolidate the datasets into our one main CSV file ‘20-24_Voting_Data_Gulf.csv’, we used several series of merges

(pd.merge) to ensure the data was correctly aligned and organized. Some column names, though, had to be altered in order to create this final dataset, which will be discussed in the following section.

4. Data Cleaning & Preprocessing

The data cleaning process for collecting this data was relatively tame, due to the fact that a significant portion of the data came from the same source, (the Alabama Secretary of State), and that over the years the datasets have remained uniform and readable. However, when merging that data with the Redistricting Data Hub, more elements were introduced, which required cleaning. The following steps were taken for each file:

- 2020 and 2024 General Election Results - Precinct Level
 - Ensuring vote totals were numeric instead of strings
 - Dropping all columns besides Contest Title, Candidate, and Total Votes
 - Dropping all rows that did not include the Presidential and 2nd Congressional District Contests
 - Removing extra spaces after contest title names
- Voter Registration Statistics 2020 and 2024
 - Ensuring vote totals were numeric instead of strings
 - Dropping data for inactive voter registration
 - Dropping rows for non-Gulf Coast counties
- 2020 and 2024 Demographics (Age, Race, Gender)
 - Ensuring vote totals were numeric instead of strings
 - Dropping rows for non-Gulf Coast counties
 - Renaming “Total_Female” and “Total_Male” to “Female_Votes” and “Male_Votes”
- Alabama County CVAP Data 2020 and 2024
 - Ensuring population totals were numeric instead of strings
 - Dropping all data columns without the CVAP prefix
 - Adding a “Year” column to each dataset to ensure the time of each estimate is known
 - Renaming all data columns to match the style from previous datasets
 - CVAP_TOT20 → CVAP_EST_TOTAL
 - CVAP_AIA20 → CVAP_EST_AMERICAN_INDIAN
 - CVAP_ASN20 → CVAP_EST_ASIAN
 - CVAP_BLK20 → CVAP_EST_BLACK
 - CVAP_WHT20 → CVAP_EST_WHITE
 - CVAP_2OM20 → CVAP_EST_TWO_OR_MORE

■ CVAP_HSP20 → CVAP_EST_HISPANIC

5. Variable Definitions

Following all of these transformations, the final dataset remained with 38 columns. Here are the column names and definitions:

- YEAR
 - The election year the data in the row pertains to
- COUNTY
 - The Gulf Coast county to which the data in the row pertains
- Contest Title
 - The election contests the data in the row that pertains to
- TOTAL_BALLOTS_CAST
 - The total amount of ballots cast in that specific election contest
- AGE_18-29, AGE_30-39, AGE_40-49, AGE_50-59, AGE_60-69, AGE_70-79, AGE_80-89, AGE_90-99, AGE_100+
 - The number of people in each age bracket who cast a ballot in that specific election contest
- FEMALE_VOTES, MALE_VOTES, OTHER_GENDER_VOTES
 - The number of males, females, unidentified, and others who cast a ballot in that specific election contest
- TOTAL_ASIAN, TOTAL_AMERICAN_INDIAN, TOTAL_BLACK, TOTAL_HISPANIC, TOTAL_OTHER, TOTAL_WHITE, TOTAL_NOT_IDENTIFIED
 - The number of people in each demographic who cast a ballot in that specific election contest
- REG_TOTAL, REG_ASIAN, REG_AMERICAN_INDIAN, REG_BLACK, REG_HISPANIC, REG_WHITE, REG_OTHER, REG_NOT_IDENTIFIED
 - The number of people registered to vote in each demographic at the time of that specific election contest
- CVAP_EST_TOTAL, CVAP_EST_AMERICAN_INDIAN, CVAP_EST_ASIAN, CVAP_EST_BLACK, CVAP_EST_WHITE, CVAP_EST_TWO_OR_MORE, CVAP_EST_HISPANIC
 - The number of citizens in each demographic estimated to be living in that county at the time of that specific election contest who were 18 years or older

6. Turnout Calculations

Based on conversations held with the clients at the beginning of the semester, for the typical turnout calculations, we used the CVAP population estimates as the denominator. For example, when calculating something like the voter turnout for Hispanics in the 2020 Presidential Election in Mobile, Alabama, it would look like this: $\text{TOTAL_HISPANIC (1198)} / \text{CVAP_EST_HISPANIC (6520)} = \text{Estimated Hispanic Turnout of 18.4\%}$.

7. Assumptions

The most important assumption made throughout the course of this analysis is that the data being collected is accurate. While a significant portion of the data comes from the Alabama Secretary of State's Office, we are unable to independently verify the statistics, especially considering the lack of documentation and data collection transparency on the website. However, based on the authoritative position of the Alabama Secretary of State, we believe that we can safely assume the data is accurate, or as close to it as possible. Regarding the CVAP data, we also have to assume the data is accurate for the purposes of our analysis, however, we are aware that due to the source of the data (the American Community Survey), calculations involving CVAP data have to presume a small margin of error.

Another important assumption made through our analysis once again relates to the Secretary of State's demographic data. We must assume that the "total ballots cast" figure in each data set refers to the presidential contest of that year, as it is not explicitly stated in the file. We can confidently make this assumption by comparing the figures for "total ballots cast" from the general election results dataset to the demographics dataset and seeing that they are almost perfectly aligned. For example, when adding the number of female (60724), male (50049), and other gender votes (149) ballots cast in the 2020 Baldwin presidential contest, it totals 110,922. We could then cross-reference the 2020 Baldwin election data set and see that 110,922 total ballots were cast, leaving us to make that assumption.

An additional necessary assumption made during the course of our analysis is that the classification system used to describe race in the American Community Survey estimations and the Alabama Secretary of State's data are the same, or incredibly similar. Without making this assumption, it would not be possible to calculate turnout in the form we did, as we have to refer to two different sources of data to complete the calculations. In both cases, the racial data is self-reported with the intention of reflecting a more social definition of race rather than a genetic one. However, that does not mean there is not a potential for inconsistencies.

8. Limitations

The first major limitation is that of using CVAP Data from the Redistricting Data Hub, which, as previously mentioned, is based on the American Community Survey. Despite the survey being conducted by the United States Census Bureau, the data is still an estimate. Only 3.5 million households receive the survey, and from that, the Census Bureau makes the calculations it deems to be most accurate to create the data that we used in our analysis to substitute Voting Eligible Population. This ultimately results in each of our calculations involving CVAP data having some level of inherent margin of error.

The second major limitation of the data is the fact that the demographic datasets do not differentiate between participation in specific election races. What this means is that there is no specific demographic data for the 2nd Congressional District contests, which were a target of this analysis. There is significant data available on the presidential elections, but not on those contests. In addition, in some counties, there is a significant discrepancy between the total number of ballots cast in the Presidential and 2nd Congressional District elections, despite the fact that both contests were on the same date and the same ballot. For example, in Clarke County, 11,975 Presidential ballots were cast, but only 2,650 for the 2nd Congressional District. An additional barrier to making estimations is that because of the level of discrepancies, it can be assumed that different demographic groups had different total shares of participation for that contest. Going back to the Clarke County reference, the Female and Male split in the Presidential election was 6779 to 5168, respectively, an approximately 57% share for Females. Hypothetically, if the main representative's platform ran on being pro-choice as a significant pillar of their campaign, then perhaps that could have led to more women being inclined to participate, therefore increasing the percentage share of female voters, even if the total number was less than in the Presidential contest. Scenarios like this are possible for each demographic group, which makes attempting to extrapolate the demographic data from the Presidential contests to the 2nd Congressional District races irresponsible in our opinion, given the current data we possess. When taking all of this into consideration, this led us to decide as a group to leave the demographic columns for the 2nd Congressional District Contests blank. If we were able to find more specific data, we would have gladly added it to the database, but ultimately do not feel comfortable making assumptions in this case due to the aforementioned considerations.

9. References

Alabama Secretary of State. (n.d.). Elections Data Downloads. Alabama Secretary of State.
<https://www.sos.alabama.gov/alabama-votes/voter/election-data>

Redistricting Data Hub. (2022, March 29). Alabama County CVAP Data (2020).

<https://redistrictingdatahub.org/dataset/alabama-county-cvap-data-2020/>

Redistricting Data Hub. (2026, February 12). Alabama County CVAP Data (2024).

<https://redistrictingdatahub.org/dataset/alabama-county-cvap-data-2024/>

[Interactive Dashboard \(Deliverable 4\)](#): A Tableau interactive dashboard that visualizes voter turnout trends across the Gulf Coast Region using various types of visualizations. The visualizations support our analysis by allowing users to interactively explore the turnout differences across counties, election cycles, and demographic groups. There are 4 different parameters the user can adjust and 3 primary calculations based on those. The 4 different parameters and their representations are the following:

- Total Votes by Race - Total number of votes by each racial category
- Total Votes by Race 2 (Total) - Total number of votes by each racial category
- Total Eligible Population by Race - Total eligible voters by each racial category
- Total Eligible Population by Race 2 (Total) - Total eligible voters by each racial category

The reason for 2 repeating parameters is the dashboards were generally designed in a layout to support viewing items by racial category against the total count. For example, if looking at white ballots cast, a user would compare the ballots cast by white racial demographics and we would take that and compare it against the total ballots cast among all racial demographics. However, in the name of giving the user more choice and allowing additional future updates if wanted by the client, this second parameter is manipulatable if they want to directly compare 2 racial categories. The 3 calculation names along with the calculations are the following:

- Total Votes By Race Percentage - $\text{Total Votes by Race} / \text{Total Votes by Race 2 (Total)} * 100$
- Voter Eligible Population Percentage Difference - $\text{Total Voter Eligible Population By Race} / \text{Total Voter Eligible Population By Race 2 (Total)} * 100$
- Total Ballots By Race vs Voter Eligible Population Percentage - $(\text{Total Votes by Race} / \text{Total Votes by Race 2 (Total)}) / (\text{Total Voter Eligible Population By Race} / \text{Total Voter Eligible Population By Race 2 (Total)}) * 100$

Total Votes By Race Percentage shows the percentage of a chosen racial demographic's votes versus the total number of ballots. *Voter Eligible Population Percentage Difference* shows the percentage of the racial demographic's total eligible voter population versus the overall voter population of each county. *Total Ballots By Race vs Voter Eligible Population Percentage* shows the difference between

the percentage of the ballots by racial demographic vs the voter eligible percentage by racial demographic. If a race is overrepresented by the ballots they submit they will have over 100%, if they are underrepresented it will be beneath 100%.

Figure 1a:

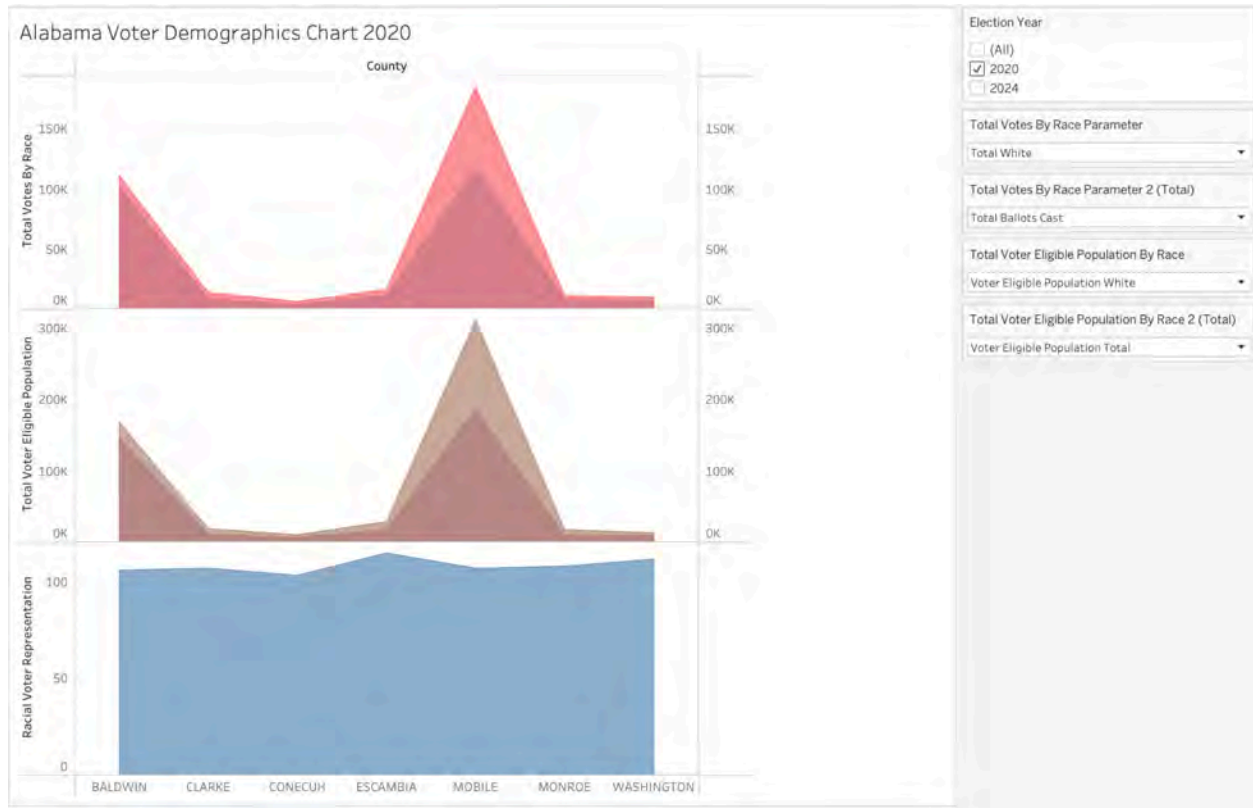
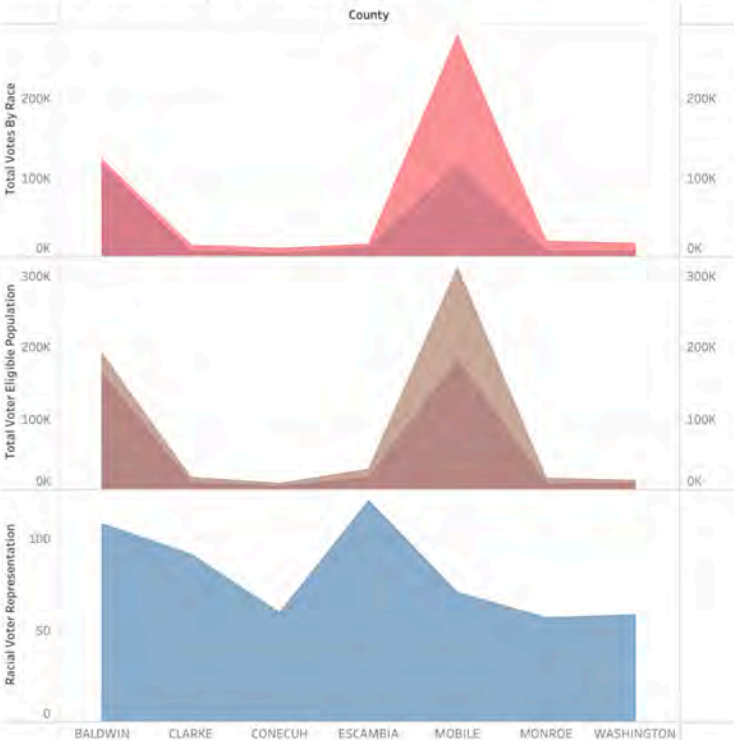


Figure 1b:

Alabama Voter Demographics Chart 2024



Election Year

☐ (All)

☐ 2020

☒ 2024

Total Votes By Race Parameter

Total White

Total Votes By Race Parameter 2 (Total)

Total Ballots Cast

Total Voter Eligible Population By Race

Voter Eligible Population White

Total Voter Eligible Population By Race 2 (Total)

Voter Eligible Population Total

Figure 2a:

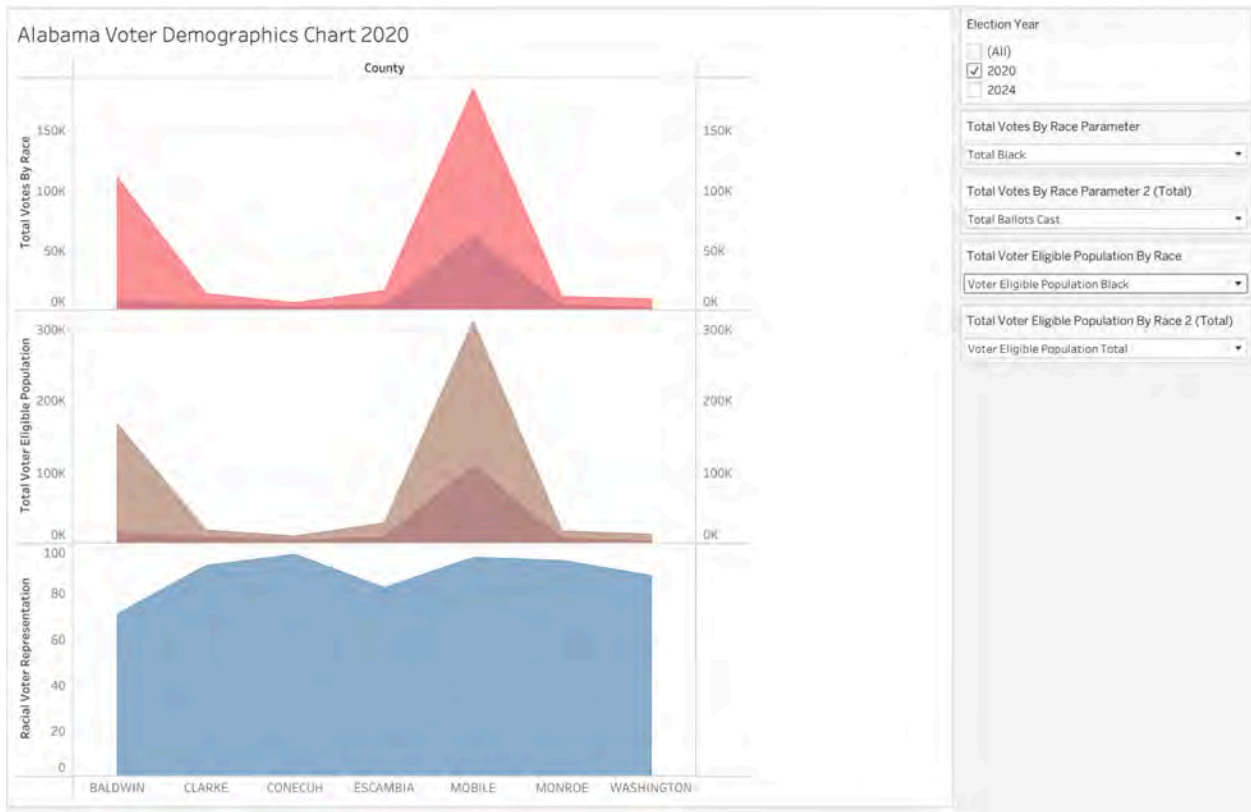
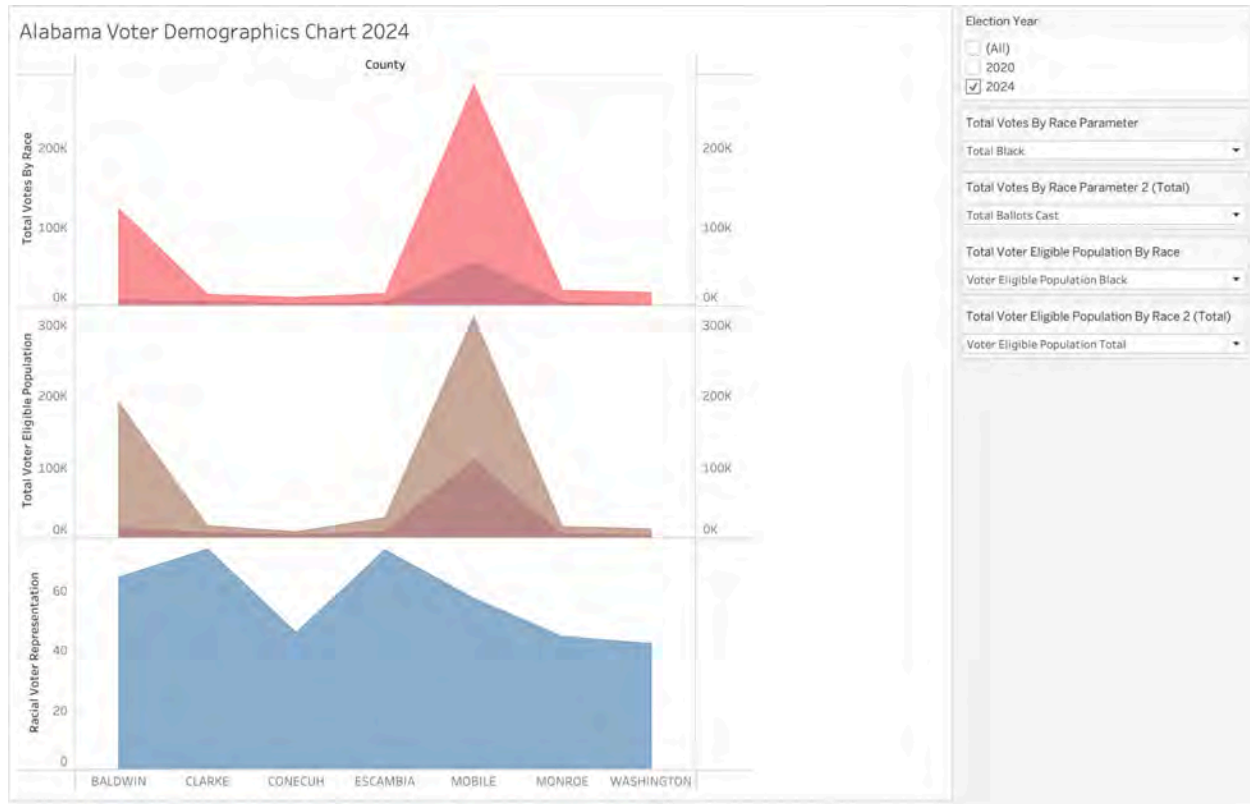
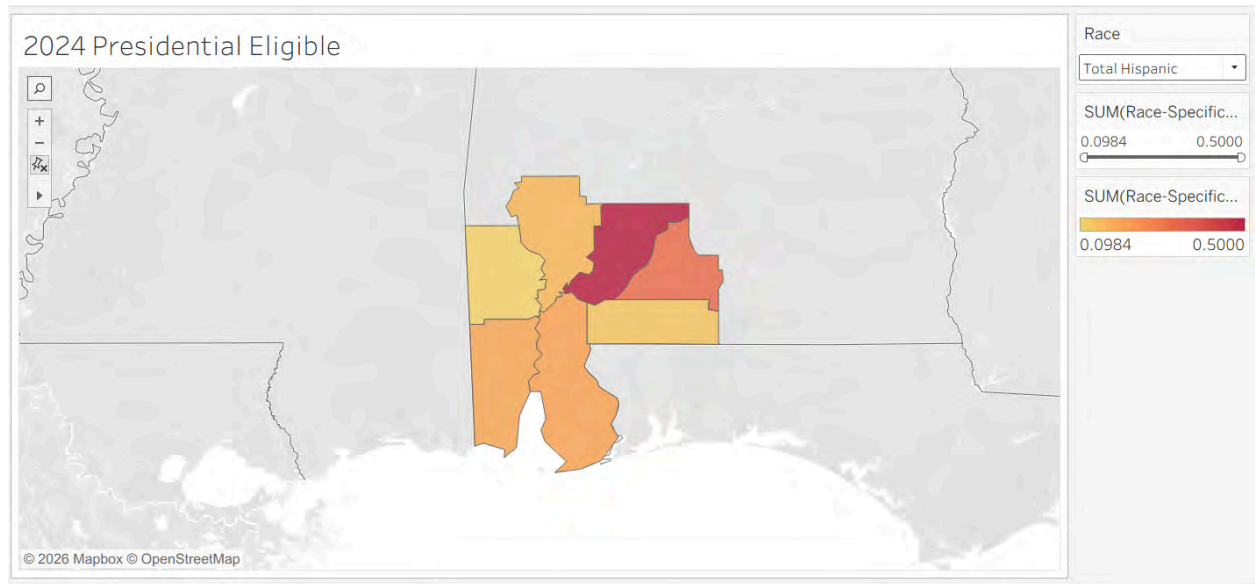


Figure 2b:



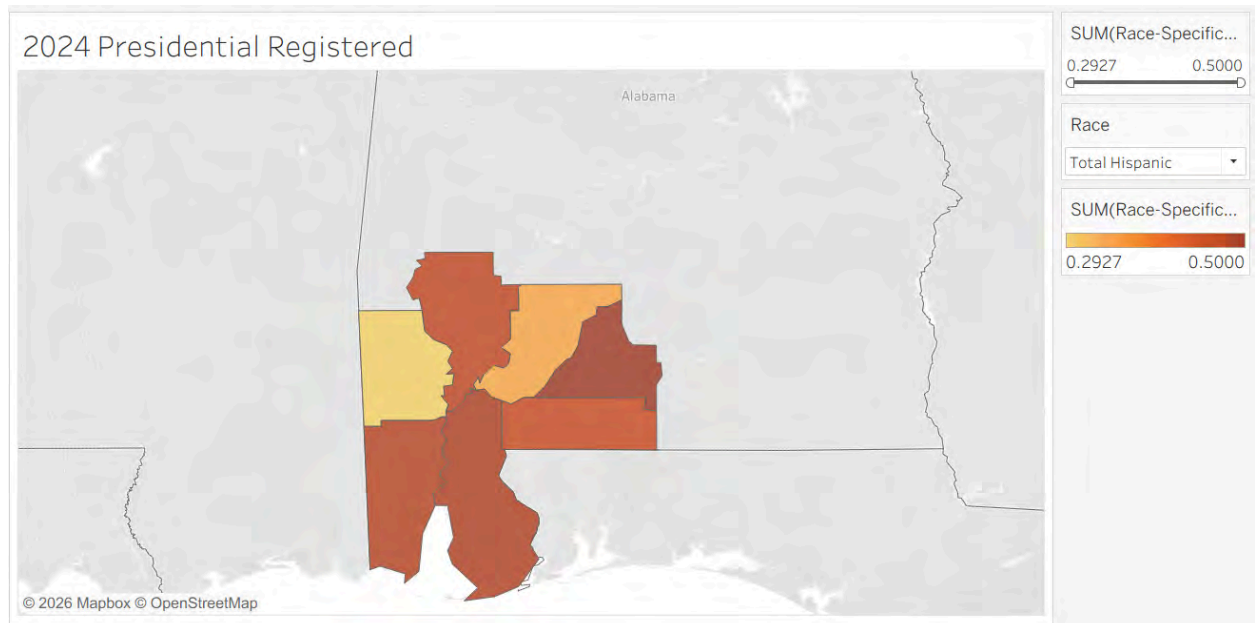
Figures 1 and 2 (1a, 2a show 2020, 1b, 2b show 2024) show charts displaying the ballots by racial demographic and the voter eligible population by demographic for each county. The smaller darker colors within each item show their relative percentages for each demographic. These also display the numbers as tooltips within Tableau. *Racial Voter Representation* displays whether these numbers are over or underrepresented compared to their voter eligible populaces.

Figure 3a:



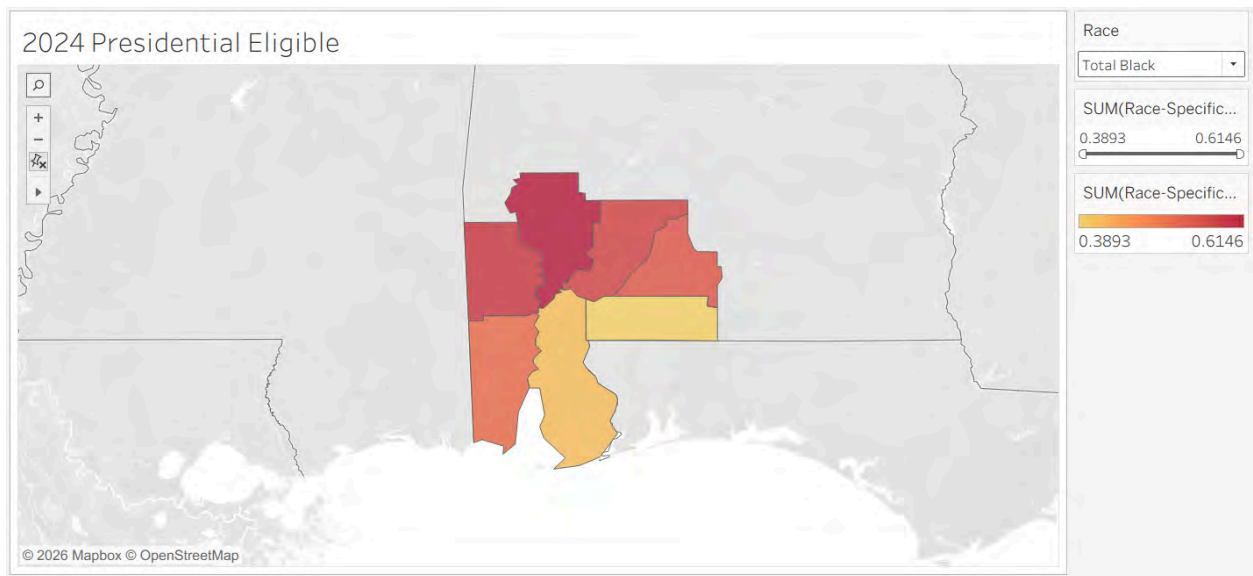
This Tableau heatmap depicts the proportion of eligible Hispanic voters that actually cast ballots in the 2024 Presidential and Vice Presidential Election. Washington county has the lowest proportion of 0.0984. There is a dropdown menu that depicts similar proportions for other races, including Black, Asian, and White eligible populations by county in Alabama.

Figure 3b:



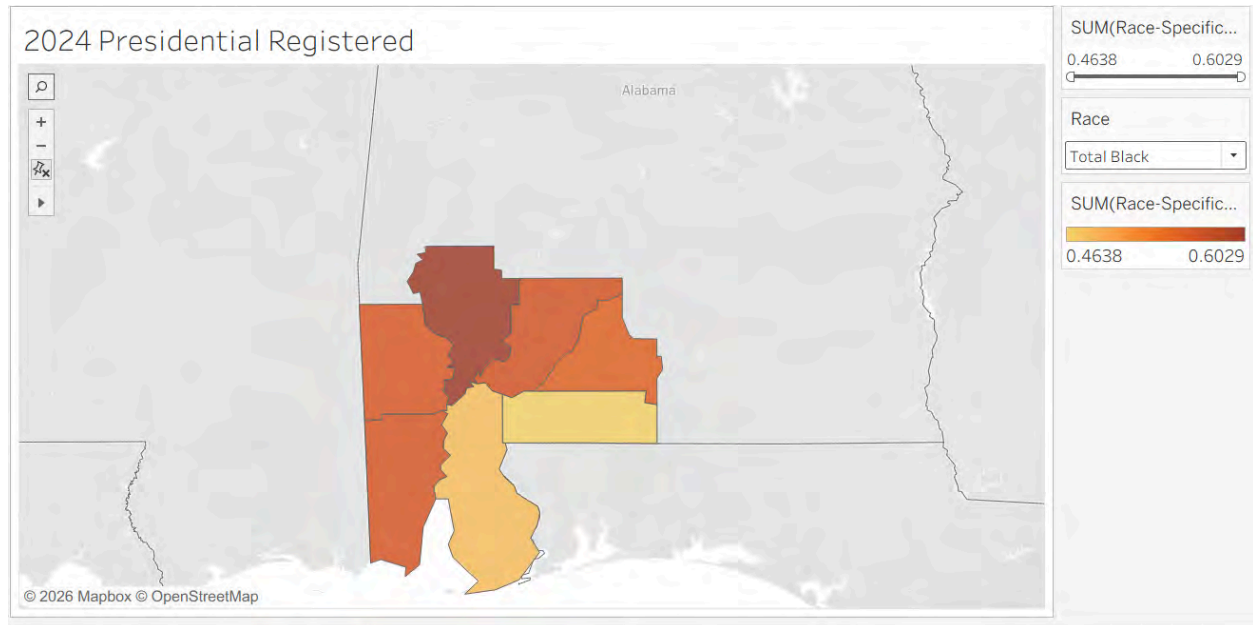
This Tableau heatmap depicts the proportion of registered Hispanic voters that actually cast ballots in the 2024 Presidential and Vice Presidential Election. There is a dropdown menu that depicts similar proportions for other races, including Black, Asian, and White eligible populations by county in Alabama.

Figure 3c:



This Tableau heatmap depicts the proportion of eligible black voters who actually cast ballots in the 2024 Presidential and Vice Presidential Election. There is a dropdown menu that depicts similar proportions for other races, including Hispanic, Asian, and White populations by county.

Figure 3d:



This Tableau heatmap depicts the proportion of registered black voters that actually cast ballots in the 2024 Presidential and Vice Presidential Election. There is a dropdown menu that depicts similar proportions for other races, including Hispanic, Asian, and White eligible populations by county in Alabama.

[Slide Deck Presentation \(Deliverable 5\)](#): A Google Slides presentation that displays the summarized findings and methodologies across the course section. Additionally, this presentation provides potential recommendations for SPLC to address some of the concerns discovered through the analysis.

[Final Report \(Deliverable 6\) \(This document\)](#): A document containing a comprehensive report that documents the project process from start to finish. It includes all of the necessary context, descriptions, methods, analysis, visualizations, findings, and recommendations from the project. This file allows any reader to understand the project, process, and conclusions made by our team.

Recommendations

Recommendations for the client (SPLC)

From the results of the turnout analysis and the dashboards, we recommend that the SPLC focus their efforts in 2 distinct buckets. One focusing on overall participation from different racial groups, for instance there was relatively low participation of Hispanic voters in Washington County compared to the number of Hispanic individuals eligible to vote during the Presidential Election. This number was low in both 2020 and 2024. Therefore, this demographic in this county is worth a deeper dive. The second bucket is the trend data between 2020 and 2024 for voter turnout. A notable instance here is the sharp dropoff throughout all the major demographic groups in Conecuh county from 2020 to 2024.

Looking at the first bucket, there was relatively high participation in voting by the Hispanic individuals who were registered voters, the significantly low turnout compared to the total Hispanic population that is eligible for voting indicates that the real challenge might not be with turnout, but with voter registration. Consequently, SPLC can benefit from focusing on outreach work aimed at registering Hispanic voters in Washington County. The results additionally suggest that Escambia and Baldwin counties should also be examined concerning black voter turnout. In most parts of the data, the participation rate of black individuals was almost identical to their representation among the population of voting age. Yet in Escambia County, a certain inconsistency between the proportion of black people who are qualified to vote and the proportion of the ballots received can be seen. Moreover, Baldwin and Escambia counties had one of the highest discrepancies between demographics and ballots among both Hispanic and white communities. Potential reasons for these discrepancies can potentially be sourced through our other project teams.

Viewing the second bucket, Conecuh was not the only county to see a decrease in voter turnout. Monroe and Washington counties also saw decreases in voter turnout. We have not done analysis on how this compares to the country's overall voter turnout in these elections, maybe this was something seen everywhere, but it does warrant a closer look by the SPLC on why there was such a large decrease in these counties.

The visuals generated by this project will assist the SPLC in geographical resource allocation. The SPLC is enabled to determine which counties have problems based on ballot votes in relation to voter registration as opposed to voting eligible population. These visuals help target which counties have voter registration and turnout issues. This may influence decisions on the nature of interventions that need to be undertaken.

It is imperative to note that the results have been obtained while taking into consideration various limitations with respect to the available data. The turnout rates used were mostly based on CVAP numbers calculated from the American Community Survey, which means that it can only be considered an estimate and not an accurate measure. Similarly, the fact that no demographic participation data was available for congressional races meant that we had limited capacity to analyze demographic turnouts during other elections other than those for the presidency.

Recommendations for future project teams

Future project teams must try to start their data collection and validation process as early as possible because one of the biggest hurdles for this project involved trying to find fully formatted election data for the required years. While the data collected from the Alabama Secretary of State were found to be somewhat consistent and reliable across the different years, substantial effort was spent cleaning, merging, and formatting the files before analysis could take place. Future project teams would do well to first establish a dataset inventory and document file naming issues, inconsistencies, and assumptions made during data processing.

The process that was used during this project involved Python and pandas software and was highly effective in managing the large data sets associated with elections, and should therefore be maintained as a base for future analysis. Creating a standardized database early was key, and is recommended to whomever undertakes a similar process in the future.

A key takeaway from this assignment would be how demographic turnout analysis can require using datasets which were created for different objectives and under different systems of classification. The next groups working on the assignment need to keep track of the assumptions made, especially in relation to comparison between Alabama Secretary of State categories of demographics and the CVAP of American Community Survey. Our team had to make some

assumptions to make turnout analysis possible, and this can limit further studies. Finding more detailed demographic participation data at the contest level should be a priority in future teams' efforts. In addition to other weaknesses, one of the biggest challenges in the project was that our team was unable to find demographic participation data in order to analyze the 2nd Congressional District races, as the data was available only for the overall election participation data. Instead of speculating and making up estimates, the team deliberately did not include any data under those demographics in order to ensure analysis integrity.

Conclusions

The GC-VOTE project focused on collecting, cleaning, organizing, and analyzing voter turnout data across Gulf Coast counties in Alabama for the 2020 and 2024 election periods. Using publicly available election datasets from the Alabama Secretary of State and Citizen Voting Age Population (CVAP) estimates from the Redistricting Data Hub, the team created a structured dataset that enables analysis of voter turnout patterns by county and demographic group. This work contributes to the broader regional analysis conducted for the Southern Poverty Law Center (SPLC) by providing county-level election participation data and demographic turnout estimates for the Gulf Coast region.

Throughout the course of the project, the team worked extensively to clean, transform, standardize, and merge data from multiple sources into a consistent and usable format. In addition to creating the final analytical dataset, the project carefully documented the methodologies, assumptions, limitations, and data transformations used throughout the process in order to improve transparency, reproducibility, and interpretability. Particular attention was given to explaining the assumptions necessary to compare datasets collected under different classification systems and to documenting the limitations associated with using estimate-based CVAP data for turnout calculations.

The analysis and accompanying visualizations identified several counties that may warrant additional investigation regarding voter participation disparities. For example, Washington County demonstrated notably low Hispanic voter participation relative to its estimated voting-eligible Hispanic population, suggesting potential opportunities for voter registration and outreach efforts.

Similarly, Escambia and Baldwin counties displayed larger discrepancies between demographic population shares and actual ballot shares for certain racial groups, indicating that additional research into local participation barriers or political engagement patterns may be valuable. In addition to racial county trends, the difference in counties for voter turnout for each election cycle warrants further investigation, notably Conecuh county dropped significantly in voter turnout across all major racial demographics. Monroe and Washington did not show quite as extensive of a drop, but it was a notable drop nonetheless. While the project does not establish causal explanations for these trends, it provides a structured framework for identifying counties, demographic groups, and trends that may benefit from further analysis or community engagement efforts.

The project also highlighted several broader challenges associated with election data analysis. The lack of contest-specific demographic participation data for congressional races limited the team's ability to extend demographic turnout analysis beyond presidential elections. Additionally, the process of locating reliable, county-level datasets that were complete and consistently formatted across election years proved to be one of the most time-consuming aspects of the project. These limitations reinforced the importance of careful documentation, standardized workflows, and transparent assumptions when working with large public datasets. Despite these challenges, the project demonstrates how publicly available election and demographic data can be transformed into a more accessible and interpretable resource for regional voting analysis. Through collaboration, data standardization, visualization development, and methodological documentation, the GC-VOTE team produced deliverables that can support future research, policy discussions, and ongoing regional election analysis by SPLC and future student teams. The cleaned datasets, Tableau dashboard, and methodological framework established through this project also provide a strong foundation for future election-cycle analyses and expanded research into voter participation trends across Alabama's Gulf Coast counties.

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Deliverables Checklist

Deliverable Folder:

https://drive.google.com/drive/folders/14Jjd4M5omuRy5wSltrKWKDI7-L_PpFv0?usp=sharing

- ☐ Deliverable 1 - Raw Original Data: [GC-VOTE RAW DATA.zip](#)
- ☐ Deliverable 2 - Cleaned and Standardized Dataset: [GC-VOTE 20-24 Voting Data.csv](#)
- ☐ Deliverable 3 - Methodology/Assumptions: [GC-VOTE Methodology Assumptions File.pdf](#)
- ☐ Deliverable 4 - Interactive Dashboard: [GC-VOTE Tableau Interactive Dashboard.twbx](#)
- ☐ Deliverable 5 - Slide Deck Presentation: [INST490-105 SPLC Gulf Coast Counties Final Presentation Template Final](#)
- ☐ Deliverable 6 - Final Report: [INST490-105 SPLC Gulf Coast Counties Final Report](#)

Polling Place Access & Distance Analysis

GC-PAD

Casey Pham (Project Manager), Ana Salazar, Keon Feizy-Marandy, Khadija Wane, Nhi Pham

Southern Poverty Law Center (SPLC)

INST490: Integrated Capstone for Information Science

Section 0105

May 18th, 2026

Key Findings

- The distance analysis and statistical testing found that Census tracts with higher combined Black/Latine population shares generally had shorter average driving times to assigned polling places, contrary to the original hypothesis.
- Statistical analyses, including Pearson correlations, Spearman correlations, linear regressions, Welch's t-tests, and Wilcoxon rank-sum tests, consistently showed a statistically significant negative relationship between Black/Latine population share and polling place drive time.
- The strongest inferential support came from the log-transformed regression model, which explained approximately 22.9% of the variance in drive time and better satisfied normality assumptions.
- Supplemental GIS visualization layers suggested that the observed negative relationship was likely influenced by the concentration of Black and Latine populations in more urbanized areas of Mobile County, where polling places and precincts are geographically closer together.

Deliverables Produced

Deliverable: 2020 Decennial Census demographic profile

File name & link:  [GCPAD-DistanceAnalysis-2020DecennialCensus-v1.xlsx](#)

Description: 2020 Decennial Census demographic data

Deliverable: County subdivision's Black population

File name & link: [GCPAD-DistanceAnalysis-CCDBlackPop-v1.csv](#)

Description: Black population for each county subdivision according to the 2020 Decennial Census

Deliverable: County subdivision's Hispanic population

File name & link: [GCPAD-DistanceAnalysis-CCDHispanicPop-v1.csv](#)

Description: Hispanic population for each county subdivision according to the 2020 Decennial Census

Deliverable: Cleaned 2020 Decennial Census data

File name & link: [GCPAD-DistanceAnalysis-Clean2020DecennialCensus.csv](#)

Description: Cleaned 2020 Decennial Census demographic data. Output from Python script.

Deliverable: County subdivisions

File name & link: [GCPAD-DistanceAnalysis-CountySubdivisionBoundaries-v1.zip](#)

Description: County subdivisions layer used for ArcGIS online visualization. From the 2020 Decennial Census

Deliverable: Demographic data preprocessing script

File name & link: [GCPAD-DistanceAnalysis-DataCleaning-v1.ipynb](#)

Description: Jupyter Notebook that includes Python scripts. Used to clean census data from the 2020 Decennial Census. Used to create schema.ini file required for census data to be read by ArcGIS as strings to maintain zero padding.

Deliverable: Centroids linked to precincts

File name & link: [GCPAD-DistanceAnalysis-MobileCentroidsSpatialJoin-v1.zip](#)

Description: Mobile Centroids spatially joined with precincts. Centroids from the 2020 Decennial Census, precinct boundaries from Mobile County's GIS division.

Deliverable: ArcGIS Pro project & ModelBuilder/Network Analyst model

File name & link: [GCPAD-DistanceAnalysis-ModelBuilder&NetworkAnalyst-v1.aprx](#)

Description: ArcGIS Pro project including the ModelBuilder w/ Network Analyst model. Model outputs distance analysis data, which is then joined to cleaned demographic data output from Python script.

Deliverable: Network analysis & GIS data processing report

File name & link:  [GCPAD-DistanceAnalysis-NetworkAnalysis&GISDataProcessingReport-v1](#)

Description: Network analysis and GIS data processing report, including methods, findings, and conclusions.

Deliverable: Mobile County 2020 polling sites

File name & link: [GCPAD-DistanceAnalysis-PollingSites-v1.zip](#)

Description: Polling site locations in Mobile County in 2020 according to Mobile County's GIS division

Deliverable: Precincts with demographic information

File name & link: [GCPAD-DistanceAnalysis-Precincts&Demographics-v1.csv](#)

Description: 2020 Precinct boundaries from Mobile County's GIS division joined w/ Black and Latine demographic data from the 2020 Decennial Census

Deliverable: schema.ini

File name & link: [GCPAD-DistanceAnalysis-SchemaForCleanCensusData-v1.ini](#) [schema.ini]

Description: Output from Python script. Needed so that cleaned demographic data from Census can be read as strings in ArcGIS pro (maintain zero padding).

Deliverable: Statistical analysis script

File name & link: [GCPAD-DistanceAnalysis-StatisticalAnalysis-v1.R](#)

Description: R script to perform statistical analysis

Deliverable: Statistical analysis output

File name & link: [GCPAD-DistanceAnalysis-StatisticalAnalysisOutput-v1.csv](#)

Description: Dataset output by statistical analysis on R Studio

Deliverable: Statistical analysis report

File name & link:  [GCPAD-DistanceAnalysis-StatisticalAnalysisReport-v1](#)

Description: Statistical analysis report, including methods, findings, and conclusions.

Deliverable: Mobile County 2020 voter precincts

File name & link: [GCPAD-DistanceAnalysis-VoterPrecincts-v1.zip](#)

Description: Voter precincts in Mobile County in 2020 according to Mobile County's GIS division

Deliverable: ArcGIS Visualization Layers

File name & link:  [GCPAD-ArcGIS-Visualization-Layers-v1](#)

Description: Static screenshots for each visualization layer in the Mobile County project in ArcGIS

Challenges

The most significant obstacles involved using ArcGIS ModelBuilder and Network Analyst to automate precinct-level travel time analysis. These tools required troubleshooting complex workflow issues like connecting correct layers, filtering precincts, and generating an origin-destination matrix. These challenges were addressed by consulting online documentation and getting support from the University of Maryland's (UMD) GIS Data and Service Center.

Connections to Other Teams

While drive time did not provide the primary mechanism for electoral inequities, it is important to consider other factors that impact civic participation. Other variables like transportation barriers (explored by GC-TRAC) and Americans with Disabilities Act (ADA) accessibility (explored by GC-AACS) contribute to civic participation. The Gulf Coast Composite Civic Participation Risk Index (CRPI), created by GC-CPRI, establishes a county-level risk score by combining these indicators to assess what communities have the highest risk of decreased civic participation.

Final Report and Deliverables

Project Context

This project evaluated polling place access in Mobile County, Alabama by analyzing the relationship between Black/Latine population share and modeled driving time to assigned polling

places. The project combined Geographic Information Systems (GIS) based distance analysis with statistical reasoning to create a tract-level accessibility dataset and evaluate whether historically marginalized communities experienced longer travel burdens when voting. The work contributed to the border regional analysis by providing both a reproducible ArcGIS workflow and a statistical framework for evaluating polling place access using spatial and demographic data. The project depended on Census demographic data, Mobile County GIS election datasets, and ArcGIS Network Analyst's origin-destination matrix output. The resulting tract-level dataset and statistical findings support future investigation in civic participation, travel time-based access, and spatial inequities.

Methods

The project used ArcGIS Pro ModelBuilder and Network Analyst to calculate realistic driving times between Census-derived population centroids and assigned polling places within Mobile County precincts. The workflow iterated through precincts individually, selected population centroids and polling places spatially, generated OD Cost Matrix layers, solved for least-cost travel routes along the road network, and appended the resulting travel records into a cumulative feature class.

Census demographic data from the 2020 Decennial Census were processed separately in Python using pandas and re libraries. Census tract identifiers were extracted, reformatted, standardized with leading zeros, and converted into join-ready GEOID and TRACTCE fields. These demographic tables were then joined to the OD COST Matrix output in ArcGIS Pro.

The final joined dataset was imported into RStudio for statistical analysis. Drive time variables were aggregated to the Census tract level and analyzed using descriptive statistics, Pearson and Spearman correlations, linear regression, Welch's two-sample t-tests, and Wilcoxon rank-sum tests. Because raw drive time distributions were positively skewed, a natural log transformation was applied to improve normality assumptions for inferential testing.

Several challenges were encountered during the workflow. The distance analysis required the use of ModelBuilder, which introduced technical complexity. This tool was used so that precinct lines could be incorporated as boundaries used to determine assigned polling places. ArcGIS formatting inconsistencies also required additional Python preprocessing to preserve tract identifiers. Additionally, three precincts had two polling places within their limits, which required averaging travel time values during aggregation. The project also addressed quality control through centroid-count failsafes, solve-success preconditions, and permanent export of temporary joins to improve reproducibility and reduce invalid outputs.

Deliverables and Findings

1. ArcGIS Pro ModelBuilder Workflow

The project established a reproducible ArcGIS Pro ModelBuilder workflow that estimated network-based driving access between Census tract population centroids and assigned polling places. This workflow improved upon straight-line distance methods by incorporating realistic road-network constraints, travel costs, and routing behavior, while simultaneously ensuring that polling places were assigned based on precinct boundaries.

2. Final Tract-linked Travel Time Dataset

The final analytical dataset joined network travel time results with 2020 Census demographic variables, including Black and Hispanic/Latine population counts and percentages. This dataset enabled tract-level statistical analysis while preserving spatial relationships between population distribution and polling place access. This dataset drove the statistical analysis and may support future analyses involving transportation access, civic participation, electoral geography, or voting equity.

3. Statistical Analysis Report

Assumption Checking and Performing Transformations

The statistical analysis report includes a summary of descriptive statistics, correlations, regression models, and two distinct tests that were utilized in our project (Wilcoxon rank-sum and Welch t-tests). The intended purpose of this analysis was to examine whether Census tracts that include higher Black and Latine population percentages encountered longer average driving times to their assigned polling place location in the year 2020. Assessing the relationship between driving time to polling locations and the target population of Black/Latine voters provides an opportunity to display if marginalized and underrepresented communities endure longer travel times, which can directly impact their motivations towards voting, being politically involved, and being able to exercise their right to vote fairly, without any preventable barriers.

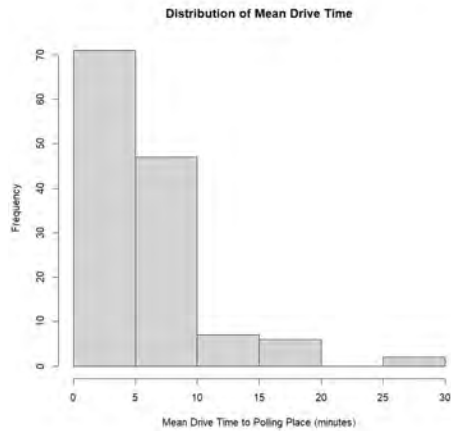


Figure 1. Histogram distribution of the Mean Drive Time to Polling Place (in minutes) by Frequency (Y).

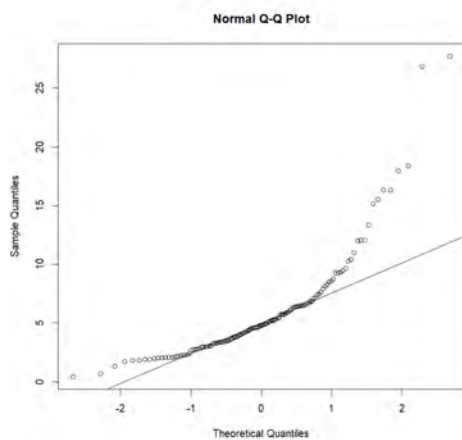


Figure 2. Q-Q Plot of the Theoretical Quantities (X) by Sample Quantities (Y). (Source: INST490 GC-PAD 2026) (Source: INST490 GC-PAD 2026)

The histogram displays the distribution of the mean drive time to the polling place (x) in minutes, with the frequency aligned to the right. A majority of the tracts had relatively short mean drive times (0 to 10 minutes), with a few tracts having times above that range (15 to 30 minutes). The Q-Q plot was created to assess normality between the theoretical quantities (x) and the sample quantities. Based on the curve, there is a distinct deviation in the upper tail. The mean drive time variable was not normally distributed based on the results of the Shapiro-Wilk test, with $W = 0.772$ and $p < 0.001$. From these two graphs, the mean drive time was positively skewed.

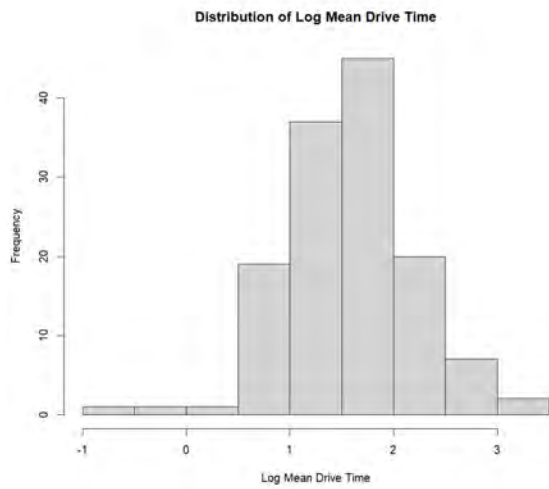


Figure 3. Transformed Log Histogram Distribution of Log Mean Drive Time (x) by Frequency (Y). (Source: INST 490 GC-PAD 2026)

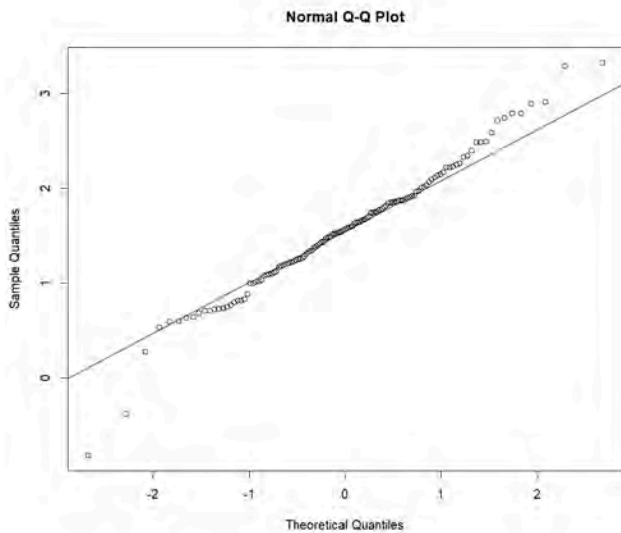


Figure 4. Transformed Log Q-Q Plot of Theoretical Quantities by Sample Quantities.(Source: INST490 GC-PAD 2026)

To apply a natural log transformation, the data was cleared of any zero or negative mean drive time values, as only positive values are needed. A new variable was created:
 $\text{log_mean_drive_time} = \log(\text{mean_drive_time})$. The Shapiro-Wilk test for the transformed log variant of the mean_drive_time variable was not statistically significant, as $W = 0.982$ and $p = 0.083$, meaning the log_mean_drive_time variable did not deviate, as visually noted by the transformed log

histogram and Q-Q plot. Using the log-transformed drive time for tests that are more sensitive to the normality assumption is supported.

Regression assumptions were evaluated using diagnostic plots and residual normality tests, and assessed through comparing residuals to fitted plots, Q-Q plots, scale-location plots, and residuals vs leverage plots. From the raw drive time variable modeled on the regression, it did not have a normal distribution with a Shapiro-Wilk of $W = 0.812$, $p < 0.001$. The residuals that were produced from the log-transformed regression model had no significant deviations from normality, with a Shapiro-Wilk of $W = 0.984$, $p = 0.125$, making this model more assumption-oriented. The raw drive time results provide audiences to understand size and results, and the transformed log results better suit model assumptions than not.

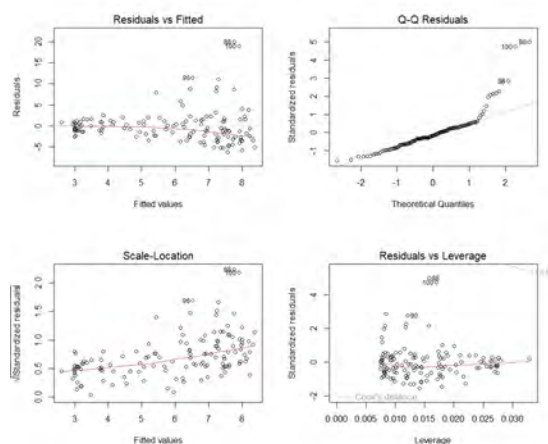


Figure 5. Regression for Diagnostic Plots for Raw Mean Drive Time. (Source: INST490 GC-PAD 2026)

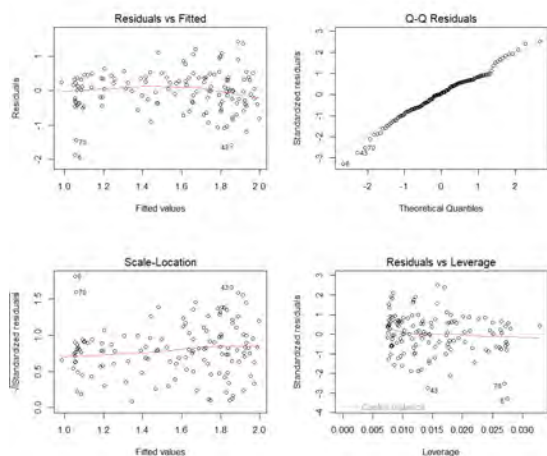


Figure 6. Regression for Diagnostic Plots for Log Transformed Mean Drive Time. (Source: INST490 GC-PAD 2026)

Descriptive Statistics

Across 133 Census tracts analyzed, the raw mean drive time variable had a range of 0.44 minutes to 27.21 minutes. The mean was 5.89 minutes and the standard deviation (sd) of 4.40 minutes. The median was 4.80 minutes, leading to a right-skewed distribution. The transformed log mean drive time had an average and median of 1.56 and a standard deviation of 0.65. Tracts in a group-based analysis were categorized using a 38.49% combined Black and Latine share threshold, which represents Mobile County and its wide percentage of residents who identify as Black or Latine. Tracts that include a combined Black/Latine population percentage greater than or equal to the number above (38.49%) were classified as lower Black/Latine tracks, while higher Black/Latine tracks served as a proxy for said communities.

An R Script using R Studio software was created for this grouping variable, and the factor order was established so our target group (Black/Latine) group was compared against the lower Black/Latine group in directional testing. This produced 65 higher Black/Latine tracts and 68 lower Black/Latine tracts. Observations of 30 were exceeded for both groups, emphasizing the use of large-sample procedures for our methodology (such as Welch's t-test), though the raw outcome was skewed, causing normality to be assessed. The higher Black/Latine group that had a higher share threshold had a lower raw mean drive time, with $M = 4.34$ minutes, $SD = 2.42$ minutes, while the lower Black/Latine group had a higher raw mean drive time ($M = 7.38$ minutes, $SD = 5.28$). For the transformed log scale, the higher Black/Latine group had a lower mean and standard deviation value ($M = 1.32$, $SD = 0.59$), whilst the lower Black/Latine group ($M = 1.80$, $SD = 0.63$).

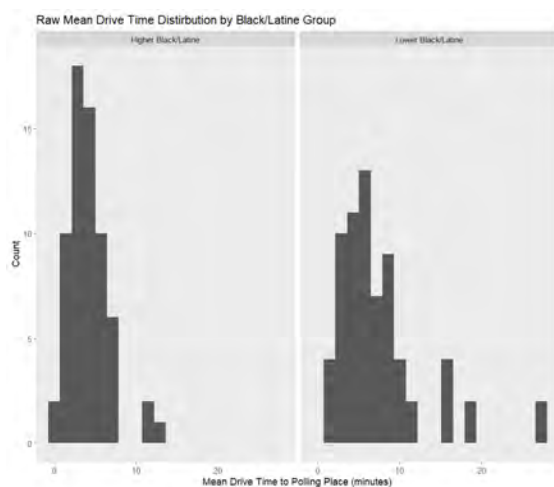


Figure 7. Raw Mean Drive Time Distribution by Black/Latine Group. (Source: INST490 GC-PAD 2026)

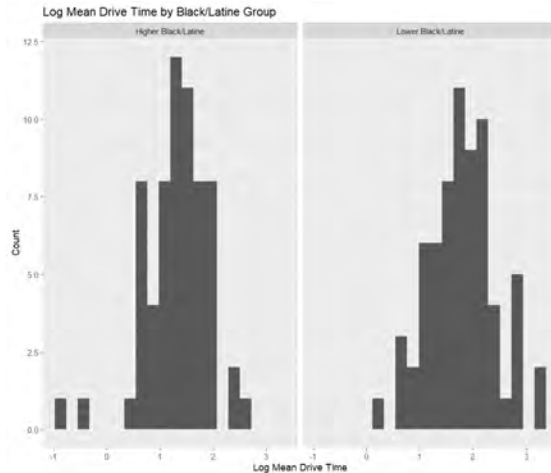


Figure 8. Log Transformed Mean Drive Time by Black/Latine Group. (Source: INST490 GC-PAD 2026)

Pearson Correlation

To assess the linear relationship between the combined Black/Latine population share and the raw mean drive time, a Pearson product-moment correlation was applied. Our null hypothesis states that there is no relationship between the combined Black/Latine population share and mean drive time, and the alternative hypothesis assumes there is a relationship between both variables.

The correlation was statistically significant and negatively correlated, with $r(131) = -0.408$, $p < 0.001$, 95% CI $[-0.541, -0.255]$, meaning the relationship between these two variables is a negative linear. The raw mean drive time was positively skewed, which provides an estimate of the relationship and confirms the negative linear relationship, but should be interpreted with caution.

Pearson Correlation (Log Transformed Mean Drive Time)

Using the variable, `log_mean_drive_time`, another Pearson correlation was made, as the log-transformed data have, to this point, fulfilled normality and suited model assumptions. The relationship between the combined Black/Latine population share and log-transformed mean drive time was statistically significant and negative, $r(131) = -0.479$, $p < 0.001$, 95% CI $[-0.600, -0.336]$, a stronger result than the raw Pearson correlation with the negative linear relationship continuing. Both the raw and log-transformed correlations show the negative association and overall statistical significance.

Spearman Rank-Order Correlation

A Spearman rank-order correlation was conducted between the combined Black/Latin population share and mean drive time, as a way to enact a nonparametric test and assess the monotonic relationship. This was utilized as Spearman correlation is less sensitive to non-normality

outcomes and outliers compared to Pearson. This test was performed on the raw mean drive time data without violating assumptions. Tied ranks were present in the data and required an approximate p-value. In the R Script, the line "exact = FALSE" was used. The Spearman correlation was statistically significant and negative, with an exact p value of -0.451, $p < 0.001$. This relationship is negative and monotonic, as Black/Latine population share increased, the mean drive time tended to decrease in ranked order. The results from Pearson correlation were supported, as the data still illustrated a significant negative relationship between the combined Black/Latine population percent and travel time.

Linear Regression

Raw Mean Drive Time

To determine if the combined Black/Latine population share predicted the raw mean drive time, a simple linear regression was conducted. The null hypothesis was that the regression slope = 0, indicating that Black/Latine population share did not predict mean drive time. The alternative hypothesis was that the slope would differ from being equal to 0.

The raw regression model had statistical significance, $F(1, 131) = 26.16$, $p < 0.001$, $R^2 = 0.166$. The combined Black/Latine population share significantly predicted mean drive time, $b = -0.056$, $SE = 0.011$, $t(131) = -5.115$, $p < 0.001$, 95% CI [-0.079, -0.035]. The equation for regression was $y = 8.58 - 0.056x$, where y is the mean drive time and x is Black/Latine percentage. Each one percentage-point increase in the combined Black/Latine population share was associated with an estimated 0.056 minute decrease in driving time, meaning a 10 percentage-point increase in the combined Black/Latine population share was associated with 0.56 fewer minutes driving (~34 seconds shorter travel time).

The raw regression residuals were not normally distributed, as Shapiro-Wilk $W = 0.812$, $p < 0.001$. This model is useful for interpreting minutes, but is not strong enough for inferential analysis.

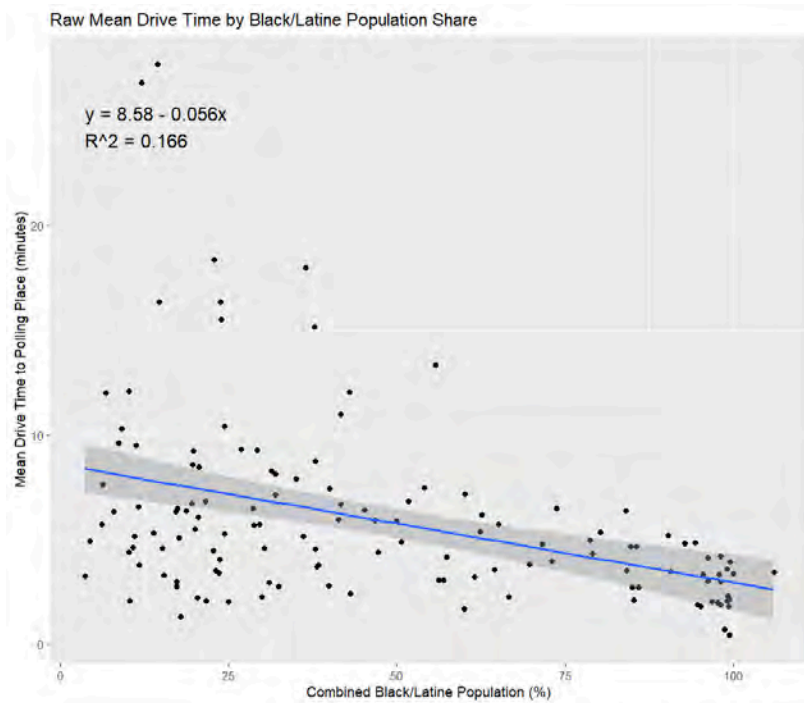


Figure 9. Raw Mean Drive Time to Polling Place (in minutes) by the Combined Black/Latine Population (%). (Source: INST490 GC-PAD 2026)

Log Transformed Mean Drive Time

A second linear regression model using the log-transformed mean drive time variable, this time as the dependent variable (Y). As log-transformed models and model residuals better meet normality assumptions, this was appropriate to utilize. The log-transformed regression model was statistically significant, $F(1, 131) = 39.01$, $p < 0.001$, $R^2 = 0.229$. The combined Black/Latine population share significantly predicted log-transformed mean drive time, $b = -0.0098$, $SE = 0.0016$, $t(131) = -6.246$, $p < 0.001$, 95% CI $[-0.0013, -0.007]$. The regression equation was $\log(y) = 2.03 - 0.0098x$, where y is the mean drive time and x is Black/Latine percentage. The negative coefficient means that a higher combined Black/Latine population share was associated with lower expected drive time.

As the outcome was log transformed, the coefficient can be interpreted as a percentage change, where each one percentage point increase in the combined Black/Latine population share was associated with a 0.98% decrease in mean drive time. The model displayed approximately 22.9% of the variance in the log-transformed mean drive time.

The residuals from this model did not significantly deviate from normality, the Shapiro-Wilk $W = 0.984$, $p = 0.125$. This model has far stronger inferential support than the raw regression model above.

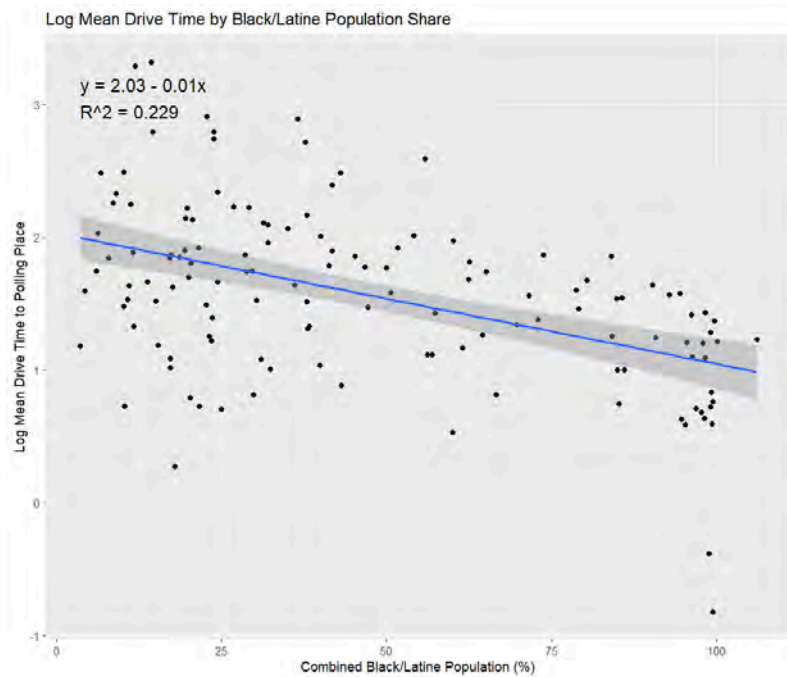


Figure 10. Log Transformed Mean Drive Time to Polling Place (in minutes) by the Combined Black/Latine Population (%). (Source: INST490 GC-PAD 2026)

Right-Tailed Welch's Two-Sample t-test

A right-tailed Welch's two sample t-test was conducted to assess whether the higher Black/Latine tracts had greater raw drive time for higher Black/Latine tracts. The null hypothesis was that the mean drive time for higher Black/Latine tracts was less than or equal to the mean drive time for lower Black/Latine tracts. The alternative hypothesis was that higher Black/Latine tracts had greater mean drive times.

The raw Welch's test hypothesized direction was not statistically significant, as $t(94.85) = -4.30$, $p = 1.00$. The higher Black/Latine group had a lower mean drive time ($M = 4.34$ minutes) than the lower Black/Latine group ($M = 7.38$ minutes). The one sided 95% confidence interval for the difference was $[-4.22, \infty]$, which aligns with the observed difference not supporting the predicted direction.

The raw mean drive time was non-normal and right-skewed; this raw test was used primarily as a descriptive and sensitivity comparison. The raw mean is helpful as it communicates the group difference in minutes, though the log-transformed Welch's test is more appropriate for inference purposes.

Log Transformed Drive Time

The log-transformed mean drive time was also assessed using the right-tailed Welch's two-sample t-test. This was the primary parametric group comparison, as log-transformed drive time better met normality assumptions.

The log-transformed Welch's test was not statistically significant in the hypothesized direction, $t(130.89) = -4.53$, $p = 1.00$. The higher Black/Latine tracts had lower log transformation mean drive times ($M = 1.32$) than lower Black/Latine tracts ($M = 1.80$). The one-sided 95% confidence interval for the difference was $[-0.66, \infty]$. After transformation, no evidence shows that higher Black/Latine tracts had longer drive times; the observed result is that the difference was opposite the predicted direction.

Wilcoxon Rank-Sum Test

A right-tailed Wilcoxon rank-sum test that included continuity correction was conducted as a nonparametric sensitivity analysis. This test does not require data to be normally distributed and examines which group may have larger values than the other. In this case, the null hypothesis is that higher Black/Latine tracts did not have greater drive times than lower Black/Latine tracts. The alternative hypothesis was that higher Black/Latine tracts tended to have greater drive times. This test was not statistically significant, with $W = 1281$ and $p = 100$. It was, however, consistent with Welch's test and provided more evidence for higher Black/Latine tracts not having greater drive times.

Summary of Findings

In all of the analyses conducted, the outcomes failed to reject the null hypothesis. No evidence from each available model or test used supported the hypothesis that higher Black/Latine tracts experienced longer drive times to assigned polling place locations. The tests indicated a significant negative association or a group difference opposite the predicted direction.

The Pearson correlation displayed a higher combined Black/Latine population share equated to a shorter drive time for both the raw, $r(131) = -0.408$, $p < 0.001$, and log-transformed drive time $r(131) = -0.479$, $p < 0.001$. The Spearman correlation included a significant negative association that was monotonic, with $n, p = -0.451$, $p < 0.001$. The regression analysis indicated that the combined Black/Latine population share significantly predicted shorter drive times. The log-transformed regression model was the most appropriate, as it showed that Black/Latine population share explained approximately 22.9% of the variance in log-transformed drive time.

The group comparison tests failed to support the directional hypothesis, as the raw and log-transformed right-tailed Welch's test had p-values of 1.00 because the observed group averages were opposite the hypothesized direction. The Wilcoxon rank-sum test had also failed to support the null hypothesis.

4. ArcGIS Visualization Layers

Visualization layers in ArcGIS were meant to supplement our analysis by demonstrating how Black/Latine voters are represented throughout Mobile county. Each layer was created in ArcGIS using census data and gradated accordingly. The layers themselves contain relevant data that contributes to the project, such as above threshold centroid layers (as used in the Welch's t-test), polling sites, population density, precinct lines, census tracts, and Black/Latine population by county subdivision.

Population Density Gradient

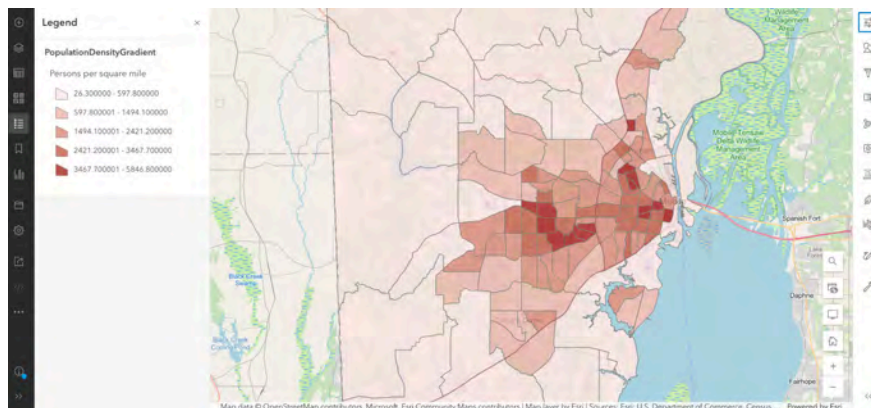


Figure 11. Population Density Gradient Map. (Source: INST490 GC-PAD 2026)

Above Black/Latine Threshold

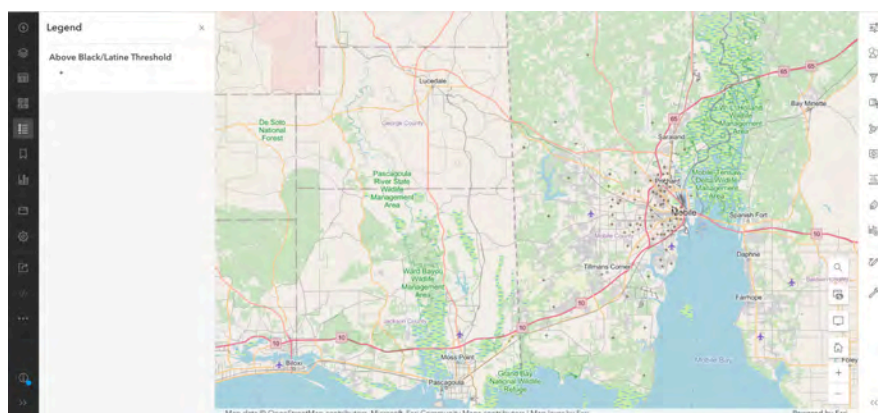


Figure 12. Above Black/Latine Threshold Layer. (Source: INST490 GC-PAD 2026)

Polling Sites

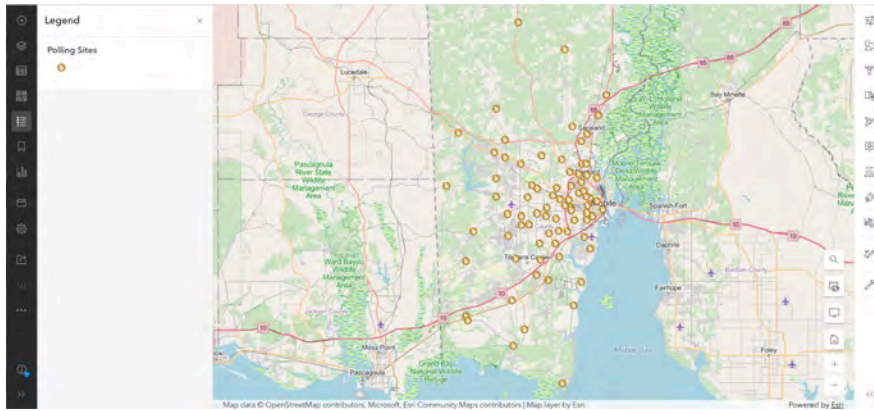


Figure 13. Polling Sites Map Layer. (Source: INST490 GC-PAD 2026)

Centroids

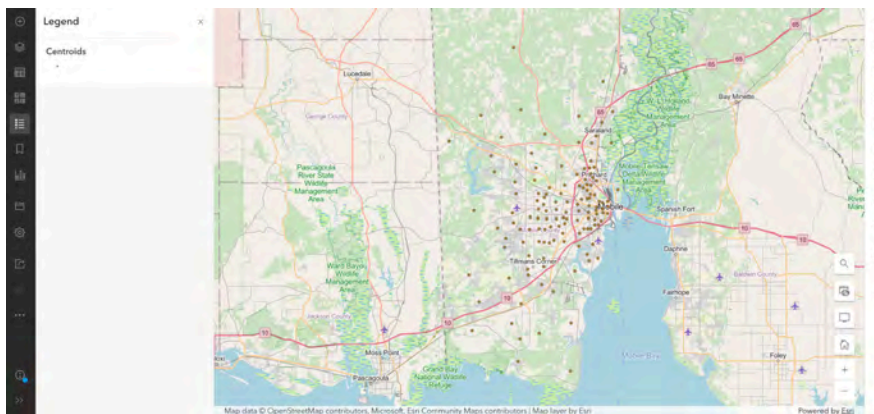


Figure 14. Centroids Layer. (Source: INST490 GC-PAD 2026)

Centroids to Polling Locations

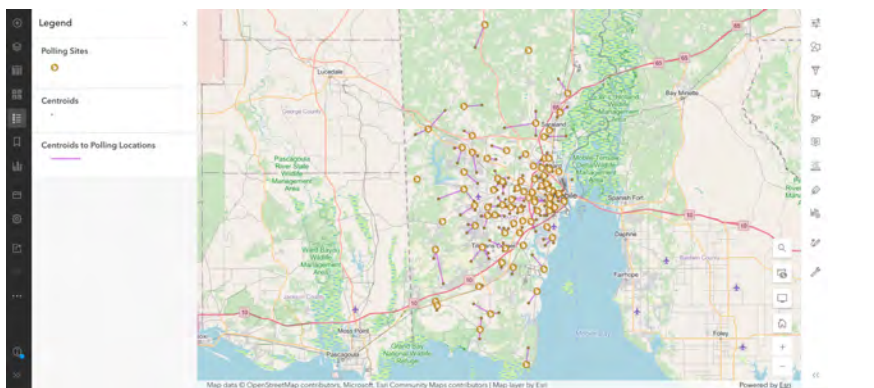


Figure 15. Centroids to Polling Locations Layer. (Source: INST490 GC-PAD 2026)

Tract Boundaries

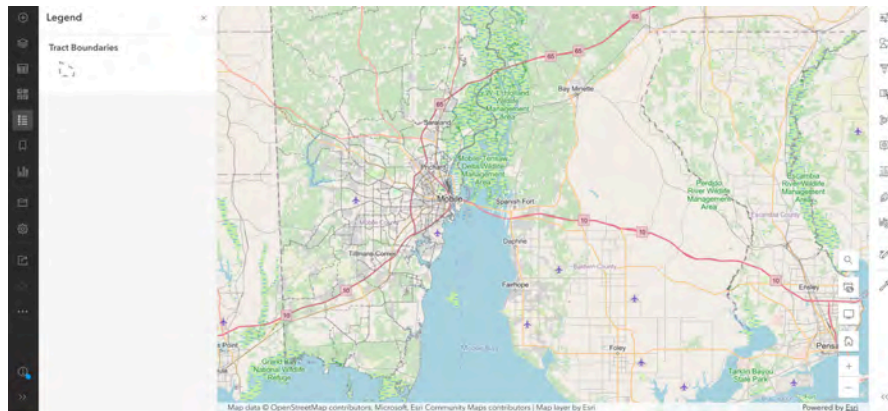


Figure 16. Tract Boundaries Layer. (Source: INST490 GC-PAD 2026)

Precinct Boundaries



Figure 17. Precinct Boundaries Map Layer. (Source: INST490 GC-PAD 2026)

CCDHispanic Gradient

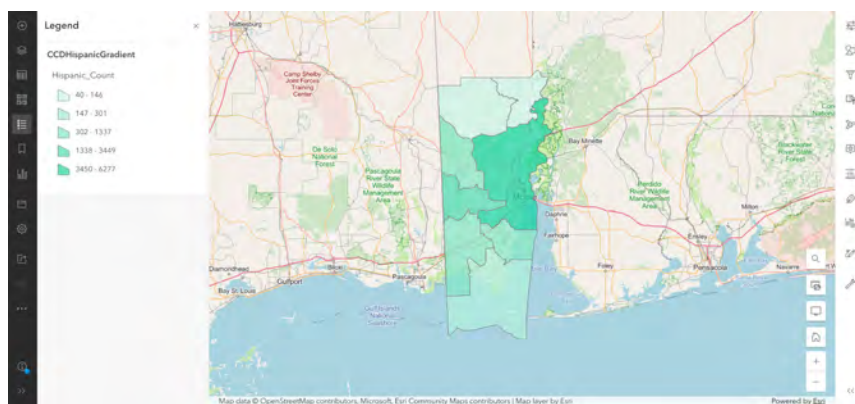


Figure 18. CCDHispanic Gradient Map Layer. (Source: INST490 GC-PAD 2026)

CCDBlack Gradient

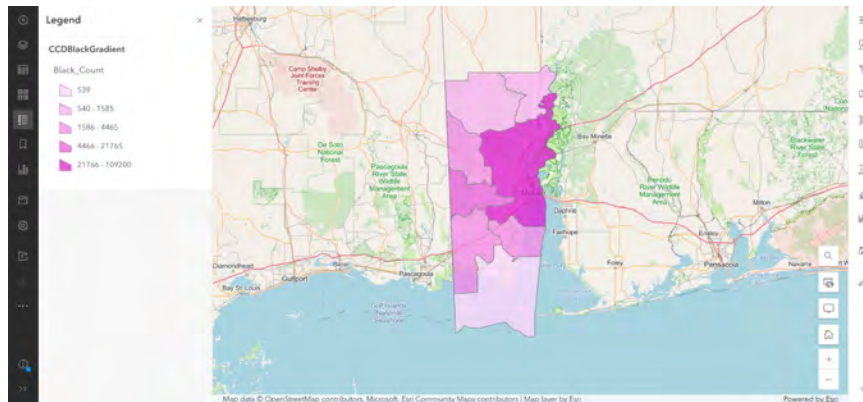


Figure 19. CCDBlack Gradient Map Layer. (Source: INST490 GC-PAD 2026)

TractsByColor

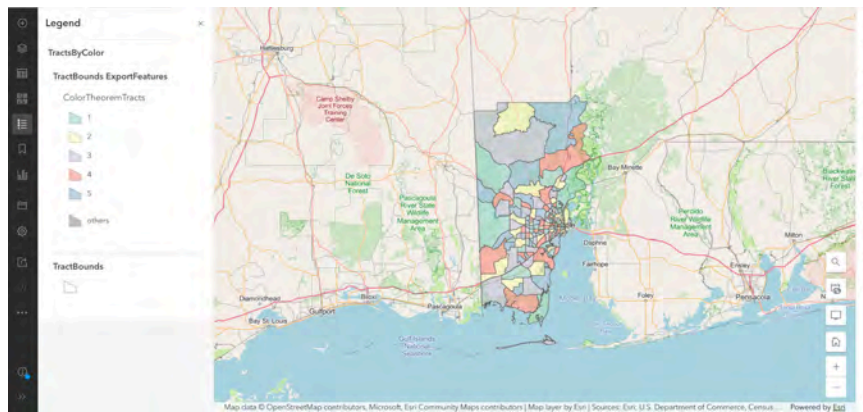


Figure 20. TractsByColor Map Layer. Source: (Source: INST490 GC-PAD 2026)

5. Final ArcGIS Map

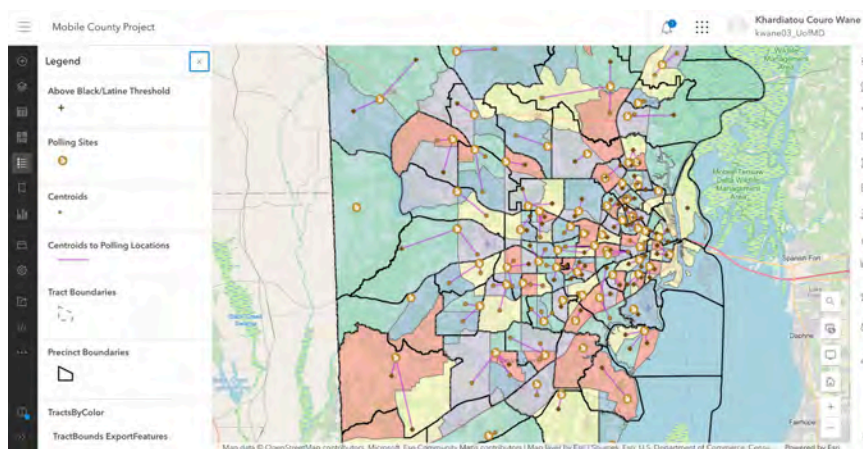


Figure 21. Final ArcGIS map containing polling place accessibility, distribution of the demographic data of Black/Latine voters, and urban density patterns in Mobile County, Alabama. (Source: INST490 GC-PAD 2026)

Recommendations

Recommendations for the client (SPLC)

This analysis suggests that polling place drive time alone is not the primary mechanism responsible for civic engagement patterns in Mobile County. It is important to recognize that the findings do not support a lack of barriers, but rather that polling place access for Black and Latine communities was not significantly negatively impacted by drive time. Geographic context helps explain these patterns; many higher Black and Latine population tracts were concentrated in denser urban areas. These densely populated locations consist of precincts with narrower boundaries and, therefore, inherently have shorter drive times from population centers. Based on this, SPLC should investigate other mechanisms impacting civic engagement, being careful to consider confounding variables like geographic context in this specific case.

Future work in expanding this methodological approach to assess polling place access across other areas of interest may also provide valuable insight into other geographic trends. This methodology is not limited to counties nor racial/ethnic demographic data; the project can be adapted to locations of varying sizes or measure the impact of other community trends like vehicle ownership or socioeconomic status.

Recommendations for future project teams

Future project teams should use tools like Network Analyst when handling distance analysis, as the road-network approach produced more realistic travel estimates than straight-line distance methods. Use of ModelBuilder is also helpful for complex methodology and automation. However, teams should allot time for pre-processing and troubleshooting when working with GIS and handling census data. Our team encountered a number of potential discrepancies on Mobile County's GIS division's website. We also found that census data was often not uniformly formatted and required cleaning before it was usable. Furthermore, our team had the advantage of utilizing Decennial Census data, which is not available for every year. Therefore, future project teams should plan on modifying our methodology for use with other data sources like the American Community Survey (ACS).

One of the most important steps in this project was standardizing primary and foreign keys, like tract identifiers, early in the workflow. Inconsistencies regarding leading zeros and field types created repeated join issues between the census and ArcGIS output datasets. Future teams should

create a fully automated preprocessing script in Python, similar to what our team has provided in this project, before importing data into ArcGIS Pro. Teams should also maintain permanent exports of temporary joins and intermediate outputs for documentation.

We recommend that future teams aspire to expand the scope of analysis beyond drive time alone. While this project found that higher Black and Latine population shares were associated with shorter average drive times in Mobile County, the results likely reflect urban density patterns rather than the absence of voting barriers. Future research could incorporate additional factors or explore potentially confounding variables such as vehicle ownership, public transportation access, voter turnout, polling place wait times, or socioeconomic indicators to better capture the full scope of voting accessibility. Teams can also assess urban and rural counties separately or conduct multi-county analyses to determine whether the patterns observed in Mobile County are consistent statewide.

It is equally important that future teams allocate time for quality assurance, demonstrating “good statistical practices” (e.g., testing assumptions, avoiding p-hacking), and exploring underlying contexts that may influence the described phenomena.

Conclusions

This project developed a GIS and statistical framework for evaluating polling place access in Mobile County, Alabama, based on data from 2020. Using ArcGIS Pro’s Network Analyst and ModelBuilder, census demographic data, Python preprocessing, and statistical modeling in RStudio, our team created a tract-level dataset that measures estimated drive times between tract population centroids and assigned polling locations (polling place within the same precinct). The analysis found a statistically significant negative relationship between combined Black/Latine population share and polling place drive time, suggesting that tracts with larger Black and Latine populations generally experienced shorter travel times to polling locations. However, this does not dismiss civic participation barriers altogether and is better understood in a larger geographic context. Additional GIS visualization layers indicated that this pattern was likely influenced by the concentration of Black and Latine populations within denser urban areas of Mobile County.

The project produced several deliverables, including a reproducible ArcGIS ModelBuilder workflow, a tract-linked travel time dataset, a statistical analysis script and corresponding documentation, and multiple GIS layers for visual supplementation. The project also highlighted important methodological considerations surrounding geographic context, data preprocessing, and statistical assumption testing when examining voting access. This work contributes to the broader

regional analysis conducted across the Gulf Coast project teams and provides a foundation for future investigations into transportation access, civic participation, electoral geography, and voting equity

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Deliverables Checklist

(DELETE AFTER SECTION COMPLETE: A summary checklist of all team deliverables, with links to each file in the team's organized, client-facing final deliverables folder.

Deliverable Folder:  INST490 GC-PAD

- ☐  GCPAD-DistanceAnalysis-2020DecennialCensus-v1.xlsx
- ☐ [GCPAD-DistanceAnalysis-CCDBlackPop-v1.csv](#)
- ☐ [GCPAD-DistanceAnalysis-CCDHispanicPop-v1.csv](#)
- ☐ [GCPAD-DistanceAnalysis-Clean2020DecennialCensus.csv](#)
- ☐ [GCPAD-DistanceAnalysis-CountySubdivisionBoundaries-v1.zip](#)
- ☐ [GCPAD-DistanceAnalysis-DataCleaning-v1.ipynb](#)
- ☐ [GCPAD-DistanceAnalysis-MobileCentroidsSpatialJoin-v1.zip](#)
- ☐ [GCPAD-DistanceAnalysis-ModelBuilder&NetworkAnalyst-v1.aprx](#)
- ☐  GCPAD-DistanceAnalysis-NetworkAnalysis&GISDataProcessingReport-v1
- ☐ [GCPAD-DistanceAnalysis-PollingSites-v1.zip](#)
- ☐ [GCPAD-DistanceAnalysis-Precincts&Demographics-v1.csv](#)
- ☐ [GCPAD-DistanceAnalysis-SchemaForCleanCensusData-v1.ini](#) [[schema.ini](#)]
- ☐ [GCPAD-DistanceAnalysis-StatisticalAnalysis-v1.R](#)
- ☐ [GCPAD-DistanceAnalysis-StatisticalAnalysisOutput-v1.csv](#)
- ☐  GCPAD-DistanceAnalysis-StatisticalAnalysisReport-v1
- ☐ [GCPAD-DistanceAnalysis-VoterPrecincts-v1.zip](#)
- ☐  GCPAD-ArcGIS-Visualization-Layers-v1

Transportation Barriers to Civic Participation

GC-TRAC

Basirat Gbadegesin, Client Liaison & Data Visualization,

Amisiyas Seyoum, Data Cleaning & ArcGIS Visualization,

Andy Nguyen, Data Collection & ArcGIS Visualization,

Ayush Jain, Data Collection & Research,

Robert Tzou, ArcGIS Visualizations

Southern Policy Law Center(SPLC)

INST490 - Integrated Capstone for Information Science

Section 0105

May 18th, 2026

Key Findings

- Finding 1: Transportation access differed between Mobile County and Baldwin County in both 2020 and 2024. Mobile County had more ZIP codes with higher no-vehicle rates, while most ZIP codes in Baldwin County showed lower transportation barriers and higher levels of vehicle access.
 - Finding 2: Public transportation coverage in Mobile County was limited outside of Mobile City. Only 36 of 73 polling locations were located within a 30-minute walk of an active bus stop, leaving many polling sites without direct public transit access.
 - Finding 3: Transportation barriers affected both urban and rural communities, but the causes differed by location. Urban areas showed higher rates of households without vehicles, while rural communities faced longer travel distances and fewer available transportation options.
 - Baldwin County and Mobile County used different public transportation systems during the study period. Baldwin County relied primarily on the BRATS on-demand transportation system, while Mobile County operated a fixed route bus system that served a smaller geographic area.
 - Short distances did not always mean that polling locations were easy to reach. In some ZIP codes, residents without personal vehicles still faced access challenges because of limited sidewalks, limited walking routes, and gaps in public transportation coverage.
-

Deliverables Produced

Github - [Datasets](#)

- Contains all datasets collected and used throughout the transportation accessibility and civic participation analysis

HeatMap - [Interactive Heat Map](#)

- Displays transportation barriers across Mobile County and Baldwin County using no vehicle rates, polling locations, and accessibility measures

ArcGis Visualizations: [ArcGIS Transportation Map](#) [HouseholdAvgVehicles2020](#)
[HouseholdAvgVehicles2024](#)

- Contains ArcGIS maps and visualizations related to transportation access, polling location accessibility, bus route coverage, and zip code transportation barriers.
-

Challenges

Our team encountered a few obstacles. One obstacle was collecting and organizing transportation, polling location, and vehicle registration datasets from multiple public sources. The datasets used different formats, geographic boundaries, and reporting methods, so as a team, to handle the obstacle by cleaning and standardizing the data in Excel and ArcGIS before beginning the analysis. Another challenge was determining which types of transportation should be included in the analysis. As a team, we evaluated multiple transportation factors, including personal vehicle access, public bus routes, and on-demand transportation services, before deciding to focus primarily on vehicle ownership rates and public transportation accessibility because those measures had the most consistent available data across both counties. The team also encountered obstacles when analyzing transportation accessibility because short travel distances did not always reflect real-world transportation conditions. Some rural communities had limited transit coverage, while some urban areas lacked safe or practical walking routes. To address this, the analysis included additional context related to bus route access, vehicle ownership rates, and walking accessibility instead of relying only on geographic measurements.

Connections to Other Teams

Team 1 – GC-DIG provided shared datasets and project documentation that supported our team’s transportation analysis. Their work included voter registration, demographic, geospatial, and infrastructure data that were used throughout the project.

Team 3 – GC-PAD contributed polling place access and drive-time findings that overlapped with our focus on transportation barriers and access to polling locations. Their spatial analysis provided additional context for understanding how travel and location accessibility may affect civic participation.

Team 8 – GC-VIZ supported our team by sharing design standards used across the final presentation. These standards helped improve the consistency and readability of our maps, charts, and visual materials.

Team 10 – GC-PORT supported our team by developing communication frameworks and presentation templates for translating technical findings into materials for different audiences. Their work helped organize how transportation access findings could be presented in reports and stakeholder-facing deliverables.

Final Report and Deliverables

Project Context

GC-TRAC focused on studying transportation access and its relationship to civic transportation in Alabama communities, with a focus on rural and underserved areas. The team examined transportation factors such as public transit availability, vehicle ownership rates, road access, walkability, and the distance between residential areas and polling locations. Using geographic and demographic datasets, the team created maps, charts, and visualizations to identify areas where transportation limitations may affect access to voting locations and other forms of civic participation.

This work supports the overall project by providing data-driven analysis of physical access to civic resources. The team’s findings help identify geographic patterns related to transportation

accessibility and polling location access across the Alabama Gulf Coast region. The project also organizes transportation and civic participation data into visual formats that are easier for technical and non-technical audiences to interpret. These materials can support future research, planning, and discussions related to transportation infrastructure and civic accessibility.

Methods

Data Sources

At the start of the project, datasets related to road infrastructure, transportation, polling sites, and other relevant information regarding voting in Alabama from 2020 to 2024 were collected before beginning the analysis process. The datasets and information were gathered from various government and transportation sources, as these sources generally provide more accurate and reliable data that are publicly available for research purposes. After collecting the datasets, we proceeded to clean and organize the data before beginning our analysis. The datasets we gathered were mainly focused on Baldwin and Mobile County.

Voter Registration Dataset:

Datasets containing the number of registered voters for Baldwin and Mobile County of both 2020 and 2024 from the Alabama Secretary of State. This dataset is useful for helping determine the number of people registered to vote, as well as identifying which voters are active and inactive. Transportation barriers may be one possible factor contributing to why some registered voters remain inactive.

- <https://www.sos.alabama.gov/alabama-votes/voter/election-data>

Vehicle Registration Dataset:

Datasets containing the number of vehicles registered from the State of Alabama Department of Revenue Motor Vehicle Division.

- <https://www.revenue.alabama.gov/motor-vehicle/registration-statistics/>

Road Networks Of Both Baldwin And Mobile County:

Datasets and files containing road networks of both Baldwin and Mobile County, which are used for our ArcGIS map, taken from the Data.gov published by the U.S. Department of Commerce, U.S. Census Bureau, Geography Division.

- Mobile County -
<https://catalog.data.gov/dataset/tiger-line-shapefile-2024-county-mobile-county-al-all-roads>
- Baldwin County -
<https://catalog.data.gov/dataset/tiger-line-shapefile-2024-county-baldwin-county-al-all-roads>

Alabama Polling Stations dataset

The polling location dataset was taken directly from the client-provided documents to ensure that polling locations remained consistent across all project teams. We were also required to convert the data into CSV format so the information could be used for the Google Maps-based heatmap visualizations.

Mobile City Bus stop and route dataset

The datasets containing the Mobile City Bus stop and routes for the ArcGIS map are taken from Transitland.

- <https://www.transit.land/operators/o-wave~mobile~al~us#map>

Zipcodes

The zipcodes of households of Baldwin and Mobile County used to make the Google Heatmap are taken from the U.S. Census Bureau's American Community Survey.

- [https://data.census.gov/table/ACSST5Y2023.S0801?g=040XX00US01\\$1600000,01\\$860000_050XX00US01003,01097](https://data.census.gov/table/ACSST5Y2023.S0801?g=040XX00US01$1600000,01$860000_050XX00US01003,01097)

Analytical Approaches

Data Cleaning:

Once the necessary datasets and information were gathered, we cleaned and refined the data so that it primarily focused on Baldwin and Mobile Counties, as these are the main areas of focus in our analysis of transportation barriers. This process included removing any incomplete or duplicate data, standardizing location formats, and filtering out data that was unrelated to the two focus counties.

Data Analysis:

The datasets that were gathered were analyzed to identify potential causes of transportation barriers affecting people in Baldwin and Mobile Counties when traveling to polling stations. The analysis focused on factors such as vehicle ownership rates, bus transportation availability and routes, and distances to polling locations.

Visualizations:

For the visualizations, we've used ArcGIS and Google Maps to showcase and explain transportation barriers to the polling station. For the ArcGIS, it's used to visualize maps to show the road networks, the bus stops and routes, and the polling station. For Google Maps, it's used to create heatmaps to estimate No Vehicle Rate and Barrier Score to the polling stations using the zip codes of the household area of Baldwin and Mobile County.

Tools Used:

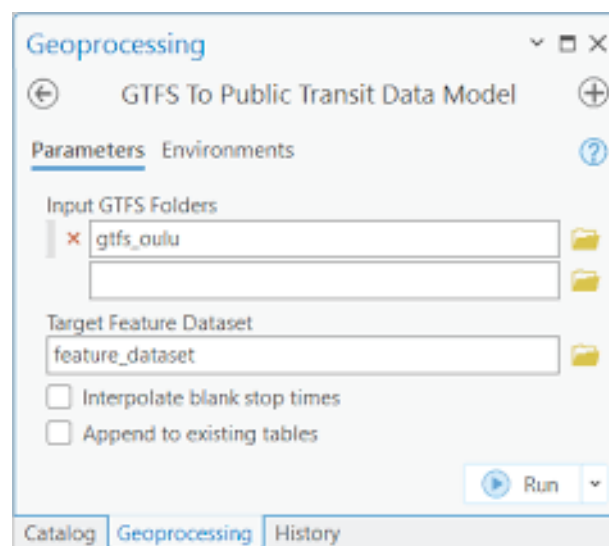
For tools used for analysis and visualization, we decided to use tools such as Excel, ArcGIS, and Google Maps to create charts, graphs, and maps that display the locations of transportation barriers affecting access to polling stations.

GTFS ArcGIS Instructions:

The General Transit Feed Specification (GTFS) is a standardized public transit data format that packages schedule frequencies, route geometries, and geographic stop coordinates into a series of relational text files, while Transitland serves as an open-data registry that aggregates and archives

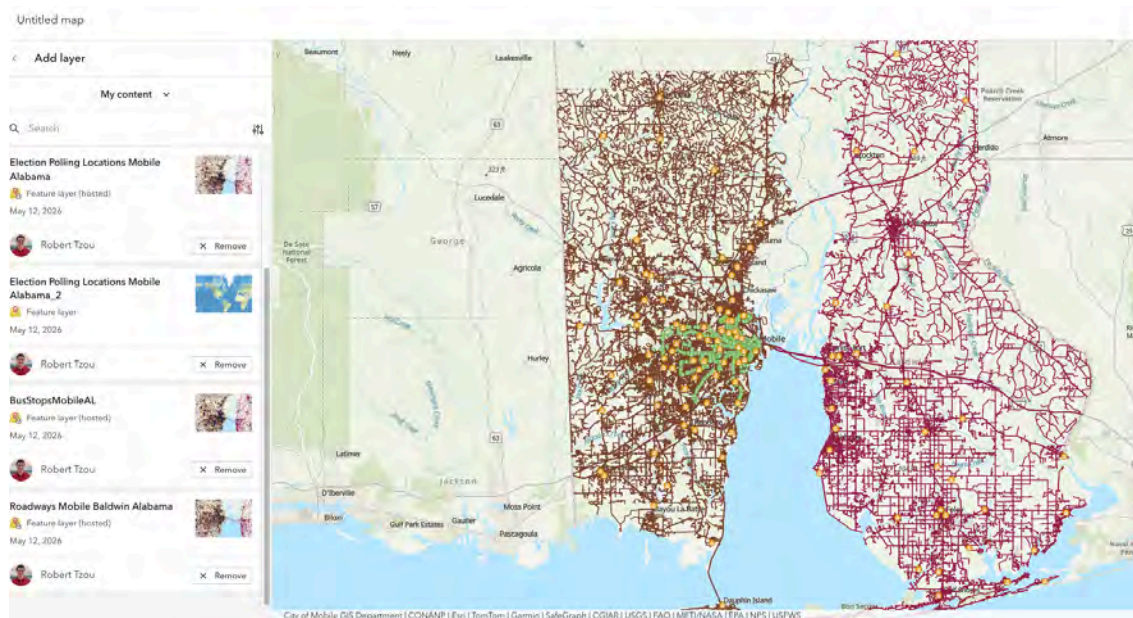
these feeds from transit agencies worldwide. To replicate or update the spatial analysis for this project, researchers must access the complete, static GTFS data for the Wave Transit System, available either via Transitland or directly through the agency's developer resources. Accessing the dataset correctly requires downloading the intact, compressed zip archive containing all core relational components. Most importantly, stops.txt for physical bus stop coordinates, shapes.txt for mapped route pathways, and trips.txt and stop_times.txt for schedule tracking. These files must remain organized together within a single directory so that ArcGIS Pro's network analyst tools can properly ingest the relational database and construct a navigable public transit network.

It must be noted that Transitland requires the creation of an email account to access the datasets needed for the analysis. To import the GTFS dataset into ArcGIS Pro on Windows for this project, begin by opening the Catalog pane, right-clicking your project's File Geodatabase (.gdb), and creating a new Feature Dataset configured with a Projected Coordinate System. Specifically, NAD, 1983 StatePlane Alabama West (Meters) to ensure accurate spatial and distance calculations. Next, navigate to the Geoprocessing pane, search for the "GTFS To Public Transit Data Model" tool, and set the input folder parameter to the directory containing your complete set of GTFS text files, verifying that crucial files like stops.txt, shapes.txt, and trips.txt are all present in the folder. Finally, designate your newly created Feature Dataset as the target destination, check the box for "Make LVEShapes" to cleanly generate the true physical geometries of the bus routes, and run the tool to translate the raw tabular schedules into fully mappable spatial layers. Here's a screenshot of "GTFS to Public Transit Data Model" for reference:



To upload your finalized spatial files and project datasets to ArcGIS Online, navigate to the University of Maryland's organizational portal at umd.maps.arcgis.com and ensure you log in using your official university enterprise credentials through the Central Authentication Service (CAS) screen. Once authenticated into the UMD cloud environment, click the Content tab at the top of the interface and select the New item button on the upper left. From the menu options, select Your device and choose the compressed zip folder containing your shapefiles or file geodatabase. Ensure that you leave the option checked to publish the data as a hosted feature layer, add the appropriate metadata, summary text, and tracking tags for organizational clarity, and click Save to successfully generate the interactive, web-accessible feature layer within your institutional directory.

Once all steps have been completed, the map itself should look like this:



To recreate the spatial analysis between voting centers and the transit network, verify that both datasets are active in your map viewer, open the Analysis pane, and select the Join Features tool. Designate the Polling Locations as your target layer and the Bus Stops as your join layer, selecting Within a distance as the spatial relationship with a threshold specified at 1.5 miles. Crucially, set the tool to Join one to one and ensure the option to Keep all target features remains checked; this prevents the exclusion of disconnected locations, ensuring that polling places outside the transit range are preserved in the final output with a match value of zero. Once the tool executes, open the resulting layer's attribute table and review the newly generated Join_Count field to extract your final metrics, where rows containing a value of one or greater signify accessible polling locations and

rows containing a value of zero explicitly isolate the underserved locations past the 1.5-mile transit threshold. The final outcome should look like a mark over the polling locations within range. Here's a link to the final outcome: [\[Link to Map\]](#)

Challenges

There have been several challenges throughout the project, including difficulties in gathering certain datasets, information, and files, as well as determining which tools and visualization methods would be best to use during the early stages of the project, in which we eventually decided to use ArcGIS and Google Maps. Some of the information took time to locate, and the datasets and files we downloaded were not always in the file formats we needed. This required us to convert them into compatible formats before they could be used. For example, the Transitland files containing Mobile City bus transportation data were not initially in a format compatible with ArcGIS. We had to use conversion tools to transform the files into the correct format before importing them into ArcGIS for analysis. Our clients have also been very helpful in providing guidance on the information we should include in our analysis. In addition, they supplied valuable resources for us to use, which have been very beneficial to our group.

Deliverables and Findings

HeatMap Analysis

Figure 1: Transportation barriers and polling location accessibility across Mobile and Baldwin Counties using no vehicle rates and polling location mapping data. Red markers indicate higher transportation barriers, while green markers indicate lower transportation barriers. Blue markers represent polling locations across both counties. (Source: U.S Census Bureau and Capitol Impact Government Gateway)

Interactive version: [\[Heat Map\]](#)

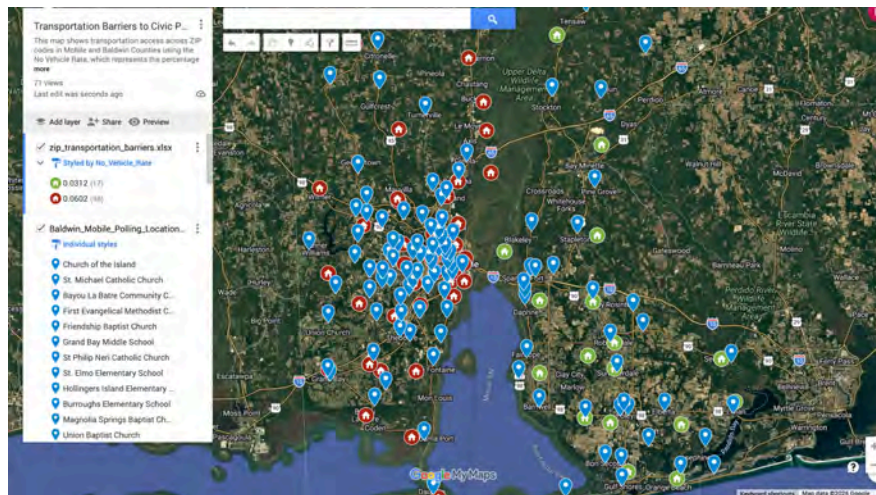
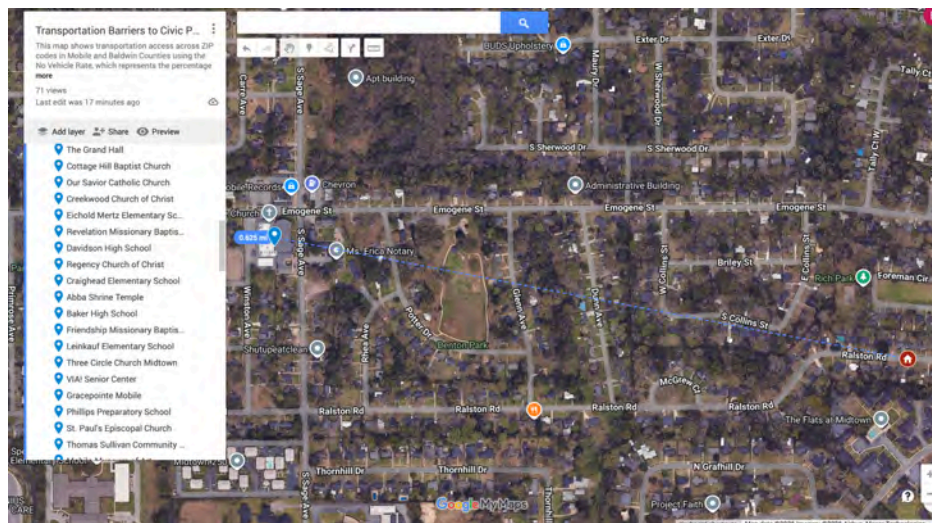


Figure 2: Estimated travel distance between ZIP code 36606 and the Three Circle Church Midtown polling location in Mobile County. The map illustrates how nearby polling locations may still present transportation challenges for residents without personal vehicles or safe walking routes (Source: Google Maps, U.S Census Bureau, and Capitol Impact Government Gateway)

Interactive version: [\[Heat Map\]](#)



The dataset used for this analysis includes two main measures: the No Vehicle Rate and the Barrier Score. These measures are based on publicly available data from sources such as the U.S. Census Bureau. The No Vehicle Rate shows the percentage of households in each ZIP code that do not have

access to a personal vehicle. It is calculated as: $\text{No Vehicle Rate} = \frac{\text{Households without a vehicle}}{\text{Total households}}$. For example, a value of 0.0602 means that approximately 6.02% of households in that ZIP code do not have access to a car. This measure is important because residents without reliable transportation may face greater difficulty reaching polling locations, especially in areas with limited public transit options. The Barrier Score estimates the number of people who may be affected by transportation limitations. It is calculated as: $\text{Barrier Score} = \text{Population} \times \text{No Vehicle Rate}$. This measure estimates how many residents in a ZIP code may lack reliable vehicle access. For example, ZIP code 36560 in Mount Vernon, Mobile County, has a population of 2,858 and a no-vehicle rate of 0.0602. This results in a barrier score of approximately 172, meaning that an estimated 172 residents in that ZIP code may experience transportation barriers related to vehicle access. Together, the No Vehicle Rate and Barrier Score show both the percentage of households affected and the estimated number of people affected within each area.

The map shows transportation accessibility across ZIP codes in Mobile County and Baldwin County using the No Vehicle Rate. Each ZIP code is represented with a different color to show the level of transportation access. Red indicates higher no-vehicle rates and greater transportation barriers, while green indicates lower no-vehicle rates and better transportation access. Blue markers on the map represent polling locations and help show the relationship between transportation access and voting accessibility. The analysis identified clear differences between Mobile County and Baldwin County. Mobile County contains more ZIP codes with higher transportation barriers, particularly in both urban and rural communities such as Mobile, Citronelle, and Wilmer. These areas showed higher percentages of households without access to a vehicle. In contrast, Baldwin County contained more ZIP codes with lower transportation barriers. Areas such as Daphne, Fairhope, Foley, and Gulf Shores showed lower no-vehicle rates and greater levels of vehicle access.

The map also shows that transportation barriers exist in both urban and rural communities, although the causes differ by location. In urban areas, households may lack vehicles because of financial limitations. In rural communities, transportation challenges are more closely related to longer travel distances and fewer transportation options. Several higher-barrier ZIP codes also appeared farther from polling locations, which may create additional challenges for residents without reliable transportation.

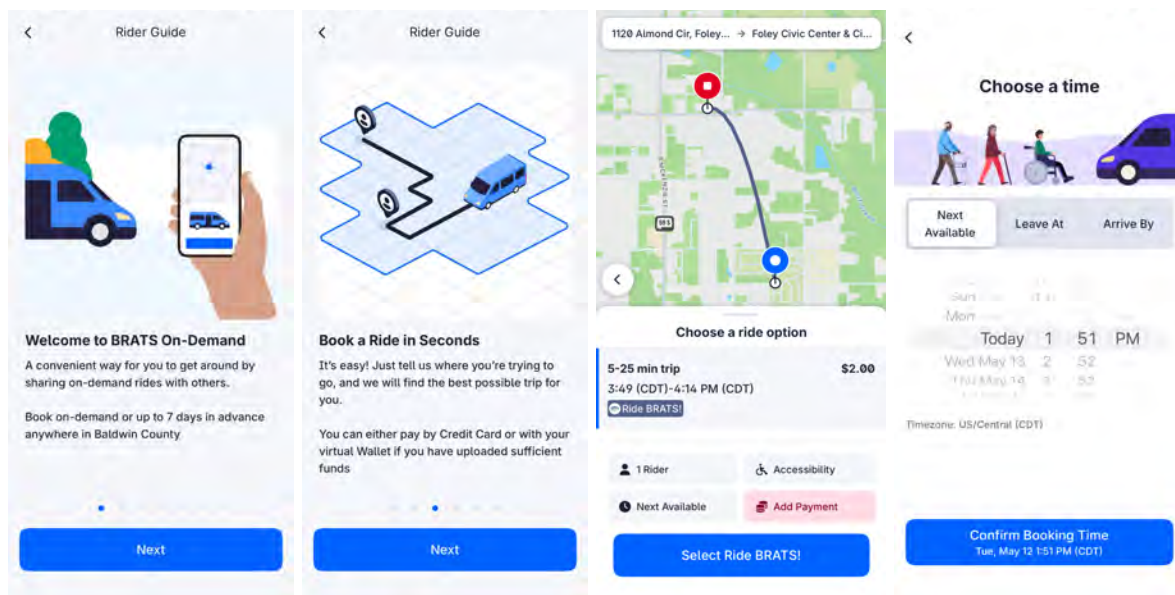
Transportation access is closely connected to civic participation because residents must be able to travel to polling locations in order to vote. In ZIP codes with higher no-vehicle rates, residents may

face greater difficulty reaching polling places, may rely on others for transportation, or may have fewer transportation options available on Election Day. The analysis shows that transportation access is not distributed evenly across Mobile and Baldwin Counties, and that some communities experience greater transportation limitations than others.

Overall, the findings show that transportation barriers vary significantly across the Alabama Gulf Coast region. ZIP codes with higher no-vehicle rates and limited transportation access may face greater challenges accessing polling locations and other civic resources. The map helps identify where these barriers are concentrated and provides additional context for understanding transportation accessibility across the region.

BRATS App – Baldwin County

Unlike Mobile County, Baldwin County does not operate a more traditional public bus transportation system. As a result, Baldwin County relies on the Baldwin Rural Area Transportation System (BRATS), which is an on-demand service that provides discounted transportation access to residents across the county, costing around \$2-\$5 depending on distance traveled. This indicates a major difference in how residents in each county access public transportation, especially when it comes to comparing urban areas to rural areas.



Mobile County's current transit system has faced challenges related to cost efficiency and declining ridership in recent years. According to reporting from [FOX10 News](#), ridership on The Wave Transit System decreased from approximately 836,000 riders in 2019 to around 628,000 riders in 2024, while city transportation costs continued to rise. These trends have contributed to discussions surrounding the future of public transportation in Mobile County.

In response, Mobile city officials have proposed a major overhaul of the public transit system through a partnership with Via Transit LLC, which would transition portions of the current fixed-route bus system toward a more modern on-demand rideshare model. The proposed system would include app-based ride booking and expanded citywide on-demand transportation services. This proposed change resembles Baldwin County's BRATS system and reflects a broader movement toward a more flexible transit model.

These developments show how transportation accessibility is actively changing across the Alabama Gulf Coast. While fixed route bus systems may provide a more predictable service, on-demand systems may better address transportation gaps in lower-density or rural communities where traditional bus routes are less effective. However, these systems may also introduce challenges related to technology access, scheduling, and service awareness.

Voter and Vehicle Registration

Voter Registration

2020: There have been around 491,571 registered voters in both Mobile County and Baldwin County. 452,685 of them were active voters, while 38,886 were inactive in both counties.

Baldwin had around 176,642 registered voters, 165,848 of them were active, while 10,794 of them were inactive.

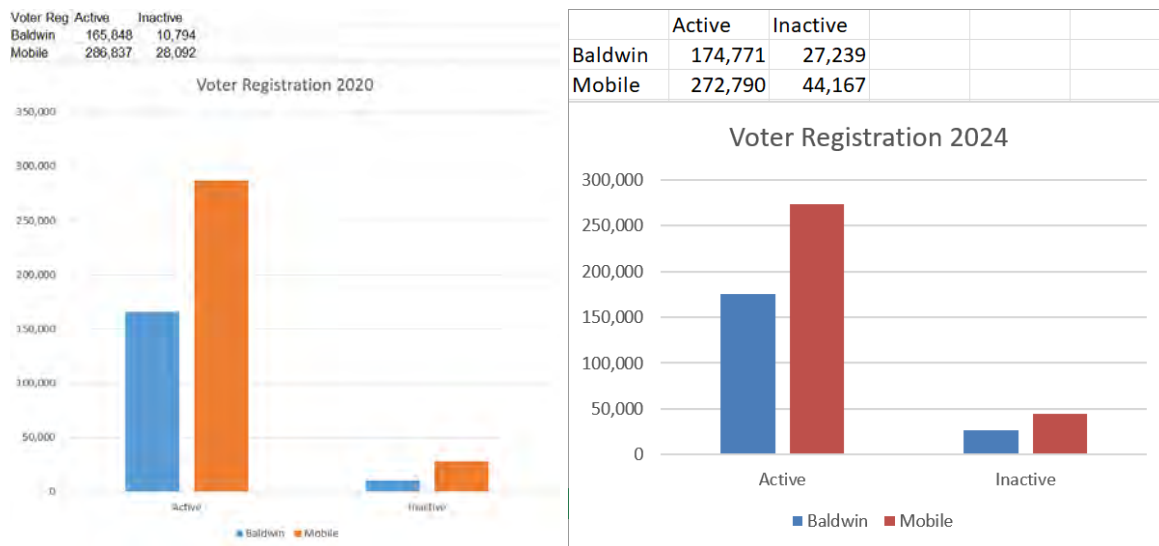
Mobile had around 314,929 registered voters, 286,837 of them were active, while 28,092 were inactive.

2024: There have been around 518,967 registered voters in both Mobile County and Baldwin County. 447,561 of them were active voters, while 71,406 were inactive in both counties.

Baldwin had around 202,010 registered voters, 174,771 of them were active, while 27,239 of them were inactive.

Mobile had around 316,957 registered voters, 272,790 of them were active, while 44,167 were inactive.

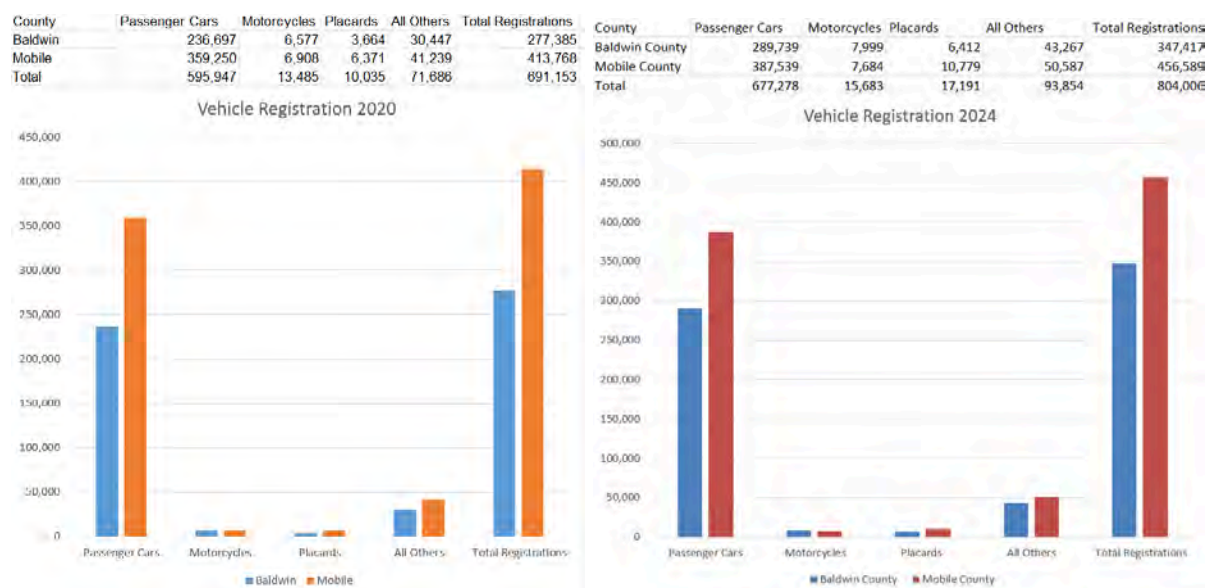
Note: Both Mobile and Baldwin seem to have double the number of inactive voters in 2024 than in 2020.



Vehicle Registration

2020: There were 277,385 registered vehicles in Baldwin, 413,768 registered vehicles, totaling around 691,153 of all vehicles registered in both counties.

2024: 289,739 vehicles registered in Baldwin, 387,539 registered vehicles in Mobile, totaling around 804,006 total number of registered vehicles in both counties.



Voter registration data helps us determine the number of people registered to vote and identify how many registered voters were active or inactive between 2020 and 2024. Although voter inactivity can be influenced by many different factors, transportation barriers may be considered one possible contributing factor that affects whether some registered voters participate. Vehicle registration data, on the other hand, helps provide context regarding transportation access. However, the number of registered vehicles does not necessarily mean that each voter owns a vehicle. Some individuals may have multiple vehicles registered under their name, while other registered vehicles may belong to businesses or government agencies rather than individual residents.

ArcGIS Map

Vehicle Ownership Analysis

Figure 1: Average minimum household vehicle ownership across Mobile and Baldwin Counties per Zip Code Tabulated Area in 2020 using census zip code tabulation areas mapping data and household vehicle ownership data. Green represents a high number of vehicles owned per zip code tabulated area, light green represents a comfortable number of vehicles owned, orange represents a fair number of vehicles owned, and red represents a low number of vehicles owned. (Source: U.S Census Bureau)

Interactive version: [[Heat Map](#)]

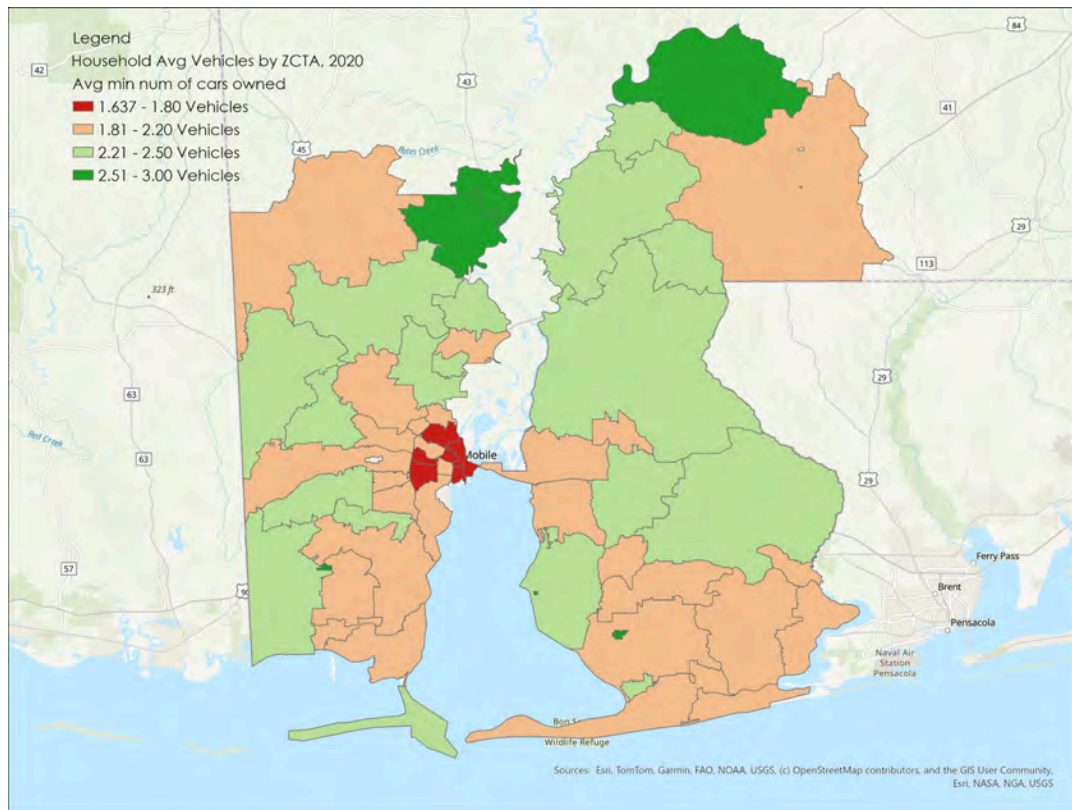
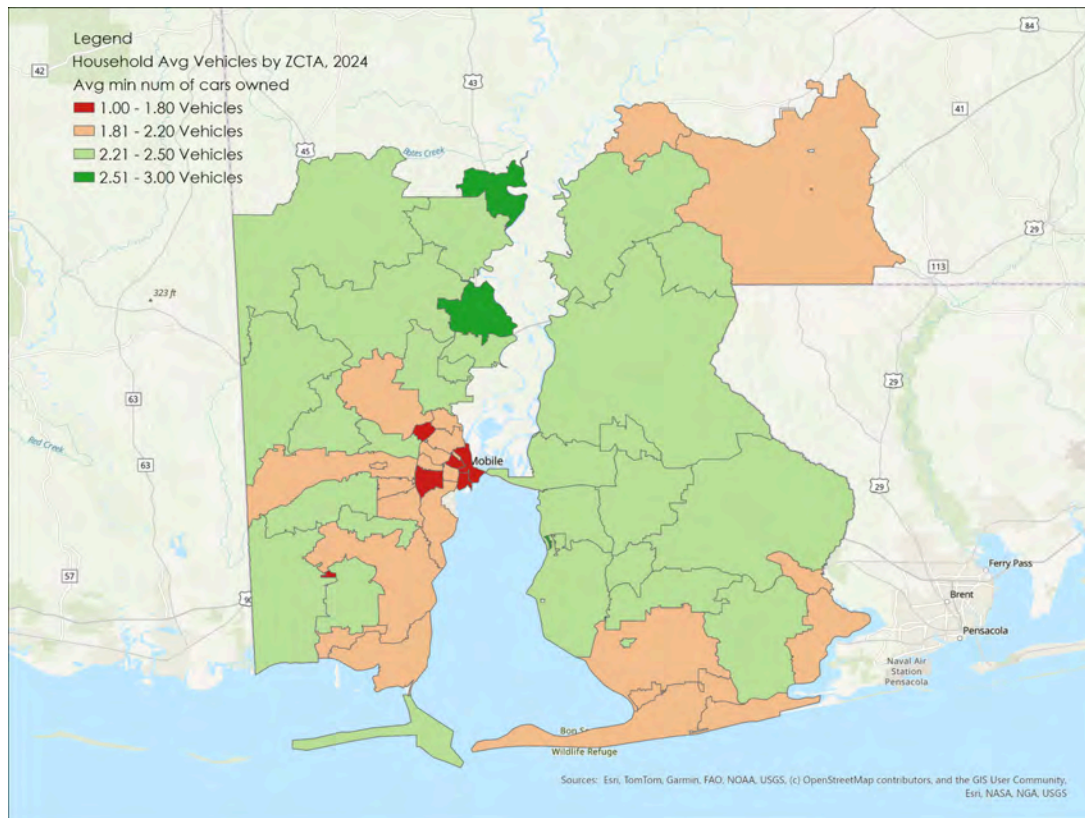


Figure 2: Average minimum household vehicle ownership across Mobile and Baldwin Counties per Zip Code Tabulated Area in 2024 using census zip code tabulation areas mapping data and household vehicle ownership data. Green represents a high number of vehicles owned per zip code tabulated area, light green represents a comfortable number of vehicles owned, orange represents a fair number of vehicles owned, and red represents a low number of vehicles owned. (Source: U.S Census Bureau)

Interactive version: [[Heat Map](#)]



The datasets used for these visualizations include the final cleaned versions of the [U.S Census Zipcodes of households of Baldwin and Mobile County dataset](#) for 2020 and 2024, found within our [GitHub](#) repository titled “combined_2020_mob_bal_zctas.csv” and “combined_2024_mob_bal_zctas.csv” respectively. The maps draw from one main data point: “Avg_min_num_cars_owned”, which is how each area within the ArcGIS map is shaded. The legend includes information regarding the criteria for what each area needs to meet in order to be shaded the color that they are currently shaded in. Relevant columns that lead to the calculation of “Avg_min_num_cars_owned” include:

Pct_no_vehicles: percentage of total individuals over 16 in households with no vehicles owned

Pct_1_vehicle: 1 vehicle owned

Pct_2_vehicles: 2 vehicles owned

Pct_3_vehicles: 3+ vehicles owned

Total: total number of individuals over 16 in each Zip Code Tabulated Area (ZCTA).

For the sake of simplicity, 3+ will be calculated as just 3, and we assume that the final result is a minimum and could be slightly higher. To calculate the average minimum number of vehicles owned per ZCTA, multiply the percentage of 0-3 vehicles owned by the total number of households, add those totals together, and divide by the total number of households. For example, "pct_no_vehicle" = 0.2, "pct_1_vehicle" = 0.2, "pct_2_vehicles" = 0.4, "pct_3plus_vehicle" = 0.2, and "total" = 5500:

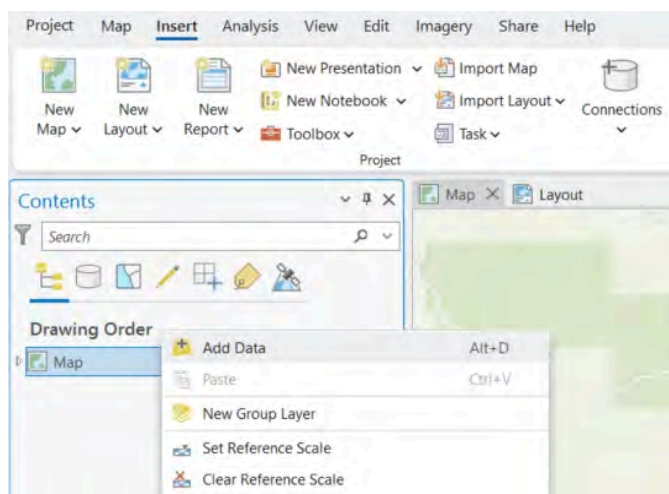
$$0.2 \times 5500, 0.2 \times 5500, 0.4 \times 5500, 0.2 \times 5500 = 1100, 1100, 2200, 1100$$

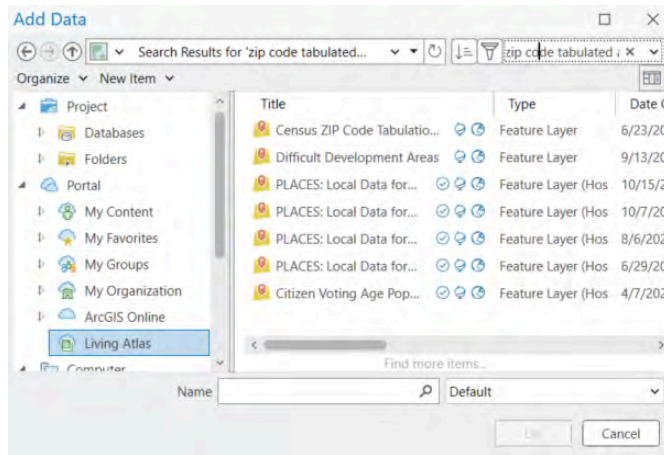
This provides households that own a certain number of vehicles. Now, multiply those numbers by how many vehicles each household owns, add them together, and divide by the total number of individuals over 16 years old in each ZCTA :

$$(1100 \times 0) + (1100 \times 1) + (2200 \times 2) + (1100 \times 3) = 8800$$

$$8800 / 5500 = 1.6 = \text{Avg_min_num_cars_owned}$$

This datapoint is calculated across all ZCTAs in Mobile & Baldwin counties, as mentioned above, and paired with ArcGIS's own Census ZIP Code Tabulation Areas mapping data. This can be accessed by inserting a new map, right-clicking the map, selecting add new data, selecting the portal, selecting the living atlas, and searching for "zip code tabulated area," as seen in the image below.





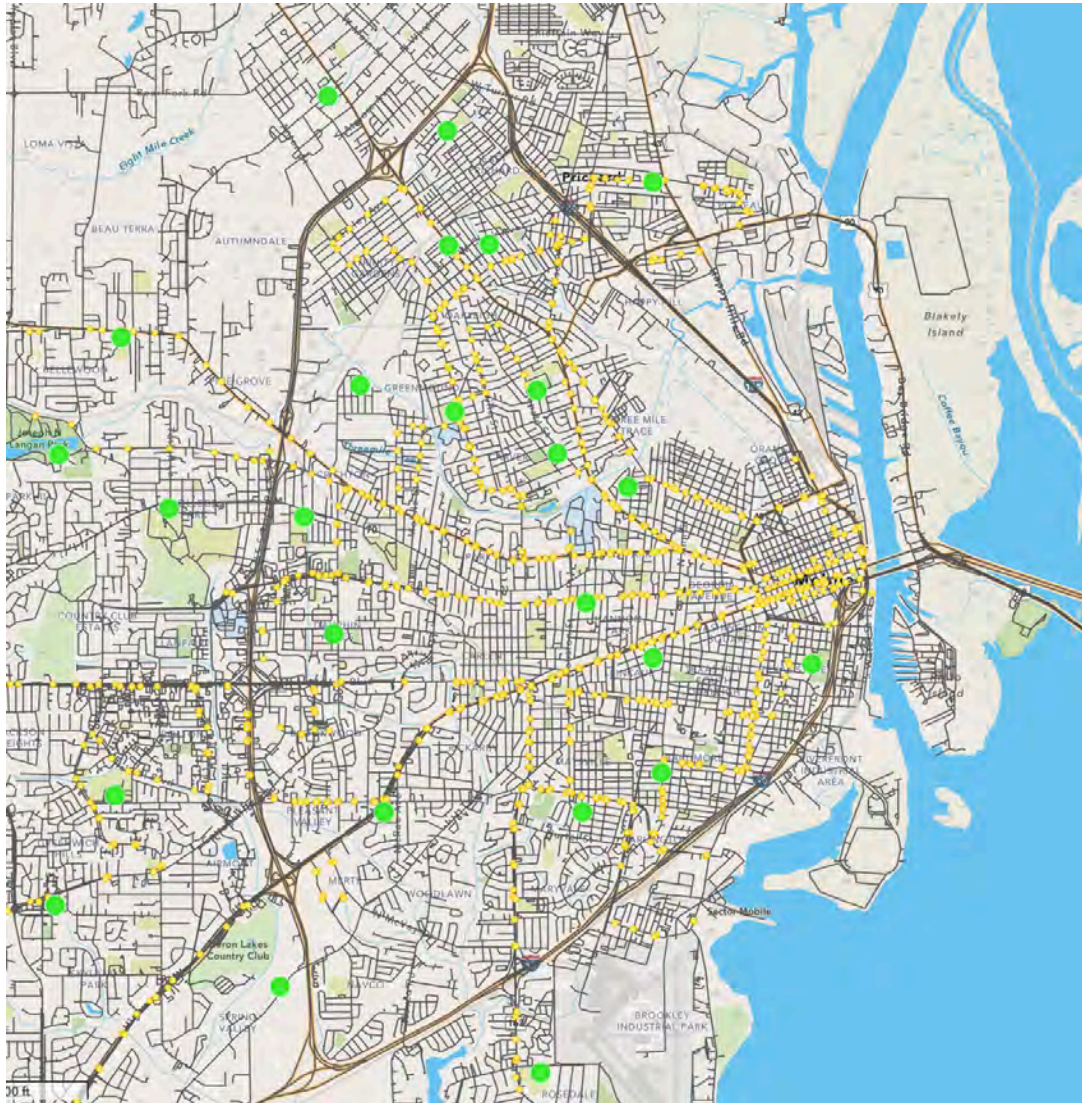
The maps highlight a few key details that draw back to the heat map analysis, most notably, the limited vehicle ownership in Baldwin County, specifically in the South, as polling locations start to thin out. Baldwin County's public transportation is less active compared to Mobile, which doesn't help those individuals with limited vehicle access. Though Mobile County does a great job in targeting households in ZCTAs with low vehicle access, as can be seen in both the 2020 & 2024 maps, as well as the heatmaps.

Overall, these maps highlight the change in vehicle ownership over time and show which areas are mostly affected by this barrier. Households in zip codes with low vehicle ownership may face challenges accessing polling locations.

Mobile County Transportation system

Figure 3: Map of Mobile City with the combination of road networks, bus stops and routes, and the polling stations. The yellow circle represents the bus stops and the route it takes while the green circle represents the polling station.

Interactive version: [\[Map\]](#)



Mobile City has a public bus system that provides residents with access to transportation throughout parts of the city. This system can help individuals travel closer to polling stations; however, not all bus stops and routes are located near polling locations. This can create inconvenience for people who rely on public transportation and may make it more difficult for them to reach polling stations, especially those located far from bus stops. Additionally, not all areas of Mobile have access to bus transportation. Some parts of the city do not have nearby bus stops or routes, making transportation access more difficult for residents, particularly those who do not own a vehicle. This issue becomes even more significant in areas outside of Mobile City, where public transportation options are limited or nonexistent.

Recommendations

Recommendations for the client (SPLC)

Additional analysis regarding vehicle ownership per ZCTA will need to be conducted, as the U.S. Census Bureau only stops at 3+ vehicles. As mentioned in the ArcGIS visualization section, that column was capped to 3 vehicles, and a minimum number of vehicles was used; that number could change if SPLC can get their hands on data that represents those households who own more than 3 vehicles, as every detail matters.

SPLC communities in Mobile County have higher no-vehicle rates and limited access to public transportation, especially in areas outside of Mobile City, where fixed-route bus coverage is weaker, with only 36 out of 73 polling stations in Mobile County being within a 30-minute walk from an active bus stop, suggesting that many polling stations may be difficult to reach for individuals without a reliable form of transportation.

Rural areas in Baldwin and Mobile counties should receive priority, as limited transit options and fewer available polling stations lead to residents needing to cover longer distances just to reach their designated polling station. SPLC should focus more on Baldwin County's BRATS transportation system's ability to decrease individuals' walking distance from their house to the bus stop.

SPLC could use the maps and heatmaps to identify high-barrier ZIP codes and low household vehicle ownership where voter outreach, transportation assistance, or further analysis could be useful. Areas with higher no-vehicle rates, higher barrier scores, and fewer nearby polling stations should be prioritized.

Recommendations for future project teams

Our recommendations for the next team would be to focus more on the disparity between the rate of vehicle ownership and voter registration. One metric we were unable to calculate was the percentage of voters to vehicles within the Gulf Coast counties. Include any and all metrics relating to car ownership. Metrics like these would be accessible within data from the American Community Survey from the U.S. Census. Most of the time spent on this project was finding the right data and framing the right story. Sifting for hours between data points on the Census or watching ArcGIS tutorials on how to import unfamiliar data types. For future teams working with transportation data,

it's important to remember what you are building: heatmaps, line graphs, and bar charts. Heatmaps will predominantly require shape files (.shp) or area-related data that can be filtered easily with a primary key like zip codes. Line graphs and bar graphs are meant to display the difference between comparisons with two-dimensional data.

As stated earlier, it took many weeks of iteration between formulating testable hypotheses and finding the right data to prove or disprove those research questions. A word to the wise is to first understand the scope of the data available from various sources. Once that scope is understood and the limitations are set, questions can be raised and tested quickly and manageably.

Conclusions

Team 4 (GC-TRAC) analyzed transportation accessibility and its relationship to civic participation across Mobile County and Baldwin County in the Alabama Gulf Coast region. The team collected, cleaned, and analyzed transportation, vehicle ownership, polling location, and public transit data to identify areas where transportation barriers may affect access to voting locations and other civic activities. Through the use of ArcGIS maps, heatmaps, and transportation accessibility analysis, the project highlighted geographic differences in transportation access between urban and rural communities and identified areas with higher no-vehicle rates and limited public transportation coverage.

The project produced several visual and analytical deliverables, including transportation accessibility maps, polling access analysis, and interactive visualizations designed to support technical and non-technical audiences. These findings provide additional context for understanding transportation accessibility across the Alabama Gulf Coast region and contribute to the larger project examining factors that may affect civic participation.

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Deliverables Checklist

- Background Research & Analysis:  Background Research and Project Context Analysis
- GitHub Repository & Datasets: [GitHub Repo](#)
- Presentation Slides:  INST490-105 SPLC Gulf Coast Counties Final Presentation Templa...
- Client Requirements Document:  Client Requirements Document Final
- Visualization Drive:  4 GC-TRAC – Transportation Barriers to Civic Participation
- Final Report:  INST490-105 SPLC Gulf Coast Counties Final Report Template_Draft2

ADA Accessibility & Compliance at Polling Locations

GC-AACS

Prince Osei, Sirisha Koirala, Sungjae Pae, Elizabeth Jarvio-Barahona, Kediell Lam

Southern Policy Law Center (SPLC)

INST490 - Integrated Capstone for Information Science

Section 0105

May 18th, 2026

Team Brief

Team 5: GC-AACS

Key Findings

- Of the 77 active 2024 polling locations in Mobile County assessed for ADA parking accessibility, 21 sites (27%) were confirmed to lack a van-accessible parking space, the most basic ADA parking requirement, while another 24 sites (31%) could not be assessed due to image quality limitations, meaning fewer than half of all reviewed locations could be confirmed compliant on this criterion.
 - Of the 23 polling locations confirmed to have at least one parking accessibility failure, 9 sites failed both the van-accessible parking requirement (A1) and horizontal clearance requirement (A2), indicating that nearly 40% of confirmed non-compliant locations present compounding physical barriers rather than a single isolated gap.
 - Slope compliance (A4) and surface stability (A5) could not be assessed at any of the 77 reviewed locations due to image resolution limitations, revealing a structural gap in what remote imagery review can determine and underscoring the need for in-person field verification at high-priority sites.
 - Only 22 of the 77 reviewed locations, fewer than 30%, could be confirmed compliant across all four accessible parking criteria, meaning the majority of Mobile County polling locations either have confirmed accessibility gaps or could not be fully evaluated with available data.
 - Three polling locations could not be found through available imagery and were excluded from analysis entirely, suggesting that public records and satellite imagery alone are insufficient to account for all active polling sites and that ground-level verification would be needed to produce a complete countywide accessibility picture.
-

Deliverables Produced

[Mobile County ADA Parking Accessibility Checklist](#)

- A structured spreadsheet recording ADA parking accessibility indicators for all 77 active Mobile County 2024 polling locations across six criteria.

[ADA Parking Accessibility Findings](#)

- A written analysis summarizing the methodology, findings, and limitations of the accessibility review.
-

Challenges

Evaluating ADA parking features through Google Earth imagery presented a major challenge, as many polling locations had blurry, obstructed, or outdated views that led to a high number of “INCONCLUSIVE” classifications in our dataset. Additionally, many ADA checklist criteria require in-person evaluation, such as slope measurements, parking stability, indoor clearances, and entrance measurements, which cannot be assessed through remote imagery. The team addressed these limitations by standardizing criteria and recording only what could be confidently verified from the available imagery.

Connections to Other Teams

Team 5 - GC-AACS conducted its ADA parking accessibility analysis independently and did not require direct data handoffs from other groups. Our work relied solely on publicly available polling location information, Google Earth imagery, and the ADA accessibility checklist for polling locations.

Team 3 - GC-PAD: Their analysis of travel-time and geographic access complements our findings by highlighting how distance-based barriers and on-site accessibility barriers may intersect for voters with disabilities

Team 4 - GC-TRAC: Their work on transportation access provides context for understanding how limited transit options may compound accessibility challenges at polling locations with unclear or insufficient ADA parking features

Team 8 - GC-VIZ: Our ADA accessibility classifications can be incorporated into their dashboards and maps to visualize accessibility disparities across counties.

Team 10: GC-PORT: Used our findings and visualizations to create a policy brief for the Executive Director of Alabama Disabilities Advocacy Program (Shannon Shelley-Tremblay). They simplified and connected different aspects of our research to create one brief to highlight the potential accessibility barriers in Alabama.

Final Report and Deliverables

Project Context

Our team's work focused on evaluating ADA accessibility and compliance indicators at polling locations within the broader 11-team regional analysis conducted in partnership with the Southern Poverty Law Center. While other teams examined areas such as transportation access, demographic trends, voting barriers, and regional voting patterns, our team concentrated specifically on physical accessibility indicators that could affect voters with disabilities. The team relied on polling location datasets, county election information, ADA-related guidance documents, and accessibility criteria developed throughout the larger project structure. In turn, findings from our team contributed accessibility-focused insights that other teams could incorporate into broader discussions about equitable voter access and regional disparities. The project also depended on coordination across teams to maintain consistency in terminology, data formatting, and reporting standards so that all final deliverables could be integrated into a unified analysis for the client.

Methods

Data Sources:

Our team used a combination of public datasets provided by the Southern Poverty Law Center (SPLC), primarily using the [Alabama Selected Counties 2020 and 2024 Precincts Google Sheets file](#), county election websites, and publicly available polling location information. These sources provided the foundational list of polling locations and associated metadata needed to conduct our review.

Google Earth satellite and street-level imagery served as the primary tool for visually assessing external accessibility features, while the ADA parking accessibility checklist, developed as part of the broader regional project, provided standardized criteria for evaluating each site. Additional supporting information came from demographic and voting-access resources used across the larger 11-team analysis, ensuring that our work aligned with the broader project context and terminology.

Analytical Approach:

Analytically, the project relied on qualitative classification and comparative review rather than statistical modeling. Each polling location was evaluated using the ADA parking criteria, and each criterion was categorized as YES, NO, or INCONCLUSIVE based on visible evidence from satellite and street-level imagery. Our team identified patterns across polling locations and grouped sites into categories based on the amount of visible accessibility evidence observed during the review process. Findings were summarized through tables, charts, and written observations to support the final report and presentation deliverables.

Data Cleaning:

Data cleaning involved standardizing polling location names, correcting county-level inconsistencies, removing duplicate entries, and ensuring uniform formatting across the spreadsheet. The team also standardized how ambiguous cases were labeled, particularly when imagery quality prevented verification.

Data Analysis:

Our team developed a standardized review process focused specifically on parking-related accessibility indicators visible through satellite and street-level imagery. Team members reviewed each polling location using six parking accessibility criteria: presence of van-accessible parking spaces and signage, horizontal clearance requirements, vertical clearance requirements where applicable, slope compliance, parking surface stability, and whether accessible parking appeared to be located along the shortest accessible route to the entrance. Each location was categorized based on the level of visible accessibility evidence available through imagery review. The team then organized findings into spreadsheets to maintain consistency across all reviewed locations.

Tools Used:

The primary tools used during the project included Google Earth for visual inspection, Microsoft Excel for organizing and categorizing findings, and Google Drive for collaboration, documentation, and reporting results. Google Earth provided both satellite and street-level imagery, enabling reviews to examine parking lots, signage, and accessible routes from multiple angles when available. Excel was used to maintain the structured checklist, apply standardized labels, and generate summary tables. Team communication, and coordination were maintained through regular meetings, shared documents, and collaborative review sessions to ensure consistency across evaluations.

Challenges and How They Were Addressed:

One of the primary challenges faced during the project was the limited ability to fully verify ADA accessibility requirements through online imagery alone. In many cases, the polling locations had poor image quality, outdated imagery, obstructed views, or unavailable street-level perspectives, making it difficult to confidently evaluate certain parking criteria such as signage visibility, slope, and parking surface conditions. To address these limitations, our team focused on identifying visible evidence rather than making definitive compliance determinations when evidence was unclear, and clearly labeled such cases as INCONCLUSIVE.

A second challenge involved maintaining consistency across multiple reviewers. Because remote imagery can be interpreted differently depending on angle, lighting or resolution, the team established shared evaluation criteria, held collaborative discussions on uncertain cases, and standardized how findings were recorded within the dataset. This collaborative approach helped

ensure that all reviewers applied the ADA criteria consistently and that the final dataset reflected a unified interpretation of the available data.

Deliverables and Findings

Mobile County ADA Parking Accessibility Checklist:

<https://docs.google.com/spreadsheets/d/1aAYjhpvhLtsLn-rFk3YgmqlRuDg99wA3JiEyD4pSqIE/edit?gid=1082227518#gid=1082227518>

We created a structured spreadsheet recording ADA parking accessibility indicators for all 77 active Mobile County 2024 polling locations across six criteria: van-accessible parking space presence (A1), horizontal clearance (A2), vertical clearance (A3), slope compliance (A4), surface stability (A5), and shortest accessible route to entrance (A6). Each criterion was coded as YES, NO, or INCONCLUSIVE based on available satellite and street-level imagery, and three locations were flagged as non-existent and excluded from analysis. Prior to this review, no centralized accessibility dataset existed for Mobile County polling locations, and this checklist fills that gap directly. It is structured to support future updates as additional imagery or in-person verification becomes available, making it a reusable foundation for continued work rather than a one-time snapshot

ADA Parking Accessibility Findings:

https://docs.google.com/document/d/1q6-wNOD_FtD8wp2UhNK_2lf5kMuhPlkvAkAn6fj93Rw/edit?tab=t.0

We created a written analysis summarizing the methodology, findings, and limitations of the accessibility review that translates the raw checklist data into conclusions accessible to a non-technical audience, including SPLC staff, legal teams, and policy partners. The document contextualizes the checklist data, identifies which criteria produced the most confirmed failures, and explains the constraints of remote imagery review, particularly the inability to assess slope compliance and surface stability, so that SPLC and future teams understand what the findings can and cannot support. Without this document, the checklist alone would require the reader to interpret the data without methodological context or analytical framing, limiting its usefulness as a standalone deliverable.

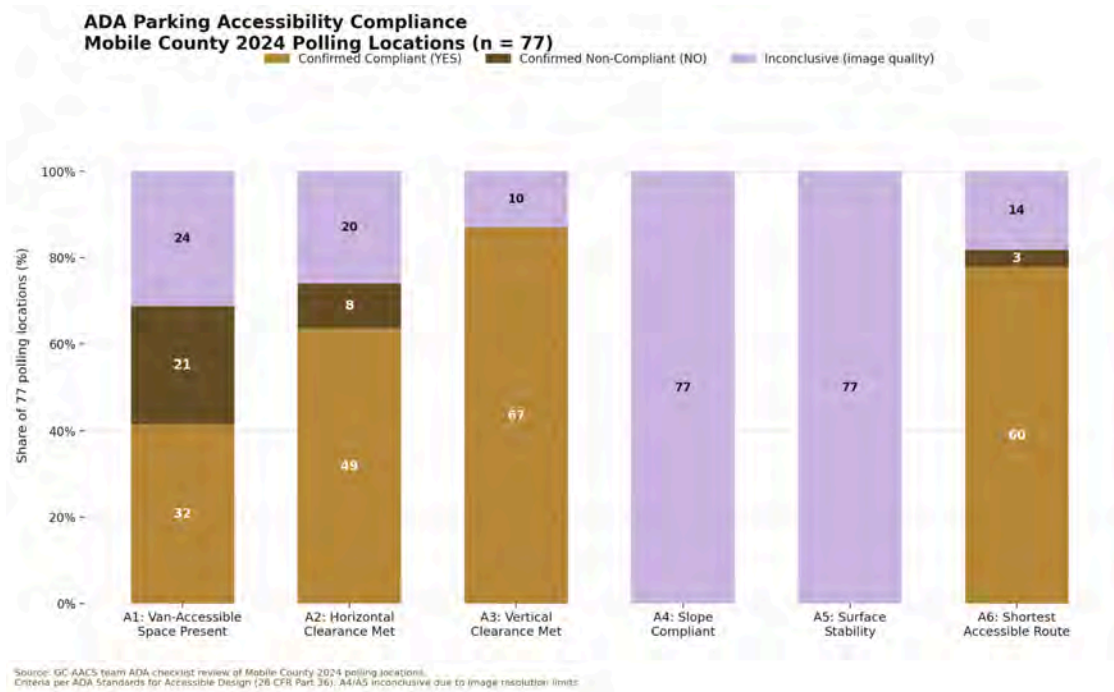
A	B	C	D	E	F	G	H
UNIQUE_ID	COUNTYFP	County	Precinct	Parking A1: Van-Accessible Space Present Is there at least one designated van accessible space with signage with the International Symbol of Accessibility and designated "van accessible"? (§§208.2, 208.2.4, 502.6)	Parking A2: Horizontal Clearance Requirements Met Are the designated van accessible spaces at least 96" wide with a 96" wide access aisle, or 132" wide with a 60" wide access aisle? (§§502.2, 502.3)	Parking A3: Vertical Clearance Requirements Met For van accessible spaces (particularly in a garage or parking structure), is there vertical clearance of at least 98" for the vehicular route to the parking space, in the parking space and access aisle, and along the vehicular route to the	Parking A4: Slope Compliant Are designated accessible parking spaces and the access aisles serving them on a level surface, with slopes not exceeding 1:48 in all directions? (Note: Curb ramps may not be part of an access aisle since they include slopes greater than 1:48.) (§502.4)
Mobile--1ST BAPTIST CH OF WILMER	97	Mobile	1ST BAPTIST CH OF WILMER	YES	INCONCLUSIVE (blurry im...	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--1ST BAPTIST CH SATSUMA	97	Mobile	1ST BAPTIST CH SATSUMA	INCONCLUSIVE (bl...	INCONCLUSIVE (blurry im...	INCONCLUSIVE (blurr...	INCONCLUSIVE (blurry image, not visible)
Mobile--1ST EVANGELICAL METHODIST	97	Mobile	1ST EVANGELICAL METHODIST	N/A (Non-exist Locat...	N/A (Non-exist Location)	N/A (Non-exist Location)	N/A (Non-exist Location)
Mobile--ABBA SHRINE TEMPLE	97	Mobile	ABBA SHRINE TEMPLE	YES	INCONCLUSIVE (blurry im...	INCONCLUSIVE (blurr...	INCONCLUSIVE (blurry image, not visible)
Mobile--AZALEA CITY CH OF CHRIST	97	Mobile	AZALEA CITY CH OF CHRIST	YES	INCONCLUSIVE (blurry im...	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BAKER HIGH SCH	97	Mobile	BAKER HIGH SCH	YES	INCONCLUSIVE (blurry im...	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BAYOU LA BATRE COMM_CTR	97	Mobile	BAYOU LA BATRE COMM CTR	YES	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BC RAIN HIGH SCHOOL	97	Mobile	BC RAIN HIGH SCHOOL	YES	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BETHEL BAPTIST CH	97	Mobile	BETHEL BAPTIST CH	INCONCLUSIVE (bl...	INCONCLUSIVE (blurry im...	INCONCLUSIVE (blurr...	INCONCLUSIVE (blurry image, not visible)
Mobile--BISHOP STATE	97	Mobile	BISHOP STATE	YES	INCONCLUSIVE (blurry im...	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BLOUNT HIGH SCH	97	Mobile	BLOUNT HIGH SCH	YES	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BOYS & GIRLS CLUB OF MOBILE	97	Mobile	BOYS & GIRLS CLUB OF MOBILE	INCONCLUSIVE (bl...	INCONCLUSIVE (blurry im...	INCONCLUSIVE (blurr...	INCONCLUSIVE (blurry image, not visible)
Mobile--BURNS MIDDLE SCHOOL	97	Mobile	BURNS MIDDLE SCHOOL	YES	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--BURROUGHS ELEM_	97	Mobile	BURROUGHS ELEM	YES	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--CHURCH OF THE ISLAND	97	Mobile	CHURCH OF THE ISLAND	YES	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--CLEARWATER CHRISTIAN CH	97	Mobile	CLEARWATER CHRISTIAN CH	INCONCLUSIVE (bl...	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--COLLIER ELEM	97	Mobile	COLLIER ELEM	INCONCLUSIVE (bl...	YES	YES	INCONCLUSIVE (blurry image, not visible)
Mobile--COLLINS RHODES ELEM	97	Mobile	COLLINS RHODES ELEM	INCONCLUSIVE (bl...	INCONCLUSIVE (blurry im...	YES	INCONCLUSIVE (blurry image, not visible)

Caption: First half of the GC-AACS ADA Parking Accessibility Checklist for Mobile County polling locations (2024). This table shows the initial portion of the team's structured dataset, including unique IDs, precinct names, and ADA parking criteria classifications based on Google Earth imagery.

J	K	L	M	N
Parking A6: Shortest Accessible Route to Entrance Are the designated accessible parking spaces located on the shortest accessible route to the accessible entrance? (§208.3.1)	Within the Voting Area F1: Floor Surface Stable/Firm/Slip-Resistant Are floor surfaces stable, firm, and slip resistant? (§302.1)	Within the Voting Area F2: No Wall Protrusions > 4" (27"-80") Is the route free of wall mounted objects that protrude more than 4" into the path of travel and are between 27" and 80" high? (§307.2)	Within the Voting Area F3: No Post Protrusions > 12" (27"-80") Is the route free of post mounted objects that protrude more than 12" into the path of travel and are between 27" and 80" high? (§307.3)	Within the Voting Area F4: Overhead Clearance ≥ 80" Are objects that hang over the route 80" or higher, including the underside of stairs? (§307.4)
NO	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
INCONCLUSIVE (bl...	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
N/A (Non-exist Locat...	N/A (Non-exist Loc...	N/A (Non-exist L...	N/A (Non-exist L...	N/A (Non-...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
INCONCLUSIVE (bl...	N/A (Non-exist Loc...	N/A (Non-exist L...	N/A (Non-exist L...	N/A (Non-...
NO	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
INCONCLUSIVE (bl...	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...
YES	N/A (Interior Feature)	N/A (Interior Fea...	N/A (Interior Fea...	N/A (Interi...

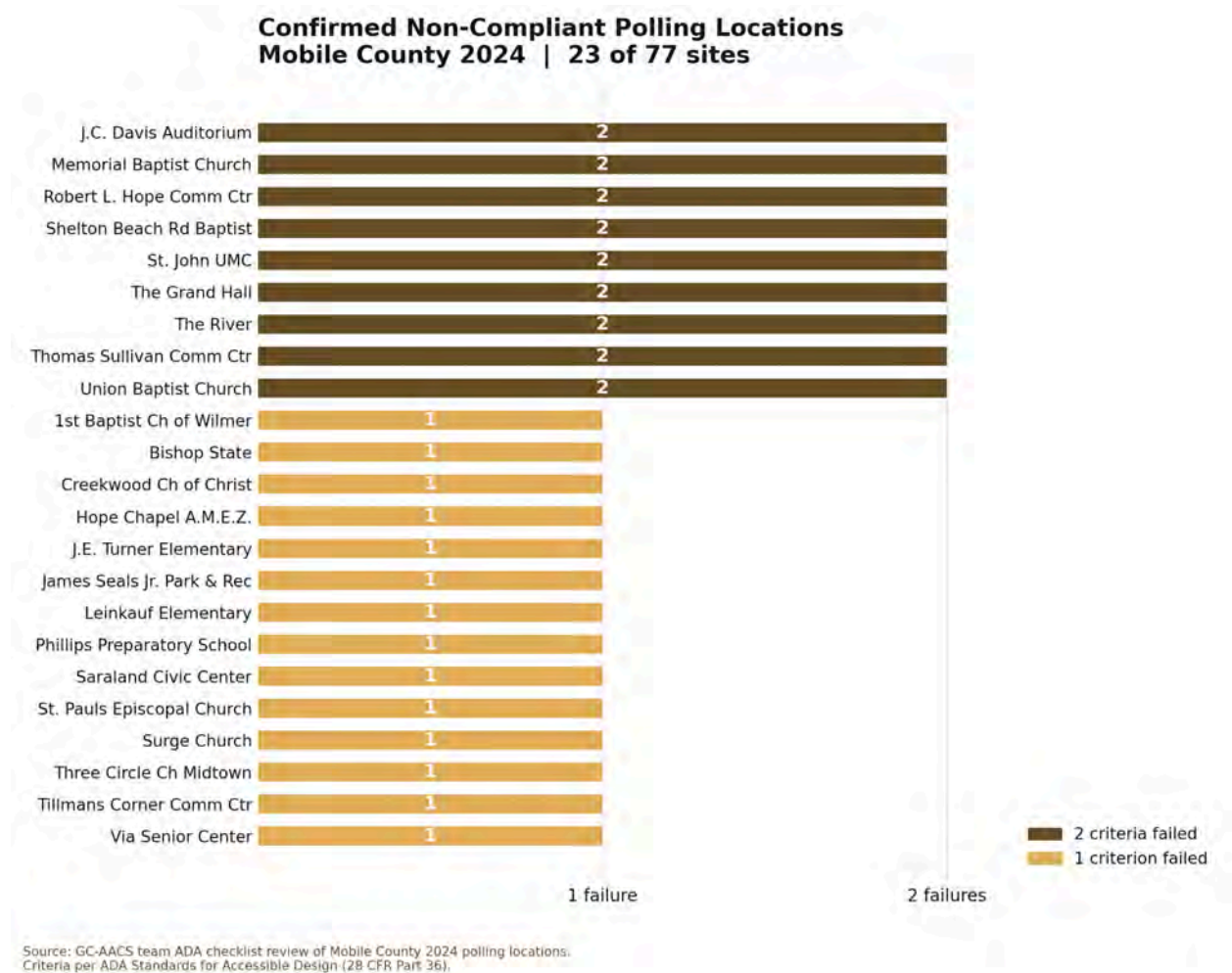
Caption: Second half of the remaining portion of the team's structured dataset.

Figure 1



Caption: ADA parking accessibility compliance results across six criteria for all 77 active Mobile County 2024 polling locations, based on GC-AACS team observational review using ADA Standards for Accessible Design, 2026.

Figure 2



Caption: Mobile County 2024 polling locations with at least one confirmed ADA parking accessibility issue, grouped by number of failed criteria, based on GC-AACS team observational review using ADA Standards for Accessible Design, 2026.

Recommendations

Recommendations for the client (SPLC)

Based on the findings of the Mobile County accessibility review, we recommend that the Southern Poverty Law Center (SPLC) use this project as a preliminary accessibility triage framework to help prioritize future in-person verification initiatives, rather than a complete ADA compliance audit. The analysis found high levels of uncertainty regarding external parking accessibility at several polling locations, with more than half of the sites analyzed lacking visible evidence from public images to prove accessible exterior parking features. This shows that accessibility uncertainty could be a useful indicator for prioritizing future field review activities.

One of the most obvious recommendations is that SPLC prioritize follow-up verification at voting places where accessible parking conditions cannot be verified with available images. Churches, schools, and community centers were the most often reported locations with minimal or unclear accessibility evidence. Because these facility types are among the most prevalent polling locations in Mobile County, they could be the most important targets for further accessibility research.

When field review time or resources are limited, we suggest that accessible parking facilities be verified first, then followed by more accessibility components. Parking accessibility was one of the few ADA-related areas that could be evaluated through a visual assessment, and it is also one of the first physical obstacles that voters with disabilities face. Particular attention should be given to the availability of van-accessible parking spaces, visible access aisles, horizontal clearance, and the route between parking lots and polling places. These components were among the most easily visible and comparable across locations.

At the same time, the data show that certain ADA criteria cannot be reliably evaluated using remote imagery inspection alone. Slope compliance, interior circulation paths, door force, and entrance thresholds could not be reliably checked with publicly available imagery. Because of these limitations, we advise SPLC to avoid classifying areas with unclear imagery as inaccessible or "ADA noncompliant." Instead, areas with insufficient visual evidence should be labeled as requiring further verification.

The project also emphasized broader gaps in publicly accessible accessibility information related to polling locations. The SPLC could benefit by supporting or requiring more consistent accessibility reporting methods from local election districts, particularly in terms of accessible parking and voter

assistance accommodations. Standardized reporting has the potential to significantly improve future accessibility evaluations while also reducing the requirement for extensive manual review.

Recommendations for future project teams

Future teams working on ADA accessibility at polling locations should build off both what worked well in this project and what didn't. One of the biggest takeaways is that using tools like Google Earth is actually a really helpful way to get started, especially when you're dealing with a large number of polling locations. It allows you to move quickly and get a general sense of what accessibility might look like across different sites. At the same time, this project made it clear that there are real limits to what you can figure out just by looking at images, so future teams will need to think carefully about how they approach that balance.

Starting with some kind of remote "triage" system is definitely worth continuing. Being able to go through locations and sort them into groups based on what you can see, like clearly accessible, unclear, or not enough information, makes the whole process more manageable. It helps avoid wasting time trying to fully analyze locations where the data just isn't there. In our case, this approach made it easier to focus on patterns instead of getting stuck on individual sites. Future teams could take this further by making the categories even more consistent or by agreeing early on what counts as "enough" visible evidence.

At the same time, it's important to be realistic about what remote analysis can and can't do. A lot of the things that actually matter for accessibility, like how steep a ramp is, how heavy a door is, or whether there's enough space inside to move around, just can't be measured from satellite or street view images. We ran into this problem pretty often, where a location might look fine from above but there wasn't enough detail to say anything for sure. Because of that, future teams should plan from the beginning to combine remote work with some level of in-person verification if possible. Even visiting a smaller number of locations would go a long way in confirming what the images suggest.

If teams do have the chance to do field visits, it would make the most sense to focus on the areas that are hardest to evaluate online but most important for voters. Parking is a big one, since it's often the first point of access and not always clearly visible in images. Things like whether there are marked accessible spaces, where they're located, and how close they are to the entrance are all really important. The route from parking to the entrance is another key area, since even if parking is

technically accessible, it doesn't help much if there isn't a clear path to the building. Entrances themselves are also worth checking in person, especially things like door width, handles, and thresholds. These are small details, but they can make a big difference for someone trying to vote.

It would also help to think more about how this work connects with what other teams are doing. Accessibility is just one part of voting access, and it becomes more meaningful when you look at it alongside other factors. For example, a location that's hard to reach and also not very accessible is going to be a much bigger issue than one that only has one of those problems. Future teams should try to connect their findings with data on transportation, demographics, or turnout so that the bigger picture is clearer.

Another area for improvement is using more than just visual data. Google Earth was useful, but it definitely wasn't enough on its own. There are likely other sources out there, like county records, accessibility reports, or even local organizations, that could help fill in some of the gaps. Even if those sources aren't perfect, combining them with visual observations would make the overall analysis stronger and more reliable.

It could also be helpful for future teams to develop a more structured way of comparing locations. In this project, we mostly grouped sites into general categories, which worked given the limitations, but it didn't always make it easy to compare one place to another. Creating some kind of basic scoring system could make the results easier to interpret and more useful for decision-making. It wouldn't have to be overly complicated, just something that allows for clearer comparisons across locations.

One really important thing is being honest about uncertainty. There were a lot of cases where we just didn't have enough information to say whether something was accessible or not. That doesn't mean those places are non-compliant; it just means we couldn't tell from the data we had. Future teams should make that distinction clear, since it affects how the findings are used. Being upfront about limitations actually makes the work more credible, not less.

Finally, future teams should think of this project as something that can keep evolving rather than something that gets completed once. Polling locations can change over time, and accessibility conditions can improve or get worse. Keeping the data organized and well-documented will make it easier for the next group to pick up where things left off instead of starting from scratch. Even small things like clear file naming, consistent formatting, and saving sources can make a big difference later on.

Overall, the next step for this kind of work is to move from identifying possible accessibility issues to actually confirming them and figuring out what to do about them. This project provides a starting point, but future teams have the opportunity to make the analysis more precise and more useful by combining different methods, improving consistency, and building on what's already been done.

Conclusions

Our team evaluated ADA-related parking accessibility conditions at polling locations in Mobile County as part of the broader 11-team regional analysis conducted with the Southern Poverty Law Center (SPLC). Using publicly available polling location data, Google Earth satellite and street-level imagery, and a structured ADA accessibility criteria checklist, Our team developed a standardized accessibility review process for assessing external accessibility features across 77 active 2024 polling locations. This approach allowed us to systematically evaluate parking-related indicators, such as the availability of van-accessible parking, horizontal clearance, vertical clearance, and accessible routes to entrances, while also documenting where image quality prevented verification.

Across the county, our findings reveal a mixed accessibility landscape. While some polling locations demonstrated clear compliance with ADA parking requirements, a significant portion either showed confirmed accessibility gaps or could not be fully evaluated due to limitations in available imagery. The number of "INCONCLUSIVE" classifications underscores a structural challenge in relying solely on remote imagery for ADA assessments. Key criteria, such as slope compliance and surface stability, could not be evaluated at any location, highlighting the need for in-person verification to produce a complete and accurate accessibility profile.

Despite these limitations, our team produced several key deliverables, including a structured accessibility checklist dataset, written findings, analysis documents, visual summaries of accessibility patterns, and recommendations for both the Southern Poverty Law Center (SPLC) and future projects. These deliverables fill a critical gap: prior to this project, no centralized or standardized data existed for ADA parking accessibility at Mobile County polling locations. Our work now provides the Southern Poverty Law Center (SPLC) with a reusable foundation for future monitoring, advocacy, and field verification efforts. The project also demonstrates the importance of collaborative and consistent documentation practices in the team's research effort. Coordination across teams

helps ensure that accessibility findings can contribute to a broader understanding of voting access barriers across the region. Also, the project established a reusable framework and organized datasets that future teams can continue to expand rather than rebuilding the process from the beginning.

Overall, this project provides a meaningful starting point for understanding polling location accessibility in Mobile County and contributes practical findings that would support future accessibility research, advocacy efforts, and election protection initiatives. More importantly, this project reinforced the value of transparent and responsible analysis when working on issues that directly affect public access. Therefore, this project can support future efforts to improve transparency, accessibility evaluation methods, and help ensure that voting remains accessible to all members of the community.

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Deliverables Checklist

- ☐  Sources Doc
- ☐  Background Research and Project Context Analysis
- ☐  Client Requirements Document.docx
- ☐  Final Presentation Contribution Brief - GC-AACS
- ☐  Mobile County ADA Parking Accessibility Checklist
- ☐  GC-AACS Client Deliverables

Broadband & Digital Access for Civic Participation

GC-BAND

Crystal Samoya: Project Manager

Amir Pourmand: Data Collector

Asia Robillard-Hill: Data Analyst

Lyndsey Bakal: Technical Writer

Garrett Bugg: Data Visualizer

Southern Poverty Law Center

INST490 - 0105

18 May 2026

Key Findings

- Clarke, Escambia, Monroe, and Washington counties consistently have very few devices (less than 5%) that have access to 1000MB download and 100MB upload speeds, compared to Baldwin and Mobile which have over 50% of devices being able to access them.
 - The counties that struggle with access to high broadband speeds are also counties that have less rates of device ownership and high rates of economic challenges.
-

Final Report and Deliverables

Project Context

Our team, GC-BAND (Broadband & Digital Access for Civic Participation), is one of eleven teams working under the Southern Poverty Law Center (SPLC), analyzing voting and civic participation across Alabama's Gulf Coast. Our team focuses on whether people in Mobile, Baldwin, Washington, Monroe, Escambia, and Clarke counties have the broadband internet access needed to look up voter registration, find polling locations, or access civic information online. Our team's work became a foundational layer for other teams, such as GC-VIZ, which needed our cleaned data and findings to build their cross-project dashboards. We benefited from GC-DIG, whose standardized FIPS codes helped us align our Census and FCC datasets with the formats other teams were using, so everything could eventually be merged. We also coordinated through Slack and the presentation working group to make sure our broadband findings complemented the work coming out of the

polling access teams, since all of us are discussing different parts of the same project for the same client.

Methods

This project used secondary data to analyze disparities in broadband internet access across gulf-coast and surrounding counties in Alabama. Due to the project being conducted entirely online, we needed access to publicly available resources to shape our decision process. We needed to identify what county characteristics correlated with broadband access gaps and determine what information would be helpful to our clients.

Data Sources

All data we obtained came from publicly available online sources.

Broadband Availability Data

The primary dataset for broadband data coverage was the FCC's Broadband Data Collection for coverage data map:

- <https://broadbandmap.fcc.gov/data-download>

This map reports available internet service by technology type and maximum advertised speed.

Some other supplementary coverage data we used were drawn from Microsoft Airband usage data which estimated actual broadband usage based on anonymized speedtests from windows

downloads. As well as Ookla Speedtest open data which provides global fixed broadband and mobile (cellular) network performance metrics.

Socioeconomic and Demographic Data

County-level demographics and economic variables were taken from:

- Census ACS 5-year estimates for demographics and internet subscriptions (tables B28002, S2801 via <https://data.census.gov>)

Analytical Approaches

Data Cleaning

Raw files were obtained from the FCC and the US Census. While everything was readable it often came with way more information than was needed, such as other states and counties. We cleaned up the data in google sheets to only include Alabama as a whole and the specific counties we were focused on.

Visualizations

We wanted to make sure that all of our reports could be easily readable by the average person with little to no broadband knowledge so we made comparisons using visuals. We used Tableau which allowed us to upload our data and analyze it with visuals and share insights.

Analysis of Counties

Based on the data we obtained across sectors, we came up with a report detailing our findings. We observed what broadband accessibility looks like in each of these counties and how it ties to the demographics. Our team set out to analyze and make connections but not assumptions.

Tools Used

- We utilised Google Sheets and Documents to organise our data into one place that was easily accessible to the group and others who may need this information.
- We used Tableau to create our visualisations
- We used Canva to develop our website

Challenges

- Reconciling different broadband metrics across sources:
 - FCC shows advertised availability
 - Census shows subscription rates
 - Microsoft shows speeds

These are all fundamentally different measurements. To reconcile these findings, we needed to determine the best way to showcase that each measure was a distinct layer to the broadband problem. FCC determines where a provider says they could serve an address. The Census shows

whether a household pays for an internet service, and Microsoft shows the real-world performance of the broadband service speed.

- Creating a community-facing map

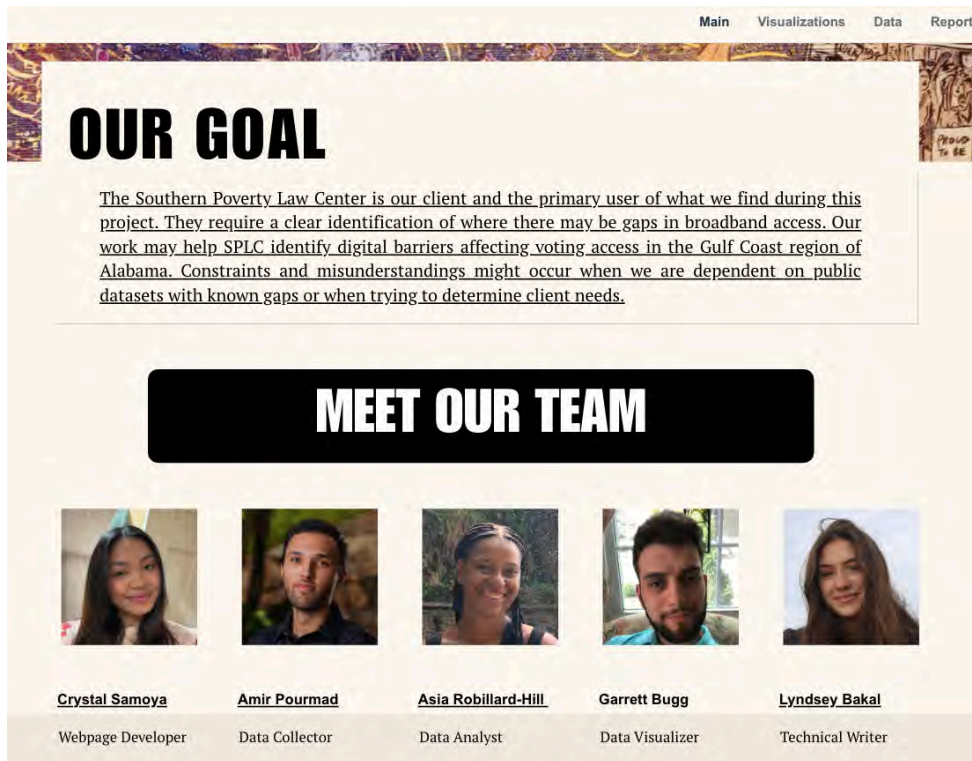
We had trouble creating a community-facing map from what type of program to use, to writing the code. In the end, we decided it was too difficult for us to create this map and be able to put in all of our effort. We ended up scrapping that portion of the project deliverables.

Deliverables and Findings

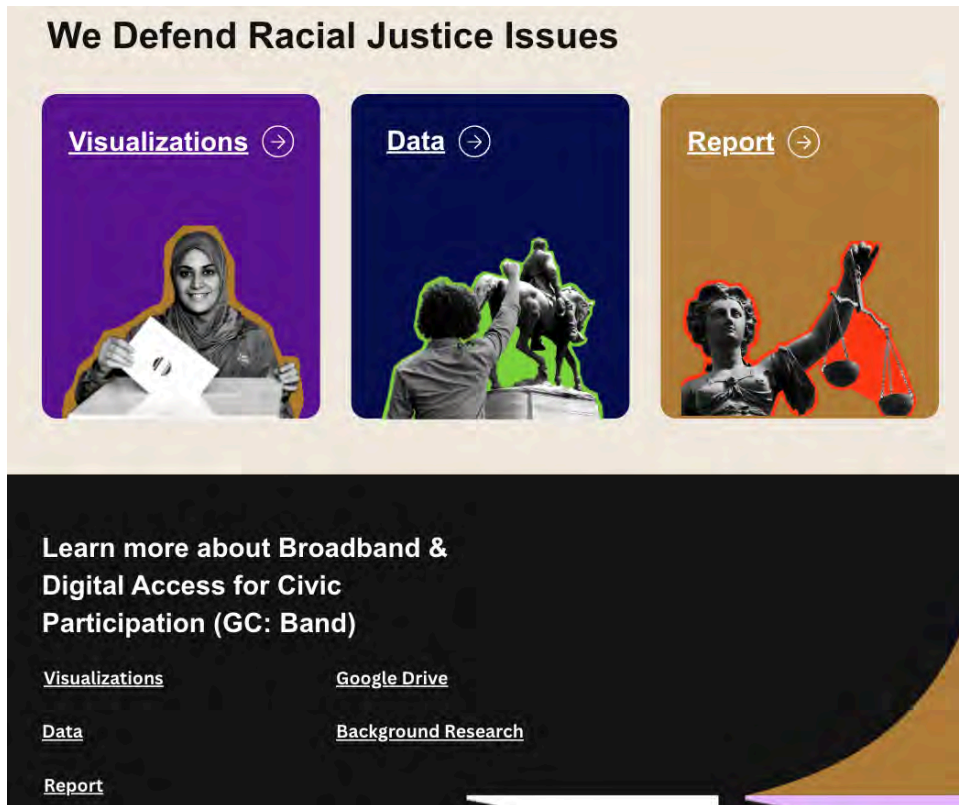
Deliverable: Community-Facing Website

Link: <https://umcp.my.canva.site/gc-band/>

Our website is organized into four main sections: a homepage with project overview and team introductions, a Visualizations page housing our Tableau charts and data graphics, a Data page documenting all sources used, and a Report page containing our full county-by-county written analysis. Each of these is described below with embedded content, rationale, and source citations. The website was designed and built by Crystal Samoya using Canva Sites. It is as the central hub tying together every other deliverable into one shareable resource for the convenience of the SPLC. The site features a branded "BAND" logo, a hero section for our project goal, team member photos and roles, and three primary navigation cards linking to Visualizations, Data, and Report pages. The footer includes links to our Google Drive folder and Background Research documents.



The SPLC client emphasized that the final product should be something you can put in front of community members, something graphical, not intimidating, and easy to focus on. A website meets all of these criteria: it requires no software to open, works on mobile devices, and can be shared with a single URL. Rather than delivering a collection of loose files, the website gives SPLC a package they can share with partners, funders, and community audiences. The homepage states our goal: "The Southern Poverty Law Center is our client and the primary user of what we find during this project. They require a clear identification of where there may be gaps in broadband access. Our work may help SPLC identify digital barriers affecting voting access in the Gulf Coast region of Alabama."

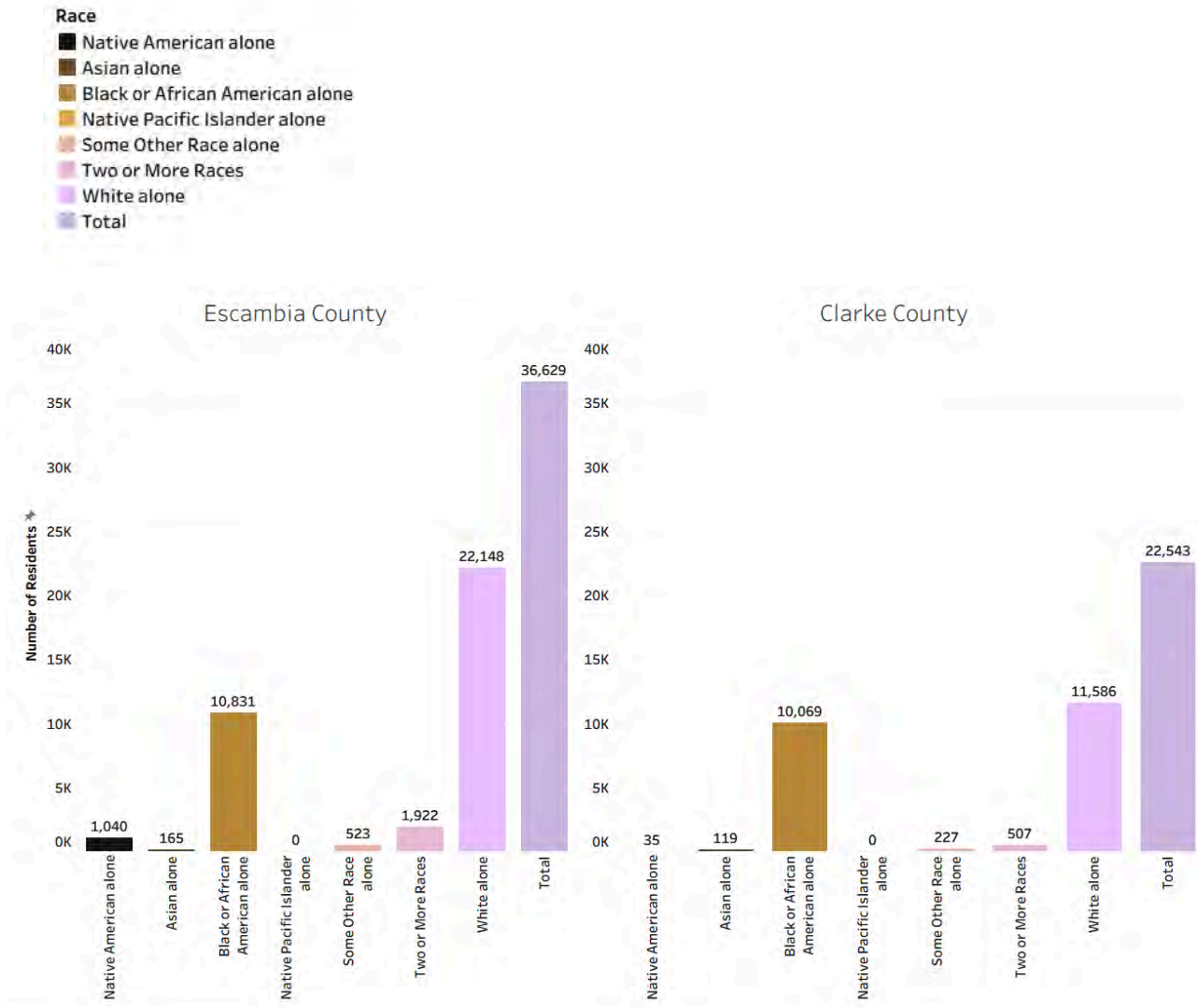


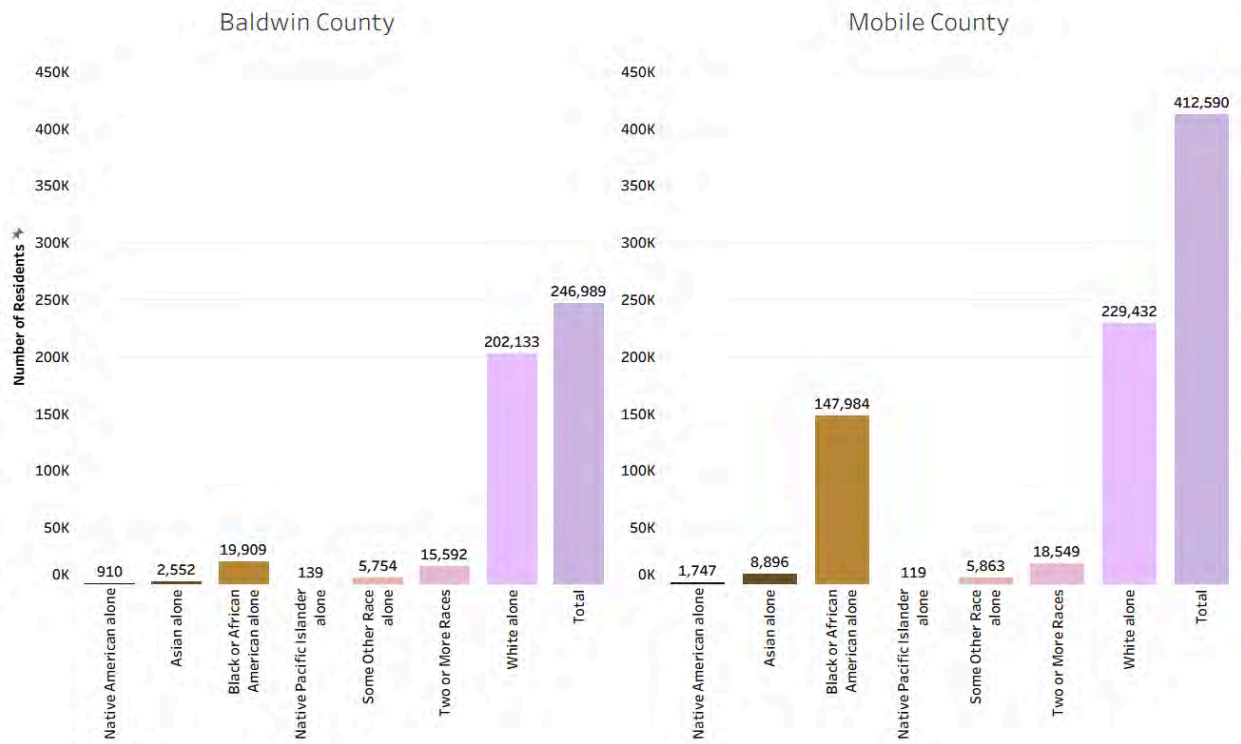
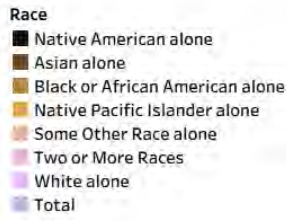
The homepage introduces the five team members and their roles: Crystal Samoya (Webpage Developer), Amir Pourmand (Data Collector), Asia Robillard-Hill (Data Analyst), Garrett Bugg (Data Visualizer), and Lyndsey Bakal (Technical Writer). Below the team section, three navigation cards with imagery related to civic justice and voting rights link to the Visualizations, Data, and Report sub-pages. Our site tagline reads "We Defend Racial Justice Issues," grounding our broadband work in the SPLC's civil rights context.

Deliverable: Visualizations

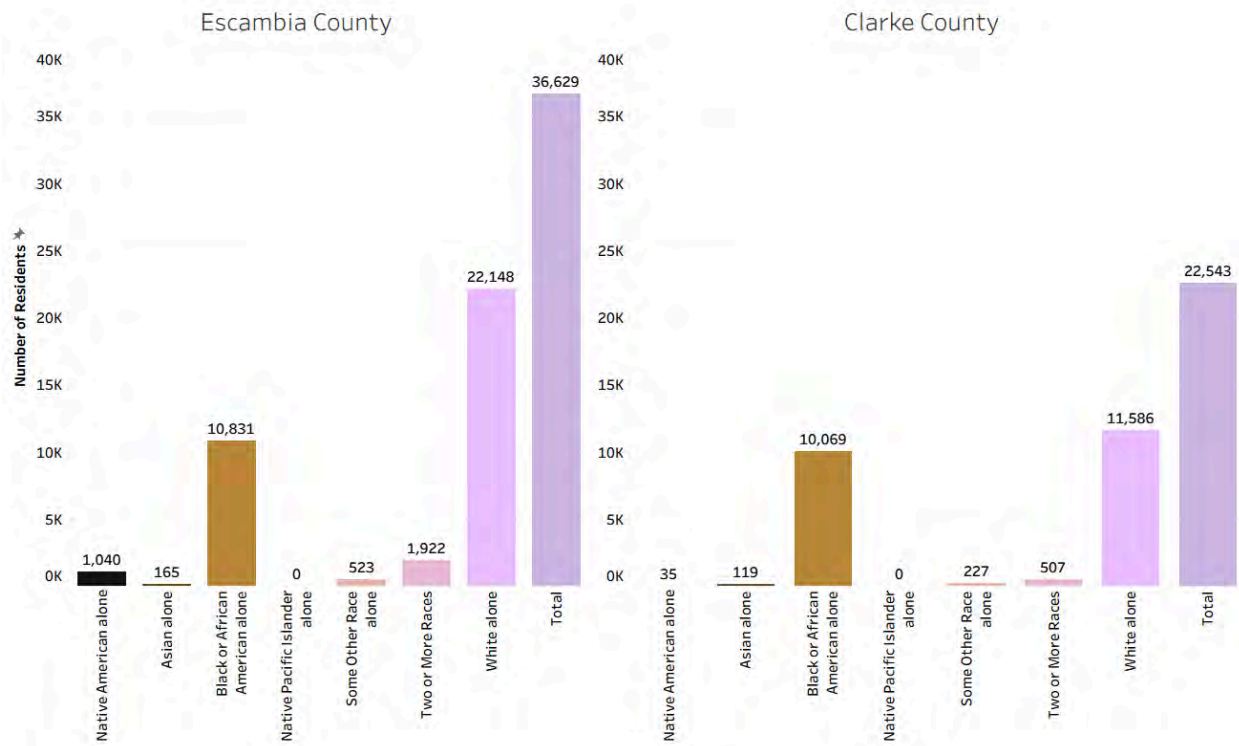
Using Tableau, I visualized the key data relevant to our research. I have stylized the visuals according to our client's recommendations.

The first visuals compare the five year race estimate statistics by county.

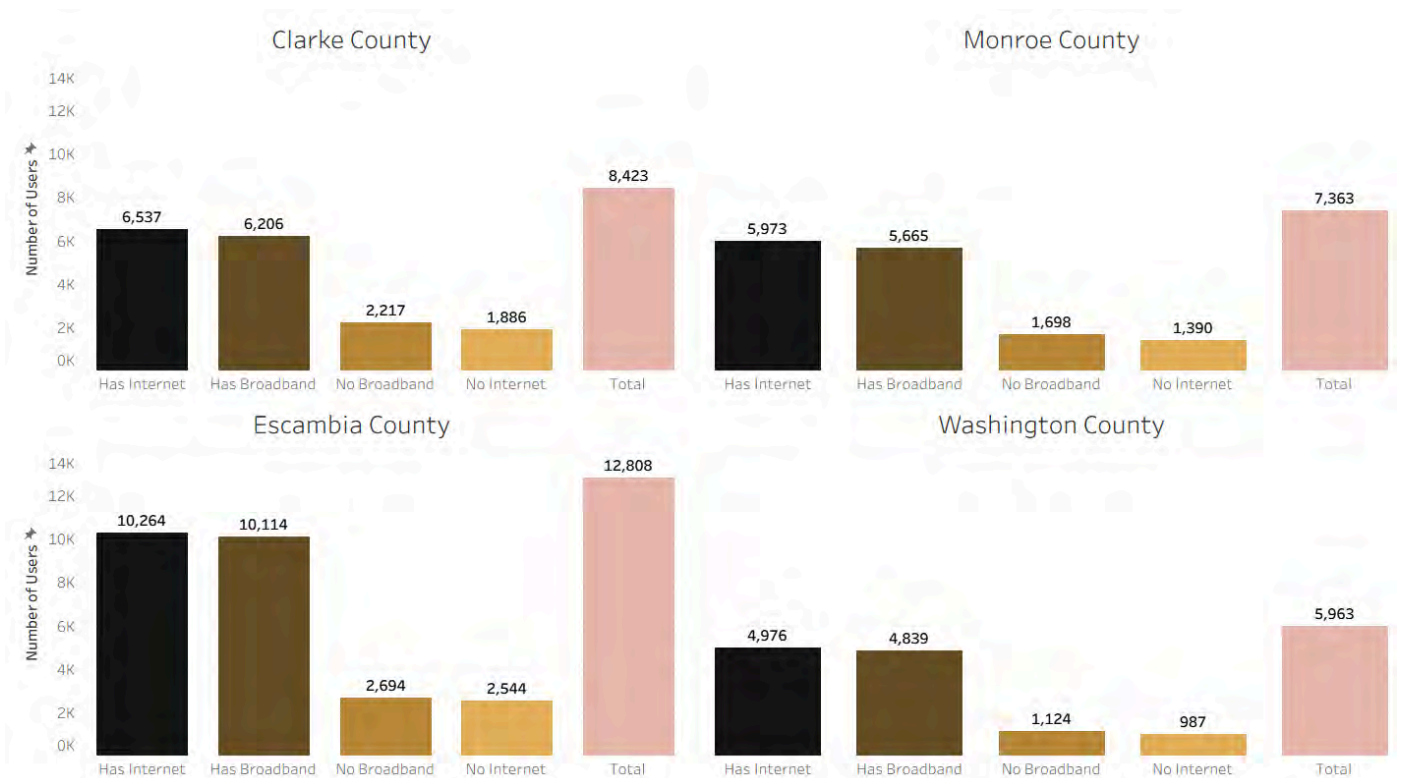
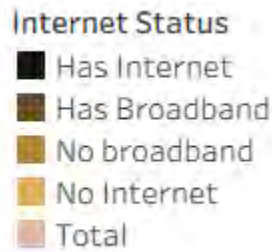




- Race**
- Native American alone
 - Asian alone
 - Black or African American alone
 - Native Pacific Islander alone
 - Some Other Race alone
 - Two or More Races
 - White alone
 - Total



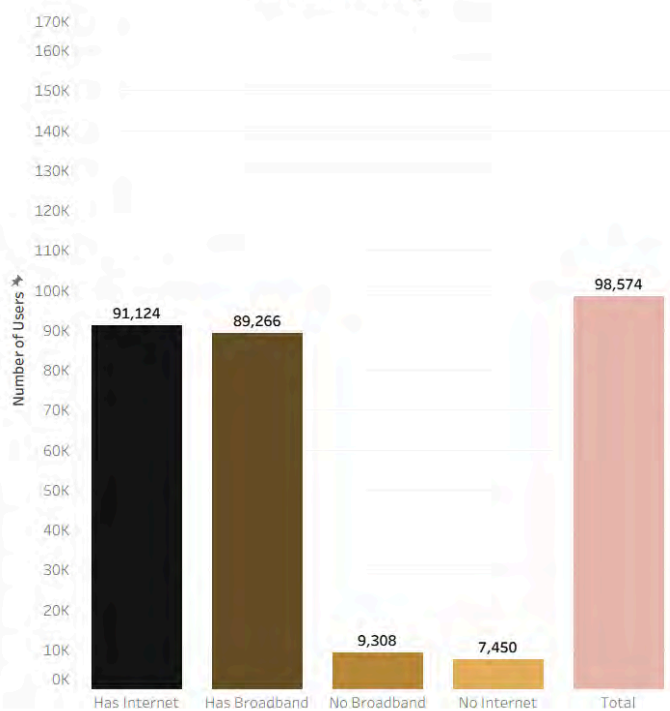
Our next visuals cover internet access statistics across our area of focus. I have grouped Mobile County and Baldwin County separately because of the difference in population scale.



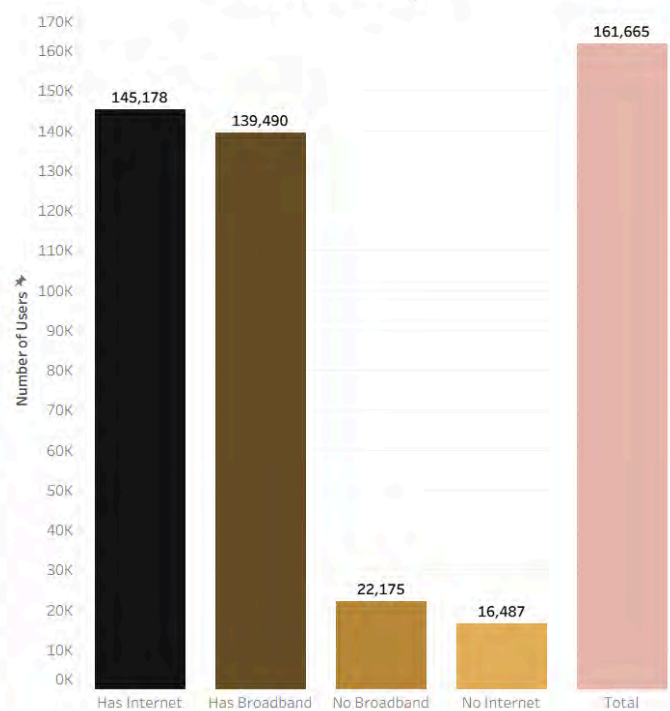
Internet Status

- Has Internet
- Has Broadband
- No broadband
- No Internet
- Total

Baldwin County



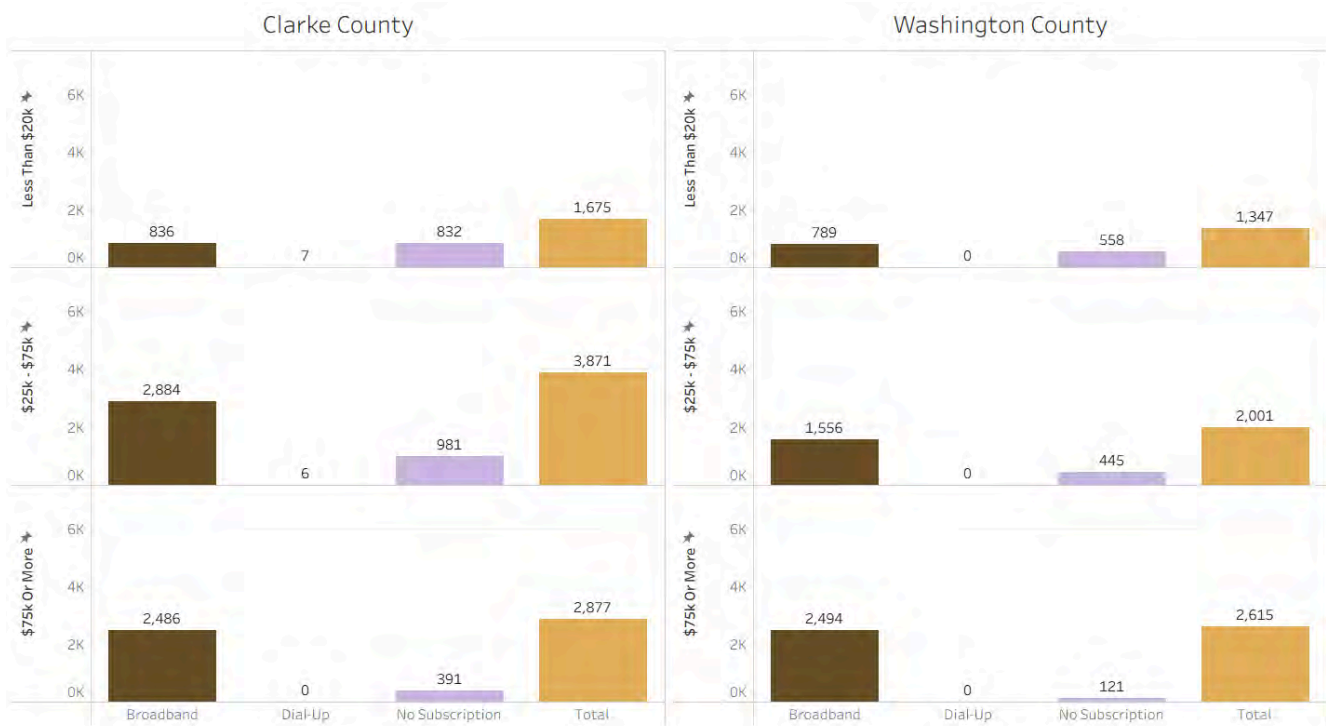
Mobile County



Next, we are visualizing the internet types compared to income for each county. This applies especially to the broadband coverage rates in comparison to income.

Internet Type

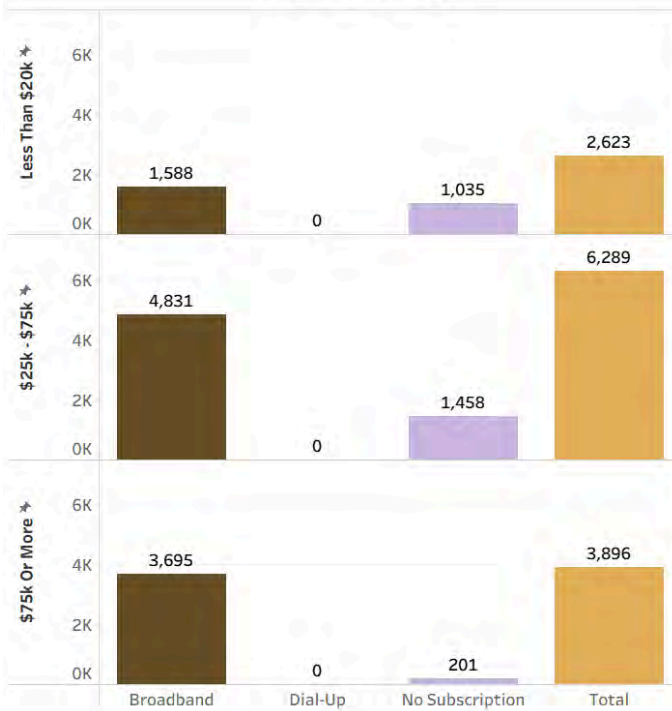
- Broadband
- Dial-Up
- No Subscription
- Total



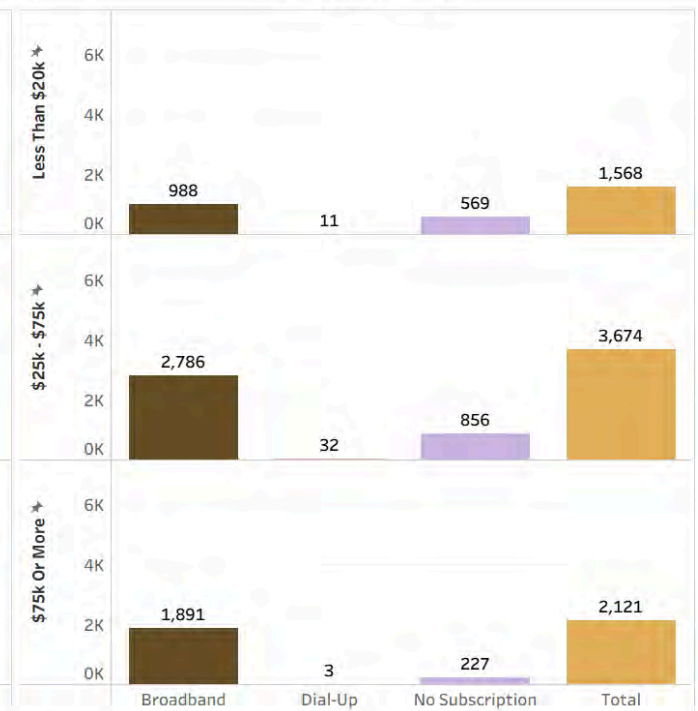
Internet Type

- Broadband
- Dial-Up
- No Subscription
- Total

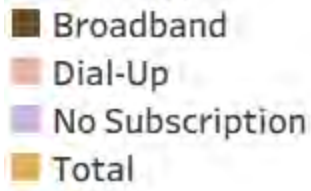
Escambia County



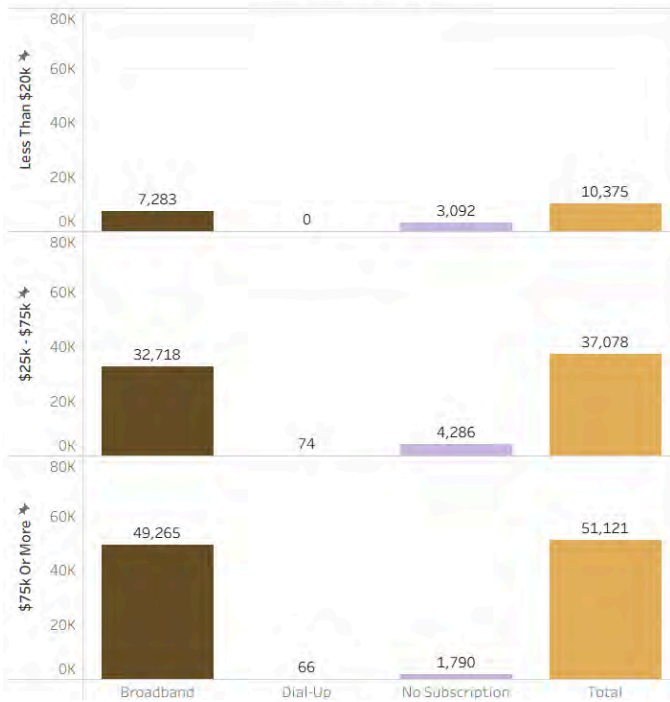
Monroe County



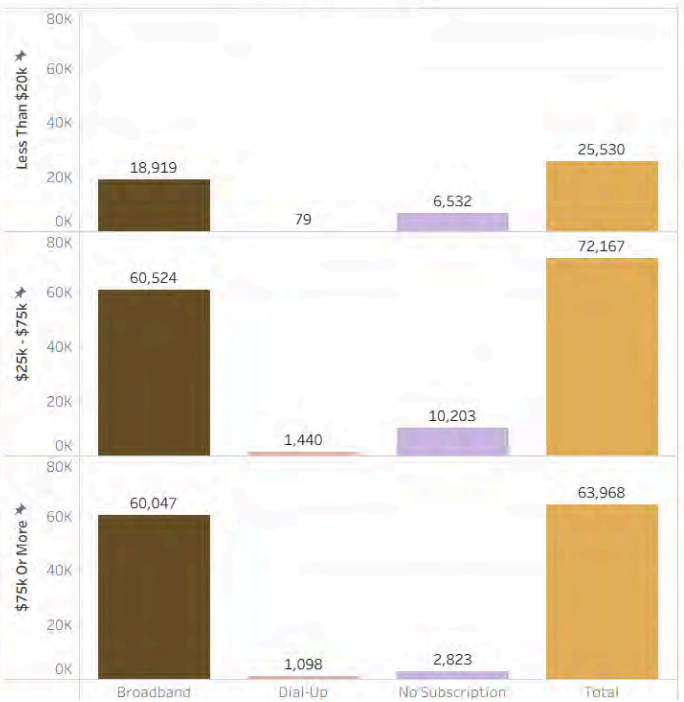
Internet Type



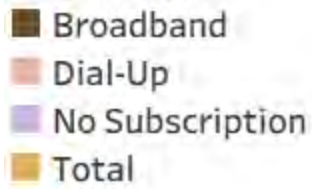
Baldwin County



Mobile County

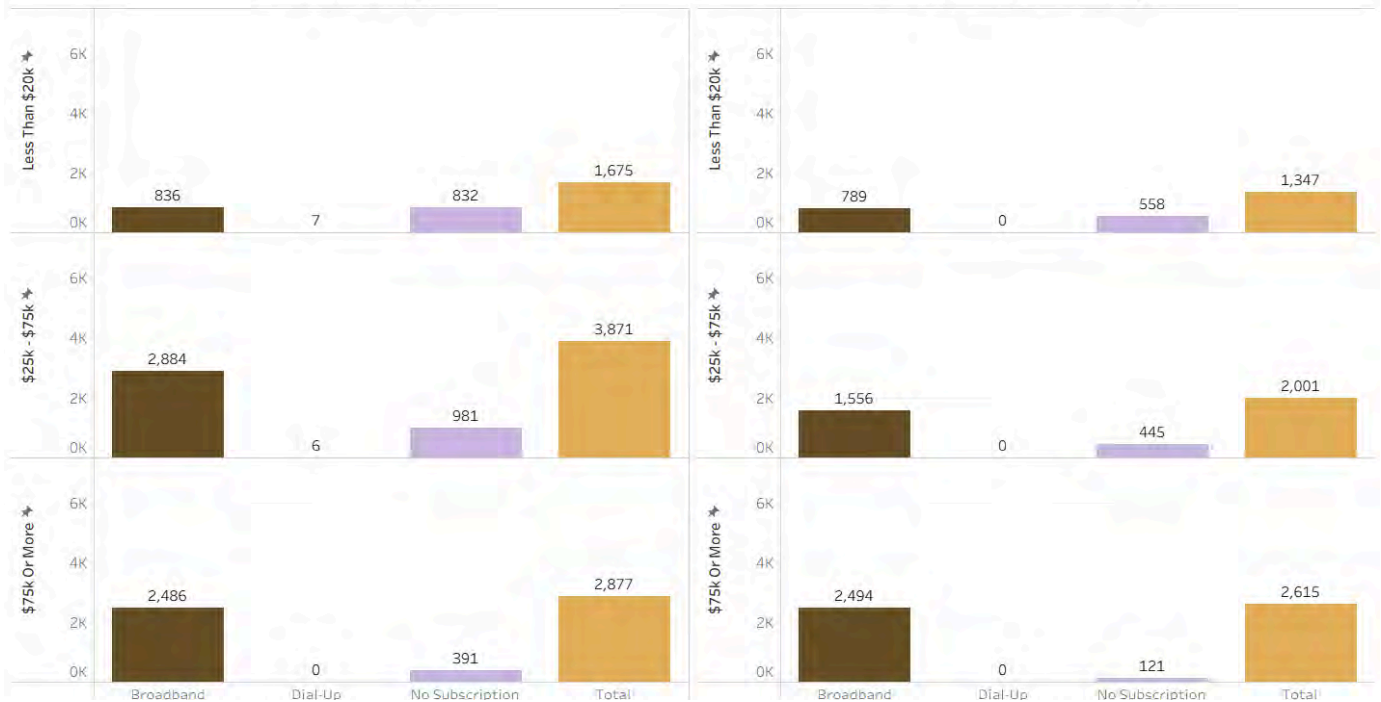


Internet Type



Clarke County

Washington County



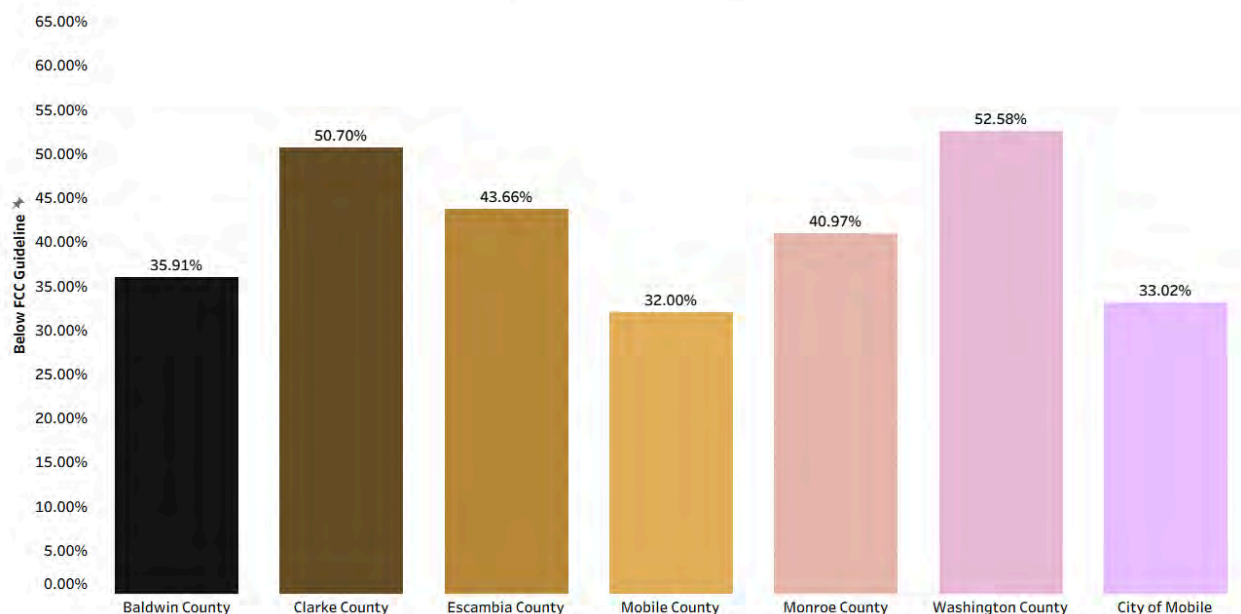
These visuals show a fairly consistent increase in broadband coverage rates as income increases. This trend generally shows in more populated areas such as Baldwin and Monroe counties as well as less populated counties such as Clarke and Washington counties.

After understanding some of the patterns around internet presence and geography, we wanted to understand internet quality. Our next visual covers a variety of different geography types, which are rural, urban, tribal, or nontribal. In these visuals, geography occasionally overlaps. For example, a nontribal area may also be a rural area. Some counties also lack certain geographical areas. For example, Clarke County had only one geographical data point. Because this geography was listed as both “Nontribal” and “Rural”, I consolidated the two identical values into “Total”. These graphs show the number of users who fall below the minimum recommendation for all internet usage, including HD video streaming.



All Areas and Locations Combined

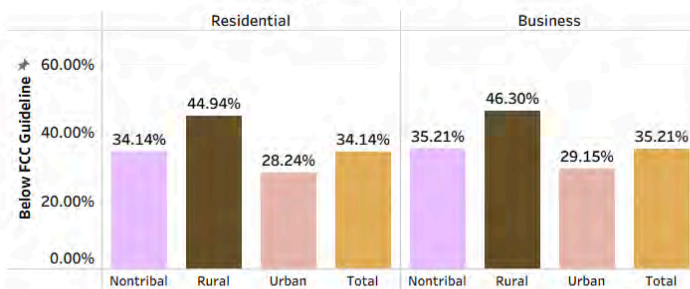
Alabama Overview



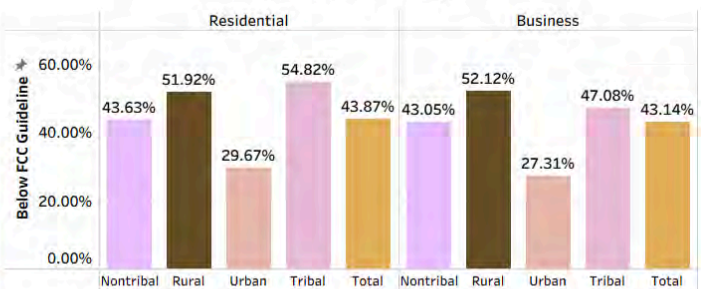
Area Data Type



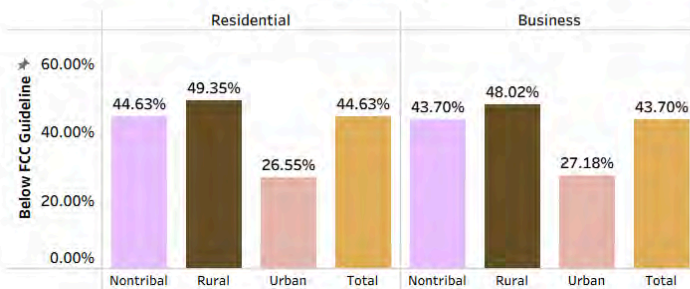
Baldwin County



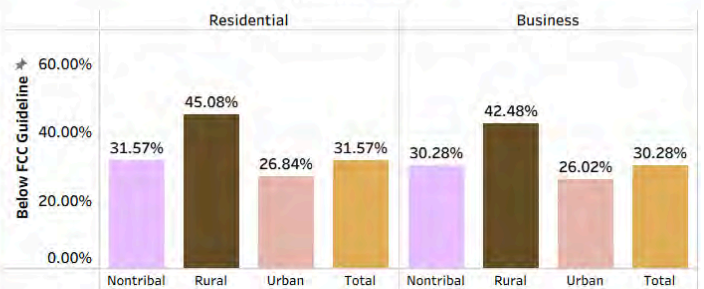
Escambia County



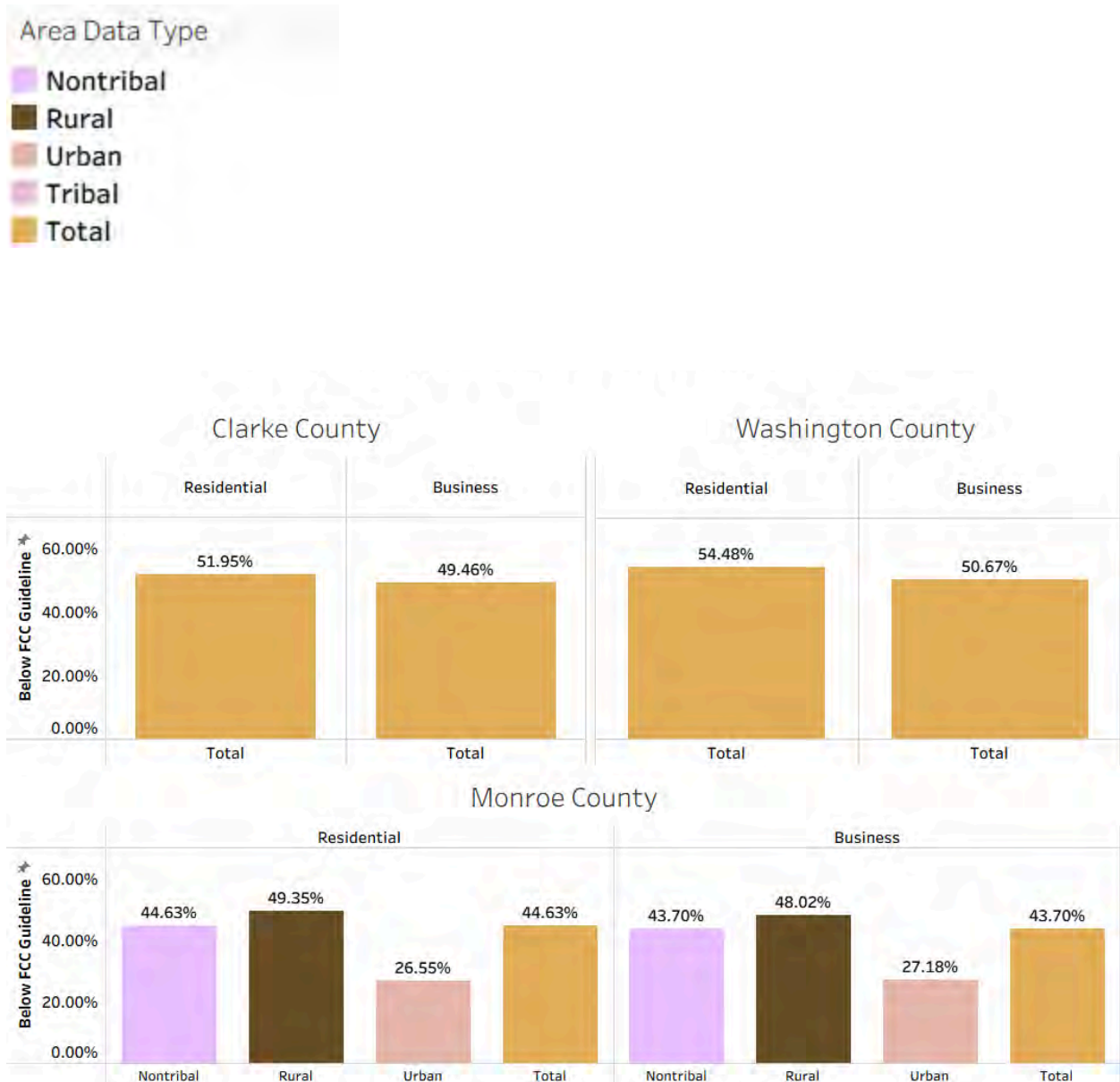
Mobile County



Mobile City



This data set contains Mobile City as well as Mobile County. This is an interesting perspective because it shows the statistics from a more specific area.

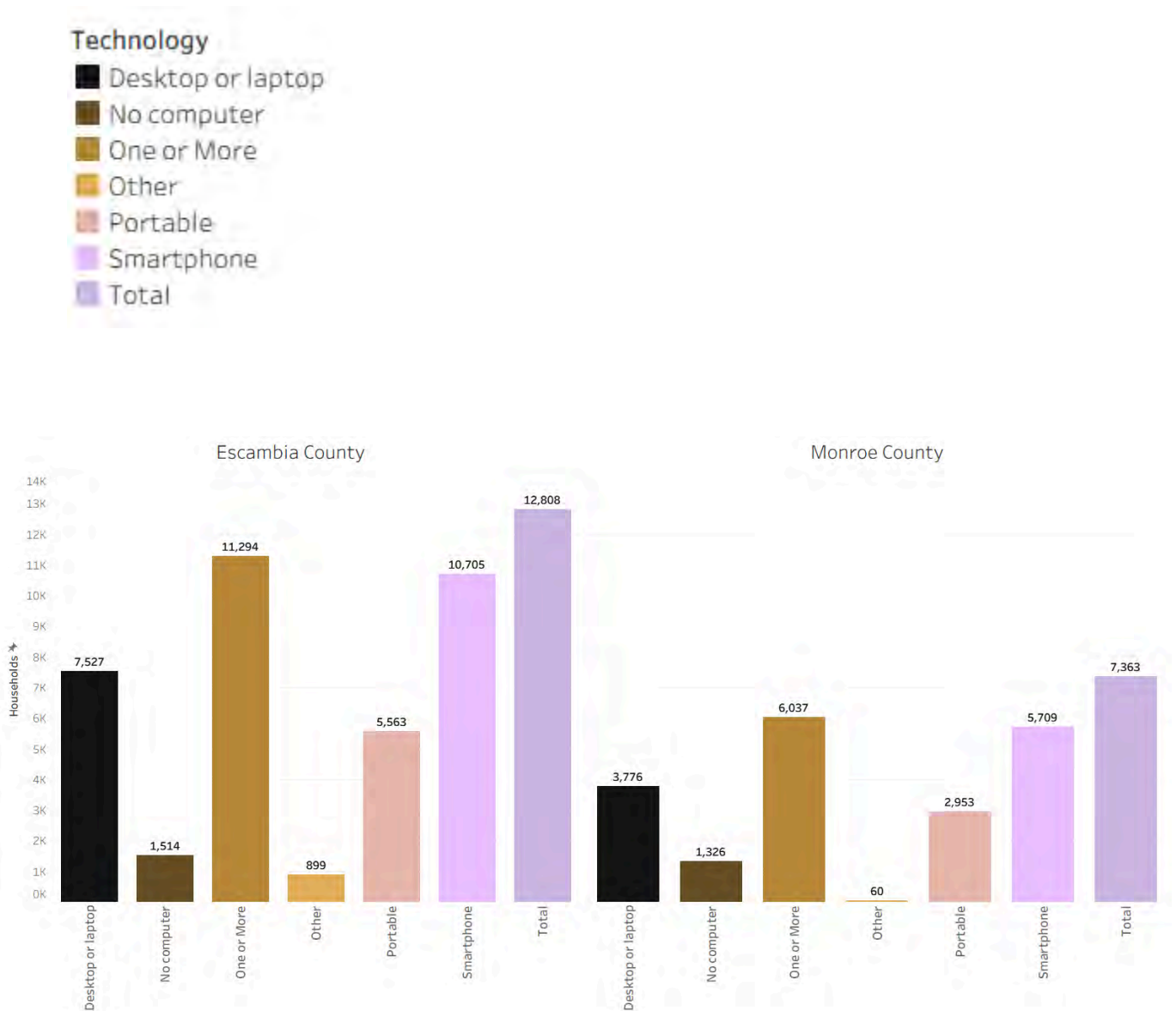


Source for FCC Recommendations:

United States Federal Communications Commission. (2022, July 18). Broadband Speed Guide. Home, Consumer, Consumer Guides. <https://www.fcc.gov/consumers/guides/broadband-speed-guide>

Our final series of visuals looks into how common individual computer types are.

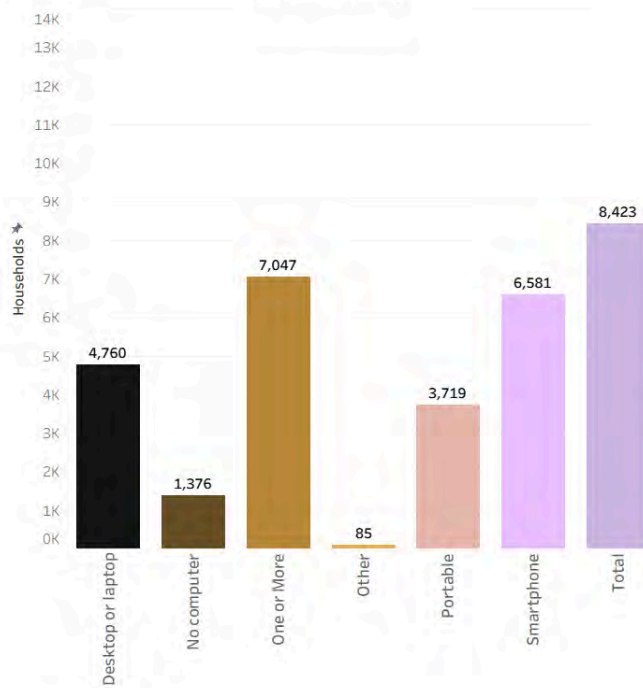
Particularly of focus here is the number of households who have no computer access.



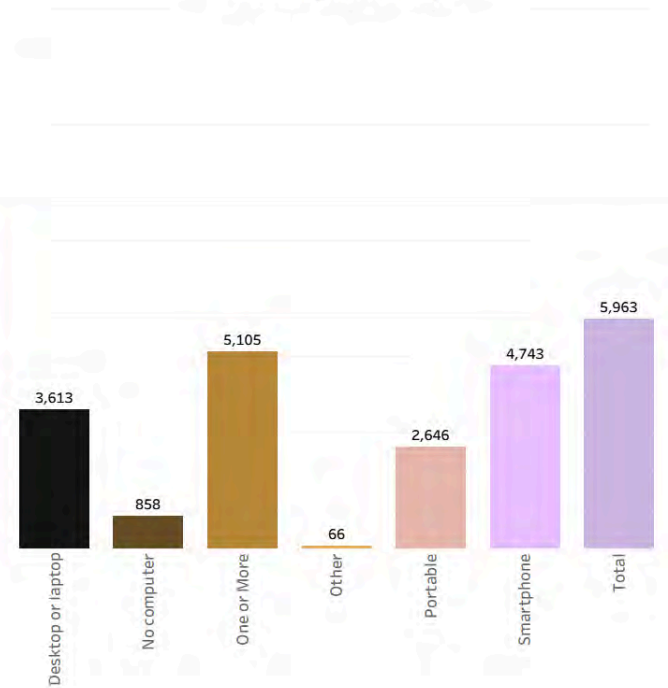
Technology

- Desktop or laptop
- No computer
- One or More
- Other
- Portable
- Smartphone
- Total

Clarke County

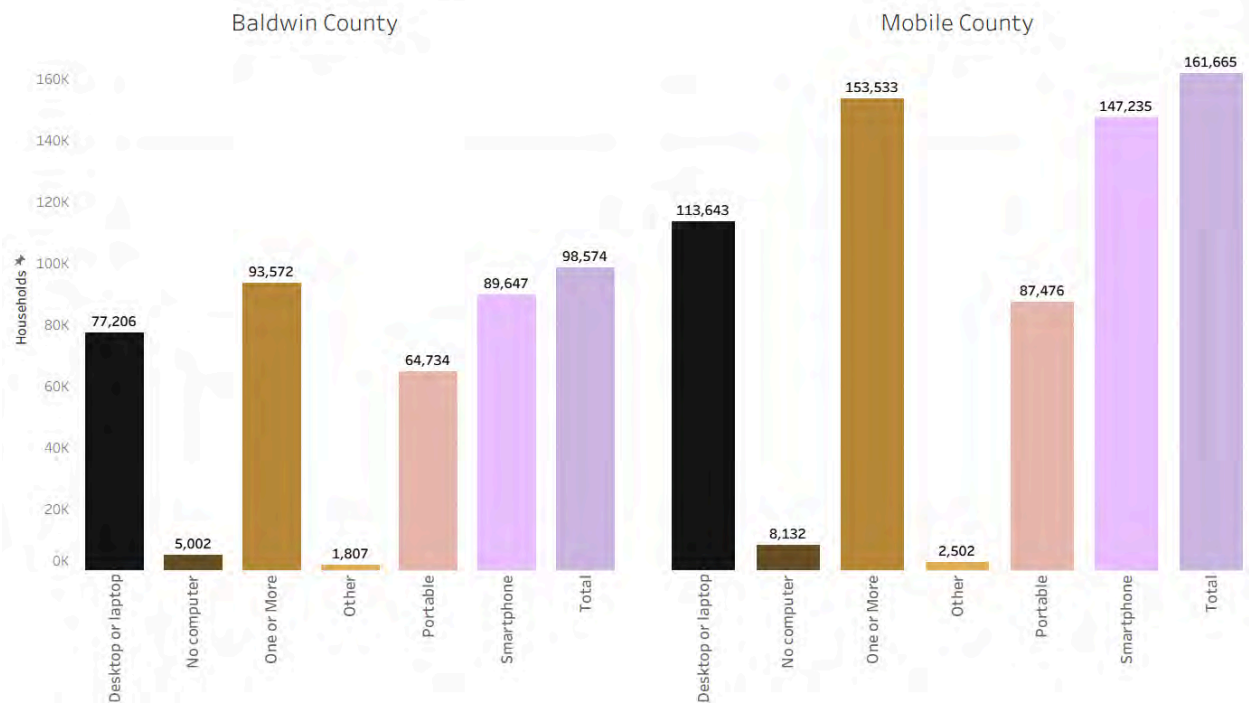


Washington County



Technology

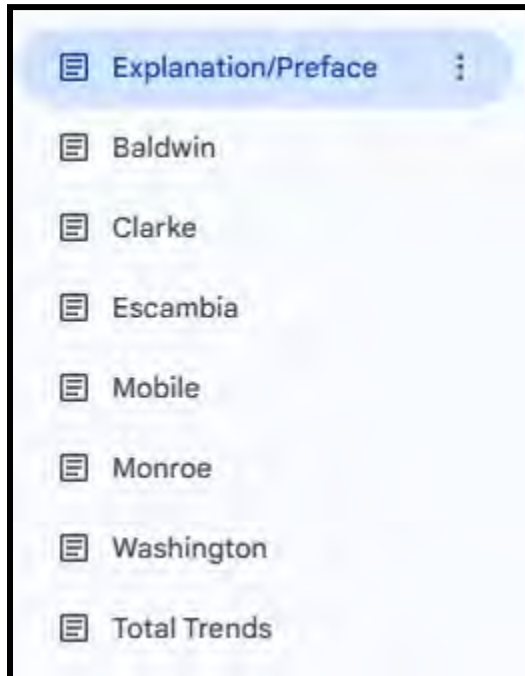
- Desktop or laptop
- No computer
- One or More
- Other
- Portable
- Smartphone



Note: I made all visualizations using Tableau.

Deliverable: Written Report on Data Gathered

Within our website, users can find a document which is a written report describing certain aspects of each of the 6 counties studied. This report is broken down into 8 sections; a preface, an analysis of each of the 6 counties, and a page where we discuss total trends we saw.



The Explanation/Preface section describes to the readers what is included in the report, as well as explanations of terms used in the report, national averages and where they were sourced from, as well as a more in-depth description of broadband speeds and what each of the levels mean.

Within the analysis of each of the 6 counties, readers will find 4 sections describing general population statistics, race statistics, types of devices that the population owns, and what broadband speeds these devices have access to. Each of these sections contains a list of the raw data from our sources as well as a summary of the data, typically comparing it to the other counties in the study or

national averages. Below is an example of what one of these sections looks like.

Population: 23,087
Median Household Income: \$49,249
Poverty Rate: 20.10%
Percentage With Bachelor's Degree or Higher: 15.00%
Employment Rate: 46.00%
Land Area: 1,238.40 square miles
Population Per Square Mile: 19
In Summary: Clarke County is one of the larger counties by area studied, though is not very densely populated compared to the other counties. The median household income is much lower than the national average, as well as the percentage of the population with a bachelor's degree or higher. In turn, the poverty rate is about double that of the national average, tied for the highest poverty rate with one other county in the study, and the employment rate is significantly lower than the national average.

Each county also has a summary that outlines the county as a whole, essentially describing the 4 sections in one concise place.

Finally, we have a Total Trends page that summarizes all of the counties studied in one place in order to analyze any trends or similarities we saw throughout the study. This is broken down into 3 sections; General Population and Race Statistics, Types of Devices the Population Owns and What Speeds the Devices Can Handle, and Trends As a Whole. In other words, readers are able to see what the population and its economic factors look like, what devices are present and what kind of speeds are accessible, and finally how the two sections just mentioned may affect one another.

This report as a whole is essential for our project's goals because it clearly defines and displays what broadband accessibility looks like in each of these counties as well as how it ties to the demographics of each of these counties. Our team set out to analyze this data and make these connections, not necessarily make assumptions on why it may be happening, which is explained in

the report. This report would probably be used mostly by SPLC themselves, while the visual deliverables would be more public-facing, though the valuable analysis of this report provides SPLC with an in-depth understanding of broadband accessibility in each of these areas.

Findings

From our datasets and analysis of this data we found that the more urban counties, or the ones with higher population densities, had the best broadband accessibility. Out of the 6 counties and 6 levels of broadband access analyzed, only 2 counties could access the highest speeds; Baldwin and Mobile. All counties were able to access the first 5 speeds, with over 99.98% of all devices being able to do so, but in counties besides Baldwin and Mobile, only 1-5% of devices were able to access the highest speed. In Baldwin and Mobile, over 50% of devices were able to access the highest speed, showing a clear gap in broadband accessibilities when considering all counties in the study. The counties that were not able to access the highest speeds, Clarke, Escambia, Monroe, and Washington, also had lower rates of device ownership while struggling with economic factors such as poverty rate, employment rate, and median household income.

For a more in-depth analysis of each county and the total trends of all of the counties, please see the report page on our website or go to this document here:

https://docs.google.com/document/d/1jDMkE5Rhp_EsfCRNtQMVPeE7cCT0QjhHGjDrUeLeCtAA/edit?usp=sharing

Recommendations

Based on our findings, we recommend that the SPLC advocate for increased internet infrastructure as a means to increase civic engagement. This applies especially to rural, low-income communities in Clarke, Escambia, Monroe, and Washington counties, as these are the places we are seeing lower broadband coverage rates. These counties consistently showed lower broadband accessibility, lower rates of device ownership, and more difficult economic conditions compared to Mobile and Baldwin counties. There are gaps that end up being created because of civic participation depending on internet access for voter registration, polling information, and access to government resources.

We also recommend that the SPLC work closely with both the Federal Communications Commission and different private internet service providers. This can help ensure that the guidelines for internet speed are kept up to date and applied across the region. FCC broadband maps and Census internet subscription data are updated regularly, and longitudinal analysis could aid in determining whether infrastructure improvements are reducing the digital divide in these counties.

In terms of recommendations for future project teams, they should begin the data collection process early because FCC broadband datasets are extremely large and require substantial cleaning before analysis can begin. This part of the project was the most time consuming because of filtering national datasets down to Alabama counties and standardizing formatting across files. Future teams might also benefit from integrating additional datasets related to voter turnout, polling place accessibility, education levels, or transportation access. It will only strengthen future analysis connecting broadband access to civic participation outcomes. In general terms, using Google Docs

was helpful because it allowed our team to quickly share documents and allow for all of us to edit the document in real time.

Conclusions

As the world becomes more and more digitized and connected through broadband networks, certain rural areas continue to be left behind in comparison to more densely populated areas that benefit from greater economic opportunity and technological access. By analyzing gulf counties in southern Alabama including Baldwin, Clarke, Escambia, Mobile, Monroe, and Washington, we see a persistent trend of areas with less population density and worse economic conditions linked to lower device and broadband accessibility.

In completing this project, we have given both the Southern Law Poverty Center and those they aim to reach a variety of sources to reference including visuals, a written report, and a website that acts as a dashboard for sources, deliverables, and information on the team. By accomplishing extensive data researching, analysis, and visualization, we hope our project will help SPLC and those they serve understand the digital divide in these southern Alabama counties and advocate for meaningful and lasting solutions and change.

Below is our team, roles, and contact information on how to reach us.

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Deliverables Checklist

Deliverable	Description	Team Member Responsible	Link
GC-BAND Website	User-centered website that is identical to the official SPLC website that contains project overview, visualizations, data, and written report.	Crystal Samoya	GC-BAND Website

Tableau Visualizations	2-D charts and graphics displaying broadband access, race statistics, internet usage, and income comparisons across the counties specified.	Garrett Bugg	*Integrated into Final Project Report*
Written Analysis Report	Full written report analyzing broadband accessibility, demographics, device ownership, and broadband speed accessibility across the counties.	Lyndsey Bakal	INST490 Report for All Counties - Google Docs
Data Sources Consolidated	Data from raw-data formatted into Google sheets and Microsoft Excel	Asia Robillard-Hill	+ Race by County + FCC_Broadban... x census_summar...
GC-Band Raw Data	All the data we found from our data sources.	Amir Pourmand	📁 GC-BAND-Raw-... ☰ Sources
Background Research & Project Context Analysis	A full analysis of our background research and project context.	Entire Team	☰ Background Res...

Gulf Coast - Composite Civic Participation Risk Index

GC-CPRI

Nathan J. Maldonado Munoz (PCA Developer), Moiz Uddin (Website Developer), Talha Sohail (Technical Writer), Akhil Ramesh (Developer & Data Wrangler), Elijah Lambros (Technical Writer & Data Wrangler)

Southern Poverty Law Center (SPLC)

INST490 - Integrated Capstone for Information Science

Section 0105

May 18th, 2026

Team Brief

Team 7: GC-CPRI

Key Findings

- Finding 1: Civic participation risk appears to be cumulative rather than isolated. The PCA results show that the strongest pattern in the data comes from several conditions moving together, including poverty, income, education, rurality, broadband access, unemployment, and demographic composition. This supports the idea that civic participation risk is not usually tied to one single barrier, but to the overlap of multiple social, economic, geographic, and infrastructure-related conditions.
 - Finding 2: The first principal component provides a strong foundation for the composite risk score. PC1 explains approximately 74% of the total variation in the selected indicators, meaning it captures the main shared pattern across the dataset. Because so much of the model's variation is explained by this first component, it provides a reasonable basis for generating the final GC-CPRI score and comparing counties on a common 0-100 scale.
 - Finding 3: The final index creates a practical way to compare counties and guide further review. The GC-CPRI score does not claim to prove causation or measure individual voter behavior, but it does provide a structured way to identify counties where multiple risk-related conditions are more concentrated. This makes the index useful as a starting point for deeper analysis, client discussion, and future updates as SPLC adds new data or refines the indicators.
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Deliverables Produced

- GitHub Repository: Completed GC-CPRI codebase containing the PCA notebooks, data folders, documentation, requirement file, and web tool files. Link: <https://github.com/lacarabela/GC-CPRI>
 - Interactive Web Tool: Client-facing web application deployed through GitHub pages that allows users to explore county-level GC-CPRI scores and indicator values. Link: <https://lacarabela.github.io/GC-CPRI/>
 - CSV Output Files: Final CSV outputs generated from the PCA workflow, including the scored county and app-ready score file. These files can be found in the GitHub repository under /data/final/ and /docs/data/. For convenience, copies of the CSV files are also available here: [📄 Final Output Files](#)
 - Supporting project documentation, explaining the PCA methodology, dataset explanations, repository handoff process, generated files, and instructions for updating or extending the model. These files can be found in the GitHub repository under /documentation/. For convenience, copies of the documentation files are also included in the shared deliverables folder available here: [📄 Documentation](#)
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Challenges

One major obstacle was defining the risk framework, since the team first had to decide which risks were relevant to voter behavior before the project could move forward. The team handled this by brainstorming a broad list of possible risks, ranking them by relevance, and narrowing the project to the strongest factors. Later, the PCA produced unclear results because too many risk factors created noise, so the team removed factors piece by piece until the output became more usable.

Connections to Other Teams

Team 1: GC-DIG - Team 7 connects to Team 1 because voter turnout and registration are directly related to voter behavior. Team 1's work provides the type of voter participation context that Team 7's risk framework is trying to explain.

Team 8: GC-VIZ - Team 7's risk framework and PCA results help identify which counties or variables may deserve closer visual attention from Team 8's deliverables.

Final Report and Deliverables

Project Context

Within the broader 11-team regional analysis, the GC-CPRI team's work serves as a comparative analytical layer that helps translate multiple county-level conditions into a single composite civic participation risk index. While other teams focused on specific issue areas, datasets, legal context, community conditions, or regional findings, this project provides a framework for bringing several relevant indicators together so counties can be compared more consistently across the Gulf Coast region. The team primarily relied on publicly available datasets rather than direct deliverables from other teams, but the index is designed to complement their work by offering a statistical and visual tool that can help identify where overlapping risks appear most concentrated. Other teams or future users could rely on GC-CPRI as a starting point for prioritizing deeper investigation, comparing county conditions, or adding new indicators based on their own findings. The index should not be interpreted as a causal model or an objectively final measure of voter risk; instead, it is a data-informed comparative tool that supports SPLC's broader mission of providing lawmakers, advocates, and community stakeholders with clearer information for understanding civic participation challenges.

Methods

Analytical Approach

In order to help create the final deliverable, a lot of different tools were used in tandem with one another. However, the most important part was that this tool was built on statistical analysis from the ground up. When the project was initially discussed among the team, it was clear that the end result would need a statistical methodology, which led the team to use Principal Component Analysis, also known as a PCA. The purpose of PCA is to take a large amount of data and reduce it numerically

into a smaller and more manageable set of information. It serves as a way to process large amounts of data especially in the context of a project like this, because of its ability to help combine related data points, such as risk factors, into something that can be more easily understood by people and used more effectively. Moreover, PCA is uniquely useful due to its ability to reveal and visualize patterns in complex datasets. In the case of GC-CPRI, the team's goal was to identify patterns across different perceived voter risks and create a single risk score as the different factors were adjusted and analyzed. The different risk factors are then fed into the analysis and calculated in relation to one another, creating a single composite score that can be utilized.

Risk Framework

To initiate the PCA modeling process, the team established a conceptual framework to define the specific variables that constitute civic participation risk. An initial comprehensive list of indicators was iteratively refined based on data availability and relevance to the regional context of the Gulf Coast. The final selection includes the following indicators:

Risks:

- Poverty Rate
- Unemployment Rate
- Median Household Income
- Adults Without HS Diploma
- Non-White Voter Share
- Rural-Urban Code
- Limited Broadband Access

These indicators were selected for their individual impact and collective relevance to the civic landscape. Voter engagement is influenced by a multi-dimensional array of factors, largely rooted in financial status, education, geography, and demographic composition. These core dimensions shape the lived experiences of individuals and their subsequent interactions with the electoral process. For the purposes of the PCA, these factors were further categorized into dimensions that capture the cumulative nature of civic risk.

Economic and infrastructure-related factors, such as median household income and broadband access, play critical roles in civic participation. Financial status directly influences voter priorities and

the ability to navigate costs associated with voting, while broadband access facilitates the acquisition of essential information regarding candidates, parties, and local issues. As civic information increasingly moves into digital spaces, the lack of reliable internet access serves as a significant barrier. Furthermore, educational attainment remains a primary determinant of civic choice; the inclusion of the share of adults without a high school diploma captures variations in foundational literacy and the capacity to synthesize complex political information. Lower educational attainment often correlates with decreased voter turnout and reduced engagement with the voting process.

Geographic and social factors also exert considerable influence. The Rural-Urban Continuum Code provides a necessary scale to account for the infrastructure and resource disparities between metropolitan and rural areas, which affect the accessibility of polling locations. Economic stability is further measured through poverty and unemployment rates, as these conditions can deprioritize voting relative to immediate economic survival or limit the physical ability to reach polling stations. Finally, the inclusion of non-white voter share addresses the specific demographic risks associated with the Gulf Coast's history and the ongoing importance of minority participation in the broader voting ecosystem. Together, these quantifiable indicators provide a logical and data-informed foundation for the comparative risk assessment.

Data Sources

The data sources used were brought into the project from a few reputable places, almost all are focused on election statistics and characteristics of the population. The authenticity of the data was confirmed, primarily due to the fact that they were taken from public officials primarily. They were then cleaned by the team to only carry what would be required to run the important analysis calculation, otherwise any other data was removed from them. Once the datasets were cleaned they were put into notebooks and were run as calculations.

One of the data sources came from the Alabama Secretary of State (SoS). They are the state's official office that handles several components of the voting process. Things such as: managing voter registration, election security, and ballot certification is routed through the Alabama Secretary of State office. They keep track of different percentages and statistics that are important voter information. For the final website and PCA their data on race was important, specifically their csv about "General Election Participation By Race". This helped create the risk index score for "Non-White Voter Share".

Another data source was the U.S. DEPARTMENT OF AGRICULTURE Economic Research Service (USDA-ERA). They are a federal agency of the United States government who are responsible for conducting surveys across the country that are related to the economic and agricultural status of different states and counties. Specifically the subunit of the USDA, Economic Research Service, are utilized to perform objective economic analysis which concerns the rural development of different areas. They are an important part of the USDA, and help provide high quality information for the economy in the counties that are being looked into. For this project, their dataset titled: "Poverty estimates for the United States, States, and counties, 2023", this helped influence the creation of the risk index "Poverty Rate". Two other risk indexes that were aided from the USDA-ERA are the "Median Household Income" and "Unemployment Rate", which both used the dataset titled "Unemployment and median household income for the United States, States, and counties, 2000–23". They also provided data for "Adults Without HS Diploma", through their dataset, "Educational attainment for adults age 25 and older for the United States, States, and counties, 1970–2023". Finally for the "Rural-Urban Code", each of the USDA-ERA sets have a "Rural_Urban_Continuum_Code_YEAR", which showcases the code over the years, depending on the data sets. They were important in giving a majority of the data needed for the PCA.

Another source that was used for PCA comes from the University of Wisconsin Population Health Institute. They have created a program referred to as County Health Rankings & Roadmaps (CHR&R), their main goal is to use evidence-based strategies to gather actionable information that can go on to help improve the quality of different communities' health. One of their studies that they conducted quantifies the percentage of households with broadband internet access. The data used is called "Broadband Access", which gives a percentage of households that have access to broadband, the raw number of households, and an error percentage. The important statistic that is used for the PCA is the percentage of access, which is easily adapted into the calculation.

Tools Used

The primary technical framework for the PCA workflow was Python, a high-level language standard for data science and statistical modeling. Python was utilized throughout the project lifecycle to clean source data, engineer county-level indicators, execute the PCA model, and calculate the final composite scores. This enabled a streamlined process for exporting the results into both the final project report and the interactive web tool.

The analysis was conducted within Jupyter Notebooks, providing an interactive and transparent environment for documenting the statistical workflow. This format ensured that each step, from data merging and scaling to visual validation, was organized sequentially and remains accessible within the GitHub repository for future updates or review.

The workflow is structured into three primary notebooks:

- 01_merge_and_feature_build: Prepares the modeling dataset by loading, cleaning, and combining source data into the core county-level indicators.
- 02_pca_analysis: Executes the core PCA modeling, including variable selection, missing value handling, and component loading review to derive the composite scores.
- 03_final_outputs: Generates the final client-ready files, including indicator metadata and the app-ready CSV/JSON files for the web tool.

This framework produced the final ranking of Gulf Coast counties, with the primary output located in `/data/final/gc_cpri_scored_counties.csv`. The critical field in this dataset is `gc_cpri_score`, which serves as the final statistically derived measure of relative civic participation risk for each county analyzed.

Challenges

One of the main challenges that came with working on the PCA was actually learning and figuring out how it works. This was new to the engineer who was working on that part of the overall project. This created a detour that had to be solved to not halt progression for the team. After reviewing some documentation, it started to make sense to be implemented. However, after being implemented the next thing to understand was the output that was coming from the PCA, this wasn't something considered initially when creating the PCA.

So when the output was generated, the next step was figuring out what to actually do with it. This required additional research and browsing online to better understand how to interpret the results. After some time, the output started to become something that could be reviewed in a meaningful way and used to determine whether the calculations were being run correctly. This was the major blocker that needed to be cleared before the team could move forward with the rest of the project.

Before the PCA was developed and worked on, there was another issue that needed to be addressed: the risk framework and the risks the team was going to choose. Without those risks, the project would not have been able to continue or progress as far as it did. At some point, the team needed to sit down and define which risks would be used. Although this was discussed about in the risk framework section, one of the key parts of identifying risks was brainstorming what could create barriers for voters and what factors could influence vote participation. A large list was defined during the team meeting, and then datasets were found that included those risks in a quantifiable way so they could be analyzed statistically. One major consideration was whether the risks had datasets that were easy to find and use, which the final selected risks ultimately did.

After the risks had been identified they were given to the PCA, so the team started with a lot of risks that would hopefully make it into the final deliverable. An issue suddenly arose, the PCA wasn't making sense. As the lead engineer who worked on the PCA put it, the large amount of factors that were being fed into the calculation was causing 'noise'. This meant that having too many factors was giving an unreliable result that was obfuscated. Almost like a fuzzy layer on top of the data, which wasn't giving something that could actually be used. To fix this, an attempt was made to cut down on the risk factors until something usable was given, this was done by cutting piece by piece until a working output was recorded. This locked the team out of a lot of risk factors that could have been done, which led to the final deliverable that is being handed over.

Deliverables and Findings

Principal Component Analysis

The GC-CPRI PCA model was developed to translate multiple county-level indicators into a single, comparative civic participation risk index. This analytical framework integrates variables spanning socioeconomic status, civic access, demographics, geography, and infrastructure. Rather than treating these factors as isolated elements, the PCA approach identifies shared patterns across the regional dataset to produce a structured composite score that supports consistent comparison across the Gulf Coast.

As illustrated in Figure 1, the selected indicators were categorized into broader risk dimensions to inform the final GC-CPRI score. Socioeconomic factors, including poverty, unemployment, median income, and educational attainment, capture the economic conditions that influence civic engagement. Additionally, indicators such as demographic composition, rurality, and broadband

access highlight geographic and infrastructure-related barriers. This methodology emphasizes that civic participation risk is cumulative, arising from the overlap of multiple social and environmental conditions rather than any single indicator.

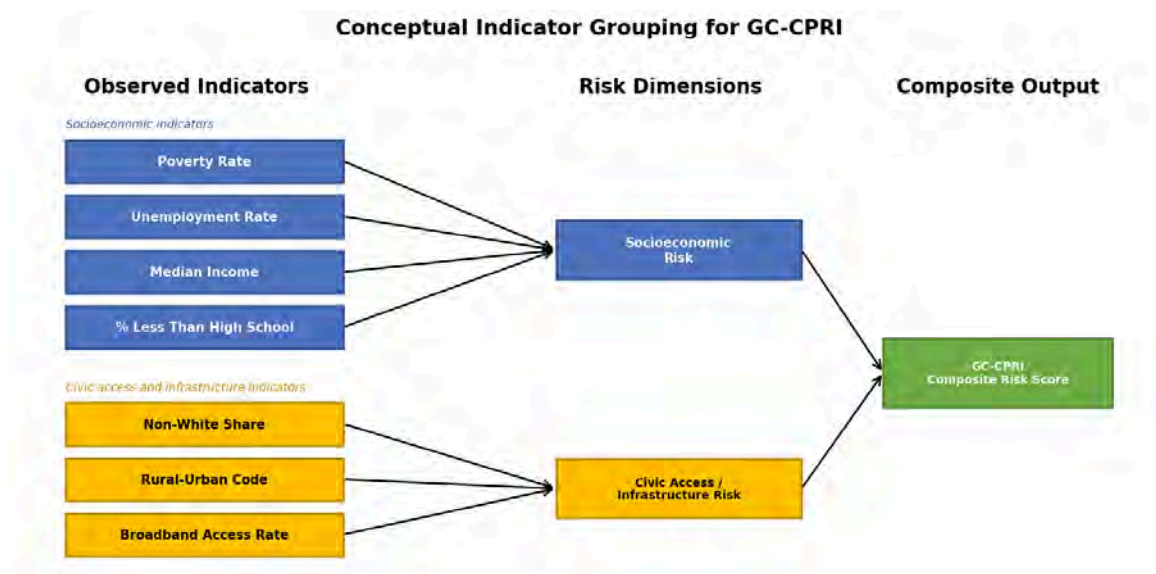


Figure 1. Conceptual Indicator Grouping for GC-CPRI. This figure shows how the observed indicators are grouped into broader risk dimensions before contributing to the final GC-CPRI composite risk score. (Source: Maldonado Munoz, N., Uddin, M., Sohail, T., Lambros, E., & Ramesh, A. (2026)).

The PCA workflow is systematically organized into three primary Jupyter Notebooks. The initial notebook facilitates the merging and preparation of the county-level indicator dataset; the second notebook executes the core PCA modeling to generate the composite risk scores; and the third notebook automates the export of final outputs for both formal reporting and the interactive web application. Together, these notebooks establish a transparent and reproducible framework that can be seamlessly updated as the client incorporates new data or refines the model's indicators.

The modeling process generates a suite of deliverables designed to support analytical review and the client-facing web tool. The primary analytical output is located at:

/data/final/gc_cpri_scored_counties.csv (Link in Drive: [gc_cpri_scored_counties.csv](#))

This file contains the final results of the statistical analysis, with the **gc_cpri_score** serving as the statistically derived measure for comparing regional risk. Scores are normalized on a 0-100 scale, where values approaching 100 indicate a higher relative concentration of civic participation risk, and those near 0 indicate lower relative risk among the analyzed counties.

To power the interactive web tool, the workflow exports additional app-ready files, including:

/docs/data/gc_cpri_scores_for_app.csv (Link in Drive: [gc_cpri_scores_for_app.csv](#))

/docs/data/gc_cpri_scores_for_app.json (Link in Drive: [gc_cpri_scores_for_app.json](#))

These specialized formats allow the web application to efficiently display scores, indicator values, and metadata. This decoupling ensures that while the notebooks maintain the analytical integrity of the PCA, the exported files remain accessible for broader stakeholder engagement.

As illustrated by the scree plot in Figure 2, the first principal component (PC1) serves as a robust foundation for the composite score, explaining approximately 74.0% of the total variation. The second component (PC2) contributes an additional 9.5%, resulting in a cumulative explained variance of 83.5%. By PC3 and PC4, the model captures 91.8% and 98.5% of the variance, respectively. This confirms that the dominant structural patterns of the dataset are captured by the primary component, validating its use as the central basis for the GC-CPRI risk score.

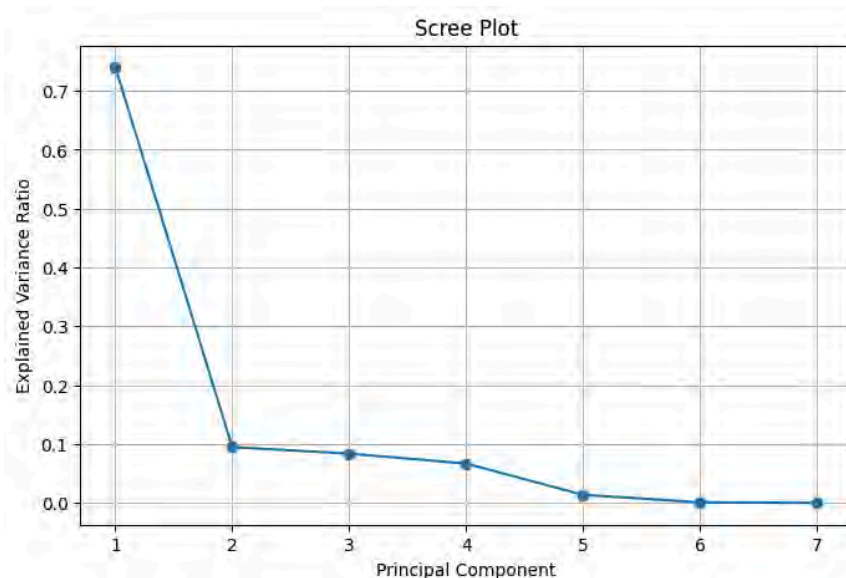


Figure 2. Scree Plot of Explained Variance by Principal Component. As illustrated by the scree plot, the first principal component (PC1) serves as a robust foundation for the composite score, explaining

approximately 74.0% of the total variation. This indicates that the primary component effectively captures the dominant structural patterns and shared variance within the regional dataset. (Source: Maldonado Munoz, N., Uddin, M., Sohail, T., Lambros, E., & Ramesh, A. (2026)).

The PC1 loading results show which variable contributes most strongly to the main risk pattern identified by the PCA. As shown in Figure 3, the strongest contributors to PC1 include median income, broadband access rate, non-white share, poverty rate, rural-urban code, unemployment rate, and percentage of adults with less than a high school education. The bar chart makes the PCA results easier to interpret by showing how each indicator contributes to the final risk index. Since these variables load in the same general direction on PC1, the model is capturing a broad shared pattern across economic conditions, infrastructure access, demographic, and rurality.

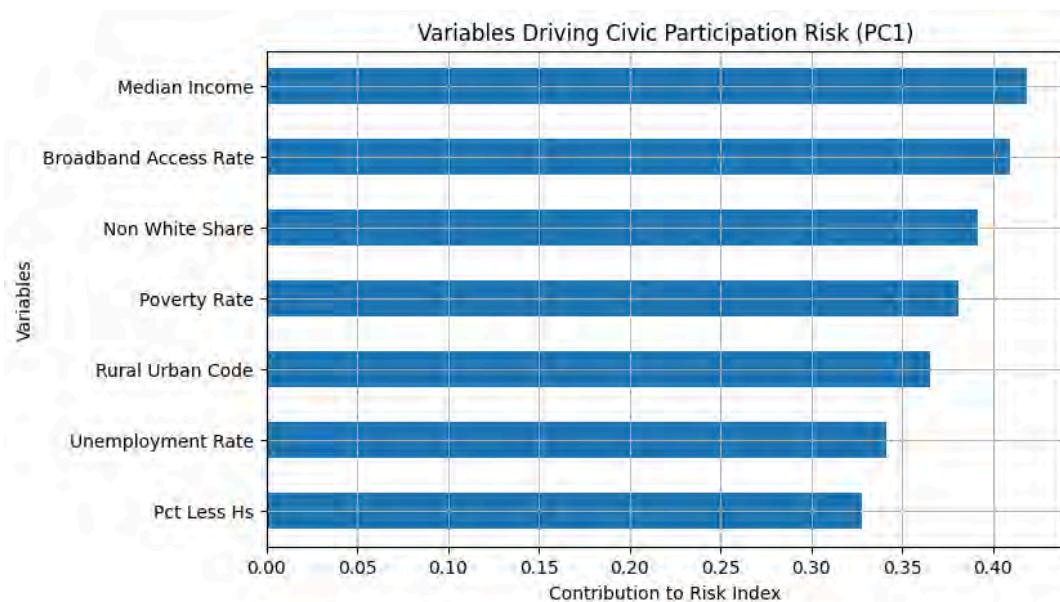


Figure 3. Variables Driving Civic Participation Risk in PC1. This figure shows the relative contribution of each input variable to PC1, which serves as the main basis for the GC-CPRI composite risk score. It helps explain which indicators most strongly influence the final risk score. (Source: Maldonado Munoz, N., Uddin, M., Sohail, T., Lambros, E., & Ramesh, A. (2026)).

The findings suggest that civic participation risk is shaped by the overlap of several conditions rather than by one individual factor. Counties with higher GC-CPRI scores generally reflect combinations of higher poverty, lower median income, lower broadband access, higher rurality, and larger non-white voter shares compared with lower-scoring counties. This pattern supports the purpose of the index: to identify areas where socioeconomic, demographic, geographic, civic, and infrastructure-related challenges appear together.

The model was also designed so it can be updated and iterated over time. The notebooks are organized in a clear sequence and documented so future users can add new indicators, remove existing variables, update source data, rerun the PCA, and regenerate the final scores and web app files. This makes the PCA framework flexible for future analysis while preserving transparency around how the index is built.

The scores should be interpreted as relative comparisons, not absolute measurements of risk or causal findings. A higher score does not prove that any one factor directly causes lower civic participation, and the model does not predict individual voter behavior. Instead, the PCA deliverable provides a transparent, data-driven framework that can help the client compare Gulf Coast counties, review overlapping barriers, and identify areas for further investigation.

Interactive Website

The website takes the Principal Component Analysis and transforms it into an interactive, accessible interface that allows SPLC staff, partners, and other stakeholders to engage with the data without needing to run any Python code. The tool is hosted at <https://lacarabela.github.io/GC-CPRI/> and is deployed via GitHub Pages from the /docs/ folder of the project repository. The website was built with vanilla HTML, CSS, and JavaScript so that future SPLC staff can maintain and extend the tool without a learning curve. This ensures a seamless and easy handoff.

The web tool loads its data from /docs/data/gc_cpri_scores_for_app.json, the output from the third PCA Jupyter notebook. This file has all the metadata, PC1 loadings, means, standard deviations and more. By exporting all these parameters, the website is able to reproduce the same scoring math from the Jupyter notebooks and can normalize the scoring into a 0-100 scale. This means that when a user adjusts the various weights and scales of the risk factors, it is consistent with the notebook outputs rather than approximating or inferring data.

When opened, the tool shows a layered view as shown in Figure 4. A header introduces the project context and summary statistics for the current model. A controls panel on the left lets the user set the various weights of each risk factor. The main map shows a county-level display of Alabama and a table that updates in real time as the user interacts with the controls.

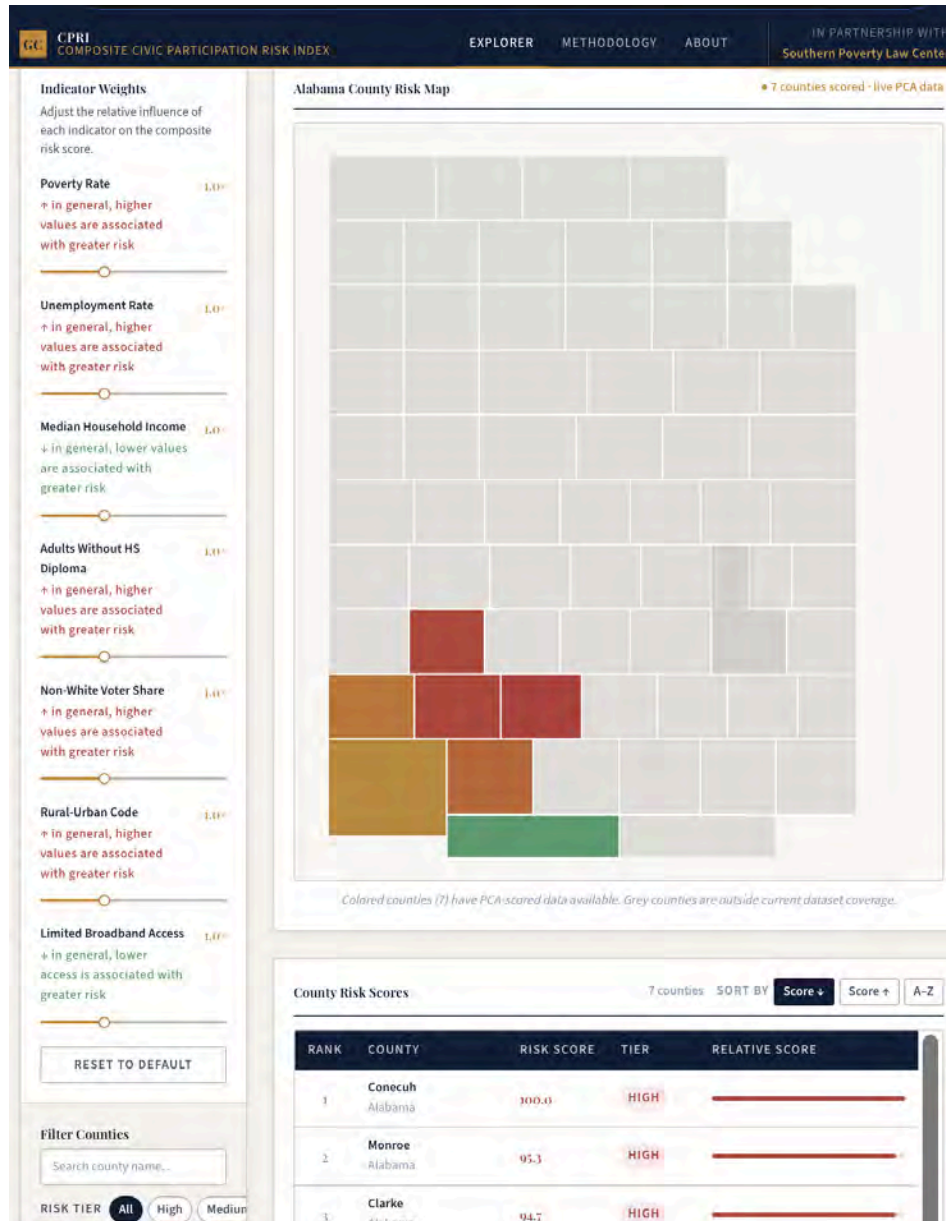


Figure 4. Full main explorer view: header, map, left control panel, and table all visible (Source: Maldonado Munoz, N., Uddin, M., Sohail, T., Lambros, E., & Ramesh, A. (2026)).

The Alabama county map shows all 67 counties in the state as SVG paths. The current 7 Gulf Coast counties which are currently included in the model are filled with a color ranging from green (low risk) through orange (medium risk) to red (high risk). Counties not included in the current workflow are grayed out to not confuse any viewer and make the map clear. If you hover over a county, you can see the county name, current score, and risk tier. If you click on a county, it opens the details

panel with the full indicator breakdown. The map serves as a birds eye view for any user to identify counties at a glance before doing a deep dive into specific indicators.

The sliders on the left allow the user to scale each individual risk factor between 0x to 3x with a default of 1x. As the user moves a slider, the map changes color in real time, and all values get recalculated in real time. This level of interactivity is important for stakeholder engagement and really allows SPLC staff to visualize what rankings would look like if a factor changed in a positive or negative way. A reset button restores all the weights to 1x and each slider has a short description of whether a high or low value is tied to a greater risk, so users can interpret values easily.

Selecting a county from the map or table opens a detailed panel on the right that slides in as shown in Figure 5. A header shows the county name, and more details of the risk score as well as a risk tier. The body has an indicator breakdown, with the z-score, rank, and contribution to overall risk. This makes the PCA level easy to understand on a deeper level and tell users why a particular county scored the way it did.

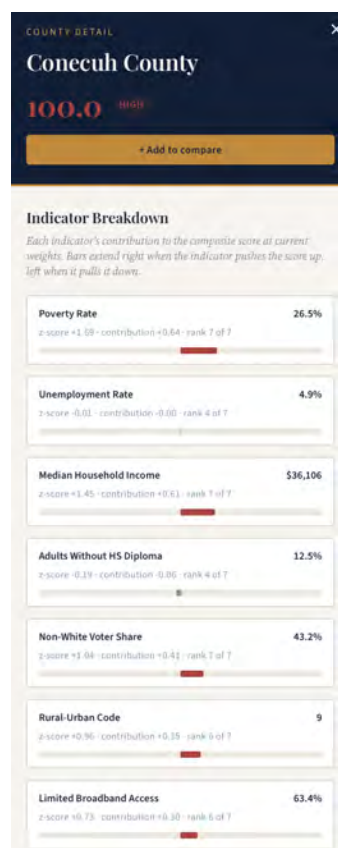


Figure 5. Detail panel open: shows detailed z-scores as well as individual county rankings for various risk factors. (Source: Maldonado Munoz, N., Uddin, M., Sohail, T., Lambros, E., & Ramesh, A. (2026)).

Users can add up to four counties for an in depth comparison. The compare cards display each county's composite score, risk, and indicator values. This feature supports prioritization discussions by allowing a seamless way to compare counties without needing to jump back and forth and keep track of values from the table.

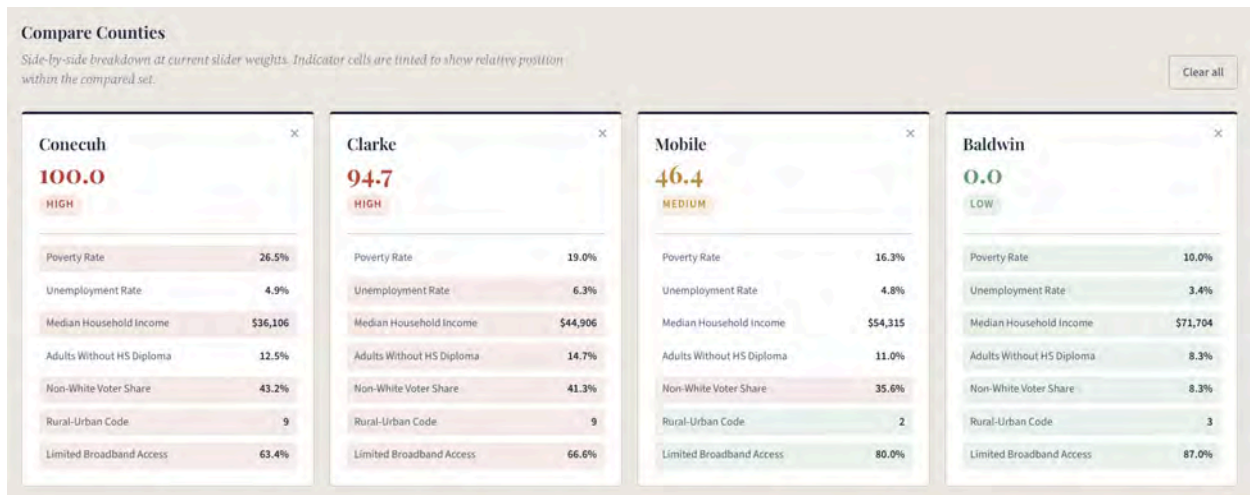


Figure 6. Compare Counties section: lets users add up to four counties and then view different factors side by side and compare them. (Source: Maldonado Munoz, N., Uddin, M., Sohail, T., Lambros, E., & Ramesh, A. (2026)).

The tool was designed to translate the PCA findings into something nontechnical stakeholders can interact with for themselves. A methodology section at the bottom of the page summarizes the composite scoring approach as well as limitations of the project, so any user can understand and delve into the methodology if they wish. The website is intended as a comparative and exploratory tool, rather than a definitive measure of civic participation risk, and the language throughout the website constantly reinforces this.

Documentation

The documentation files were originally written in Markdown so they could be easily viewed, edited, and maintained directly with the GitHub repository. The original Markdown versions can be found in the repository's documentation/ folder, while Word document versions were created for easier review and sharing with the client. The Markdown files were converted to .docx format using the Python py pandoc library, which allowed the team to preserve the structure of the original files, including headings, lists, links, and general formatting.

- **Dataset Documentation:** The dataset documentation provides an overview of the source datasets used throughout the GC-CPRI project. It explains where the data came from, why each dataset was relevant to the civic participation risk index, and what key columns or variables were included. This document is intended to help SPLC and future project teams understand the role each dataset played in the analysis and evaluate whether those sources should be reused, updated, or expanded in future versions of the model.
 - **PCA Methodology Documentation:** The PCA methodology documentation explains how the GC-CPRI model was built, including the notebook workflow, data preparation process, PCA approach, scoring logic, generated output files, and update instructions. It is designed as a technical handoff document so SPLC or future teams can understand how the composite score was created and how to rerun or modify the workflow. This documentation also clarifies how the model should be interpreted, including its use as a comparative analytical tool rather than a causal or absolute measure of civic participation risk.
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Recommendations

Recommendations for the client (SPLC)

Based on the GC-CPRI results, SPLC should use the index as a starting point for identifying Gulf Coast counties where multiple civic participation barriers appear to overlap. Counties with higher composite risk scores may warrant closer review because they tend to reflect combinations of socioeconomic, demographic, geographic, and infrastructure-related challenges. Rather than treating the score as a final conclusion, SPLC can use it to prioritize additional investigation, local context, and community engagement may be most useful.

SPLC should also consider using the interactive web tool when presenting or discussing the findings with lawmakers, advocates, or community stakeholders. The tool makes the index more accessible by allowing users to explore county scores, compare indicators, and see how different risk factors contribute to the overall model. This can help communicate that civic participation risk is not driven by one issue alone, but by several conditions that may compound one another.

The broadband and rurality indicators also suggest that outreach strategies may need to vary by county. In areas where broadband access is lower or rurality is higher, SLPC may want to consider non-digital or hybrid outreach strategies rather than relying only on online voter education, digital campaigns, or web-based resources. The index does not prove that limited broadband or rurality directly causes lower civic participation, but it does identify these as relevant access-related conditions that should be considered when planning outreach or advocacy.

Recommendations for future project teams

Future project teams should treat the current PCA workflow as a foundation rather than something that needs to be rebuilt from scratch. The notebooks, documentation, and output files were organized so that future teams can add new indicators, update existing datasets, rerun the PCA, and regenerate the final scores and web tool files. A natural continuation of this work would be to expand the model with additional access-related variables, such as polling place density, distance to polling locations, transportation access, voter registration access, early voting availability, or other Alabama specific voting policy indicators.

Future teams should also prioritize data cleaning and source evaluation early in the project timeline. Some public datasets are more straightforward to use than others, while election-related files may require more cleaning, restructuring, and documentation before they can be used in the model. Starting the data review process earlier would make it easier to identify gaps, standardize county names, confirm variable definitions, and avoid delaying the modeling phase.

Finally, future teams should continue documenting every major decision made in the index construction process. This includes why indicators were included or removed, how missing values were handled, how scores were normalized, and how outputs were generated for the web tool. Because the GC-CPRI score is a comparative analytical tool rather than a causal model, clear documentation is essential for helping SPLC and future users understand what the index can support, what it cannot support, and how it should be responsibly interpreted.

Conclusions

GC-CPRI was developed to help SPLC better visualize and understand civic participation risk across Alabama's Gulf Coast counties through a PCA-based composite risk index and an interactive web tool. This model brings together seven risk factors: poverty rate, unemployment rate, median household income, educational attainment, non-white voter share, rural-urban code, and broadband access, creating a data-informed foundation for comparing where barriers to civic participation may be more concentrated. The interactive tool makes the deliverable more useful by allowing users to explore county score, adjust risk factors, and see how changes affect the overall index.

At the same time, the index should be understood as a starting point rather than a complete or final measure of voter risk. The results do not prove causation or capture every local condition that may shape civic participation, and they should be interpreted alongside community input, qualitative research, legal context, and other regional findings. Even with these limitations, GC-CPRI provides SPLC with a transparent and flexible framework that can be updated over time as new data becomes available. Overall, the project gives SPLC a practical tool for comparing counties, identifying areas for deeper review, and supporting future conversations around voting rights and civic access in Alabama.

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Deliverables Checklist

- GitHub Repository: <https://github.com/lacarabela/GC-CPRI>
- Exported Codebase:  GC-CPRI Codebase (DON'T ALTER FILES)
- Interactive Web Tool: <https://lacarabela.github.io/GC-CPRI/>
- GitHub Codebase Handoff Instructions:  GitHub Codebase Handoff Instructions
- GitHub Important Links:  GitHub Important Links
- Dataset Documentation:  Dataset_Documentation.docx
- PCA Methodology Documentation:  PCA_Methodology_Documentation.docx
- Final Generated Composite Risk Index: [gc_cpri_scored_counties.csv](#) | [gc_cpri_scores_for_app.csv](#)

Visualizing Voter Access in Alabama's Gulf Coast

Prepared for: Southern Poverty Law Center (SPLC)

Course and Section: INST490.0105

Noah Lehrfeld (Communication Liaison), Hana Elbarony (Data UX designer), Alex Peterson (Stakeholder & Process Analysis), Jin Peng (Project Manager), Mohammad Shahid (Data Analyst), Sera Belasco (Quality Assurance)

Date: 5/18/2026

Team 8: GC-VIZ

Team Brief

GC-VIZ is the visualization team within the broader 11-team capstone analysis of voter access in Alabama's Gulf Coast and Black Belt regions for the Southern Poverty Law Center. Our team studied county-level patterns in voter registration and ballot participation across the seven Gulf Coast counties (Baldwin, Clarke, Conecuh, Escambia, Mobile, Monroe, and Washington) in the 2020 and 2024 general elections, with particular attention to how participation varies by race. Our role is to translate the voter file and turnout data assembled by upstream teams into an interactive dashboard and supporting visualizations that SPLC legal and policy staff, community organizers, and partner organizations can use to identify jurisdictions warranting closer attention.

Key Findings

- **Finding 1:** When applied to Team 1-DIG voter file data, the GC-VIZ dashboard showed that Mobile County's total ballots cast dropped by approximately 5,000 between 2020 and 2024, while the share of ballots cast by White voters rose by about 1.5 percentage points. Across the same period, Mobile's White voter share rose from 63.67% to 65.15%, and its Black voter share fell from 33.48% to 31.85%.
- **Finding 2:** Ballot participation across the Alabama Gulf Coast region in the 2020 and 2024 general elections was heavily dominated by White voters, with a substantial drop-off for Black voters and very small representation from Hispanic, Asian, American Indian, and other demographic groups.
- **Finding 3:** The four Gulf Coast counties with the highest Black voter share (Clarke, Conecuh, Mobile, and Monroe) all showed lower proportional Black voter turnout in 2024 than in 2020. Conecuh County showed the most pronounced shift, with Black voters representing 44.81% of the voting demographic in 2020 compared to 39.18% in 2024, a decline of 5.63 percentage points. Mobile County, the largest in the region, lost nearly 6,000 Black voters across the same period.

- **Finding 4:** The Hispanic share of ballots cast rose in most Gulf Coast counties between 2020 and 2024, with the largest absolute increases in Mobile and Baldwin and smaller increases in Conecuh and Escambia. Clarke, Monroe, and Washington showed slight decreases. The directional shift is consistent across five of the seven counties in scope.
-

Deliverables Produced

Interactive Dashboard

An interactive community dashboard that provides an interactive view of demographic distribution and voting patterns across Alabama counties.

Link: [Figure 1](#)

Link: [Figure 2](#)

Link: [Figure 3](#)

Link: [Figure 4](#)

Challenges

Our team faced two main challenges. The largest was dependency on upstream voter file data, which we addressed by building dashboard scaffolding against the agreed schema while finalized inputs were being prepared, allowing us to plug data in with minimal rework. Secondary challenges included Tableau's limited support for collaborative editing, which we resolved by taking turns on dashboard edits and coordinating updates over text. We selected Tableau over Power BI because Power BI is not available on macOS.

Connections to Other Teams

Team 1 - DIG: provided cleaned voter file and turnout datasets (registration counts by race, ballots cast by race, organized at the county-year level) that the GC-VIZ dashboard visualizes.

Presentation Team: Receives supporting visualizations from our group for use in the final cross-team presentation.

Final Report and Deliverables

Project Context

GC-VIZ is the visualization and output team within a broader 11-team capstone analysis of voter access in Alabama's Gulf Coast and Black Belt regions for the Southern Poverty Law Center. The Southern Poverty Law Center works to advance civil and human rights in the American South, including by expanding voter access and civic participation in communities where structural barriers limit both. The broader project investigates patterns in voter access and civic participation across these regions, with particular attention to whether and how structural conditions affect voters' ability to cast a ballot. Where other teams collected, cleaned, and contextualized voter data, GC-VIZ's role is to present that data in interactive and static formats that SPLC legal and policy staff, community organizers, and partner organizations can use to identify jurisdictions warranting closer attention.

Our team's work depends on upstream inputs, primarily a cleaned voter file and turnout dataset from Team 1-DIG, and produces downstream deliverables (an interactive Tableau dashboard, four supporting visualizations, and supporting documentation) consumed by SPLC staff and community partners. In addition to our analytical deliverables, our team also produced the cross-team visualization style guideline used by other groups for final deliverables, including the color palette and font choices that appear across the report.

Methods

Public Systems Landscape

The Elections Division of the Alabama Secretary of State sets statewide rules and regulations for federal and local elections. The individual counties themselves then administer these elections. Core operational components include voter registration, polling place operations, absentee ballot operations, provisional ballot review, and certification process. At the county level, the Board of Registrars manages voter registration records. The Judge of Probate oversees Election Day operations, the Circuit Clerk oversees absentee ballots, and the Appointing Boards certify results. In Alabama, you are required to show your photo ID for in-person voting as of 2014. In Alabama, you must also have a valid qualifying reason to vote by mail. Additionally, according to Alabama's 2017 "Moral Turpitude," many felonies disqualify individuals from being able to vote. These state laws were passed after the Supreme Court's *Shelby County v. Holder* decision in 2013, which allowed states to no longer require federal oversight when changing voting laws. Election rules and data can be sourced from the Alabama Secretary of State. The local county probate or circuit clerk has the specific details of polling locations and turnout results. US Census Bureau data can be used for population demographics and statistics.

Data Source

GC-VIZ worked with two primary tabular datasets supplied by Team 1, GC-DIG. Both are organized at the county-year level and disaggregated by race.

Registration Data

Schema: year, county, total_active_inactive, plus active and inactive counts for each of the following racial categories: Asian, American Indian, Black, federally-registered, Hispanic, Korean, White, other, and not-identified, with totals for active and inactive.

Ballots Cast Data

Schema: year, county, total ballots cast, plus ballots cast by the same racial categories listed above.

Analytical Approach

We mainly focused on comparing voting participation by county, year, and racial group to identify patterns in ballot participation. Instead of only looking at total ballots cast, we also examined racial group counts and percentages so the dashboard could show both the size and composition of voter participation.

The dashboard was designed to answer questions such as:

- Which counties had the highest or lowest number of ballots cast?

- How did ballot participation change across years?
- What racial groups made up the largest or smallest share of ballots cast?
- How does the selected racial group's participation vary across counties?

Tools Used

- **Visualization Platform:** Tableau
- **Data Preparation:** The data we received from other teams was cleaned pre-transfer.
- **Documentation:** PDF for client-facing deliverables

Challenges and How They Were Addressed

Dependency on upstream data

Our team continued working and designing the prototype to build the structure of the dashboard, while we waited for the finalized upstream data to avoid wasting time. We started by creating a prototype that goes with the structure of the project. We focused on the most important filters and visuals so the final data could be inserted with less work, just adding the data sets. This helped our plan to handle delays by setting team guidelines for dashboard edits.

Privacy and aggregation

All work was performed at the county aggregate level. No voter-level personally identifiable information was used at any stage, in compliance with Section 8 of the Requirements Document and SPLC's data handling requirements.

Nonpartisan framing

All visualizations and accompanying text describe access patterns rather than electoral outcomes. No party affiliations, candidate preferences, or partisan interpretations appear in any deliverable. This matches the nonpartisanship constraint set in the Requirements Document.

Deliverables and Findings

Description

The interactive community dashboard is powered by the Tableau platform. It provides an interactive view of demographic distribution and voting patterns across Alabama Gulf Coast counties. It is

organized into four main views: a geographic map that displays demographic percentages by county, a bar chart showing total ballots cast per county, a comparison chart that breaks down ballots by race, and a county-level comparison of Hispanic voter share across election years. At the top of the dashboard, users can interact with filters for county, election year, and race. Once users choose what they want to see, the visuals update accordingly. Additional interactivity is provided through a hover tool that reveals details on the map, including demographic breakdowns for each county. When the dashboard loads, users see the map centered on Alabama with counties color-coded based on the selected race. The bar chart shows counties like Baldwin and Mobile standing out for higher ballot counts, and the comparison chart summarizes total ballots across different race groups.

Rationale

The interactive dashboard is the primary deliverable of GC-VIZ. It directly supports OBJ-1 (4 visualizations and at least one final dashboard), OBJ-2 (interactive county-level geographic visualization), and OBJ-3 (publicly accessible and understandable for non-technical audiences) from the Requirements Document. It also supports OBJ-4 by being structured so that SPLC can update it after project completion. The dashboard uses voter file and turnout data supplied by Team 1-DIG.

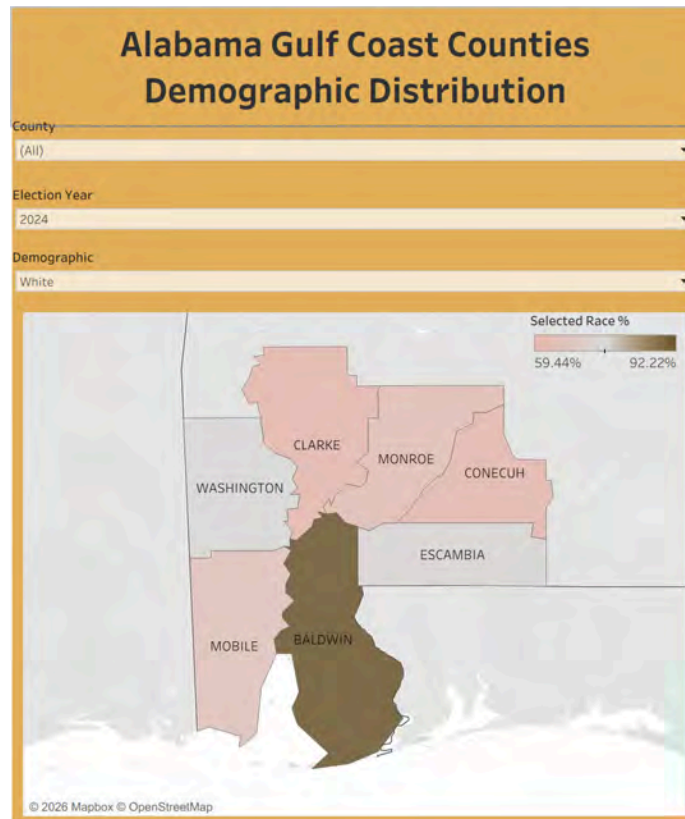
Findings

The findings below are framed as patterns that the GC-VIZ dashboard surfaced when applied to upstream voter file data. They are not the result of independent statistical analysis or causal investigation. Per section 8 of the Requirements Document, the dashboard surfaces patterns that may indicate barriers. Understanding the causes of those patterns requires further investigation that is outside of our scope.

Finding 1: Mobile County's total ballots declined, and the racial composition of its electorate shifted between 2020 and 2024

When applied to Team 1-DIG voter file data, the GC-VIZ dashboard showed that Mobile County's total ballots cast dropped by approximately 5,000 between 2020 and 2024, while the share of ballots cast by White voters rose by about 1.5 percentage points. Figure 1 shows Mobile County's White share rising from 63.67% to 65.15% and its Black share falling from 33.48% to 31.85% across the same period. As the largest county in the Gulf Coast region by ballot volume, Mobile is the jurisdiction where these shifts are most consequential in absolute terms.

Figure 1: Alabama Gulf Coast Counties Demographic Distribution



Caption: Demographic distribution of voters across Alabama Gulf Coast counties in 2024, filtered to show the White voter share. The dashboard supports filtering by county, election year, and demographic, and the comparative figures in Finding 1 (White share rising from 63.67% to 65.15% in Mobile County, Black share falling from 33.48% to 31.85%) are drawn from these filter combinations across the 2020 and 2024 views.

Source: Team 1-DIG voter file data, processed through the GC-VIZ dashboard

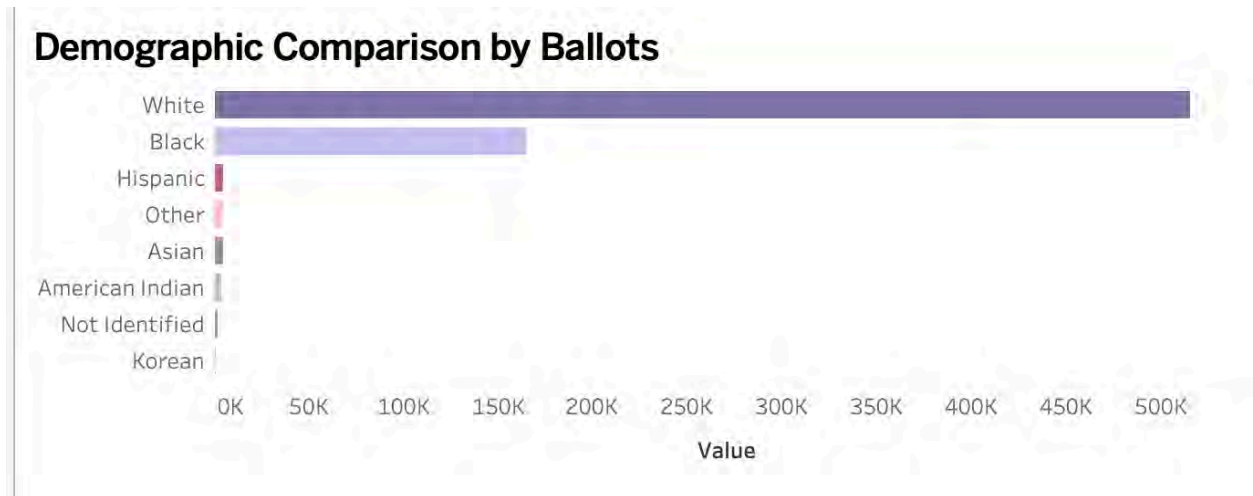
[Link](#)

Finding 2: Ballot participation across the Gulf Coast was heavily dominated by White voters in both 2020 and 2024

The dashboard's Demographic Comparison by Ballots view shows that ballot participation across Alabama's 2020 and 2024 general elections in the Gulf Coast region was concentrated among White voters, with a substantial drop-off for Black voters and very small representation from Hispanic, Asian, American Indian, and other demographic categories. This aggregate pattern frames the context for the county-level shifts documented in Findings 1, 3, and 4: the racial composition of

the Gulf Coast electorate is heavily skewed, and even small shifts in participation share translate into meaningful changes in who is represented at the ballot box.

Figure 2: Demographic Comparison by Ballots



Caption: Total ballots cast by racial demographic across the Alabama Gulf Coast region, aggregated across 2020 and 2024.

Source: Team 1- DIG voter file data, processed through GC-VIZ dashboard.

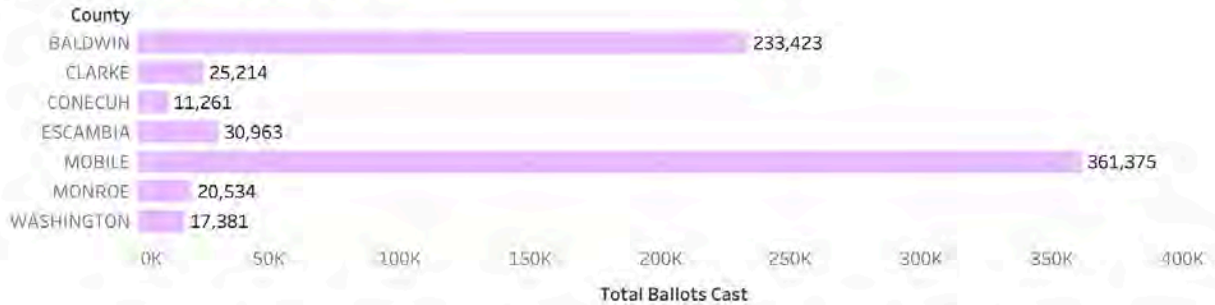
[Link](#)

Finding 3: The four counties with the highest Black voter share all showed declining proportional Black voter participation between 2020 and 2024

Within the Gulf Coast region, the four counties with the highest Black voter share (Clarke, Conecuh, Mobile, and Monroe) all had lower proportional Black voter turnout in 2024 than in 2020. Conecuh County showed the most pronounced shift, with Black voters representing 44.81% of the voting demographic in 2020 compared to 39.18% in 2024, a decline of 5.63 percentage points. Mobile County, the largest in the region, lost nearly 6,000 Black voters over the same period, indicating the pattern is not limited to small rural counties. Because the trend appears consistently across one large urban county and three smaller rural ones, it is unlikely to reflect county-specific operational conditions alone.

Figure 3: Counties by Total Ballots Cast

Counties by Total Ballots Cast



Caption: Total ballots cast by county across the Alabama Gulf Coast region, illustrating the scale difference between Mobile and Baldwin and the five smaller counties (Clarke, Conecuh, Escambia, Monroe, Washington)

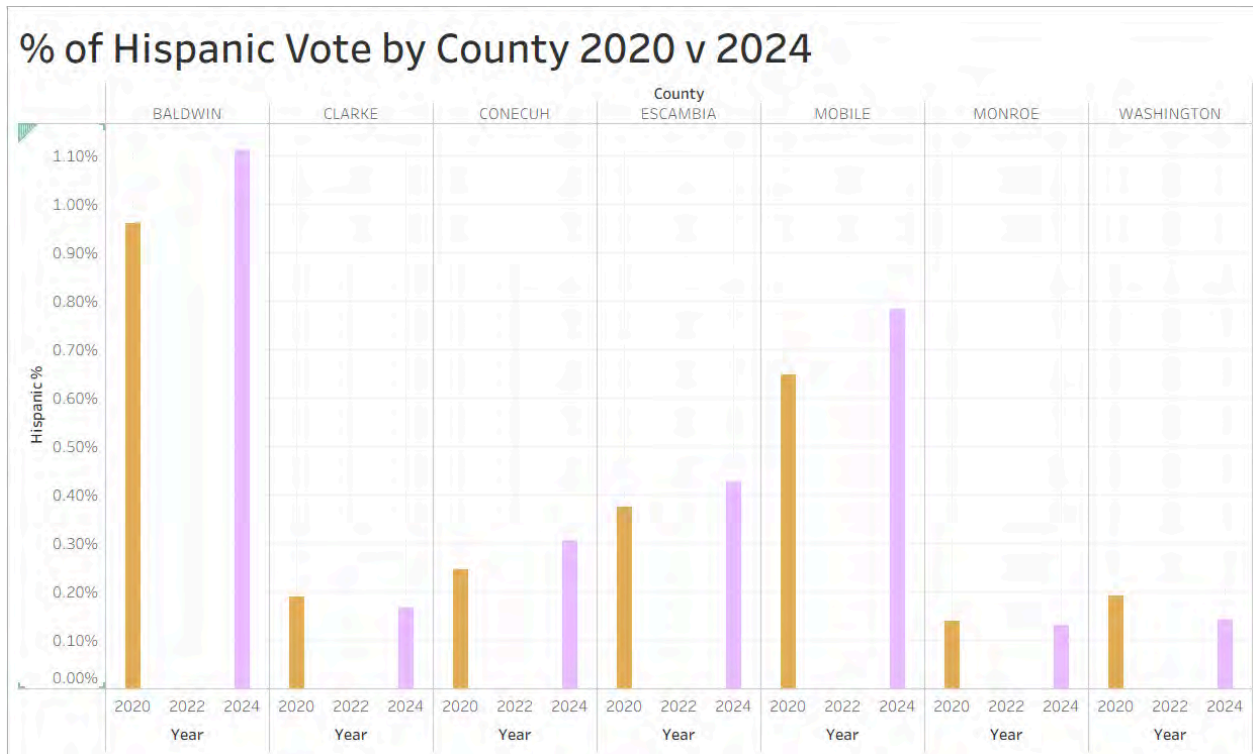
Source: Team 1- DIG voter file data, processed through GC-VIZ dashboard.

[Link](#)

Finding 4: The Hispanic share of ballots cast rose in most Gulf Coast counties between 2020 and 2024

The Hispanic share of ballots cast rose in most Gulf Coast counties between 2020 and 2024, with the largest absolute increases in Mobile and Baldwin, the two largest counties in the region. Mobile's Hispanic ballot share rose from approximately 0.65% to 0.78%, and Baldwin's rose from approximately 0.95% to 1.10%. Conecuh and Escambia counties showed smaller but consistent increases over the same period, while Clarke, Monroe, and Washington showed slight decreases. Although Hispanic voters represent a small share of the Gulf Coast electorate in absolute terms, the directional shift is consistent across five of the seven counties in scope.

Figure 4: Percentage of Hispanic Vote by County, 2020 vs. 2024



Caption: Bar chart comparing the percentage of Hispanic voters from 2020 to 2024

Source: Team 1- DIG voter file data, processed through GC-VIZ dashboard.

[Link](#)

Who uses these deliverables and how?

The dashboard and supporting visualizations are intended for three main audiences. SPLC legal and policy staff can use them to identify counties with widening disparities and target advocacy and litigation resources accordingly. Community partners, including the Black Belt Community Foundation and the United Women in Christ, can use county-level views relevant to their own service areas without requiring technical staff support. Furthermore, due to the scalable design of our deliverables, future capstone teams can extend the dashboard to additional counties, election cycles, or metrics using the same schema and design conventions.

Visualization Design and Analytical Documentation

Our team selected its chart types based on the data being measured. Differences across categorical variables are shown with bar charts, and differences across counties are shown with maps. Design decisions also considered OBJ-3 accessibility guidelines, including colorblind access, labeling, and metric clarity. This documentation supports OBJ-3 by making the design decisions transparent and

reviewable, as well as OBJ-4 by giving future analysts the context needed to update or extend the dashboard.

Recommendations

Recommendations for the client (SPLC)

SPLC can use this deliverable to identify where racial participation gaps appear in Alabama's Gulf Coast in the 2020 and 2024 general elections. The clearest case is Mobile County, where the dashboard surfaces three converging signals between the two cycles. Total ballots cast dropped by approximately 5,000 ballots, the share of ballots cast by White voters rose by about 1.5 percentage points, and the county lost nearly 6,000 Black voters. Mobile is the largest county in the region by ballot volume, and the convergence of these three patterns makes it the strongest candidate for closer SPLC attention, whether through voter education and outreach investment, records requests to the Mobile County Board of Registrars to understand list-maintenance practices, or further investigation of access conditions specific to the county. Beyond Mobile, SPLC should also examine the regional pattern surfaced by Finding 3. The four Gulf Coast counties with the highest Black voter shares, namely Clarke, Monroe, Mobile, and Conecuh, all showed lower proportional Black voter participation in 2024 than in 2020, with Conecuh's drop the most pronounced (44.81% to 39.18%, a decline of more than five percentage points). Because this trend is consistent across one large urban county and three smaller rural ones, the explanation is unlikely to be county-specific, and SPLC should consider whether a regional-level inquiry into statewide policy changes affecting Gulf Coast voters between 2020 and 2024 is warranted.

We also recommend sharing this dashboard, and future versions of it, with the community partners best positioned to act on what it shows. The Black Belt Community Foundation, whose giving stream supports educational resources in the rural Gulf Coast counties where Black voter participation declined most sharply (Clarke, Conecuh, Monroe), would benefit from access to the county-level demographic distribution views for those jurisdictions. The United Women in Christ, an official SPLC partner running a Voter Education program, would benefit from dashboard views that surface participation patterns by race within their service areas to support their outreach to women and girls. The dashboard also surfaces an increase in Hispanic ballot share across most Gulf Coast counties

between 2020 and 2024, with the largest absolute increases in Mobile and Baldwin and smaller increases in Conecuh and Escambia. SPLC and its community partners should consider whether existing voter education and outreach materials are reaching Hispanic voters effectively in these counties, and whether translated or culturally targeted outreach would build on this growing participation. This trend runs in the opposite direction of the Black-voter participation patterns documented in Findings 1 and 3, and the contrast itself warrants attention: the same regional electorate is becoming more accessible for one demographic group while becoming less so for another.

Finally, adding a polished, public-facing version of the dashboard to the SPLC website would give community members a clear, county-level view of access conditions in their own jurisdictions, which may support increased civic engagement and election participation. Per the project's nonpartisan commitment, public-facing framing should describe access patterns rather than electoral outcomes, and the dashboard's honest limits, namely that it surfaces patterns rather than establishing causation, should be visible to public users alongside the visualizations themselves.

Recommendations for future project teams

Working with other teams proved to be the most reliable way to obtain polished election data, and the next team should prioritize early communication with upstream teams to confirm which group is producing which dataset. Our team experienced a small roadbump partway through the project when it took longer than expected to identify which team(s) held the voter file data we needed. This is avoidable with a kickoff conversation in the first two weeks. When upstream data is incomplete, supplementing with publicly accessible county election commission records works well, but should be flagged in documentation so future analysts can distinguish team-produced data from public-source data. The visualization style guideline our team developed, which includes a color palette drawn from the SPLC website, consistent typography, and chart-type conventions matched to the kind of comparison being shown, can and should be reused across all future SPLC-related work to keep deliverables visually coherent with SPLC's existing brand and sources.

The most valuable next step is to extend the dashboard's time coverage by adding 2022 midterm data, which would significantly clarify whether the participation patterns we surfaced between 2020 and 2024 reflect a steady trend or are specific to presidential cycles. From there, natural continuations include adding an inactive-status view, since our current dashboard shows ballots cast and demographic distribution but not the inactive-share metric, which would let SPLC examine whether voter list maintenance is producing racially disparate outcomes. This is a question the

Conecuh and Mobile findings raise but cannot answer. Other continuations include layering polling-place location, bandwidth, or accessibility data from other teams onto the map, and extending coverage to Black Belt counties beyond the Gulf Coast to test whether the trends we surfaced are statewide or regionally bounded. On tooling, our team selected Tableau over Power BI after weighing the trade-offs, but a future team working primarily on Windows devices may find Power BI's collaborative workspace easier than Tableau's download/upload workflow. The tooling decision should be made early and based on the team's device mix and licensing access, not on which platform feels more familiar individually.

Conclusions

GC-VIZ produced an interactive dashboard, four supporting visualizations, design and data documentation, and this report. Together, these deliverables give SPLC a tool that shows voter access patterns at the county level across multiple election cycles in the Alabama Gulf Coast region.

Using the dashboard, we observed a trend across Gulf Coast counties, most notably a decline in the proportion of ballots cast by Black voters across most Gulf Coast counties between 2020 and 2024. The largest county in the region by population size, Mobile, lost nearly 6,000 Black voters between 2020 and 2024. SPLC can use this dashboard to share this information with community members and partner organizations, which may support increased voter participation among Black and other minority groups in Gulf Coast counties.

The dashboard is structured to be maintained beyond this semester, extended to new regions, and used by community partners with varying levels of technical experience. The findings it surfaces are honest patterns, not causal claims, and they identify specific jurisdictions and questions that warrant SPLC's closer attention.

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Deliverables Checklist

- [Requirements Document](#)
- [Background Research and Project Context Analysis](#)
- [Visualization Design and Analytical Documentation](#)
- Visualization links:
 - [Figure 1](#)
 - [Figure 2](#)
 - [Figure 3](#)
 - [Figure 4](#)
- [Color Style Guide](#)
- [Cleaned dataset\(s\) used by the dashboard](#)
- [Final Presentation Contribution Brief](#) (GC8-VIZ)
- [Final Project Report](#)

Community Context and Personas

Southern Poverty Law Center

Team 9: Community Context and Personas Final Report

Team Identifier: GC-CCP

Client: Southern Poverty Law Center

Course: INST490-0105

Section: 0105

Date: May 18, 2026

Team Members and Roles:

Bruktawit Mekonnen, Project Manager

Ryan Birmingham, Persona Developer

Ayanna Sterling, Barriers Researcher

Sue Chen, Persona Developer

Ifa Bato, Barriers Researcher

Team Brief

Team 9: GC-CCP

Key Findings

- **Finding 1:** Through our initial research, we found that SPLC's mission is heavily geared towards civic engagement and increasing voter turnout in black and latine voters. We developed an understanding of how the process of voting through the experience of black and latine voters was unique and created a barrier for them while practicing their right to vote and/or participating in civic engagement practices in the Gulf Coast Counties. Our personas greatly represented the demographic aimed to gather information from and we found the biggest barriers were accessibility, transportation, language, scheduling, misinformation, and fear and confusion.
 - **Finding 2:** From our found barriers, we were able to find new experiences that were caused by the challenges our personas were facing. For example, due to the lack of public transportation, senior voters felt like a burden when asking for a ride to vote. Likewise, due to the presence of language barriers in the Alabama community, our immigrant voter would have a harder time finding ways to communicate at the voting polls if she were to experience confusion and if so, having the fear of asking for help would be another obstacle she would have to overcome.
 - **Finding 3:** Through our recommendations, we were able to research further in depth into SPLC's community outreach programs and different initiatives curated for advocacy and civic engagement, specifically in Alabama. Programs such as Days of Democracy, Alabama Youth Council, and Learning for Justice. We found more clarity in how SPLC reaches out to its community and how we can help curate these programs to better fit the needs of our personas.
-

Deliverables Produced

- **Deliverable 1:** Our first deliverable's main purpose was to begin the development of our personas and other artifacts, gather information, and record our initial meeting notes in one place.
 - **Deliverable 2:** The second deliverable looks at the main barriers that affect voter turnout among African American and Latine communities.
 - **Deliverable 3:** The third deliverable examines the SPLC's current programs for their target audience and makes recommendations based on the four personas we created.
-

Challenges

After concluding our research, our main challenges were finding reliable sources to use for our barriers and challenges report as specific data for the Gulf Coast Counties proved to be niche. Thus, we utilized information that spoke to our personas in a more generalized context than anticipated. Nonetheless, we adjusted our persona development to incorporate as much research on our Gulf Coast County personas to represent our intended audience. We also found it difficult in utilizing other UX/UI platforms as our persona developers were most comfortable using Figma. Figma proved to be efficient in creating our personas but tedious in some aspects.

Connections to Other Teams

Although our team didn't use any other teams specific findings/deliverables, our team worked to represent the targeted demographic for SPLC and thus our connection with other teams is in representing who we are collecting this data for.

Final Report and Deliverables

Project Context

Our team's project is part of the larger Community Context and Personas work for SPLC, which focuses on understanding voter experiences and community needs in Alabama's Gulf Coast counties and related regions. For our team's segment of the greater project, we researched the target communities, identified likely voter barriers, developed personas that represent different groups SPLC may be trying to reach, and created recommendations on how SPLC can help reach out to these personas and their targeted demographic. Our work depended on the project requirements, SPLC meeting notes, and public datasets such as Census data, Redistricting Data Hub data, Bureau of Labor Statistics data, and election-related information to build the personas and barrier analysis. We also relied on the client's feedback that the main barriers included misinformation, distrust, transportation, fear, voter ID confusion, polling place changes, and rights restoration confusion.

Methods

As a team, we researched the Southern Poverty Law Center (SPLC) and the Alabama programs to understand how the organization helped the community within Alabama. We looked at the SPLC's policy for 2026 and the current Alabama programs that are available. These programs include the Alabama Youth Center, Days of Democracy, Town Halls, Alabama Institute, and Learning for Justice.

Majority of the information we have gathered are from:

- SPLC Official website
- SPLC article and program pages
- Research Alabama's voting rights, civic participation, and community

By gathering all of this information, we were able to determine the main issues that are happening within the communities in Alabama in order to create accurate personas.

In order to create our personas, we created fake personas based on true situations and problems that were commonly occurring within the Alabama community. Each of the personas that

were created are separated into 4 different categories that the SPLC would work with.

The four personas we made were:

- a young adult who is getting into the politics world
- An immigrant who came to the USA and is having a hard time getting information about voting
- A PTA parent
- An incarcerated person who is trying to get their voting rights back

After creating these four personas, we compared their needs to the programs provided by the SPLC and figured out what should be done to help accommodate the personas needs within the programs that the SPLC provides.

There were several tools that we used for this project. One of them was Figma. We used Figma as a tool to create the personas that we have today. Each persona has a description explaining their demographics and what they are struggling with. We also used SPLC's website to gather information and data to help form the personas. We also used google doc to document our deliverables and write down necessary information.

There were several challenges we faced when creating our deliverables. One of them was the lack of public information we could find on the internet. Due to this, we found it hard to create a persona because there wasn't much public information on what people personally felt about the barriers they faced around voting. To solve this, we looked at common experiences that the community Alabama as a whole feels when facing the voting barriers. Then we categorize the experiences to create each persona.

We also had to be cautious about making stereotypes in our personas. This is because we don't want to offend anyone. To solve this, we made sure none of the information we put down is an assumption but an actual fact.

Deliverables and Findings

[Artifact and Persona Initiation \(Deliverable 1\)](#): In this document, we outlined the scope of our project, gathered relevant information in regards to our personas, added potential resources, and documented our meeting notes. Deliverable 1 served as the backbone for our future works as it was the document that initiated our project. The document starts by highlighting the target audience and counties within Alabama. Further in the document, we included some outlines of personas and outlined our research information. Our initial persona notes included:

- African American Male in early twenties born in the U.S
- Latin American Female who immigrated to the U.S in 2010 and has citizenship
- African American Female mid 50s
- African American Male in late 30s with a felony conviction but went through the formal rights restoration process

From there, we drafted a formula of information for each persona. This includes the following: The TLDR of a persona, their demographics, goals, motivations, behaviors, visual, and narrative. For the narrative, we also included a skeleton of information we wanted to focus on: daily life, responsibilities, information and trust, community, and their environment. After, we included helpful links, possible outreach strategies, and SPLC meeting notes.

[Voter Barriers and Challenges Report \(Deliverable 2\)](#): The purpose of deliverable 2 was to highlight the main voting barriers and challenges that the African American and Latine communities face in the Black Belt and Gulf Coast regions. The document used the research we gathered to tie the numbers and facts we were learning to the personas we created. Our goal was to empathize with the possible experiences that our personas could experience. In doing so, we could highlight the key barriers that were impacting voter participation.

The barriers that we found included:

- Where to vote

Finding and knowing where to vote can be a difficult factor for many in Alabama. Between 2012 and 2018, hundreds of polling locations were closed across Southern states, which increased travel distances and made voting less accessible.

- Transportation

Transportation is one of the most direct barriers. In many rural parts of Alabama, polling locations are spread out, and some areas have experienced polling place closures over time.

- Socioeconomic factors

Work and time constraints are also important. Many people cannot afford to take time off or wait in long lines. Research has shown that longer wait times tend to occur more often in communities with higher populations of minority voters. There is also the issue of trust. Historical and systemic inequalities have contributed to a level of distrust in the voting process. Combined with constant exposure to negative political messaging, this can lead to people feeling disengaged.

- Information Credibility

Voter ID laws can also create challenges. While they are intended to regulate voting, not everyone has easy access to obtaining a valid ID. National data shows that millions of Americans do not have a current government-issued ID, and this disproportionately affects lower-income communities.

- Voting Restoration and Rights Confusion

Misinformation is another major issue. Some individuals, especially those who have been involved in the justice system, may believe they are not allowed to vote when they actually are. This leads to people choosing not to participate at all.

- Language Barriers

In Alabama, the amount of residents who are immigrants has steadily increased over the years. Currently, Alabama does not follow Section 203 of the Voting Rights Act which would require them to provide language assistance. The lack of language assistance for voters creates a language barrier for some voters.

Below is the journey of the Immigrant Voter persona we created.



STEP 1

Finding Out When and Where to Vote

Mia hears about the election at church, but the date and polling location are never confirmed.

Is it next week?
Where do I go?

¿Elección?
¿Cuándo?
¿Dónde?

- Heard about it at church
- Shared device with limited access
- Not sure of the date or polling location

! Mia wants to vote, but getting reliable information is difficult.



I registered when I became a citizen in 2019. But am I still registered?

STEP 2

Confirming Her Registration

Mia is not sure if she is still registered to vote. She can't check online, doesn't know who to call, and is afraid of being turned away at the polls.

BARRIERS

- No easy internet access
- Doesn't know voter assistance phone numbers
- Difficulty understanding phone systems in English
- Afraid of being rejected at the polls

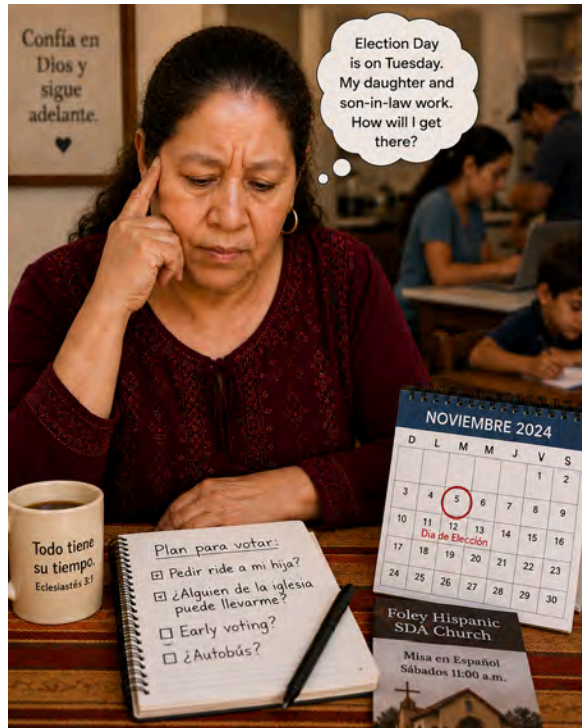
WHAT MIA NEEDS

- Human help in Spanish
- Paper confirmation of registration status
- Community-based voter help and hotlines

¿Estoy registrada?
¿Sigue activa?
¿A quién puedo llamar?

¿Cómo reviso mi registro?

! Mia wants to be prepared and make sure her vote counts.



STEP 3

Planning the Trip

Mia doesn't drive. Her daughter and son-in-law work during the day. Election Day is on a Tuesday, and she's not sure how she will get to the polling place.



BARRIERS

- No personal transportation
- Family members work and cannot take time off
- Limited public transportation options
- Church is on the weekend, not on Election Day
- Doesn't want to be a burden

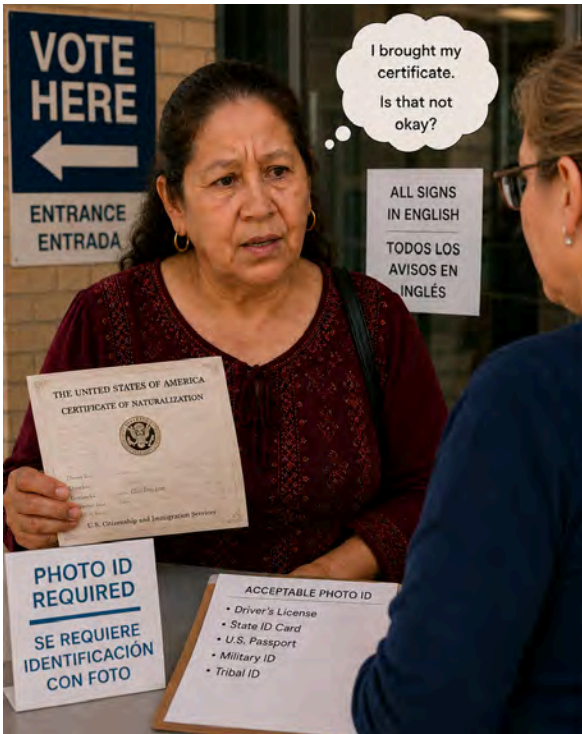


WHAT MIA NEEDS

- Community rides or carpools
- Early voting or weekend voting options
- Clear information about transportation resources in Spanish
- Church-led voter support and ride coordination



Mia is determined to vote, but needs help getting there.



STEP 4

Arriving at the Polls

Mia is not sure which building is her polling place. The address she found online may be outdated. When she arrives, the poll worker speaks only English. She brought her naturalization certificate — they ask for a photo ID instead.



BARRIERS

- Outdated or confusing polling place information
- English-only signage and communication
- Confusion about acceptable photo ID
- Limited bilingual poll workers or interpreters
- Embarrassment and stress in an unfamiliar environment

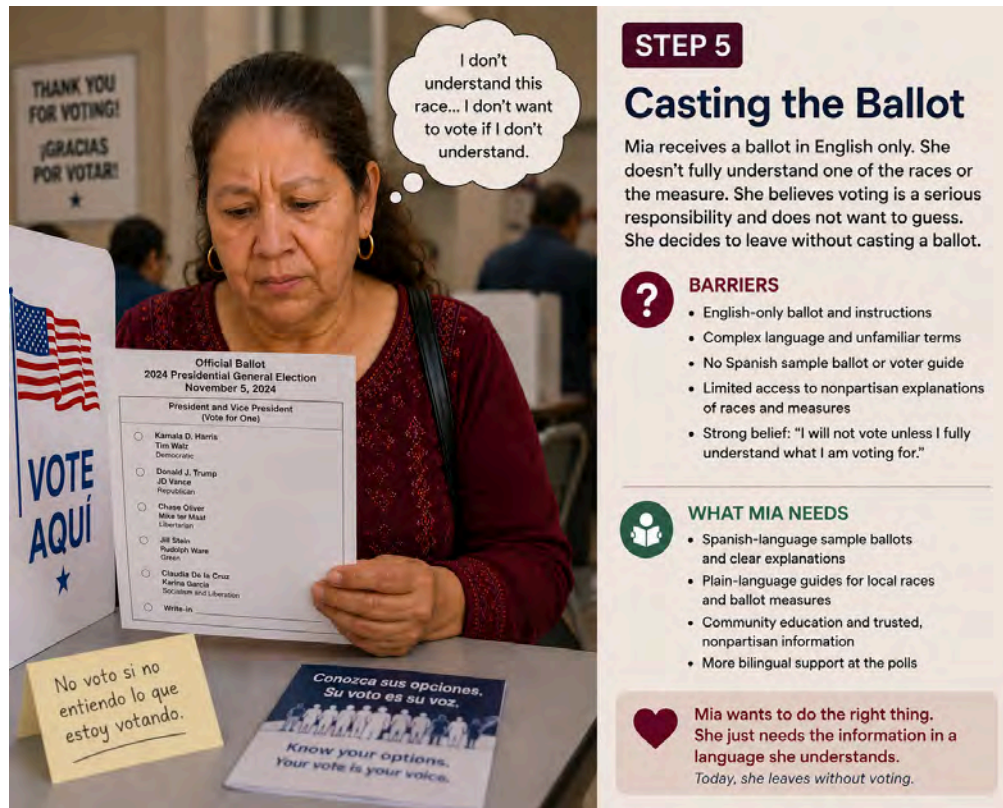


WHAT MIA NEEDS

- Accurate polling place information in Spanish
- Clear, simple ID requirements in Spanish
- Bilingual greeters or poll workers
- Welcoming, culturally responsive environment



Mia wants to vote and brought her important documents, but she feels nervous and unsure of what to do next.



[Recommendations and Personas \(Deliverable 3\)](#): Our last deliverable provided a background of the SPLC and their current initiatives, our final personas, and the recommendations that we created to help the groups our personas represent. The Southern Poverty Law Center (SPLC) works to dismantle white supremacy, strengthen intersectional movements, and advance the human rights of all people through litigation, education, and tracking hate groups. In knowing that the Southern Poverty Law Center works to address these issues in ways such as education and litigation, working to identify ways to strengthen their mission is our ultimate goal. Through identifying initiatives already set in motion by the SPLC and our research conducted to pinpoint specific personas that SPLC aims to represent, we want to take the time to propose persona-specific recommendations towards the initiatives and programs SPLC hosts in order for SPLC to continue accomplishing their mission. Looking into SPLC's current programmatic initiatives and their current 2026 policy priorities for Alabama, we can begin to understand how SPLC reaches out to its community members

In terms of 2026 policy priorities for Alabama, SPLC's priorities include the following topics:

Food Security for Children and Family

Health Care Access and Affordability
Ending Utility Fee Prosecutions
Affordable Quality Housing
Securing Democracy and Ensuring Voting Rights

In highlighting the policy priorities for SPLC's Alabama office, we can understand what the office intends to put its effort towards in the coming year. Specifically, these policies will be pulled into litigation efforts, class action lawsuits, institutional reform, and other initiatives to ensure these policy priorities are achieved.

Likewise, the SPLC Alabama office has programs aiming to achieve their overall mission of strengthening democracy and voting rights, dismantling white supremacy, ending unjust imprisonment, and eliminating poverty and economic inequality. Programs such as Alabama Youth Council, Redistricting Advocacy & Community Engagement Program (RACE), Alabama Advocacy Institute, Town Halls, and Learning for Justice.

SPLC - Alabama Current Programs:

Alabama Youth Council:

The Alabama Youth Council is a community initiative spearheaded by the Southern Poverty Law Center, Alabama Office, designed to empower teenagers and young adults (ages 19 and under) to participate in civic engagement, leadership, and community advocacy. The youth council has the opportunity to help the Alabama community through connecting with local leaders and gaining hands-on experience to develop a greater understanding of the democratic process & grassroots organizations. The program is held at regional branches in different counties in Alabama that meet bi-weekly. Each branch has its own governing policies, constitution, and committees based on the interests of its current members. Likewise, the program holds an annual Youth Summit that is a hotspot for the youth council in different regions to convene and connect.

Redistricting Advocacy & Community Engagement Program (RACE):

The Redistricting Advocacy & Community Engagement program (RACE) is an initiative looking to embolden community members to participate in map-making processes that define their political representation.

Days of Democracy:

Days of Democracy is a Alabama-state wide effort looking to bring community members together for trainings and democratic action days encouraging people to learn how the government works and how they can help make a difference. Participants of these training sessions should expect to engage in fun and hands-on lessons on how the government works and ways to get involved in their community, partake in workshops to build skills like preparing to vote and speaking on local issues, and practice what participants learned in real life. These sessions are expected to last two to three hours. Prospective participants can sign up for these trainings by county through SPLC - Alabama's Days of Democracy page.

Alabama Advocacy Institute:

The Alabama Advocacy Institute is a five-week annual program created to better equip organization leaders in advocacy. In providing leadership experience to community leaders, advocates, organizers, and smaller organizations, SPLC aims to deepen their skill set and gain training in the world of advocacy. Participants of the Alabama Advocacy Institute will be able to learn how to develop strategies in developing campaigns, collaborate with fellow community members from different areas of Alabama, and design a community campaign/action plan as a capstone project.

Town Halls:

The SPLC - Alabama office holds numerous town hall meetings encouraging community members to stay informed on issues, advocacy efforts, and overall educational resources. These meetings are held to discuss ongoing issues in Alabama. For example, student due process rights in education, voting rights protection, and localized civic engagement practices. Likewise, initiatives such as RACE and "Days of Democracy" are training series teaching community members how to work through local governance and organize advocacy efforts. Attendees for these Town Halls are advised to register in advance and can track new meetings on the SPLC website.

Learning for Justice:

The Learning for Justice (LFJ) program is an educational initiative that "cultivates and nurtures dialogue, learning, reflection and civic action from those closest to and harmed most by injustices in the South" ("About Learning for Justice"). The LFJ program includes a curriculum administered to K-12 schools that hold four key components: identity, diversity, justice, and action. The LFJ program provides resources for class rooms and communities such as the Learning for Justice Magazine, lesson plans, film kits & documentaries, and professional development seminars. Youth initiatives

have branched off from the LFJ program, primarily known as the **Youth Learning for Justice Program**. The Youth Learning for Justice Program equips young adults with toolkits, learning journeys, book reviews, and other resources to encourage and empower youth leadership.

Persona Overview:

Jay Hawk



Age: 22 **Gender:** Male
Race: African American
Location: Mobile County
Job: Early Employment

Goals

- Learn how voting effects his future
- Live up to parents legacy

★ Recent graduate from University of Alabama

Voting habits

- Gets information from parents
- Only votes during presidential elections

Motivations

- Parents participated in civil rights movement
- Wants to be more independent

He grew up learning about the civil rights movement. Both of his parents and grandparents were part of the civil rights era and understood the struggles during that time period. They would always tell him about these struggles and remind him to vote. However, he feels like the politics nowadays feel overwhelming and trying to navigate around the voting world and figure out who he believes in.

Mia Rodriguez



Age: 62 **Gender:** Female
Race: Latina
Location: Baldwin County
Job: Unemployed

Goals

- Be a active US citizen
- Support her daughter's family

- ★ Immigrated in 2010
- ★ Citizenship in 2019
- ★ Lives with daughter's family

Voting habits

- Depends on daughter's family on transportation
- aware of voting in the US but unsure how to vote
- Relies on her daughter and church/communities for reliable information

Motivations

- Better schools for grandchildren
- Opportunities for older Latin generations to immigrate to the USA
- Improving the US naturalization process
- Trustworthy communication for community members

She is in her early 60s, originally from Mexico, and lives with her daughter's family in Baldwin County. When she immigrated from Mexico to the United States, she did not know much English, how the government worked, or what day to day life was like. Her days revolve around helping her daughter's family through cooking meals, picking up her grandkids from school, helping with homework, and providing stability at home. She attends Mass on the weekend at Foley Hispanic SDA Church and feels connected to the community through church. She wants to be an active citizen but finds sourcing information to be difficult. When it comes to voting, she won't participate unless she fully grasps what she is voting for.

Lauren Moore



Age: 48 **Gender:** Female

Race: African American

Location: Mobile County

Job: Administrative
Assistant at local
healthcare facility

Goals

- Support efforts that improve education
- Keep home life stable
- Stay informed on local issues

★ PTA mom at WP Davidson High School

★ Activate at Little Flower Catholic Church

Voting habits

Communication

- Mobile County Website
- Mobile County Facebook

Schedule

- Busy schedule with kids at two different schools
- Demanding job
- Volunteers on the weekends
- Attends PTA meetings

Motivations

- Ensure opportunities and education for her children and the children in the community
- neighborhood safety

She is a 48 year old African American woman in Mobile County who participates in her community through faith and participation in WP Davidson High School's PTA. She works full-time at the local health care facility and also supports her kids' school activities. Neighbors, school parents, and church parents see her as a trusted voice in the community and as someone who organizes food drives and other community events. She votes occasionally but finds it difficult to make time to vote and to learn information about who to vote for.

**Johnson
Mcbroom**



Age: 35 **Gender:** Male
Race: African American
Location: Conecuh County
Job: Unemployed

Goals

- Regain their right to vote
- Fully rejoin society and rebuild his life
- Fair treatment from government bodies

★ *was incarcerated for non violent felony*

Voting habits

- *Does not currently vote due to an unclear procedure for becoming eligible to vote again*
- *Relies on his community organizations to get information. This mainly includes information via word of mouth and community groups*

Motivations

- *Wants to feel included in their community and wants to make an impact on what they believe in*
- *Wants to grow as a person after being incarcerated*

He is a 35 year old man who convicted of theft at the age of 30. As a result, he lost many of his civil rights, including the right to vote. After completing his sentence and returning to his community, he now wants regain his rights. He feels that society judges him for the actions of his past. To regain a sense of self and grow, he feels that regaining his right to vote is a step in the right direction. With how things are, he is currently struggling with the current system to figure out how to get his rights back.

Recommendations

After reviewing our deliverables and our findings pertaining to our SPLC community and the personas we have developed, our group has garnered a better understanding of the Alabama Gulf Coast Counties demographic. In creating these personas, we acknowledged the many different lives these personas live. It's important to note the differences of these personas as they will help us comprehend how best we can give SPLC recommendations on reaching out to their target demographic and complete their overall mission of dismantling white supremacy, strengthening intersectional movements, and advancing the human rights of all people.

As mentioned, in our deliverables and findings, our main focus is our personas. Through further analysis, we've seen the specific variables that can create an unwelcoming environment for our personas to exercise their right to vote. Each persona is different in their reasoning for inhibition, nonetheless, we find it that SPLC should prioritize looking into how to meet the needs of these individuals.

It is important to reiterate the programs that SPLC holds in the Alabama office as they have historically made a large impact on the Alabama community and how they impact the personas we have developed.

Alabama Youth Council:

The Alabama Youth Council is a community initiative spearheaded by the Southern Poverty Law Center, Alabama Office, designed to empower teenagers and young adults (ages 19 and under) to participate in civic engagement, leadership, and community advocacy.

Redistricting Advocacy & Community Engagement Program (RACE):

The Redistricting Advocacy & Community Engagement program (RACE) is an initiative looking to embolden community members to participate in map-making processes that define their political representation.

Days of Democracy:

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Alabama Advocacy Institute:

The Alabama Advocacy Institute is a five-week annual program created to better equip organization leaders in advocacy. In providing leadership experience to community leaders, advocates, organizers, and smaller organizations, SPLC aims to deepen their skill set and gain training in the world of advocacy.

Town Halls:

The SPLC - Alabama office holds numerous town hall meetings encouraging community members to stay informed on issues, advocacy efforts, and overall educational resources. These meetings are held to discuss ongoing issues in Alabama.

Learning for Justice:

The Learning for Justice (LFJ) program is an educational initiative that “cultivates and nurtures dialogue, learning, reflection and civic action from those closest to and harmed most by injustices in the South” (“About Learning for Justice”). The LFJ program provides resources for class rooms and communities such as the Learning for Justice Magazine, lesson plans, film kits & documentaries, and professional development seminars.

Recommendations for the client (SPLC)

Based on our findings, we would recommend SPLC meet the needs of those in their intended demographic. Through our personas we were able to identify challenges, barriers, and obstacles that created a deeper understanding as to why Gulf Coast voters were having a hard time in finding ways to vote before they stepped into the voting polls. Likewise, in identifying the current programs SPLC - Alabama holds, we can further develop our persona - based recommendations. Our recommendations are as follows for each respective persona:

Persona #1: The Youth Voter

Adjust the Alabama Youth Council (AYC) / Youth LFJ: The Alabama Youth Council has an age cap set at 19, we recommend that SPLC either extend the stopping age to include young adults OR create a subsection of AYC titled as **“AYC Young Professionals Network”** or mentorship program

targeting ages 20–25. Making sure that young adults are kept in the loop of discussing ongoing issues in their community.

"Legacy to Legislative" Town Hall Series: Host specialized, modern town halls in Montgomery that pair Civil Rights veterans (appealing to his respect for family legacy) with young activists discussing contemporary, tangible issues like local utility fee prosecutions and workforce housing. This connects historical struggles directly to the youth activist community.

Micro-Targeted "Days of Democracy" Workshops: Create 1-hour "Express" versions of the Days of Democracy training sessions tailored for young professionals. Focus heavily on down-ballot local elections (city council, county commissioners) to show this persona exactly how local voting impacts their daily life and career opportunities more than just presidential cycles.

Persona #2: The Immigrant Voter:

Church-Embedded "Days of Democracy" Pop-Ups: Instead of expecting this persona to travel to an SPLC county event via bus, bring the training directly to her trusted space. Partner with the **Foley Hispanic SDA Church** (and similar parishes in Baldwin County) to host bilingual, post-Mass workshops.

Bilingual Voter Navigation Toolkits: Create simplified, Spanish-language physical toolkits distributed via church bulletins. These should include step-by-step voter registration guides, sample ballots, and an explanation of rights regarding language assistance at the polls.

Persona #3: The School & Church Mom:

PTA and Church "Liaison" Program through Alabama Advocacy Institute: Recruit community pillars like this persona for an abbreviated, hybrid version of the Alabama Advocacy Institute. Give them the tools to become a "SPLC Ambassador." Since our persona is already a trusted voice, she can distribute voting guides and info on food security directly to her PTA and Little Flower Catholic Church networks.

Asynchronous "Lunch & Learn" Digital Town Halls: Because this persona's schedule is packed, traditional 2-hour evening town halls won't work. Offering 30-minute podcast-style or Zoom "Lunch & Learn" briefings specifically tracking Mobile County policy issues, which she can consume during her break at the healthcare facility will be beneficial.

Persona #4: The Reentry Voter:

“Days of Democracy” Reentry-Specific Tracks: Modify a track of the Days of Democracy curriculum to explicitly address felony disenfranchisement laws in Alabama. Train local organizers on how to speak to reentering citizens without judgment, focusing heavily on word-of-mouth outreach since he relies on community grapevine communication.

Amplifying Voices via Town Halls: Create space in SPLC Town Halls to highlight the intersection of unjust imprisonment, poverty, and civic death. Giving individuals with this persona a platform to speak safely helps satisfy his goal of showing his community that he has grown, while simultaneously educating others facing the same barriers.

Recommendations for future project teams

After conducting our research on SPLC, our personas, and developing strategies/persona-based recommendations on how best SPLC can accommodate our personas needs, it's crucial to highlight how the continuation of this research can further help SPLC in incessantly improving their outreach to their communities in increasing voter turnout. As our team worked primarily in researching the community personas in the Gulf Coast Counties in Alabama, how these personas can be further developed in other regions of Alabama is beneficial to getting a holistic review of the Alabama office's demographics. Developing new initiatives and/or updating preexisting ones held by SPLC can bring an entirely new perspective for the community that can either help or hinder voter turnout for these personas. Persona development is an ongoing process. As time moves forward, so do community members. Changes within SPLC and/or the Alabama community at large will cause a rippling effect and will bring new questions, challenges, and barriers for community members. Ongoing research towards the Alabama community is imperative for civic engagement.

Expanding on our current research findings, our team holds specific recommendations for the future of this research. Our team recommends finding reliable, up-to-date information pertaining to the current demographic of Gulf-Coast county communities. Through census data and/or conversing with individuals who live in the Gulf Coast counties, would prove to be beneficial in comprehending the community members. Holding conversations with our client helped our team to gain clarity on their demographic and how their community has looked like historically. Furthermore, asking for the specific personas SPLC was looking to gain more information on was essential and pointed us in the right direction in our research. Likewise, understanding existing laws, current government practices,

and ongoing efforts by community leaders who work specifically to increase voter turnout and overall civic engagement is valuable information. Reaching out to advocacy groups, SPLC staff members, and community leaders helped our team to find our specific path of developing our personas.

Specific online platforms were vital in our persona development. Our team utilized the online platform, Figma, to develop our personas. It was important to our team to use a platform that our persona developers were comfortable using and had prior experience working with. Our team would encourage future researchers to use Figma to further understand these personas as well as new persona development. Moreover, our team would recommend the use of platforms that are UX/UI design focused (i.e. Lunacy, Sketch) to develop personas.

Researching challenges and barriers for these personas proved to be a challenge in itself. In our research, we set out to understand the voting process and how each specific persona would interact with the process with the prior knowledge of knowing their background. We found the research process to be an uphill battle as data spanned across the nation and not in the specific region of the Gulf Coast Counties. We recommend for future teams to use reliable resources such as the database catalogs, research articles, and scholarly papers to use in finding more information on barriers and challenges pertaining to their personas. Likewise, making sure that the data is specific to the region you are researching.

Conclusions

This capstone project successfully provided the Southern Poverty Law Center (SPLC) with a comprehensive set of research based deliverables designed to better understand and address barriers to voter participation among African American and Latinx communities in Alabama, with a particular focus on the Black Belt and Gulf Coast regions. Throughout the semester, our team combined research, project management, design, and client feedback to create practical tools that the SPLC can use to strengthen their voter outreach as well as civic engagement.

In order to manage the scope of the project, our team utilized a shared Task Tracker in Google Sheets that organized priorities, deadlines, contributors, as well as progress across all deliverables - Deliverable 1, 2, and 3. This tracker allowed us to divide responsibilities while also maintaining accountability and ensuring that each section was completed on time. Although each member was

assigned ownership of certain tasks, we still collaborated by reviewing each other's work and making revisions as a group to ensure consistency and alignment with SPLC's goals.

The first deliverable, the Community Context and Personas Report, established a foundation for the project. We conducted extensive research on demographic trends, labor and socioeconomic data, voter turnout, and the historical context of voter rights in the Alabama area. These findings were then translated into detailed personas representing key voter groups such as a young african american voter, an immigrant citizen, a community oriented mother, as well as a formerly incarcerated individual navigating rights restoration. Each of the personas included demographics, goals, behaviors, and stories that showed how voting barriers affect people differently. In order to make these personas more engaging and visually accessible, our persona development team members designed and organized them in Figma, along with supporting visual assets.

Our second deliverable, the Voter Barriers and Challenges Report, synthesized insight from our persona research and outside sources to identify the most significant barriers to voters. These barriers included transportation challenges, voter ID confusion, misinformation, fear, polling place changes, time and financial constraints, and information gaps. Our analysis demonstrated that many voters face multiple barriers simultaneously, stressing the need for solutions that address not only practical accessibility issues but also deeper concerns related to civic engagement and trust.

Our third deliverable, the Recommendations Report, essentially transformed our research findings into actionable strategies tailored to each persona and connected directly to SPLC's existing programs such as the Alabama Youth council, Days of Democracy, Alabama Advocacy Institute, and Learning for Justice. These recommendations included targeted outreach initiatives and specialized programming for reentry voters. By aligning the recommendations with SPLC's current initiatives, realistic and implementable ideas that can help the organization better connect with the communication it serves were created.

We would say a particularly valuable aspect of this project was our collaboration with SPLC staff as these were useful in obtaining meaningful feedback and using the feedback as a guide to build deliverables of value to the organization.

Overall, this project showed the importance of looking beyond data to understand the lived experiences of the people behind it. By researching community contexts, creating detailed personas, identifying systemic barriers, and developing targeted recommendations, our team successfully

produced a set of deliverables that can help the SPLC design more informed and effective voter outreach strategies. This capstone project not only allowed us to apply the skills we developed throughout the Information Science/Technology and Information Design programs, but also showed how thoughtful research and design can support meaningful social impact.

Deliverables Checklist

- ☐ [Client Deliverable Folder: GC- CCP Folder](#)
- ☐ [Deliverable #1: Artifact and Persona Initiation](#)
- ☐ [Deliverable #2: Voter Barriers and Challenges](#)
- ☐ [Deliverable #3: Recommendations](#)

Policy and Organizational Translation

GC-PORT

Siddhi Kacholia (Project Manager), Yen-An Andrew Lu, Sirak Abraham, Margaret Nowell

Southern Policy Law Center (SPLC)

INST490 - Integrated Capstone for Information Science

Section 0105

May 18th, 2026

Team Brief

Team 10: GC-PORT

Key Findings

- Finding 1: Transportation access affect the ability to reach polling sites
 - Finding 2: Mobile and Baldwin counties appear to have different transportation systems
 - Finding 3: Rural areas could face longer travel distances and limited options
 - Finding 4: Reliable access of a vehicle isn't supported to some residents
 - Finding 5: Polling access can vary from locations to infrastructure
-

Deliverables Produced

1. Stakeholder Overview: We created an overview of stakeholders, why they are relevant to the research, and the best ways to communicate findings to them.
 - a. https://drive.google.com/file/d/1X1koYqdg7fhNiA4N62VN5qah-nPGmKmN/view?usp=drive_link
2. Policy Briefs: We created three different briefs focused on three polling barriers. Each brief includes an overview, key relevant findings, policy implications, and supporting visuals.
 - a. https://drive.google.com/file/d/1h9TWr4HZioqT1ddsgnhinOE2DqVJ_xSk/view?usp=drive_link
 - b. https://drive.google.com/file/d/1eyaTl0JGDUidyG5IXs3lisa9OgrM0gVZ/view?usp=drive_link

- c. https://drive.google.com/file/d/1y2MOEm4w2JVIsV5hkR67fABEpL_KkNqZ/view?usp=drive_link
 - 3. Visual Translation Presentation: We created a public-facing presentation that translated technical voter participation and transportation findings into simplified visuals and plain-language explanations for policymakers, advocacy organizations, and community stakeholders.
 - a. https://docs.google.com/presentation/d/1txk8eumklmSgMZIfvxFYasnRBHC_E7rI2INNIPGoLhQI/edit?usp=drive_link
 - 4. Maps and Visual Graphics: We created maps, charts, and visual graphics that communicate polling access findings and transportation-related barriers and accessible visuals for policymakers, advocacy groups, and community stakeholders.
 - a. https://drive.google.com/file/d/15VLGN00wO2ALA6fHjn913OMH9v1OLAQf/view?usp=drive_link
-

Challenges

One challenge our team encountered was coordinating communication across multiple project groups within an online course environment. Since most communication took place through email, it was sometimes difficult to receive timely responses, updates, or deliverables from other teams that our work depended on. Unlike in-person collaboration, email communication often slowed down discussions and made it harder to quickly clarify questions or request additional information. Different teams were also progressing at different speeds throughout the semester, which created challenges when trying to gather finalized findings, visuals, or summaries needed for our communication materials.

Another challenge was the large amount of technical data, maps, geographic analysis, and statistical findings shared across the broader project. Because our role focused on policy translation and communication rather than conducting the analysis itself, we had to carefully evaluate which findings were the most relevant, understandable, and appropriate for policymakers and community stakeholders. Some technical findings were highly detailed and difficult to translate into clear public-facing language without oversimplifying the information or losing important context. In addition, different stakeholders required different communication approaches, meaning the same findings often needed to be framed differently depending on the audience.

To help address these challenges, our team coordinated with the professor throughout the project for clarification and guidance regarding project expectations and deliverable directions. We also reviewed other teams' attached deliverables and progress check materials in order to stay informed about findings and project developments, especially during periods where our team was slightly behind schedule. This helped us continue building communication strategies and stakeholder materials even when some project outputs were still in progress. Our team ultimately focused on identifying the clearest and most actionable findings while making sure limitations, uncertainty, and context were communicated responsibly and accurately.

Connections to Other Teams

Team TRAC - GC - We used findings and visuals from their reports and other research to create a policy brief for the Director of Transportation (John R. Cooper) in Alabama. We simplified and connected different aspects of their research to create one brief to highlight the potential transportation barriers in Alabama.

Team AACS - GC - We used findings and visuals from their reports and other research to create a policy brief for the Executive Director of Alabama Disabilities Advocacy Program (Shannon Shelley-Tremblay). We simplified and connected different aspects of their research to create one brief to highlight the potential accessibility barriers in Alabama.

Team BAND - GC - We used findings and visuals from their reports and other research to create a policy brief for the Digital Expansion Division Chief, Alabama Department of Economic and Community Affairs (Maureen Neighbors). We simplified and connected different aspects of their research to create one brief to highlight the potential broadband access barriers in Alabama.

Final Report and Deliverables

Project Context

Our team supports the broader regional analysis project by translating technical findings from other teams into accessible materials for policymakers, advocacy groups, and community organizations. The project relies on geographic analysis, transportation findings, demographic research, and polling access data produced by partner teams. These findings may include maps, transportation accessibility information, demographic trends, and regional comparisons related to polling access. Our role is to develop communication strategies, stakeholder-specific summaries, policy-oriented materials, and visual explanations that make technical information easier to understand and apply for non-technical audiences. In addition, our work focuses on adapting communication styles depending on the audience, such as policymakers, transportation officials, disability advocacy organizations, or community groups.

Methods

Our team focused on translating technical transportation and voter participation findings into communication materials that could be more easily understood by non-technical audiences. Rather than conducting original GIS or statistical analysis, GC-PORT reviewed visualizations, transportation maps, turnout analysis, and polling-access findings created by partner teams including GC-DIG and GC-TRAC.

The team evaluated how policymakers, advocacy organizations, and community stakeholders might interpret technical findings and visualizations. We then adapted these materials into simplified presentation slides and public-facing visual explanations using plain-language summaries, reduced technical terminology, and audience-focused communication strategies.

To support accessibility and readability, the team emphasized concise explanations, simplified visual hierarchy, clear key-takeaway sections, and minimal technical language. Visual communication principles were used to help audiences quickly identify major findings related to voter participation trends, transportation barriers, polling accessibility, and public transportation infrastructure.

Tools used throughout the project included Google Slides, Google Docs, visual outputs provided by partner teams, and publicly available transportation and voting-access datasets included in the shared deliverables. Ongoing collaboration with other teams was necessary throughout the semester.

because many communication materials depended on finalized maps, graphics, and transportation findings from partner deliverables.

Deliverables and Findings

Stakeholder Overview (Deliverable 1): A document containing stakeholder-specific communication strategies for presenting polling access findings to policy makers, advocacy groups, and public officials. This deliverable includes explanations of who key stakeholders are, why they are relevant to the project, and the best approaches for communicating technical information to each audience. Stakeholders included in this deliverable include transportation officials, disability advocacy organizations, and economic and community development agencies. This deliverable meets the project goal by helping SPLC and partner organizations translate technical geographic and transportation findings into language that is easier for non-technical audiences to understand and apply. It also supports future policy briefs, presentations, and public-facing outreach materials by providing guidance on how findings should be framed for different audiences.

John R. Cooper

Director of the Department of Transportation

Who They Are

- Oversees transportation systems and infrastructure planning across Alabama
- Responsible for mobility access and public transportation related planning
- Helps manage how residents travel to important public services
- Plays a role in improving transportation accessibility in rural and urban areas

Why They Matter

- Transportation access affects how voters reach polling locations
- Rural communities may face longer travel distances to vote
- Limited transportation options can create barriers to civic participation
- Polling place accessibility is connected to broader mobility and infrastructure issues
- Transportation data can help explain geographic gaps in voter access

Best Approaches to Communicate

- Focus on transportation and mobility related challenges
- Use maps and visuals to show polling distance patterns
- Include regional comparisons between areas such as the Black Belt and Gulf Coast
- Present findings in a clear and non technical way
- Emphasize public access and infrastructure concerns rather than political outcomes

Shannon Shelley-Tremblay

Executive Director, Alabama Disabilities Advocacy Program

Who They Are

- Works to protect the rights of individuals with disabilities across Alabama
- Focuses on accessibility, inclusion, and equal access to public services
- Advocates for disability related policy and accessibility improvements
- Supports compliance with accessibility protections and standards

Why They Matter

- Voters with disabilities may face transportation and mobility challenges
- Physical accessibility barriers can make polling locations difficult to reach or navigate
- Accessibility concerns may affect voter participation and civic engagement
- Polling place accessibility is connected to broader disability rights issues
- Their perspective helps identify barriers related to equitable voting access

Best Approaches to Communicate

- Use clear and accessible language
- Focus on mobility limitations and accessibility concerns
- Connect findings to ADA related considerations and accessibility protections
- Keep visuals and summaries simple and easy to understand
- Emphasize equitable access to voting and public services

Maureen Neighbors

Digital Expansion Division Chief, Alabama Department of
Economic and Community Affairs

Who They Are

- Supports statewide programs related to infrastructure, development, and public access
- Focuses on improving access to services and regional connectivity
- Works on initiatives related to community development and digital expansion
- Helps address disparities in access across different regions of Alabama

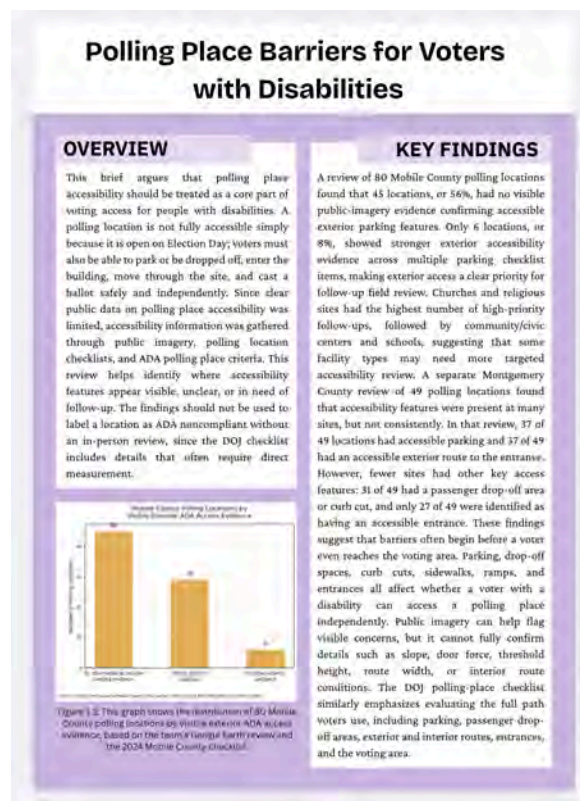
Why They Matter

- Infrastructure and service access can influence civic participation
- Geographic barriers may affect access to voting information and polling locations
- Regional inequalities may impact communities differently across Alabama
- Differences between areas such as the Black Belt and Gulf Coast may reflect broader access disparities
- Their role connects public services, infrastructure, and community accessibility

Best Approaches to Communicate

- Focus on regional equity and public access to services
- Use clear summaries and easy to understand visuals
- Include charts and regional comparisons to explain geographic patterns
- Highlight disparities in infrastructure and accessibility between regions
- Frame findings around improving access and community support systems

Policy Brief #1Policy Brief #2Policy Brief #3 (Deliverable 2): We created three policy briefs that translate technical findings into clear, non-technical takeaways for public officials, advocacy leaders, and civic access stakeholders. These briefs are built from findings produced by the transportation access, ADA accessibility and compliance, and broadband access teams. Each brief connects another team's data to a specific audience: the transportation brief uses polling access and mobility findings for the Director of the Department of Transportation; the ADA brief uses polling-place accessibility findings for the Alabama Disabilities Advocacy Program; and the broadband brief uses digital access findings for the Digital Expansion Division Chief at ADECA, whose division supports Alabama's statewide connectivity planning and broadband expansion work. These policy briefs serve as communication deliverables that help make the project's findings usable beyond the technical teams. Other teams provide the data, visualizations, and analysis, while the policy briefs explain why those findings matter, who they affect, and what policy or planning considerations they raise. We specifically used each team's main findings found in the final presentation team contributions briefs, as well as visuals and other information from the final presentation and other documents in their good drive folders. (Note: We also used the other section's GC-AACS's final report).



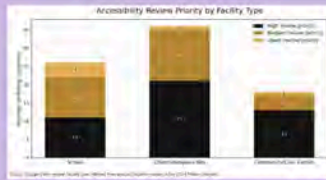


Figure 1.2: This graph shows high, medium, and lower (more-up) priority by inferred facility type, based on precinct names and visible exterior access evidence.

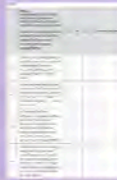


Figure 1.2: This is a portion of the DOJ's search larger rubric to assess accessibility in polling sites.

POLICY TAKEAWAYS

- **Prioritize field verification at locations with unclear accessibility.** Public imagery is useful for screening, but in-person review is needed to confirm ADA-related features.
- **Use the rubric as a triage tool.** Locations with missing or unclear parking, drop-off areas, entrances, or exterior routes should be reviewed first.
- **Focus on the full path of travel.** Accessibility should include parking, passenger drop-off areas, curb cuts, sidewalks, entrances, ramps, elevators, interior routes, and the voting area.
- **Do not describe sites as "ADA noncompliant" based only on public imagery.** Safer wording is "not verifiable from public imagery" or "requires field verification."
- **Request clearer documentation from election officials.** Updated photos, accessibility checklists, and site-specific information would help voters and advocates understand which polling places are accessible.
- **Plan temporary Election Day fixes where needed.** The DOJ notes that some barriers can be addressed with temporary measures, such as portable ramps, signage, cones, or adjusted routes.
- **Make accessibility information public and easy to understand.** Voters with disabilities should be able to know before Election Day whether a polling place has accessible parking, an accessible entrance, and a usable route to the voting area.
- **Center independence and safety.** Polling places should be evaluated based on whether voters with disabilities can vote safely, privately, and independently, with the same opportunity to participate as other voters.

Transportation Analysis for Policy Use

OVERVIEW

This brief argues that transportation access should be treated as part of voting access in Mobile County. In-person voting depends not only on where polling places are located, but also on whether residents can realistically get there. Mobile County has more ZIP codes with higher no-vehicle household rates than Baldwin County, suggesting that some residents may face greater difficulty reaching polling locations. The purpose of this brief is to identify where transportation barriers may exist and show how transportation planning can better support civic access.



Figure 1.1: This heat map shows the level of transportation barriers based on the No Vehicle Rate and Voter Turnout. Mobile County has more red ZIP codes, indicating higher no-vehicle household rates, while Baldwin County is mostly green, suggesting lower no-vehicle rates overall.

KEY FINDINGS

Transportation analysis shows that Mobile County has more ZIP codes with higher no-vehicle household rates than Baldwin County. This means more Mobile County residents may face difficulty reaching polling sites without access to a car. Specific areas of concern include parts of the city of Mobile and rural areas such as Citronelle and Wilmer. These areas may face different types of barriers. In urban areas, residents may live closer to polling places but be less likely to own a vehicle because of cost. In rural areas, residents may face longer distances, fewer transportation options, and greater dependence on private vehicles, public transportation, or rides from others. Because these barriers are not spread evenly, county-wide averages may hide ZIP codes with stronger access challenges. Drive-time analysis is helpful, but it does not fully measure access. A polling place may be a short drive away, but that does not help residents who do not own a car, cannot afford transportation, rely on limited transit, or cannot easily arrange a ride. Finally, inactive voter counts increased across Mobile and Baldwin Counties between 2020 and 2024. This does not prove transportation barriers caused voter inactivity, but it supports the need to examine transportation access alongside polling place geography, registration barriers, public transit access, and household vehicle access.

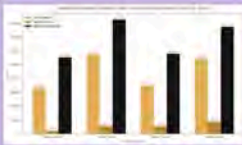


Figure 1.6: This graph shows that rural voter registration increased in both Mobile and Baldwin Counties from 2016 to 2018. However, justice sites present also increased, especially in Mobile County. Source: State of Alabama Department of Revenue, Voter Vehicle Registration Report for 2016 and 2018.



Figure 1.7: This graph compares frequent active voters, justice sites, and registered vehicles in Mobile and Baldwin Counties. It shows that Mobile County has many registered vehicles overall, but this does not mean every household has multiple access to a car. Source: State of Alabama Department of Revenue, Voter Vehicle Registration Report for 2016 and 2018.

POLICY TAKEAWAYS

- **Treat polling locations as part of the transportation network.** Voting access should be considered alongside access to other public services, such as schools, healthcare, government offices, and community centers.
- **Prioritize areas with high no-vehicle household rates.** These communities may need targeted transportation support because residents may have fewer realistic options for reaching polling places.
- **Use ZIP-code and neighborhood-level data for planning.** County-wide averages can hide local transportation gaps, so smaller geographic analysis can better identify where support is most needed.
- **Coordinate transportation planning with election planning.** Polling-place decisions should consider nearby transit routes, on-demand ride services, road connectivity, pedestrian access, and travel burden.
- **Expand the definition of access beyond drive time.** Planning should account for whether residents can actually use the trip option available to them, not just how long the trip would take by car.
- **Develop targeted mobility supports for Election Day and early voting.** This could include partnerships with local transit providers, ride-assistance programs, shuttle options, or better public information about available transportation.
- **Use clear maps and visual tools to guide decisions.** Layering polling locations with no-vehicle rates, transit routes, and rural travel distances can help identify where transportation interventions would be most useful.

Broadband Gaps and Civic Access in Alabama

OVERVIEW

This brief argues that broadband access should be treated as part of public-service access in Alabama. Internet access affects whether residents can find voting information, check polling locations, access government services, complete forms, and stay informed about civic deadlines. The findings show a clear regional divide. This matters because digital gaps may reinforce existing regional inequalities, especially in communities that already face lower income, higher poverty, and lower population density.



Figure 1.1: Speed of fixed broadband covered in urban and rural areas.

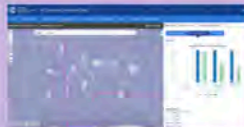
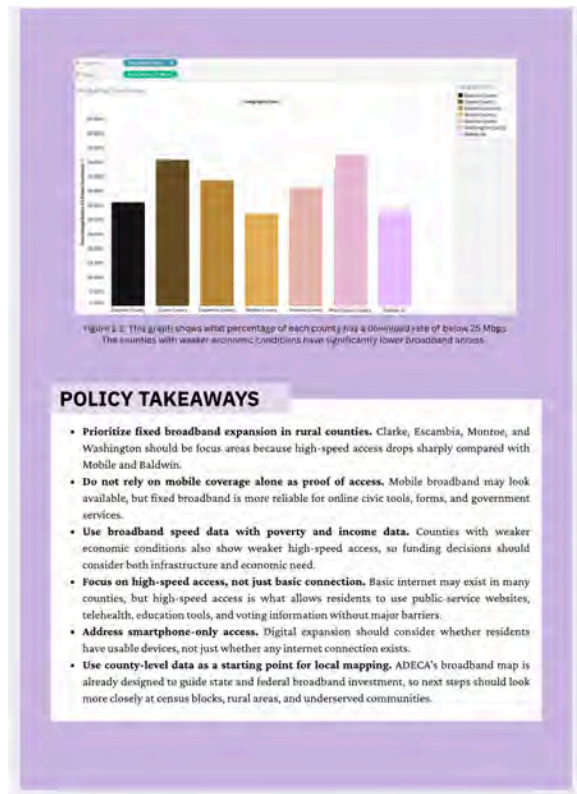


Figure 1.2: Speed of mobile broadband covered in urban and rural areas.

KEY FINDINGS

Across the counties studied, Mobile and Baldwin Counties had the strongest broadband access, especially at the highest internet speed tested. More than 50% of devices in Mobile and Baldwin had access to the highest speed level, while Clarke, Escambia, Monroe, and Washington had only about 1-5% of devices with access to that same highest speed. This shows that basic internet access may exist, but high-speed access is not evenly distributed. Fixed broadband matters most for reliable civic participation, because it supports stable home internet through cable, fiber, or DSL. Mobile broadband may appear available on maps, but in rural areas it can be slower, less reliable, or capped, making tasks like voter registration, polling location lookup, absentee ballot requests, and election information access harder to complete. The findings also suggest that broadband access is connected to economic conditions. Counties with higher poverty rates, lower median household incomes, and lower education levels tended to have weaker access to high-speed internet. Clarke, Escambia, Monroe, and Washington had some of the highest poverty rates and weaker access to the fastest speeds, while Mobile and Baldwin performed better overall. Device ownership also varied by county. Mobile and Baldwin had the highest rates of residents with at least one computing device, while rural counties had lower device ownership overall. In several rural counties, a significant share of residents relied only on smartphones, which may limit their ability to complete longer forms, access government websites, or use online civic tools comfortably.



Public-Facing Visual Translation Materials (Deliverable 3): Deliverable 3 focused on translating technical transportation and voter participation findings into simplified visual communication materials for public-facing audiences. The deliverable was designed to support policymakers, advocacy organizations, SPLC stakeholders, and community members who may not have had technical backgrounds in GIS analysis, transportation systems, or statistical interpretation.

The presentation translated visualizations created by GC-DIG and GC-TRAC into simplified explanations emphasizing voter participation trends, turnout differences across demographic groups, transportation barriers, travel distance challenges, and public transportation accessibility. Rather than focusing heavily on technical methodology, the presentation emphasized why the findings matter and how transportation and infrastructure conditions may affect civic participation across Gulf Coast communities.

Several visual communication strategies were used throughout the deliverable. Technical terminology was reduced where possible, visual layouts were simplified, and key-takeaway sections were included to help audiences quickly identify major findings. The slides also emphasized accessibility

through large headings, concise paragraphs, minimal text density, and simplified explanations alongside charts and maps.

The presentation included multiple translated visualizations, including voter turnout comparisons, turnout change analysis, transportation barrier heat maps, polling-access distance visualizations, and public transportation infrastructure maps. These visuals were adapted to prioritize audience understanding rather than technical detail.

This deliverable supported the broader goals of the SPLC project by helping transform technical transportation and voter-access findings into communication materials more appropriate for stakeholder outreach, policy discussions, and public education efforts.

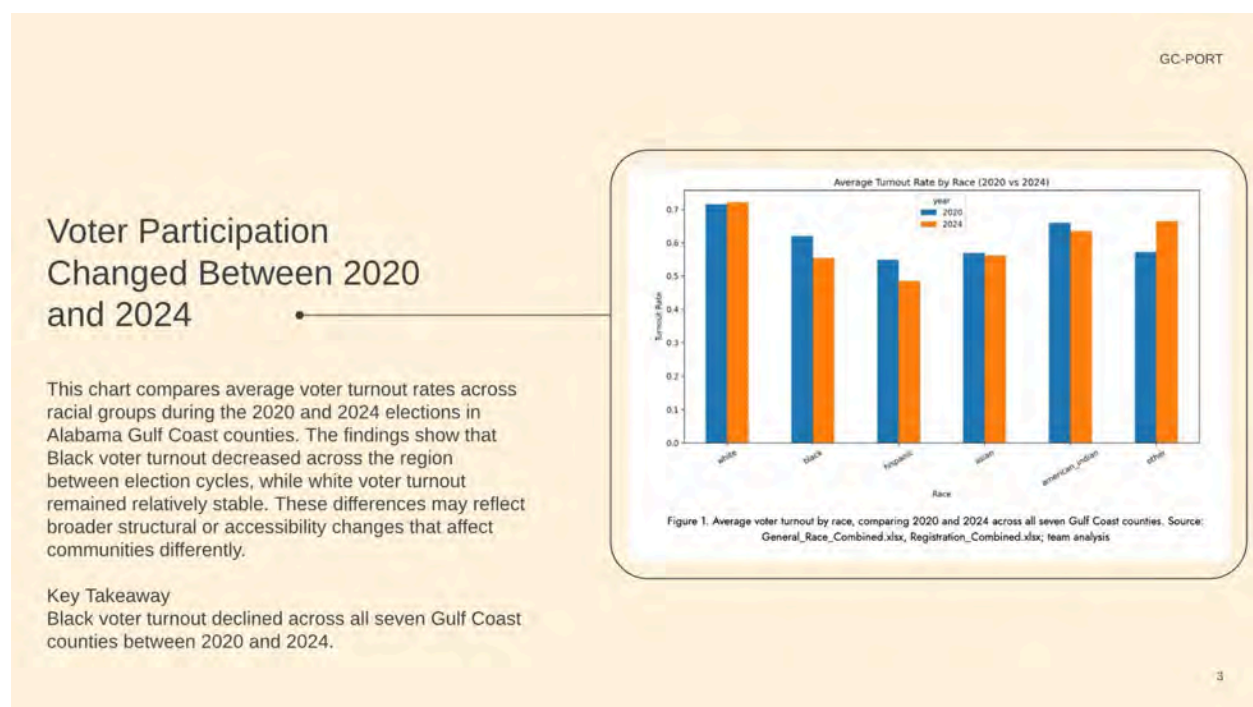


Figure 1. Public-facing adaptation of GC-DIG voter turnout findings comparing average turnout rates across racial groups during the 2020 and 2024 election cycles in Gulf Coast counties. The visualization was simplified to emphasize major participation trends and improve accessibility for non-technical audiences.

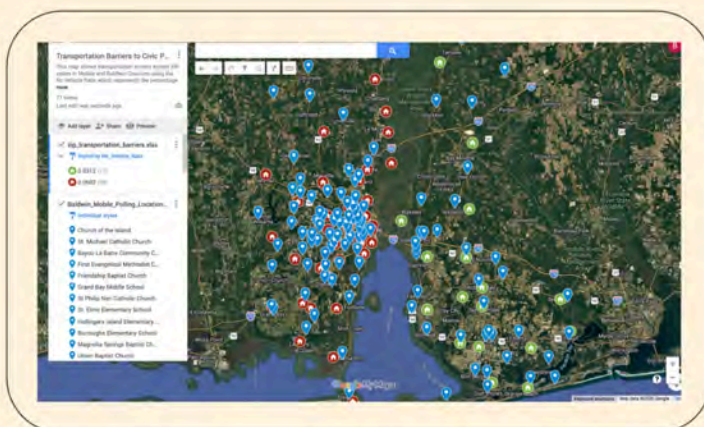


Figure 2: Estimated travel distance between ZIP code 36606 and the Three Circle Church Midtown polling location in Mobile County. The map illustrates how nearby polling locations may still present transportation challenges for residents without personal vehicles or safe walking routes (Source: Google Maps, U.S Census Bureau, and Capitol Impact Government Gateway)

Transportation Access May Affect Polling Accessibility

This heat map visualizes transportation-related barriers across Gulf Coast communities and their relationship to polling location accessibility. Areas with lower transportation access or longer travel distances may face additional challenges when residents attempt to reach polling locations during elections.

Key Takeaway

Transportation access is not evenly distributed across Gulf Coast communities.

Figure 2. Simplified transportation-access visualization adapted from GC-TRAC findings to demonstrate how transportation barriers and travel distance may affect polling location accessibility across Gulf Coast communities. The translated visual emphasizes stakeholder understanding rather than technical analysis.

Public Transportation and Voting Access Are Connected

This map combines road networks, public transportation routes, bus stops, and polling locations across Mobile County. The visualization demonstrates how transportation infrastructure may influence residents' ability to access polling locations, especially for individuals who rely on public transit instead of personal vehicles. Yellow represents the bus stops and routes while green represents the polling station.

Key Takeaway

Transportation infrastructure may influence how easily residents can reach polling locations.

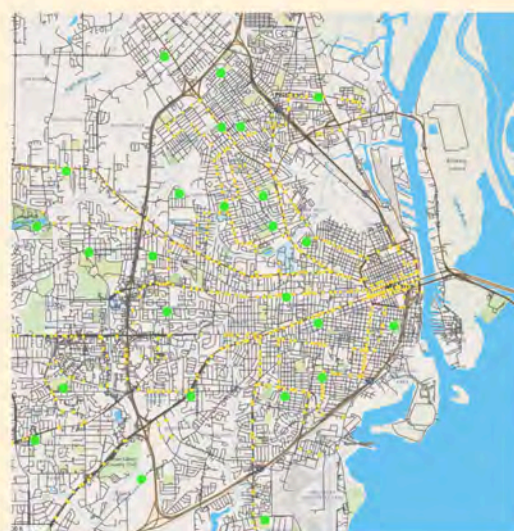


Figure 5. Mobile County transportation system map showing roads, transit routes, bus stops, and polling locations. Source: GC-TRAC.

Figure 3. Public-facing transportation infrastructure map adapted from GC-TRAC analysis showing the relationship between road networks, bus routes, transit stops, and polling locations in Mobile County. The visual was simplified to support stakeholder communication regarding transportation accessibility and civic participation.

These translated visuals demonstrate how technical transportation and voter participation findings can be adapted into communication materials more accessible to policymakers, advocacy organizations, and community stakeholders. Simplifying visual hierarchy, reducing technical language, and emphasizing key takeaways helped improve readability and audience understanding throughout the deliverable.

Recommendations

Recommendations for the client (SPLC)

SPLC should continue prioritizing communication materials that translate technical transportation and voting-access findings into formats accessible to non-technical audiences. Many GIS maps, transportation analyses, and statistical findings contain important policy implications, but the usefulness of those findings depends heavily on whether stakeholders can quickly interpret and apply the information.

Public-facing communication materials such as simplified visual summaries, stakeholder presentations, and policy-focused explanations may help improve outreach efforts and support future discussions surrounding transportation accessibility, polling access, and civic participation barriers in Gulf Coast communities.

SPLC may also benefit from continuing to develop community-focused outreach materials that simplify transportation and voting-access findings for broader public audiences. While technical analyses and GIS visualizations provide valuable insight into polling accessibility and civic participation barriers, many stakeholders may not have the background necessary to interpret detailed transportation or demographic data independently. Expanding the use of simplified visual summaries, stakeholder presentations, and public-facing communication materials could help improve awareness of voting-access challenges across Gulf Coast communities.

Recommendations for future project teams

Future project teams should establish earlier coordination between technical analysis teams and communication-focused teams. Because many communication deliverables depended on finalized maps and transportation findings from partner groups, revisions and updates occasionally delayed the development of public-facing materials.

Future teams may also benefit from designing technical visualizations with public audiences in mind earlier in the project process. Simplified legends, clearer labels, and reduced visual clutter may improve the usability of maps and charts for stakeholder communication and outreach purposes.

Future teams should also prioritize maintaining organized repositories of finalized visuals, datasets, and presentation materials throughout the semester. Because communication-focused deliverables often depend on completed analysis and visualizations from partner groups, delayed access to finalized materials can create challenges for teams responsible for stakeholder communication and public-facing outreach products. Earlier organization and standardization of deliverable files could improve collaboration efficiency across teams.

Conclusions

Our project translated technical findings from other teams into plain-language, stakeholder-focused materials for policymakers, public officials, advocacy leaders, and civic access stakeholders. The main goal was to make complex data easier to understand and use and turn research findings into accessible, decision-focused information for non-specialist audiences. We created an overview that explains the most effective ways to communicate with different policy audiences, along with a series of policy briefs focused on transportation access, ADA accessibility, and broadband access. Each brief was designed with a specific policymaker or official in mind, but the materials can also be used by other officials working in similar areas. The briefs highlight key voting barriers, explain relevant findings in simpler terms, and provide policy considerations that may help support increased voter participation. We also created a simple policy brief template that can be adapted for other officials, issue areas, or future findings. In addition to the briefs, we explained complex visuals in simpler words so that readers can quickly understand what each chart or map is showing. While the technical teams provided the data and analysis, our work turned those findings into actionable summaries and policy suggestions for the people who are positioned to use them in planning, advocacy, and decision-making.

Deliverables Checklist

Deliverable Folder:

<https://drive.google.com/drive/folders/1uqR4BsoceeTj0n4ere35DWgdvQg3Y3gm>

- ☐ Deliverable 1 - Stakeholder Overview: [Stakeholder Overview](#)
- ☐ Deliverable 2 - Policy Briefs: [Policy Brief #1](#), [Policy Brief #2](#), [Policy Brief #3](#)
- ☐ Deliverable 3 - Public-Facing Visual Translation: [Public-Facing Visual Translation](#)
- ☐ Deliverable 4 - Maps and Visual Graphics: [Maps and Visual Graphics](#)

Civic Participation and Access Design

GC-CPAD

Janet Dabu (Project Manager), Mei Pierson (Notetaker), Ismail Touray (Communicator),

Daniel Chavez Celis (Reviewer)

Southern Policy Law Center (SPLC)

INST490 - Integrated Capstone for Information Science

Section 0105

May 18th, 2026

Team Brief

Team 11: GC-CPAD

Key Findings

- Finding 1: Throughout this process, we realized that the target audience reads at a sixth grade level. This realization made it crucial to make sure our visuals and writing was clear and simplistic in manner in order to be appealing and easily digestible for the audience.
 - Finding 2: The Supreme Court's ruling on Shelby County v. Holder allowed states to reintroduce laws that previously would have been stopped by federal authorities such as discriminatory redistricting policies and ID requirements.
 - Finding 3: The governor of Alabama and the Secretary of the Alabama Law Enforcement Agency closed 31 drivers-license-issuing offices where eight of the eleven offices were in majority Black counties, restricting ways to obtain IDs, ahead of the 2016 election.
 - Finding 4: Due to the socioeconomic barriers, voting is not just a simple show up and vote process for many in Alabama especially for Black, Latinx, and other low income voters. As a result of their socioeconomic status, these voters face issues related to transportation, access to election information, and to inflexible working hours.
 - Finding 5: Absentee voting is a difficult process in Alabama and a privilege given to voters facing specific situations such as working a shift of 10 hours or more hours, having an illness, or being out of county.
-

Deliverables Produced

- Journeymap/Flowcharts: Visual process maps that showcase steps involved in voting and barriers faced by voters in the civic process:  Journeymap/Flowcharts
 - One-page Guides: Concise visual guides designed to educate voters with quickly understanding the voting process ranging from where to how :  1-Page Guides
 - Graphics Justification Document: A supporting documentation highlighting our thought process and reasons behind the design choices relevant to our visual guides and materials:  GC-CPAD: Graphics Justification Documentation
 - Audience Summary: A research paper on the target audience which includes the demographic, socioeconomic, and other factors that voter with being able to effectively participate in the civic process :  GC-CPAD: Audience Summary
-

Challenges

One of the challenges we faced was condensing the deep historic contexts in relation to civic barriers within our visuals which were limited to only one page. To address this situation, we divided the visuals into different types of guides covering the areas voters will need. Another obstacle was our reliance on secondary sources instead of getting direct feedback from members of the community which made our research less specific to the Gulf Coast.

Connections to Other Teams

Team 3 - GC-PAD: We utilized the map showcasing the polling locations provided by Team 3 which helped to narrow down our scope to Mobile, Alabama. This information allowed us to know how redistricting changes the polling location landscape depending on the voting period.

Final Report and Deliverables

Project Context

Within the broader regional analysis for SPLC, we placed our efforts on identifying what the barriers are in the civic process, audience needs, and what communication strategies will be the most relevant in educating the voters of Alabama. As far as dependencies, we relied on Team 3: GC-PAD for information on polling locations which allowed us to narrow our focus to just Mobile instead of covering the entire state of Alabama. This was in hopes to avoid dragging out the scope of the project while offering us vague insight as to what voters are really interested in learning about. We were able to use our findings to create guides that will be relevant to the needs of Black, Latinx, and socioeconomically challenged voters.

Methods

Research Approach

To satisfy the project requirements, our research team took a qualitative, audience-based approach to studying issues related to voting experience, voter information, barriers, and educational needs among Black and Latinx communities in Mobile, AL. While primary field research was not conducted, the team utilized secondary sources from historical documentations, relevant academic literature, and government sources.

The task entailed three interrelated aspects. Firstly, we drafted a summary of audience research in terms of race, ethnicity, socioeconomic factors, and civic process barriers to voting in Gulf Coast counties. Secondly, the team identified the steps and civic process voters needed to go through to understand where and how to vote, specifically in the context of precinct-based voting in Mobile County. Finally, we created voter education material according to these findings in the form of a "Where to Vote," "Absentee Voting," "How to Vote," and "How to Check Your Vote Was Counted" visual guides.

Data Sources

The following key sources were used throughout the analysis of audience needs and barriers:

- Governmental and state election information, such as Alabama voter information tools (VoterView) that allow individuals to check their registration status and learn their polling place.
- Mobile County and precinct-based voting resources, such as information related to polling places and voting geography.
- Historical documentation and accounts that describe civil rights issues in Mobile County or across Alabama as a whole.
- Information about civic process barriers is provided by SPLC and other organizations that deal with voting rights.
- Scholarly literature on voter access and socioeconomic differences in political participation (especially regarding transportation access, time constraints, work schedule, income, language access, and unfamiliarity with the voting process).

The above resources were used throughout the research phase and content creation phase. Our Audience research mostly relied on scholarly and historical sources, while voter guides drew mainly from government and election-related documents to provide voters with the actual instructions for voting.

Analytical Methods

In order to conduct audience research, we approached voter experience and barriers through the following aspects: black voters, Latinx voters, socioeconomic conditions, and civic process barriers to voting. Each of the four aspects provided information about the barriers voters faced, especially in Mobile, and the audience needs that can be met by means of voter education. For example, the socioeconomic analysis of the SPLC-identified communities highlighted transportation access issues, inflexibility of the voters' work schedule, lack of information on the voter education process, and other barriers to political participation. Likewise, the Latinx population was examined in terms of language barriers, voter process barriers, and other civic issues.

Process Mapping

Based on the information gained through the audience analysis, we identified all the actions required to be taken before and during the voting process in Mobile County. Specifically, we looked at how to check your voter registration status, which polling place is assigned to your home precinct, how to prepare the necessary identification, and which transportation options you may need. All this information made the core of the "Where to Vote," "Absentee Voting," "How to Vote," and "How to Check Your Vote Was Counted" voting guides.

One of the major civic barriers highlighted through the research is voter confusion, especially regarding their polling place. Since Mobile voters need to know where to go on the day of voting, process mapping was aimed at creating a simple step-by-step procedure for locating their voting site.

Content Translation and Voter Education Strategy Development

Having identified the main needs of the SPLC-identified communities and civic barriers to voter experience, we started developing educational material that can meet the audience's needs and eliminate some of the identified barriers. Thus, the team created content that provides voter guidance in plain language, short and clear steps, and visual instructions.

As a result, voter education materials are expected to help voters navigate through the information, learn how to find their polling place, what identification to bring to the polling place, and where to find additional voting information.

Visual and Communication Design

To develop voter education materials that can meet the needs of the SPLC-identified communities, we need to think about how they will communicate and interact with our materials. As highlighted in the audience research summary, Facebook and Instagram should be considered viable communication channels for reaching out to the SPLC-identified communities. Therefore, we designed visual materials that would fit Facebook and Instagram formats.

For that reason, we applied SPLC style guidelines for designing social media graphics and content. Specifically, our materials are based on the following color palette of dark gold, light gold, and black. Continuing with the SPLC style sheet and making sure our graphics are standardized by following Web Content Accessibility Guidelines (WCAG) which include proper color contrast, visual hierarchy, and a focus on images.

Tools Used

Throughout the project development and implementation process, the following tools were applied: online government and county election resources to verify civic process steps, registration verification, polling place lookup, and other information about precinct-based voting; shared research document that facilitated audience research; team collaboration and project management tools to draft research summary, design voter flowcharts, collaborate, and compile all research findings into one set of educational materials.

Challenges and Solutions

One of the main challenges of creating voter education materials was finding ways to address the civic barriers in a concise manner. Specifically, it may be hard to explain all the barriers and historical contexts surrounding the issue in only one page of voter instruction. To address this challenge, our team chose to create different sets of materials. Thus, audience research provided a comprehensive overview of voter issues, while voter guides and flowcharts offered more immediate solutions.

Another challenge was associated with a specific civic process in the target region. As explained previously, voters must first identify and verify their polling place, since they must stick to polling places in their assigned precincts. In other words, voters are limited in voting locations based on their address. This issue is addressed by instructing the readers on the steps required to identify polling places in advance.

It is also worth mentioning the challenge of creating evidence-based recommendations that do not become too generalized. In this regard, our team ensured that each communication strategy recommended in the report was based on a certain barrier found in the evidence. For example, recommendations concerning visual materials, social media communication, and voter guide flow charts were based on information about barriers to the voting process. Additionally, the focus on the Latinx population and Black voters was identified by SPLC. In other words, we avoided making general statements by following the research and communities identified by the client.

Finally, we were limited to using only secondary sources in our study. In fact, the project findings reflect documented patterns and issues that affect voter experience, not individual cases and opinions. Nonetheless, the team tried to highlight major concerns and gaps that can be addressed through voter education.

Deliverables and Findings

Audience Summary

A comprehensive summary of research we conducted on SPLC-identified communities in order to create purpose driven social media posts. The purpose of the audience summary is to understand our audience, those of the Black and Latinx community of the gulf coast region of Alabama. We must first understand who our audience is and by conducting research we have a well informed idea of

who we are creating informational materials for. That research is summarized into an easily accessible document in APA format with all our sources provided.

Link in Drive: [GC-CPAD: Audience Summary](#)

Journey Map and Flowcharts

The following four deliverables are resources to inform SPLC of the current civic participation processes. The deliverables are told from the perspective of the average citizen looking into voting. It will not only allow us to easily create simplified informational materials, but also find areas where this system needs to be improved upon.

The following Canva link is the client copy of the “GC-CPAD Journey Map and Flowchart” project page to allow further editing if SPLC chooses to do so: [GC-CPAD Journey Map and Flowcharts](#)

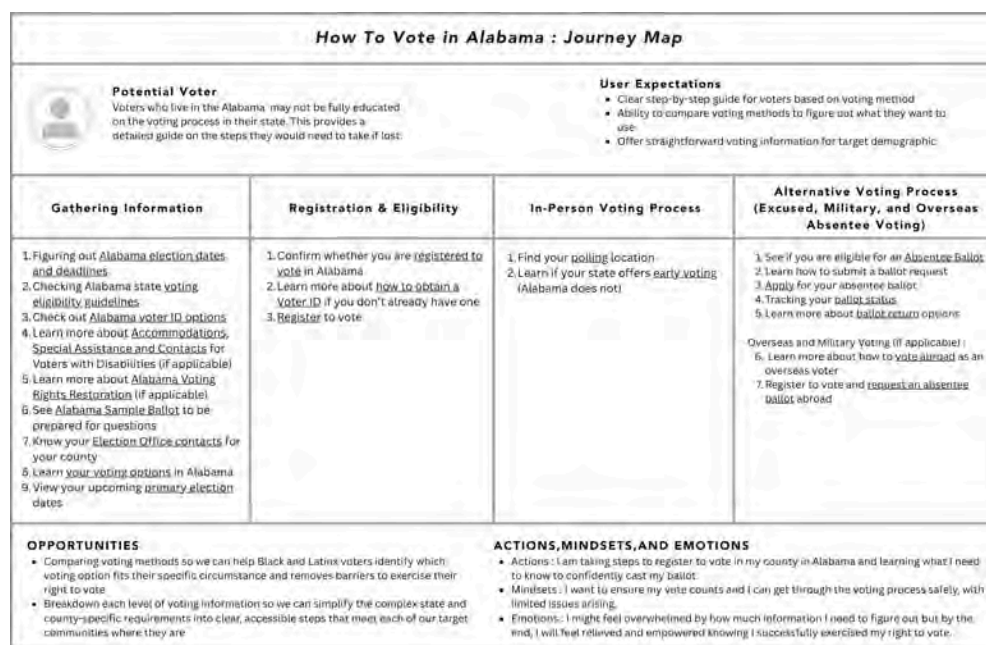


Figure 1: “How To Vote in Alabama” Journey Map

The journey map conveys the experience of the average voter in Alabama. Links to relevant resources for each part of the voting process are also included. A corresponding interactive PDF is provided below.

Link in Drive: [GC-CPAD How To Vote in Alabama - Journey Map.pdf](#)

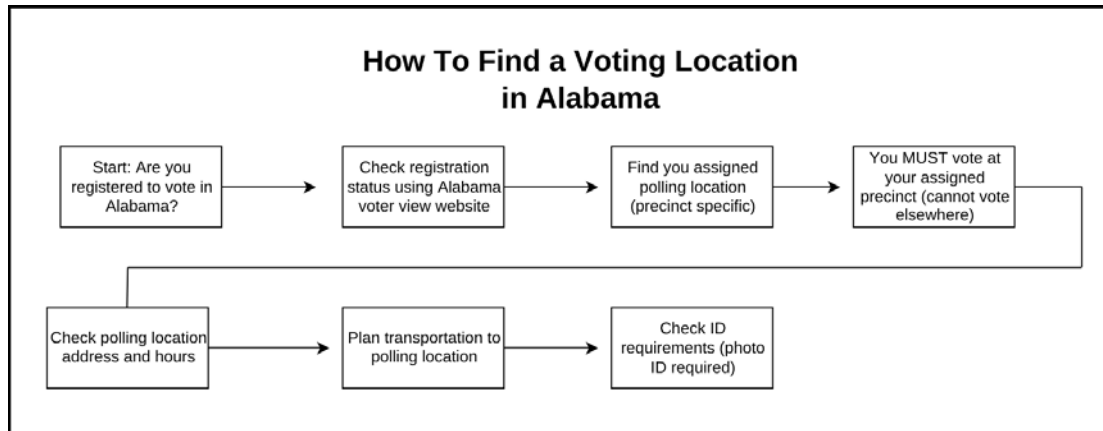


Figure 2: “How To Find a Voting Location in Alabama” Flowchart

A flowchart that shows the steps of locating a polling location where an Alabama voter may be able to vote at.

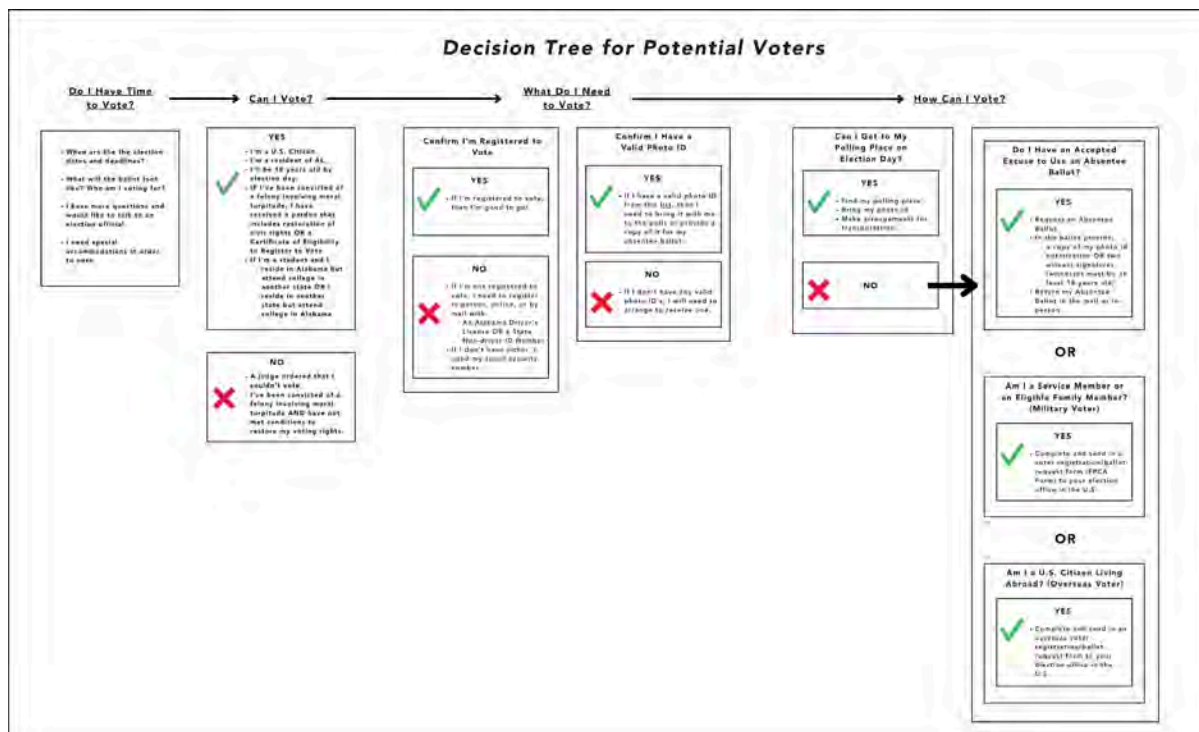


Figure 3 “Decision Tree for Potential Voters” Flowchart

A flowchart that shows the voter's experience based on the decisions they have made throughout the process. This visual provides different scenarios that a potential voter may partake in. A corresponding PDF is provided below.

Link in Drive: [PDF GC-CPAD Decision Tree for Potential Voters.pdf](#)

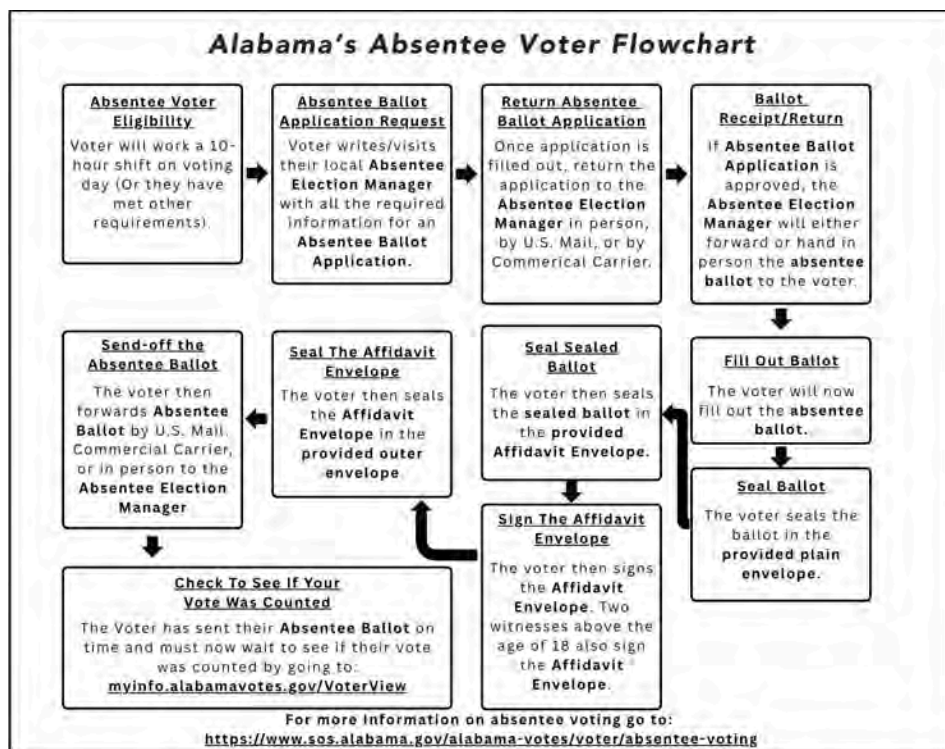


Figure 4 "Alabama's Absentee Voter" Flowchart

This flowchart shows the necessary steps in order to participate in absentee voting. The flowchart also shows how complicated the absentee voting process is. A corresponding PDF is provided below.

Link in Drive: [PDF GC-CPAD Alabama's Absentee Voter Flowchart.pdf](#)

One-Page Guides

The following four deliverables are one-page graphics that are intended for posting on SPLC social media. The purpose of these deliverables is to provide a simplified way of informing the citizens within the Black and Latinx communities of how they may be able to engage in civic participation. Posting on social media will ensure that the information stored in these deliverables will be seen by

our intended audience thus providing more access across both communities and increasing civic participation. The final version of the one-page guides were done on photoshop to meet the font requests provided by SPLC, but they were originally created on Canva.

The following link is the “GC-CPAD One Page Guides” project page to allow further editing if SPLC chooses to do so: [GC-CPAD One Page Guides](#)

Our project used images provided by Canva and the following link goes to Canva’s licensing page for an explanation on using their images for commercial use: [Canva’s Licensing Explained](#)

SP
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WHERE TO VOTE

A Guide for Voters in ALABAMA

Find your assigned polling place and make a plan to vote.

STEP-BY-STEP: FIND YOUR POLLING PLACE

- 1**
CHECK YOUR REGISTRATION
Go to myinfo.alabamavotes.gov/voterview.
- 2**
FIND YOUR POLLING STATION
Enter your name and county to find your assigned polling location.
- 3**
CONFIRM YOUR PRECINCT
You must vote at THIS location only. You cannot cross precinct lines.
- 4**
CHECK YOUR BALLOT
View your sample ballot to see who and what will be on your ballot.
- 5**
MAKE A PLAN
Know how you will get there and when to go. Polls are open 7AM-7PM.

WHAT TO BRING

- VALID PHOTO ID**
Driver's license, state ID, or passport.
- YOUR NAME AND ADDRESS**
Make sure your information is up to date.
- ANY NEEDED DOCUMENTS**
If you recently moved, be ready to show proof of your new address if required.

POSSIBLE BARRIERS PLAN AHEAD

- TRANSPORTATION**
Some polling places are far away. Plan your ride early and allow extra time.
- LONG LINES**
Lines can be long, especially during busy times. Go early or later in the day if possible.
- CONFUSION**
If you are unsure about anything, ask a poll worker. They are there to help.

Figure 5: “Where to Vote” 1-Page Guide

The first one-page guide and it depicts the steps an Alabama voter must undergo to locate a suitable polling location. The guide also shows what to bring and potential barriers the voter might encounter.

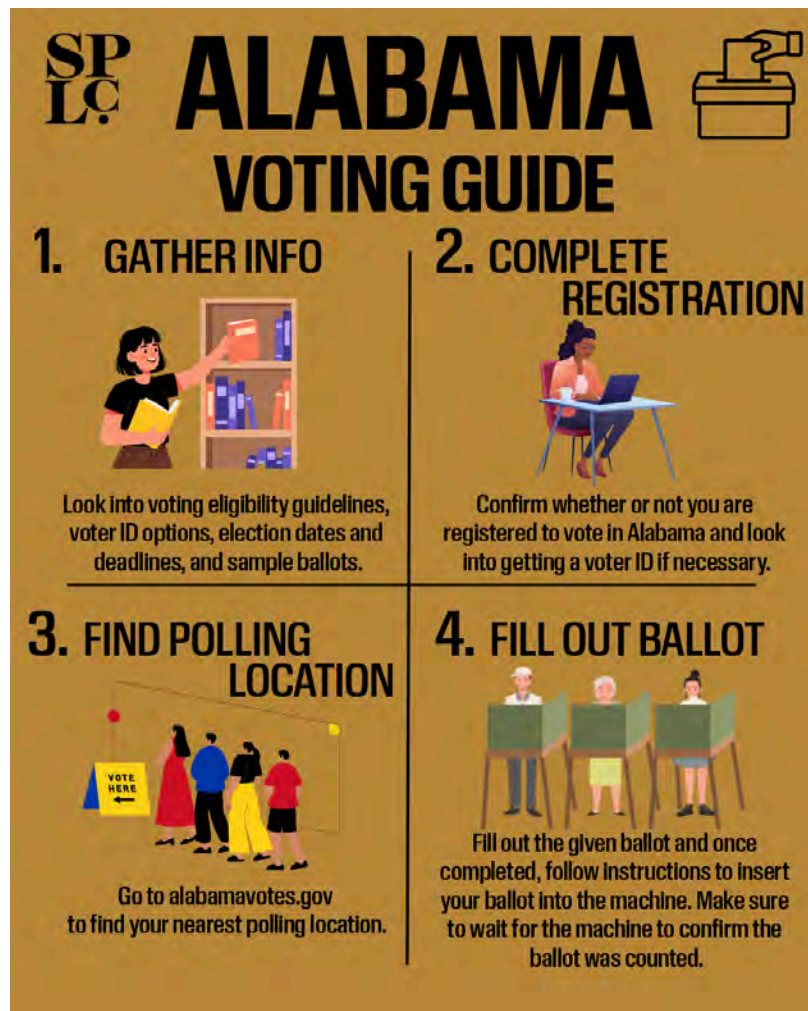


Figure 6: “Voting Guide” 1-Page Guide

The second one-page guide and it depicts the steps an Alabama voter must follow to properly be able to vote in an election. Guide provides eye popping graphics with relatable depictions of SPLC-identified communities.

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ALABAMA

How to Check Your Vote Was Counted

1. Go to myinfo.alabamavotes.gov/VoterView



2. Enter your First Name, Last Name, and Date of Birth

First Name

First Name

Last Name

Last Name

Date of Birth

Year

Month

Day

3. Click the box next to "I'm not a robot" and follow the instructions

☐ I'm not a robot



4. Click the "Look Up" button

Look Up

Figure 7: "How to Check Your Vote Was Counted" 1-Page Guide

The third one-page guide and it depicts the steps an Alabama voter must follow to see if their votes were counted. The guide shows screenshots from the Alabama VoterView website to familiarize Alabama voters when checking on their ballot's status.

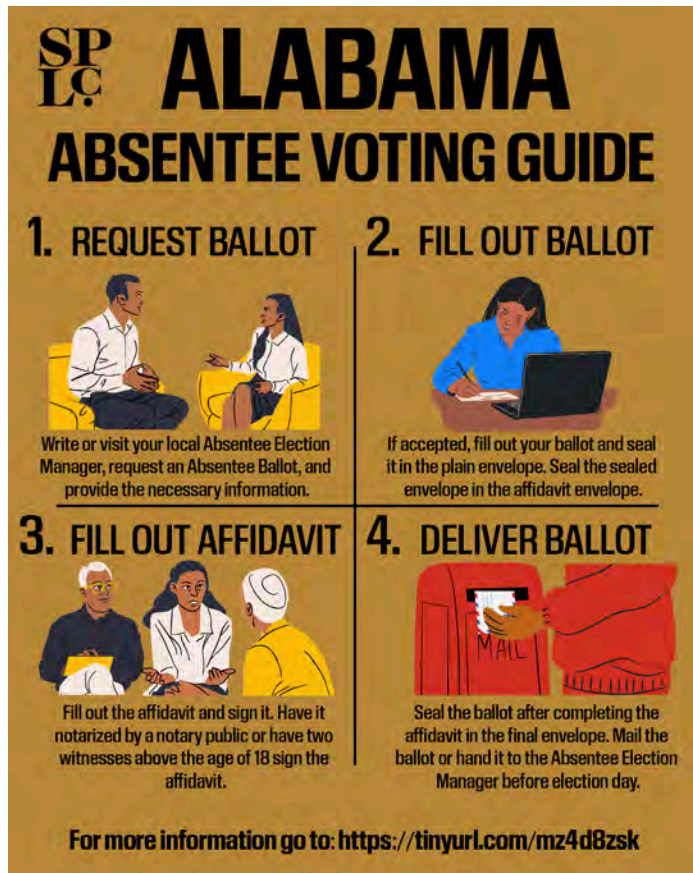


Figure 8: "Absentee Voting Guide" 1-Page Guide

The fourth one-page guide and it depicts the steps an Alabama voter must follow if they would want to vote through Absentee Voting. The guide provides a hyperlink to the Alabama secretary of state's page on Absentee Voting.

Graphics Justification Document

As we began creating our deliverables, we concluded that we needed to start at the basics when it came to the steps voters needed to go through to understand where and how to vote. We split up the voting process into four key one-page guides : "Where to Vote," "Absentee Voting," "How to Vote," and "How to Check Your Vote Was Counted." To summarize our decision-making process while creating these one-page guides, we created our graphics justification document to explain our rationale for designing each visual. This includes information on each guide such as why we chose that topic and each design decision we made. This was in hopes of emphasizing the importance of each guide topic and it would help our audience succeed in their voting journey.

Link to Drive: [GC-CPAD: Graphics Justification Documentation](#)

Recommendations

Recommendations for the client (SPLC)

While doing research for our deliverables, we found that information was fairly easy to come across. However, we noticed a lack of support on the state level for Spanish speakers to understand the civic process. This may be something SPLC could pursue in the future. While looking at census numbers for the Latinx population, there was no way to tell if those in the census were citizens or not. Regardless, it brought up the topic of undocumented immigrants. While they cannot vote, if SPLC wants to expand the rights of those who live in our country, then looking at ways to push for legislation to open pathways to citizenship could be another direction to take.

Recommendations for future project teams

Future teams should be aware that many of the stories and pertinent information is wrapped up in legal cases. Reading reports and analyses helped boil down information in a way that is easy to understand. However, these reports would bring up legal cases to support the conversation. If we were to do this project again, starting on the policy side would have let the story of communities come out more easily. We would recommend future teams construct a timeline on major court cases and smaller bills to construct the legal environment surrounding the changes around civic participation.

A natural continuation of our work could include several expansions on our current research. For example, for our 1-page guides, we chose four topics based around how to vote, where to vote, how to verify your vote, and absentee voting. Any of these four topics could be expanded. To expand on how to vote, we would recommend creating guides or supporting materials for people who were charged with a felony to walk them through the process of determining whether they are not eligible to obtain their voting rights or they can secure their right to vote through a pardon or certificate. Another guide could show all the accepted photo ids for the polls and absentee voting as well as explaining how to obtain an id if they do not already have one. For where to vote, we would recommend making specific guides for each county or district in the Gulf Coast area. These guides could specify which locations have accessible ramps and where they are, transportation options, and other quirks specific to different districts. For how to verify your vote, access to a registered Alabama voter would be helpful to show further navigation once the voter gets past the login screen. For

absentee voting, another guide could be created showing which accepted excuses are allowed to register for absentee voting or how to secure an emergency absentee ballot.

Conclusions

Following the completion of our deliverables and this project, we can confidently conclude that voting is a right that we have to continue to fight for. It is a right that is taken for granted by many and exercising it protects each individual's right to a say in their government and who represents them. Reflecting on our time looking at civic participation and access design, we focused on the barriers individuals in the Black and Latinx community face when they go to vote. We completed extensive research on current events, our target communities and the specific barriers they face, and possible solutions. We completed an audience summary that offered insight into race, ethnicity, socioeconomic factors, and civic process barriers, which helped us further understand what shortcomings we needed to address. By breaking down the steps of voting into four simple one-page guides, we hope to have increased voter education and remove some barriers they face. Overall, we are confident that we have set a strong foundation for SPLC to build on to continue fighting for minority voting rights.

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Deliverables Checklist

- Journeymap/Flowcharts :  Journeymap/Flowcharts
- One-page Guides :  1-Page Guides
- Graphics Justification Document :
 -  GC-CPAD: Graphics Justification Documentation
- Audience Summary :  GC-CPAD: Audience Summary